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(54) **METHODS FOR CASCADE
AMPLIFICATIONS OF THERAPEUTIC
PAYLOADS (CATP) & COMPOSITIONS FOR
CANCER IMMUNOTHERAPIES AND GENE
THERAPY**

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(57)

ABSTRACT

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8, 2024, provisional application No. 63/738,692, filed
on Dec. 24, 2024.

Publication Classification

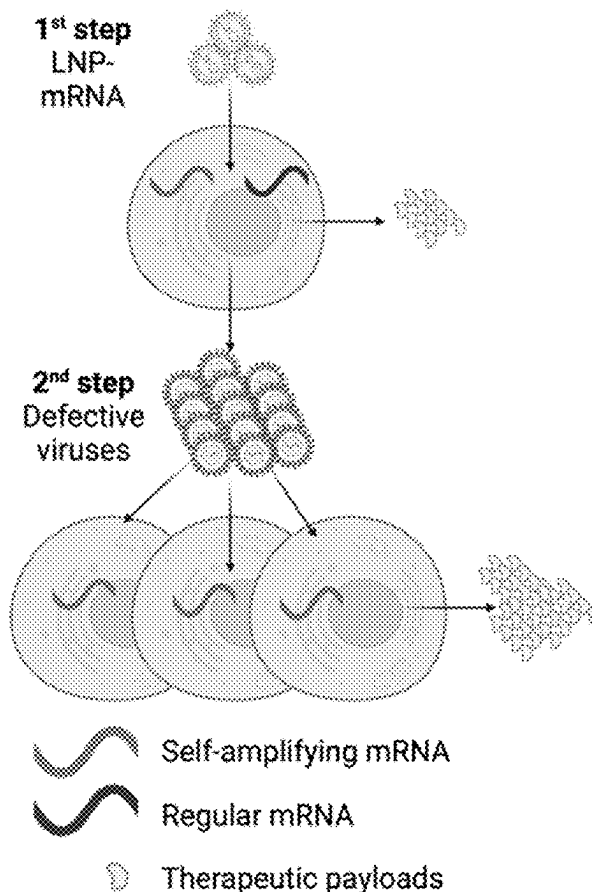
(51) **Int. Cl.**

A61K 35/768 (2015.01)

A61K 38/16 (2006.01)

The invention relates to compositions and methods for the preparation, manufacture and therapeutic use of oncolytic defective virus compositions and methods of in vivo synthesis thereof. The composition includes a first nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) encoding at least one gene of interest (GOI) or a plurality of GOIs, a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein, and at least one payload delivery system.

Specification includes a Sequence Listing.



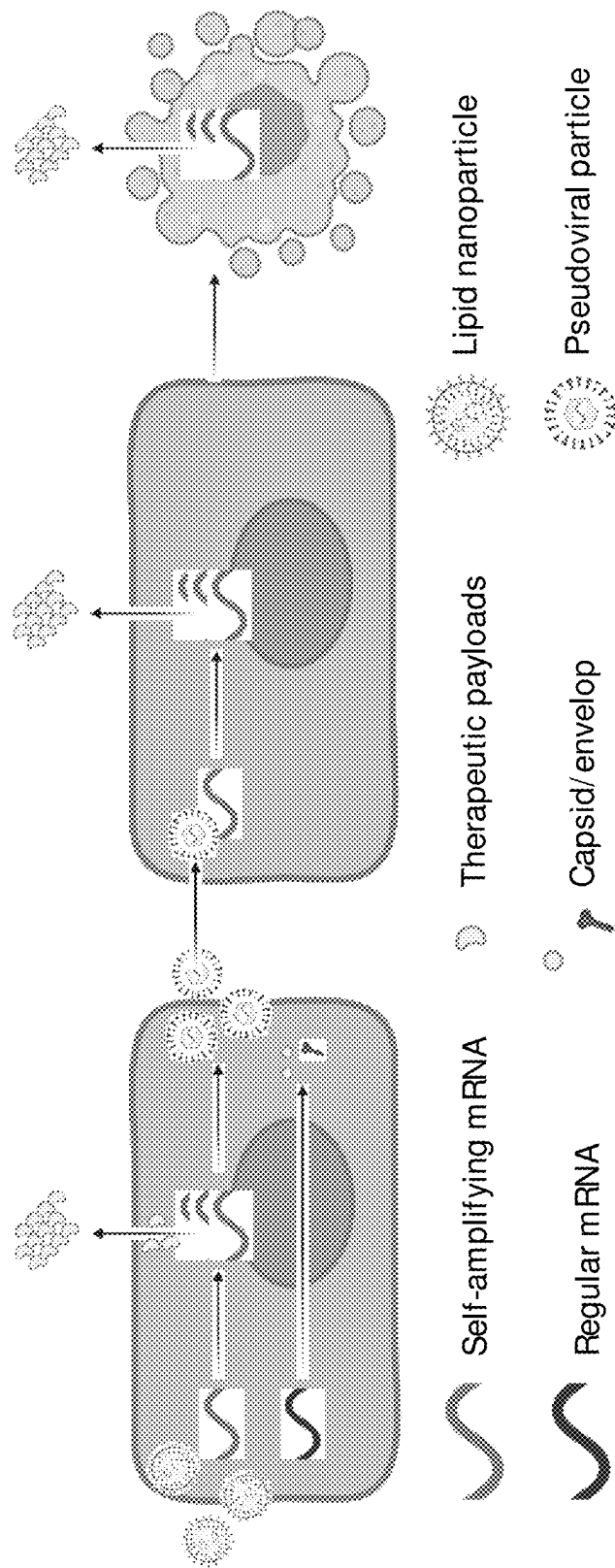


Fig. 1A

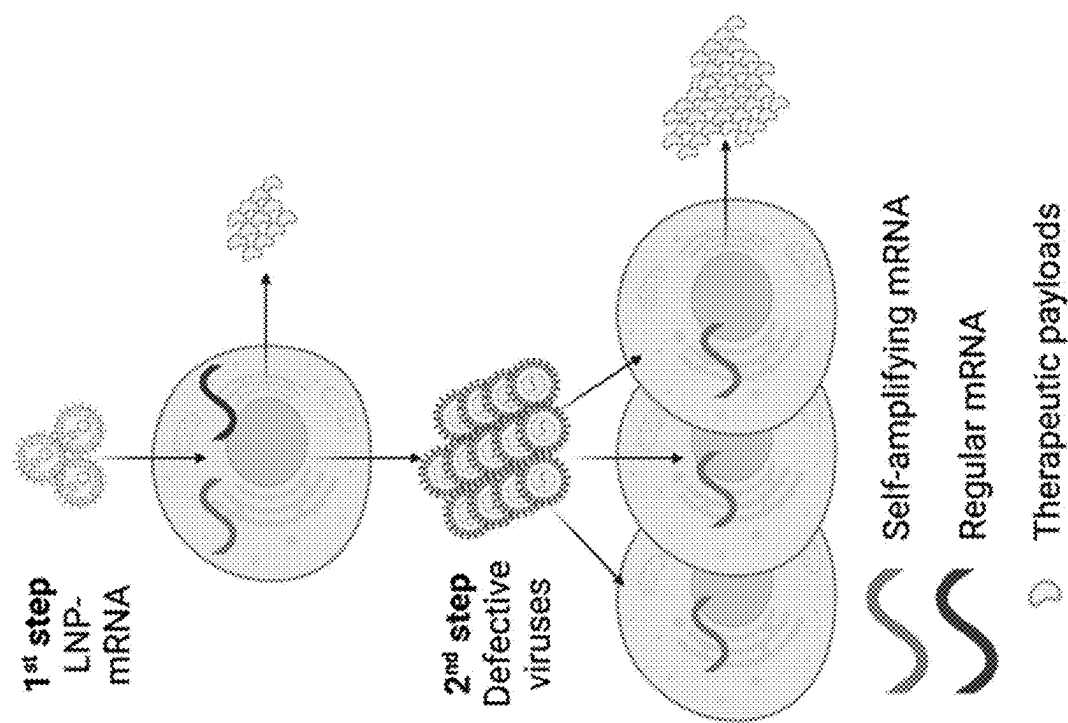


Fig. 1B

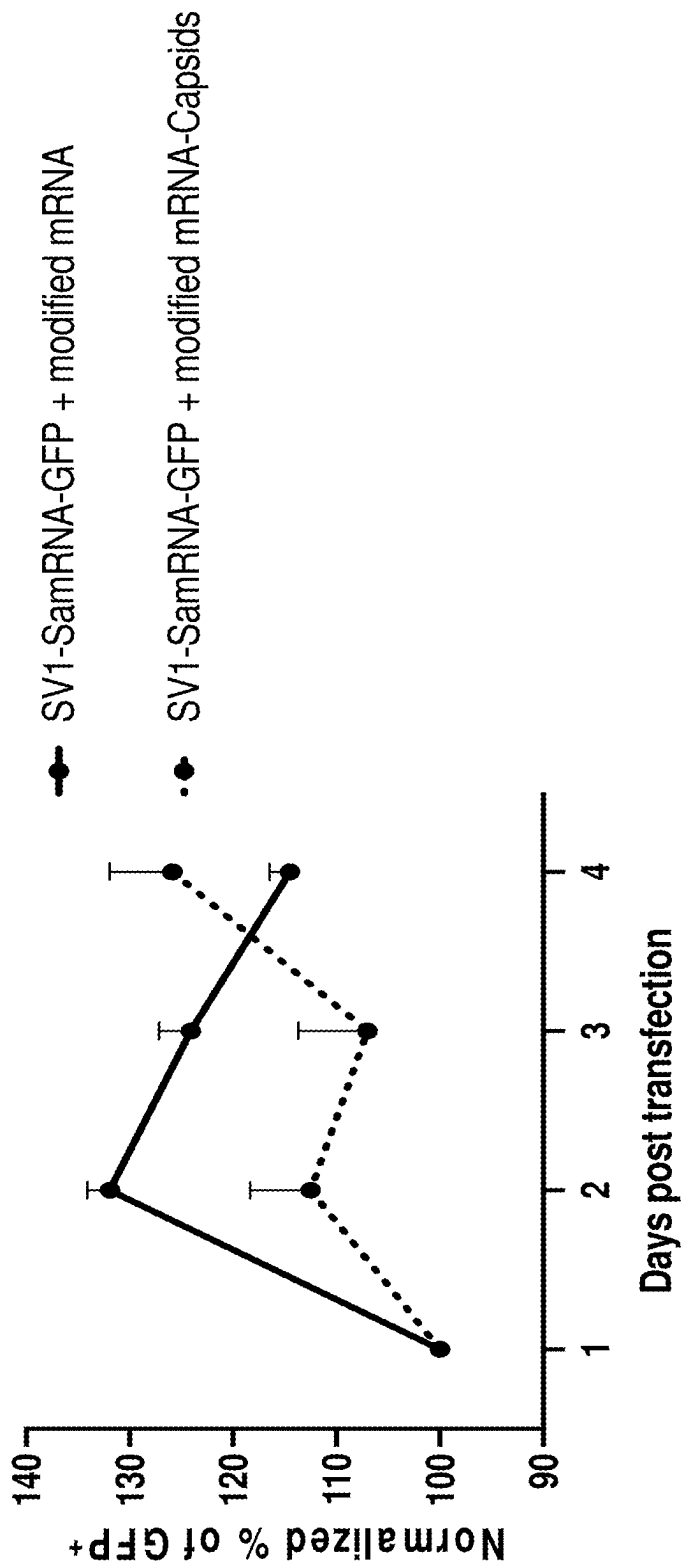


Fig. 2A

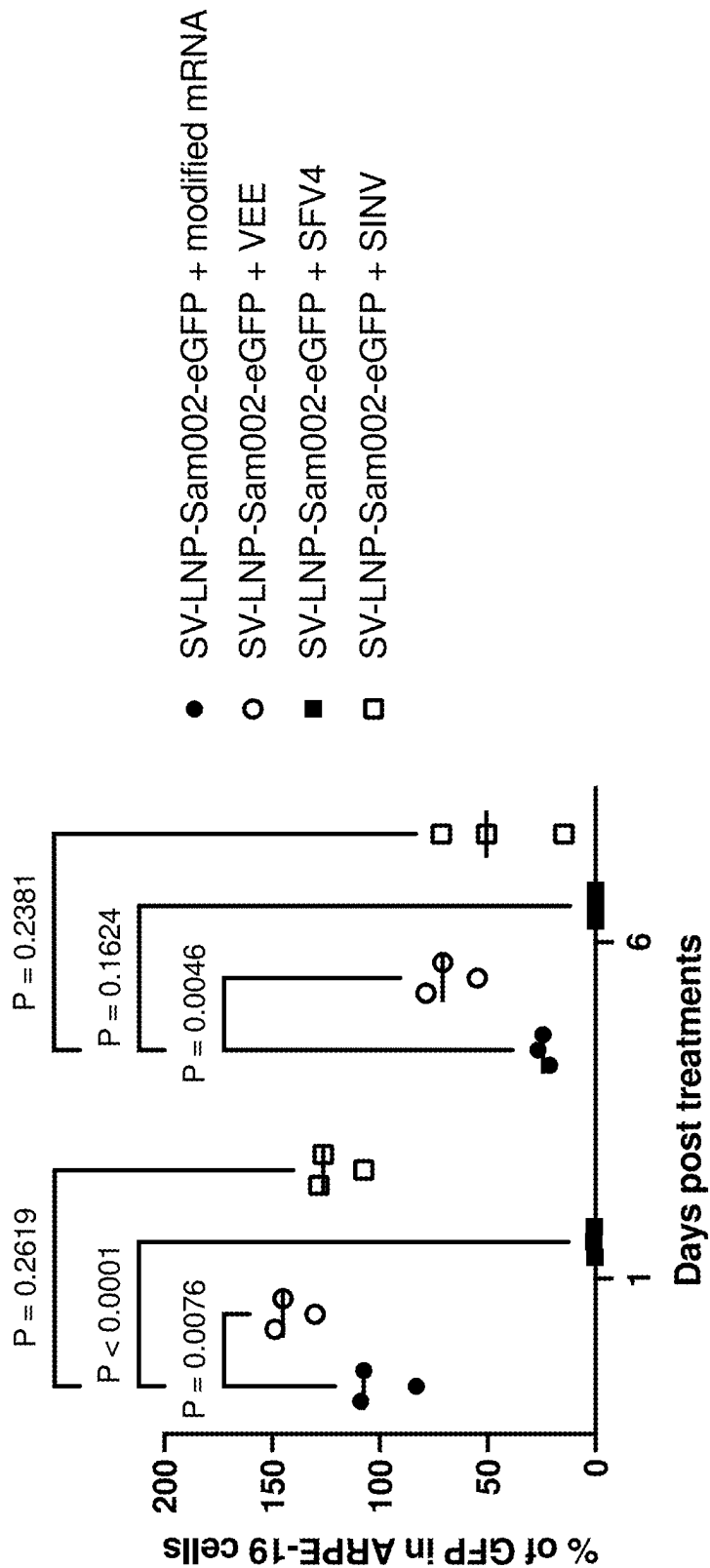


Fig. 2B

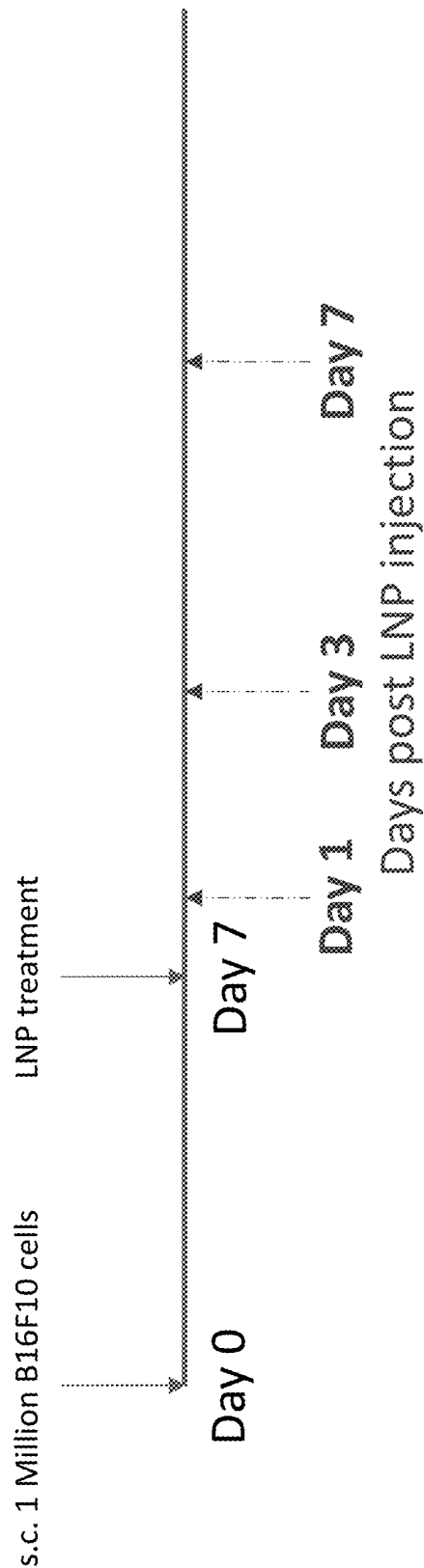


Fig. 3A

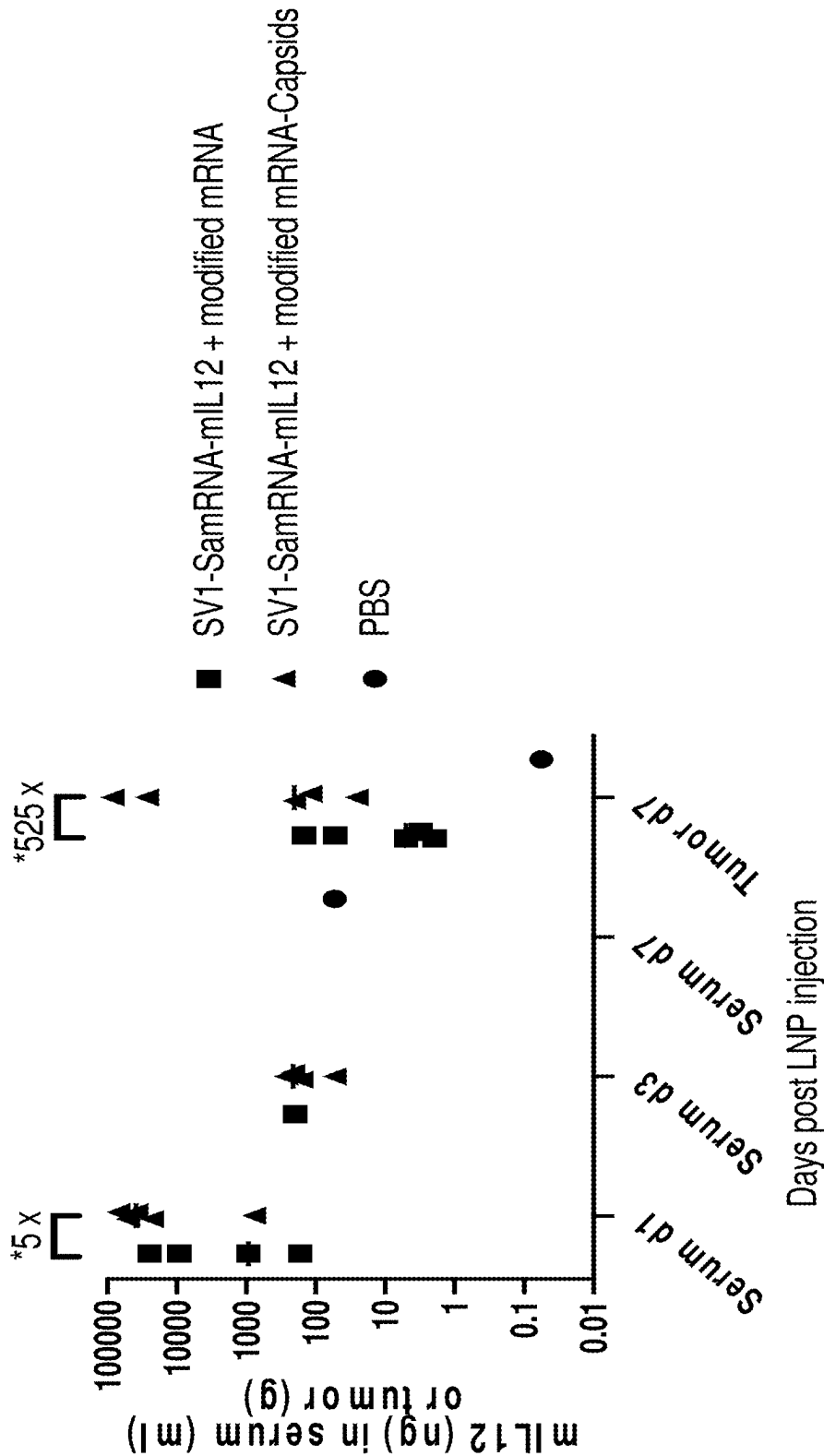


Fig. 3B

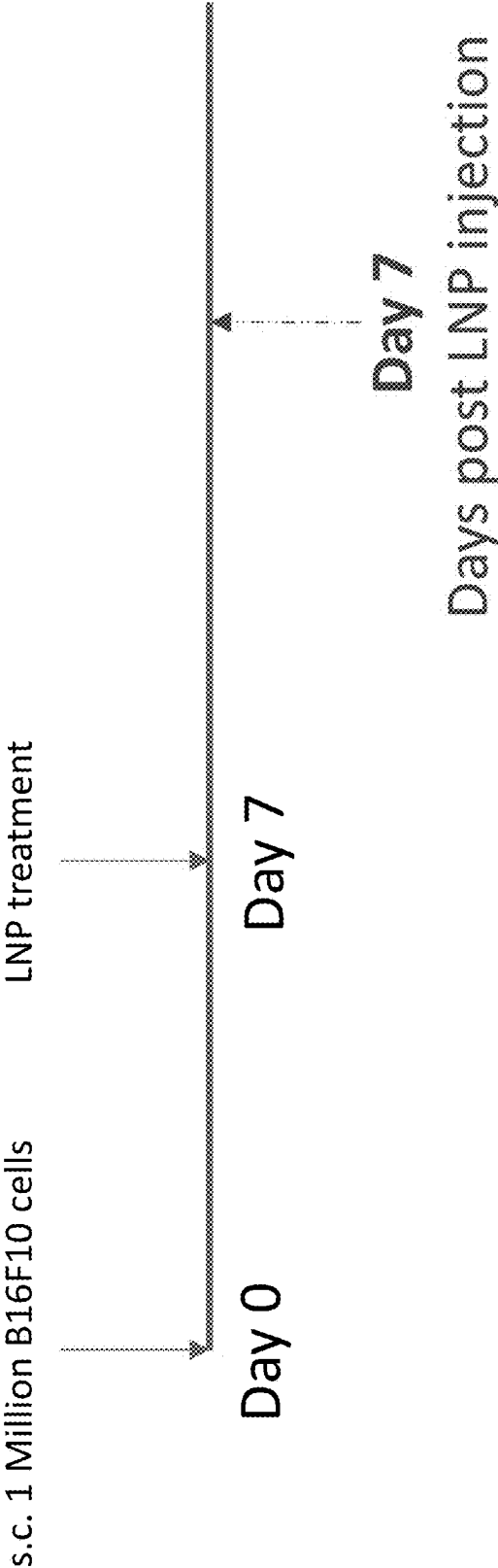


Fig. 4A

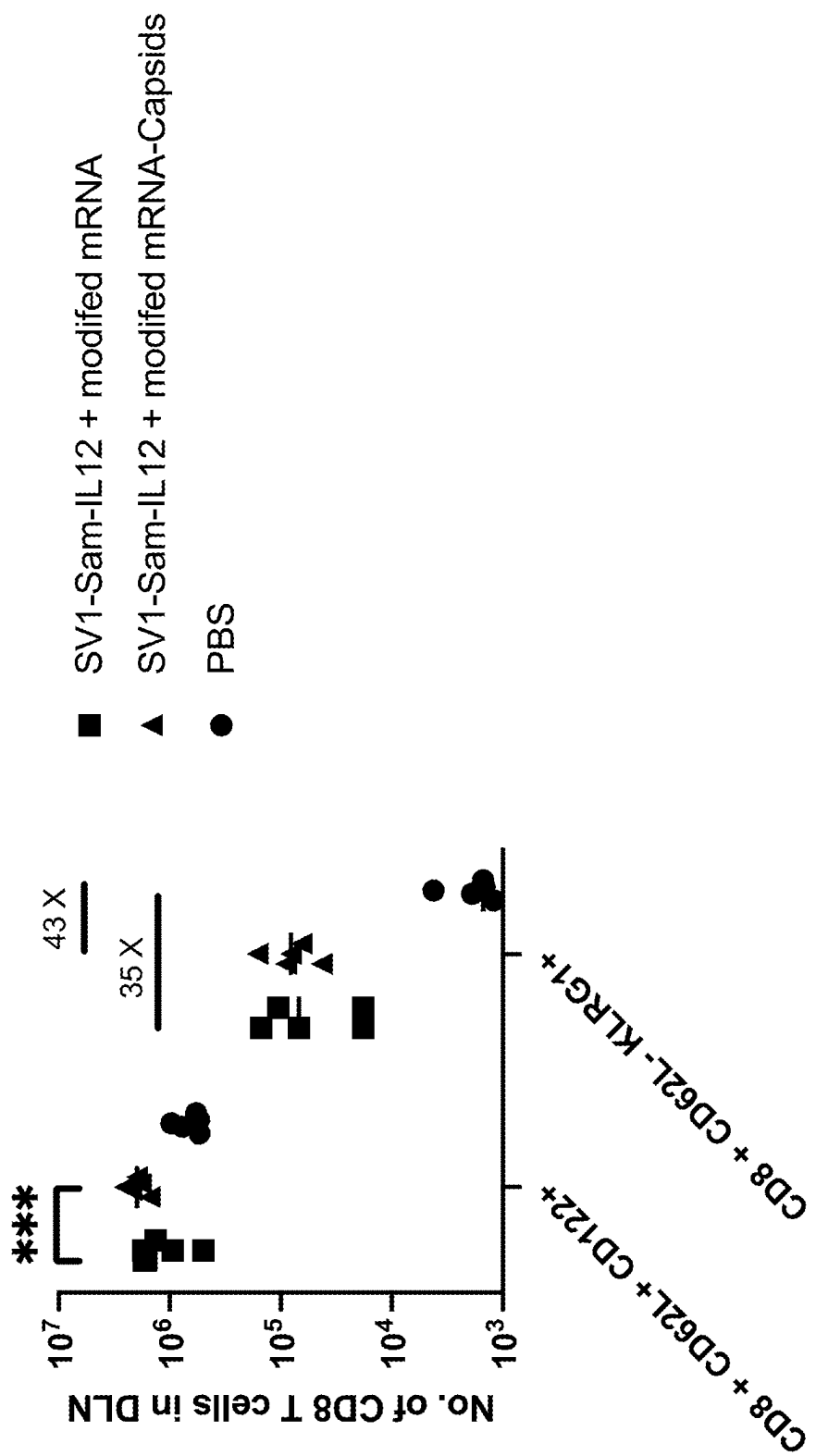


Fig. 4B

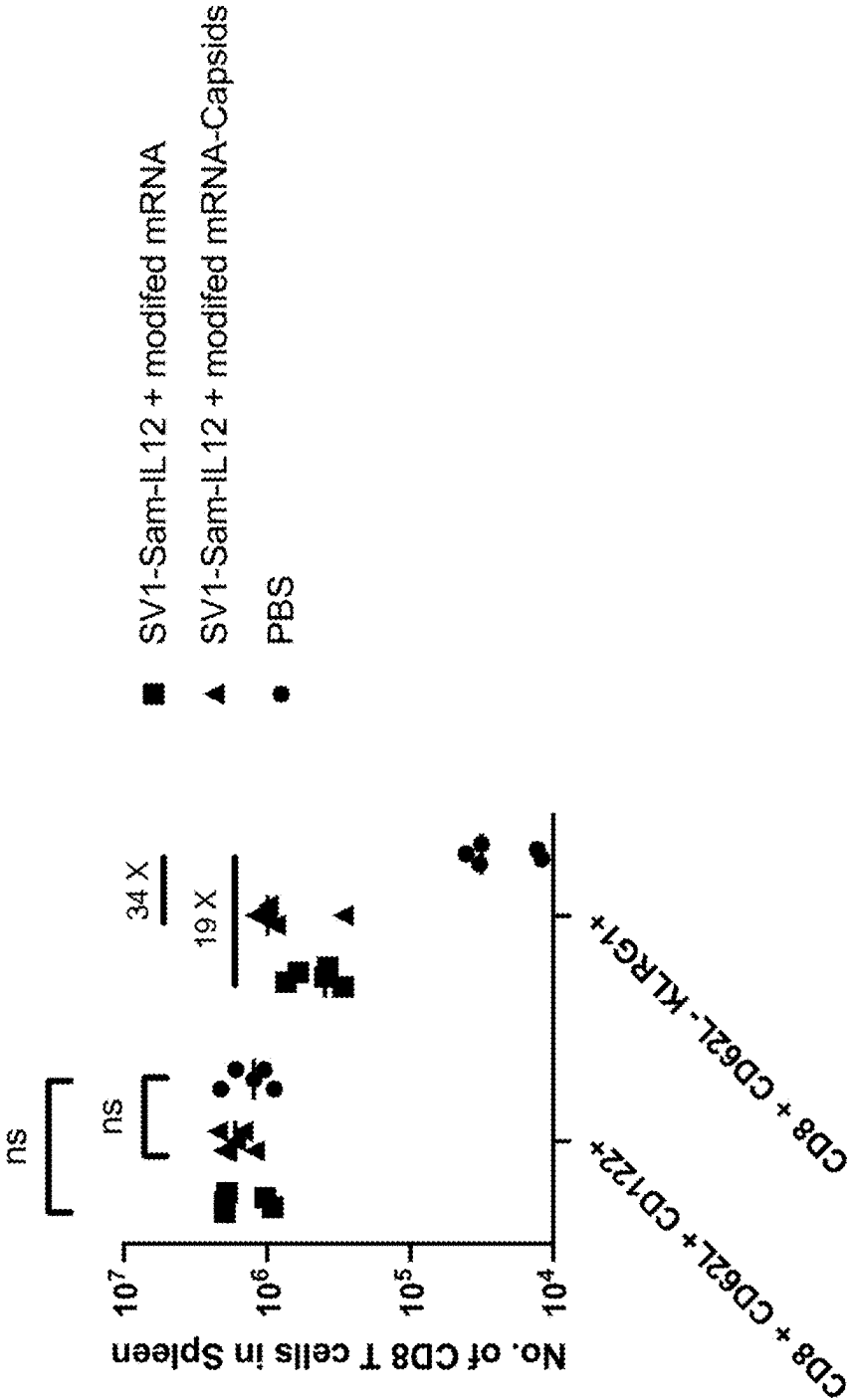


Fig. 4C

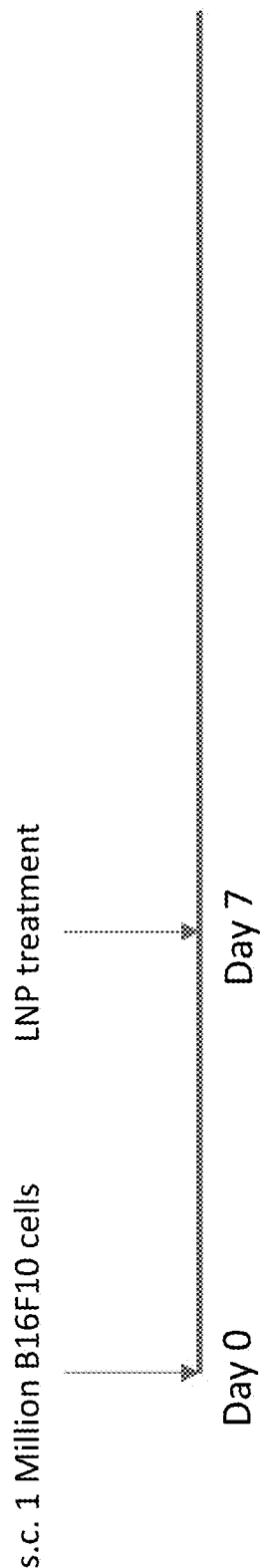


Fig. 5A

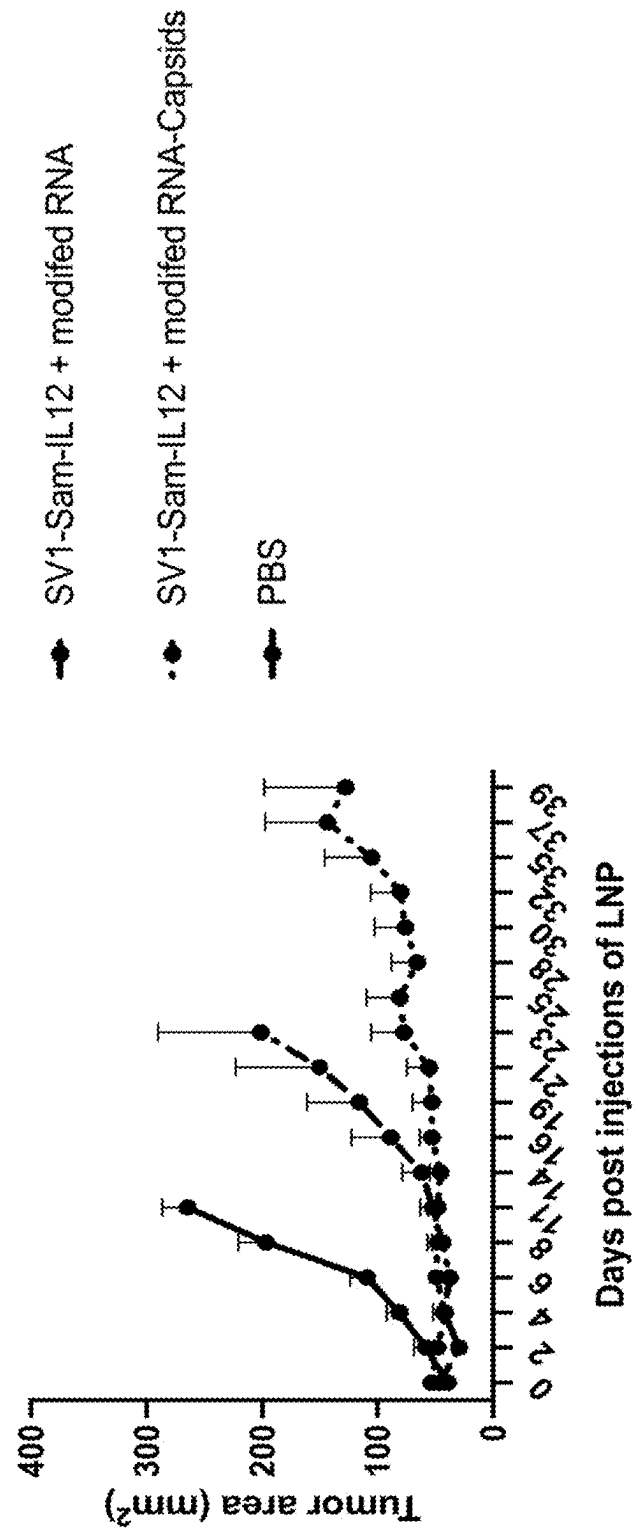


Fig. 5B

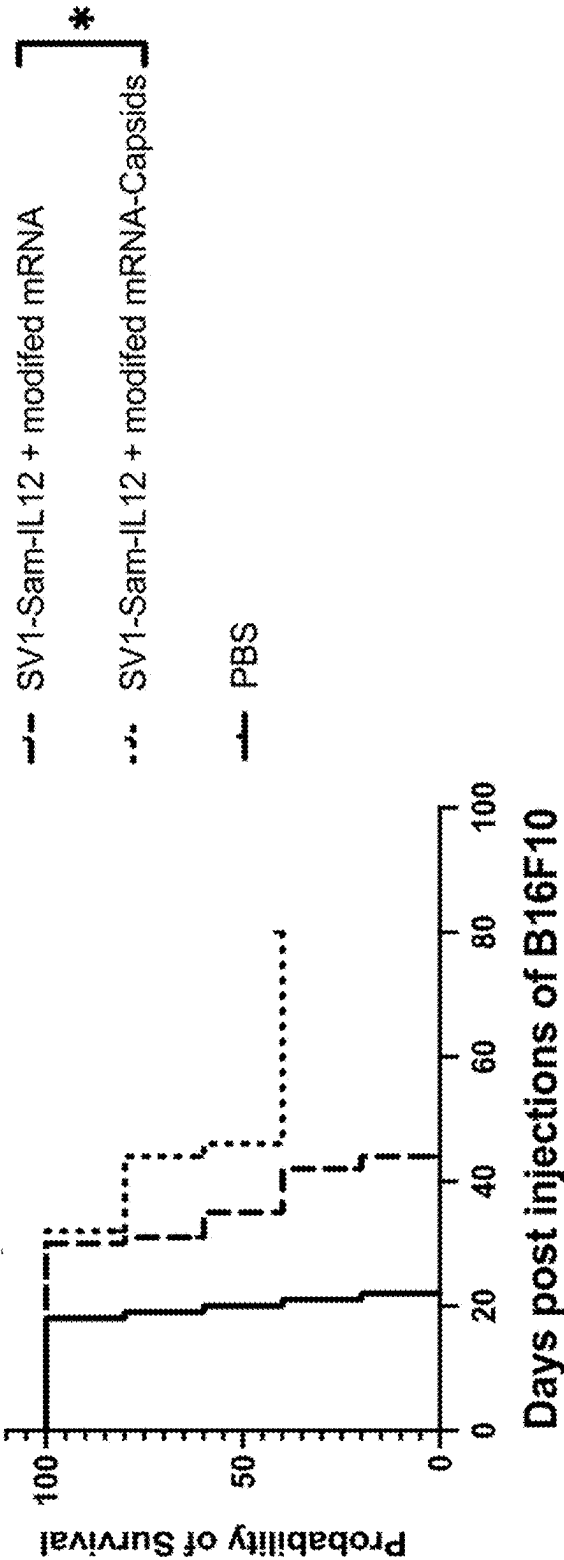


Fig. 5C

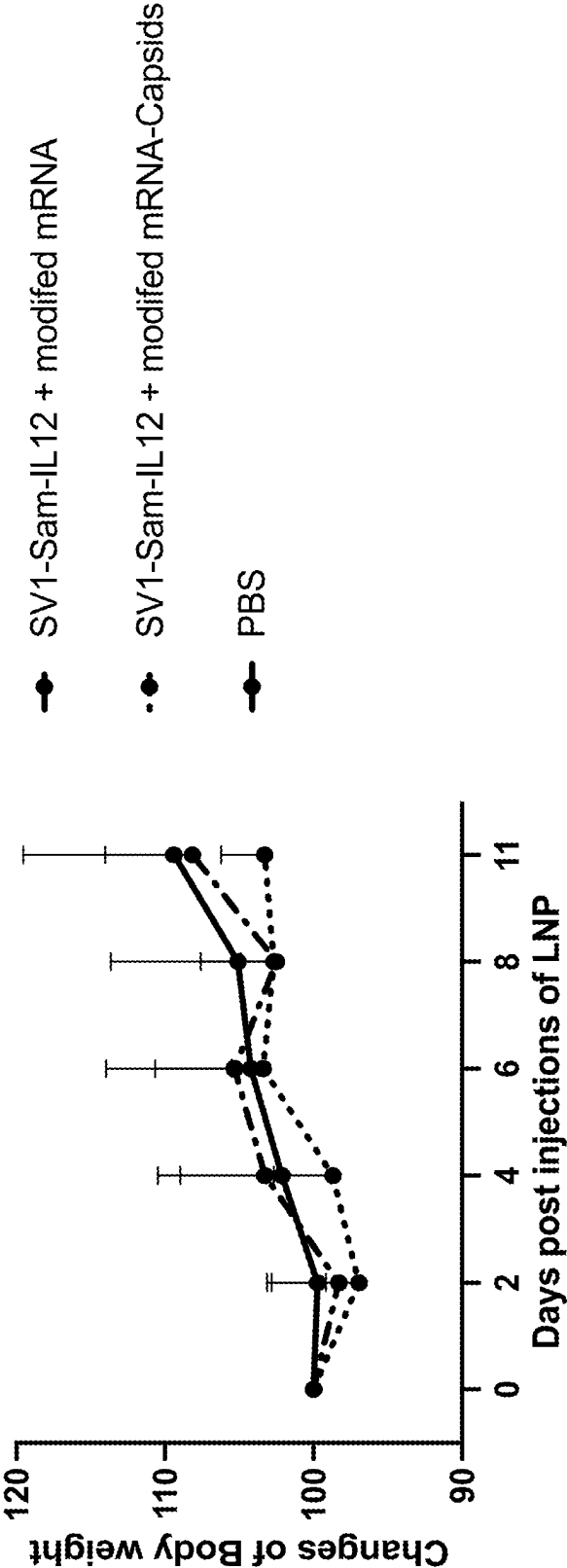


Fig. 5D

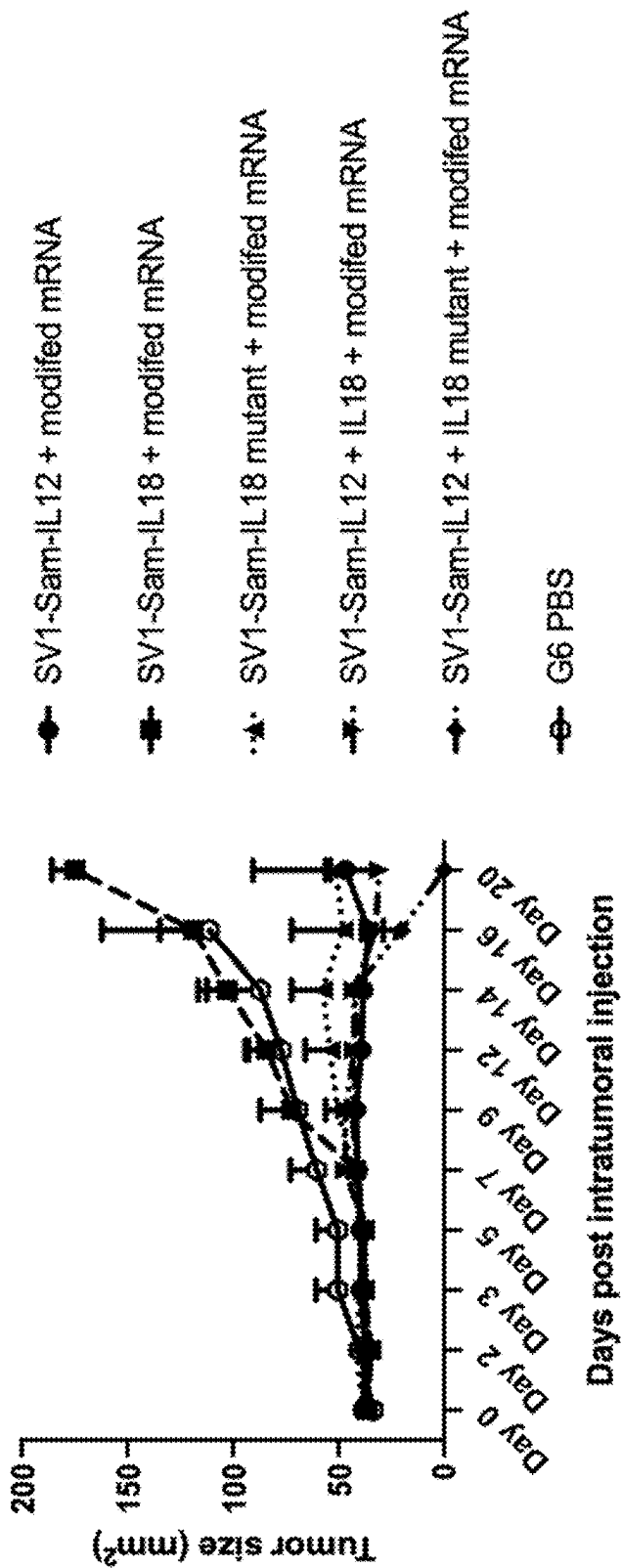


Fig. 6

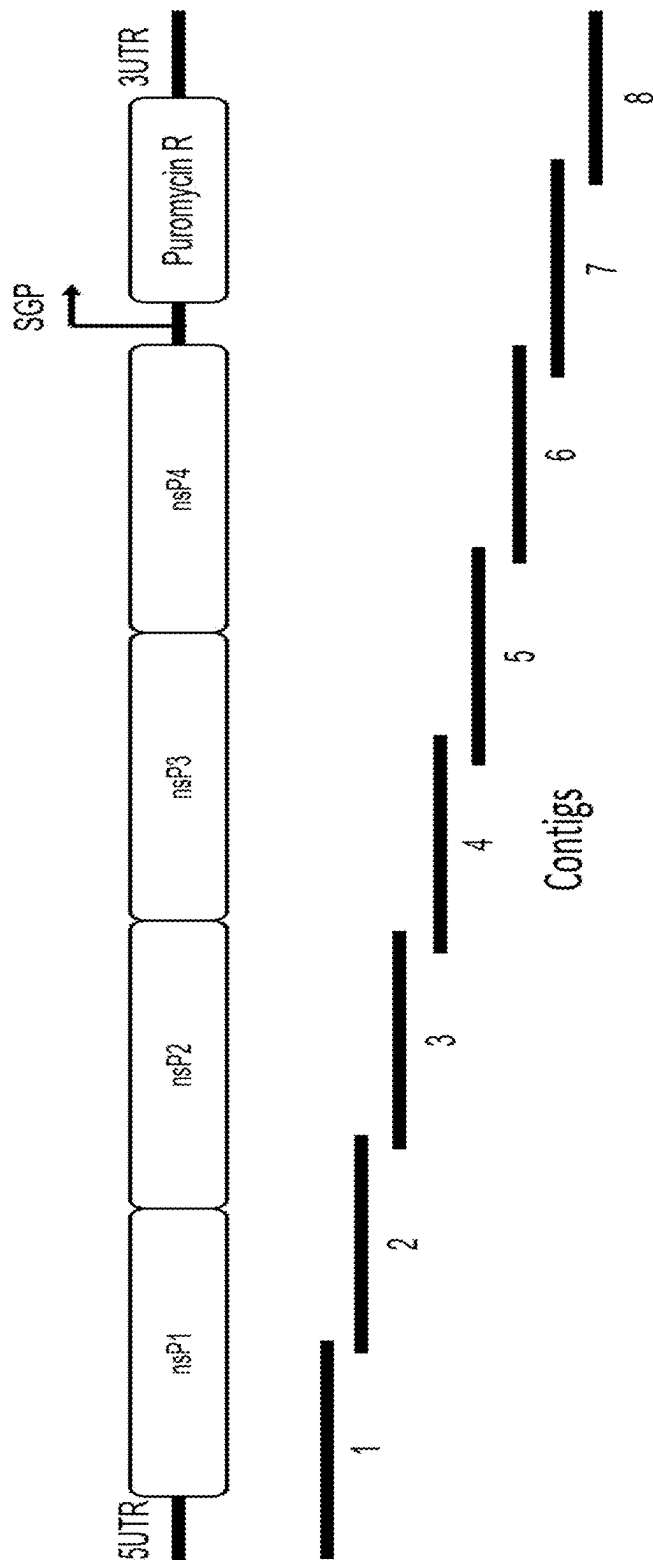


Fig. 7

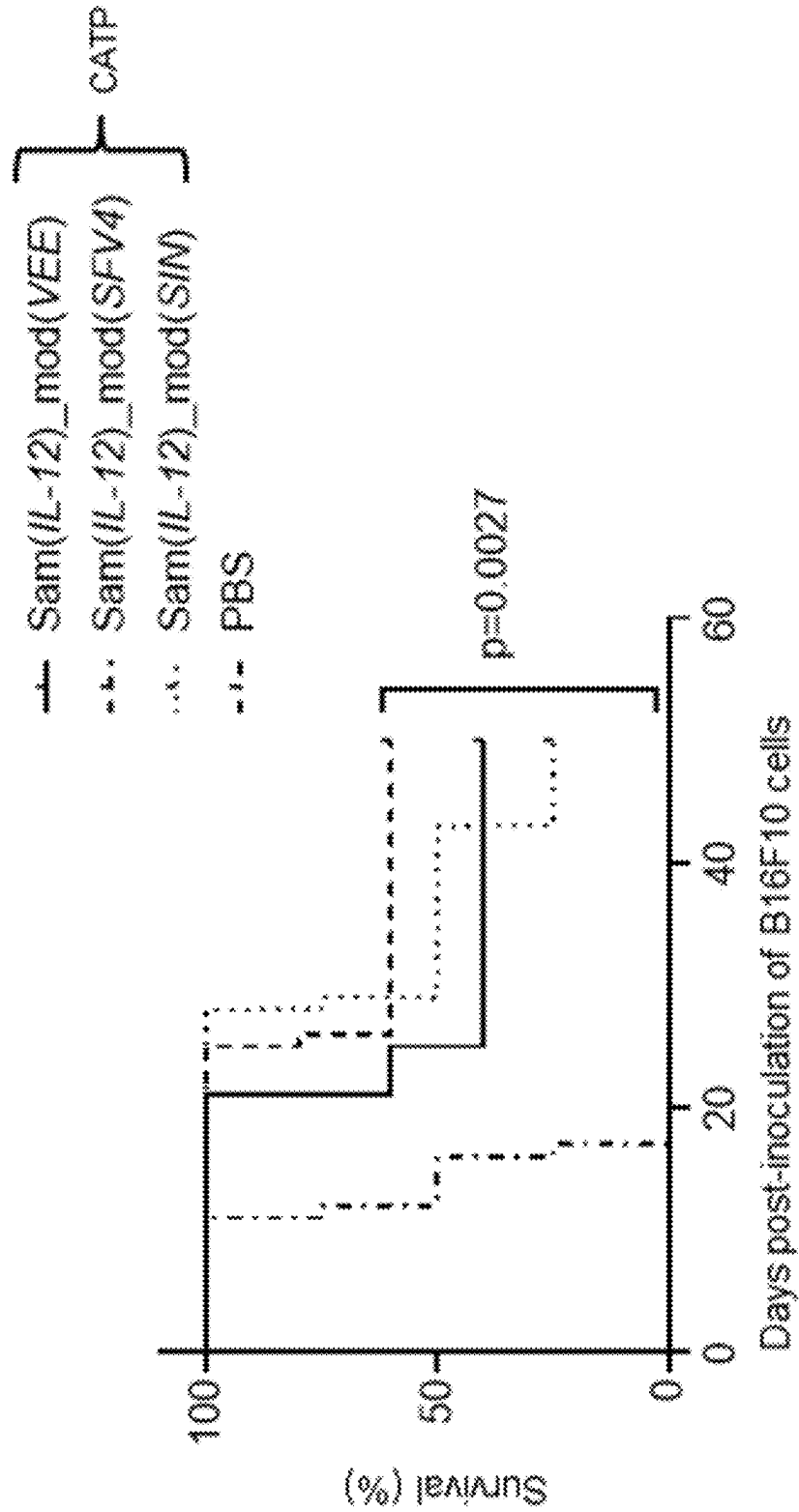


Fig. 8

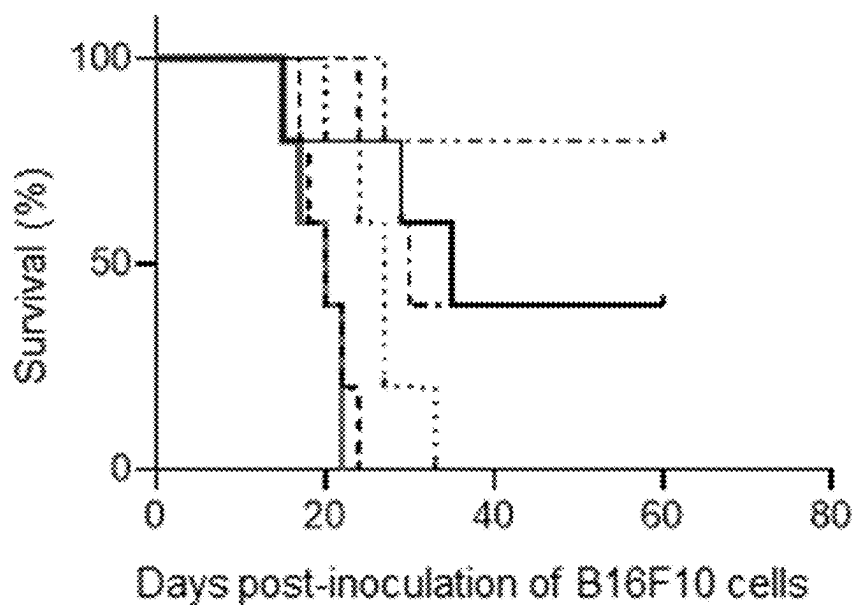
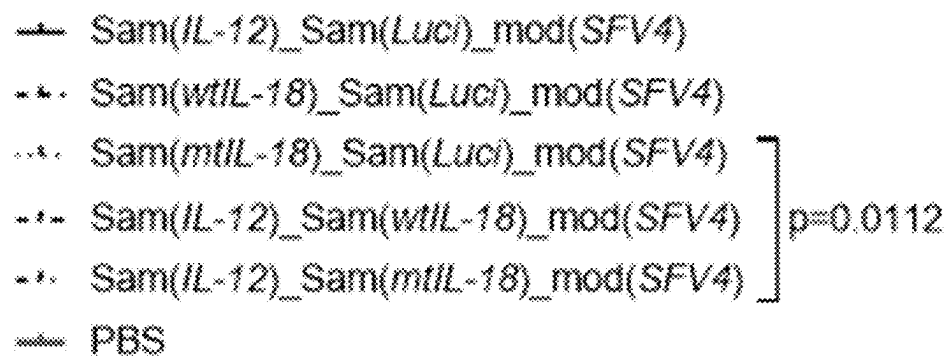


Fig. 9A

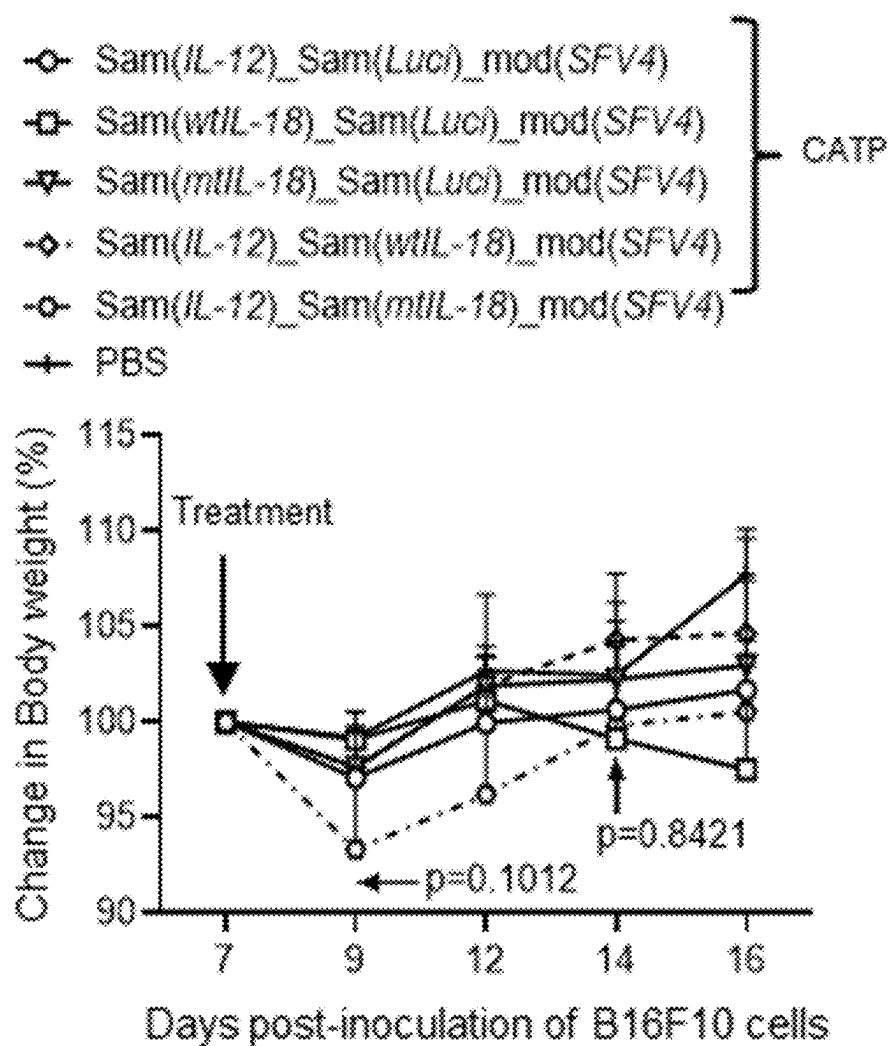


Fig. 9B

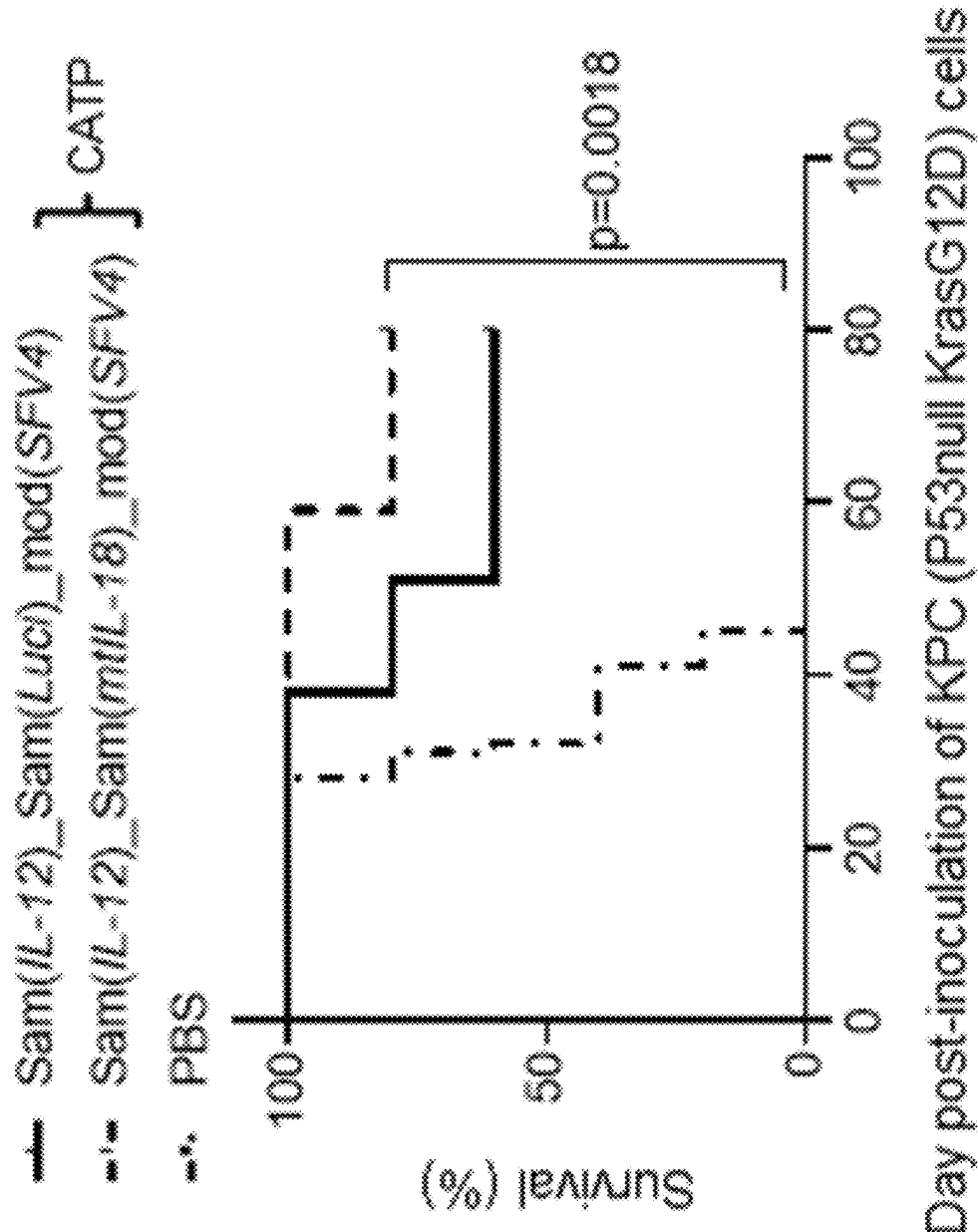


Fig. 9C

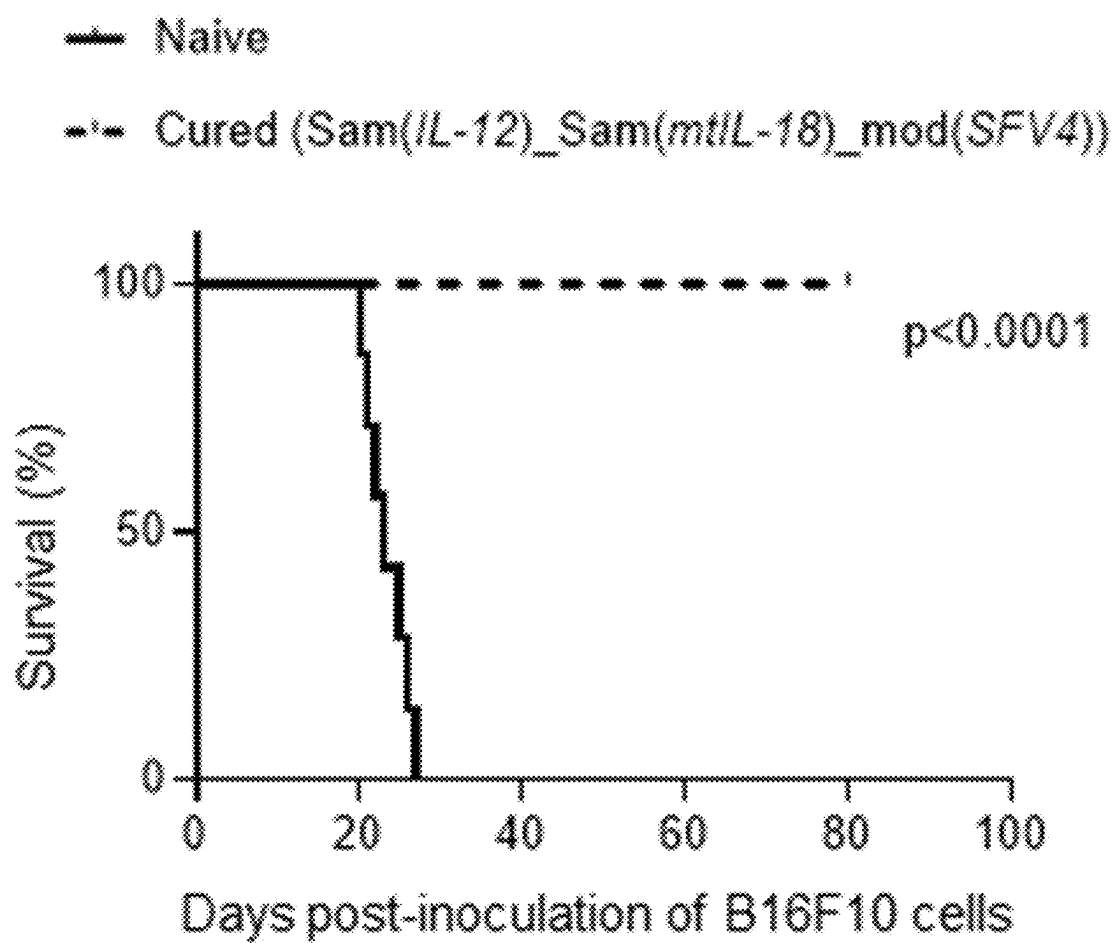


Fig. 9D

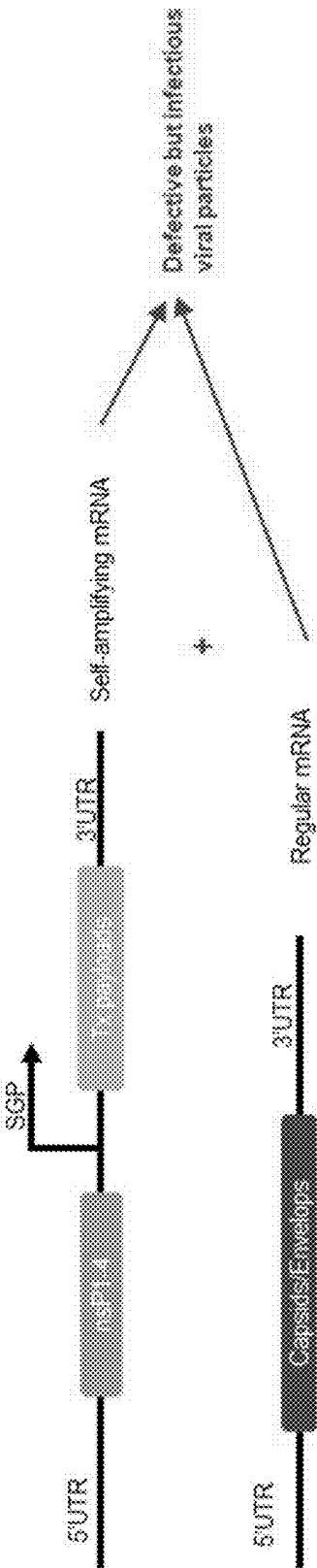


Fig. 10A

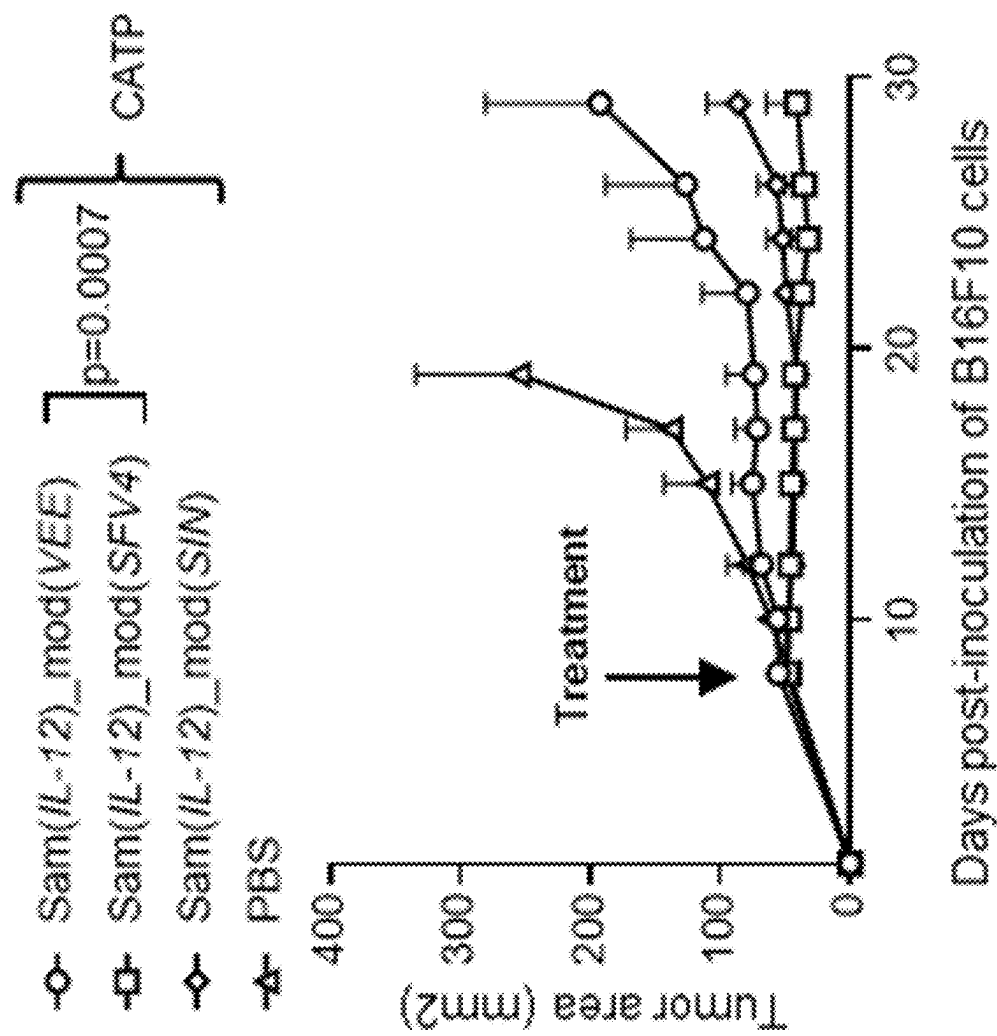


Fig. 10B

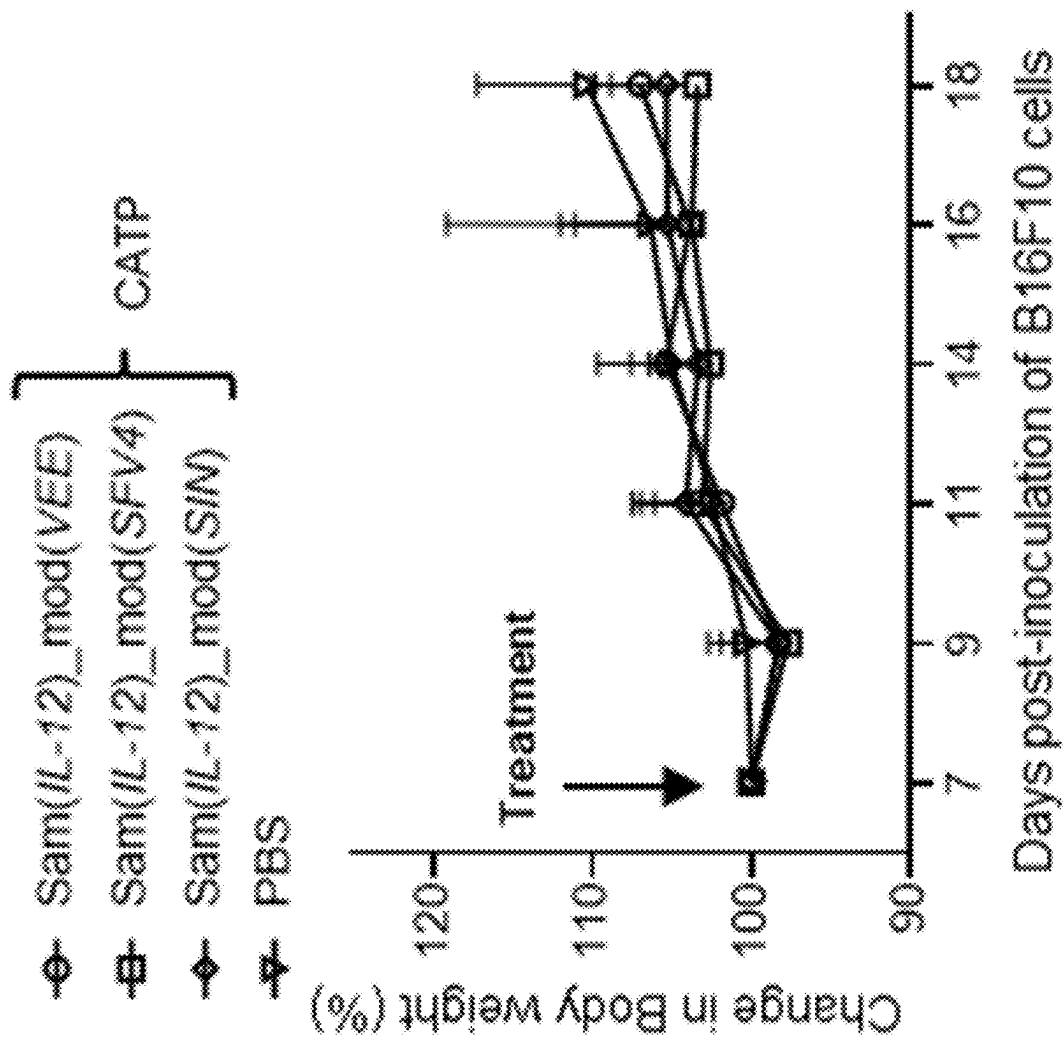


Fig. 10C

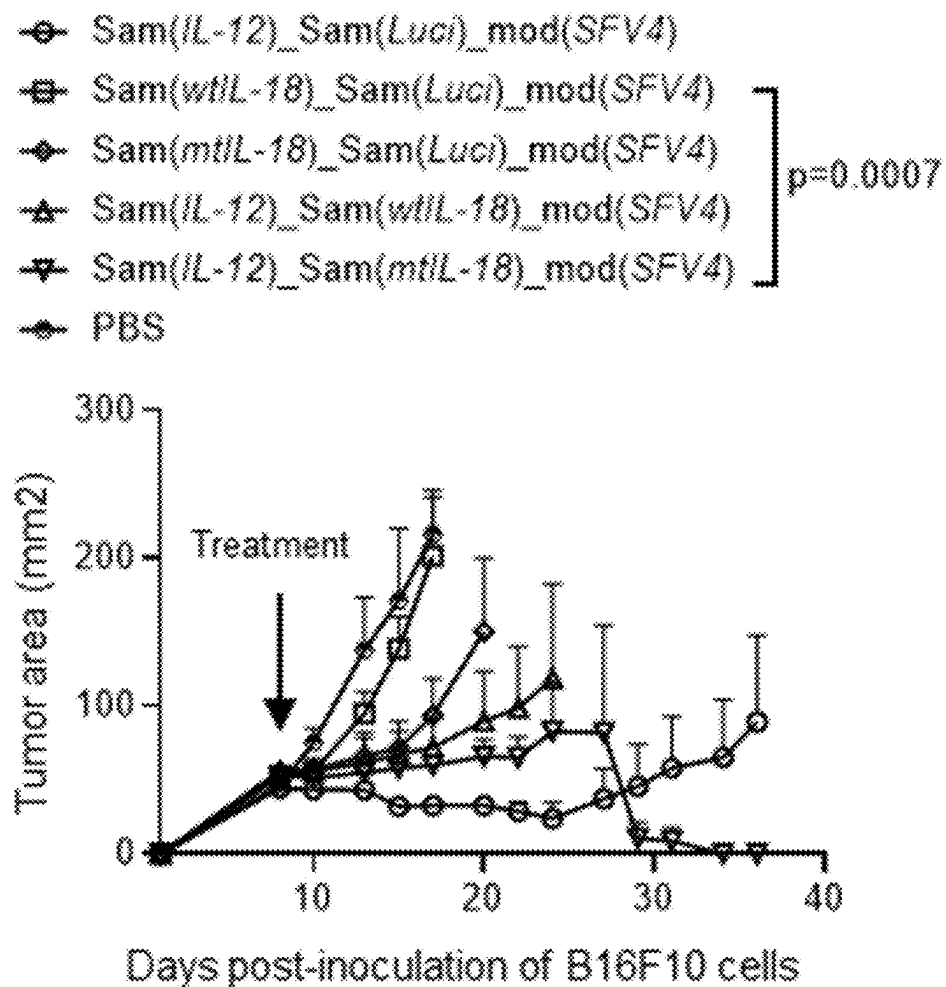


Fig. 11A

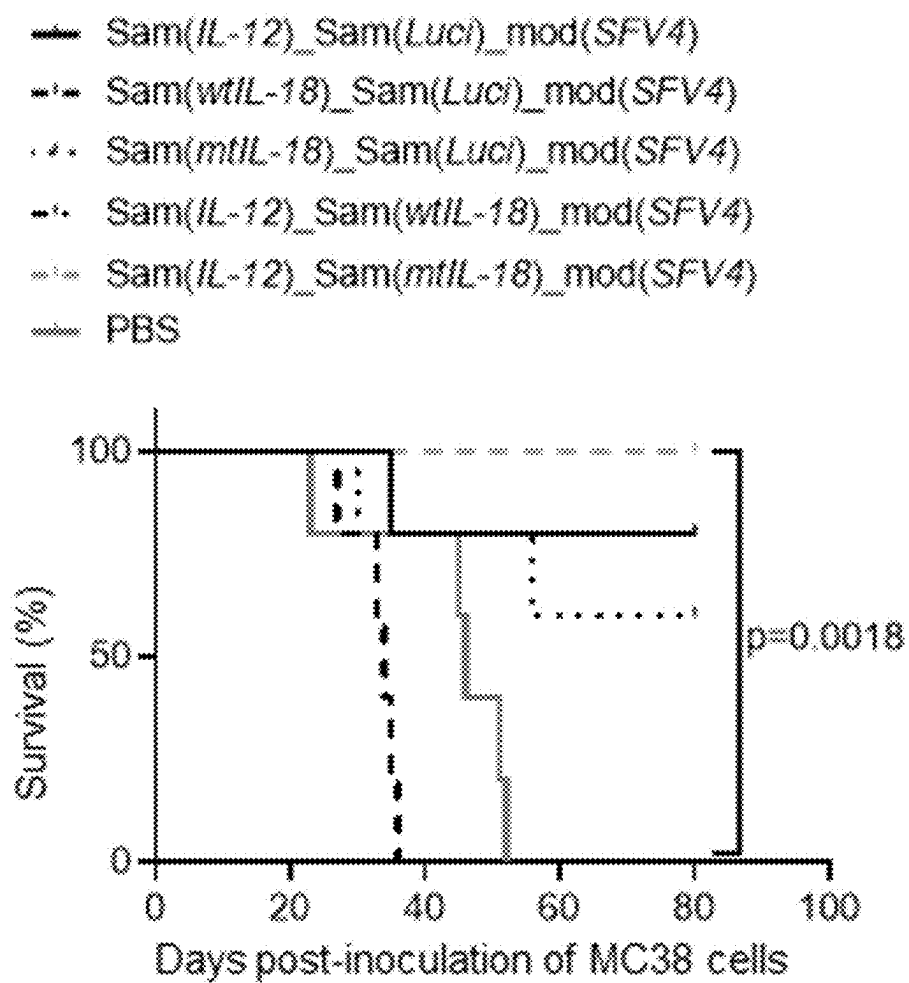


Fig. 11B

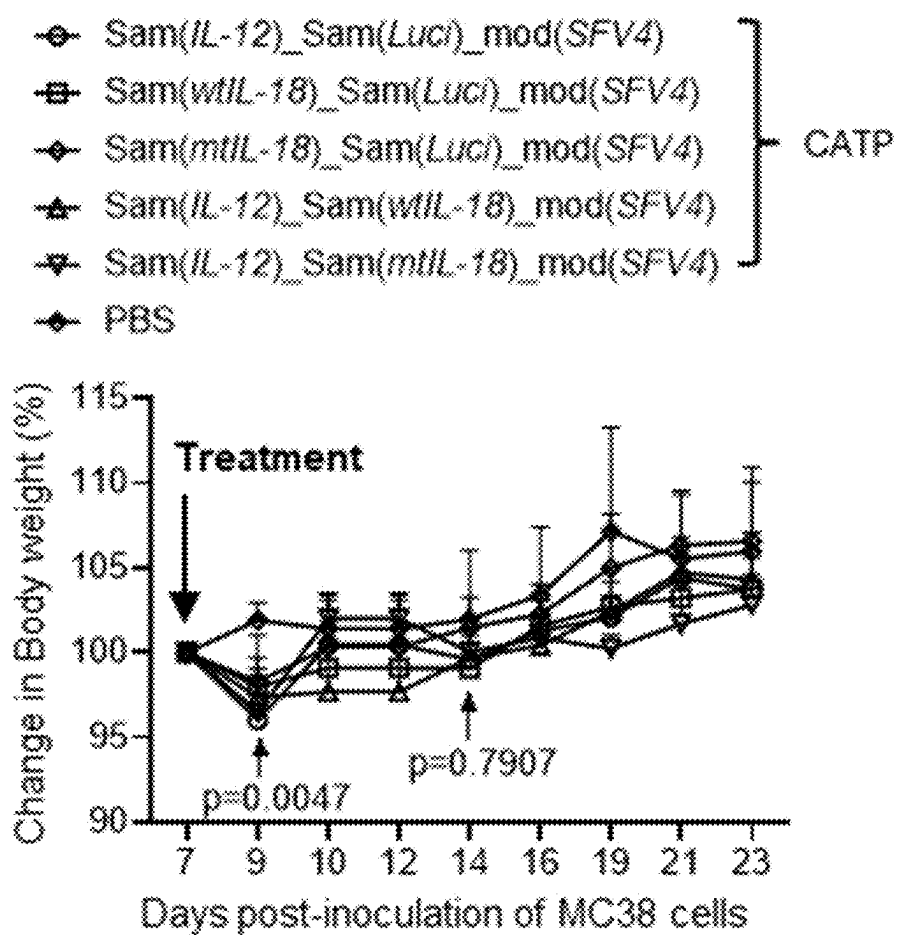


Fig. 11C

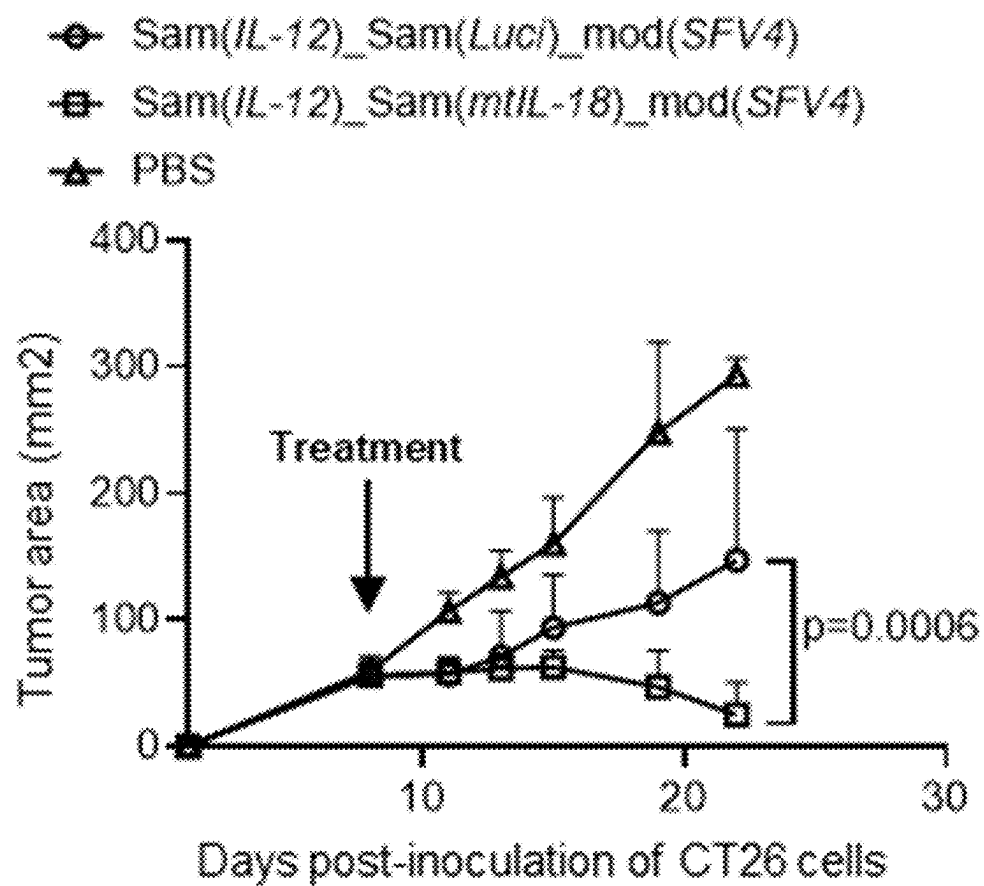


Fig. 12A

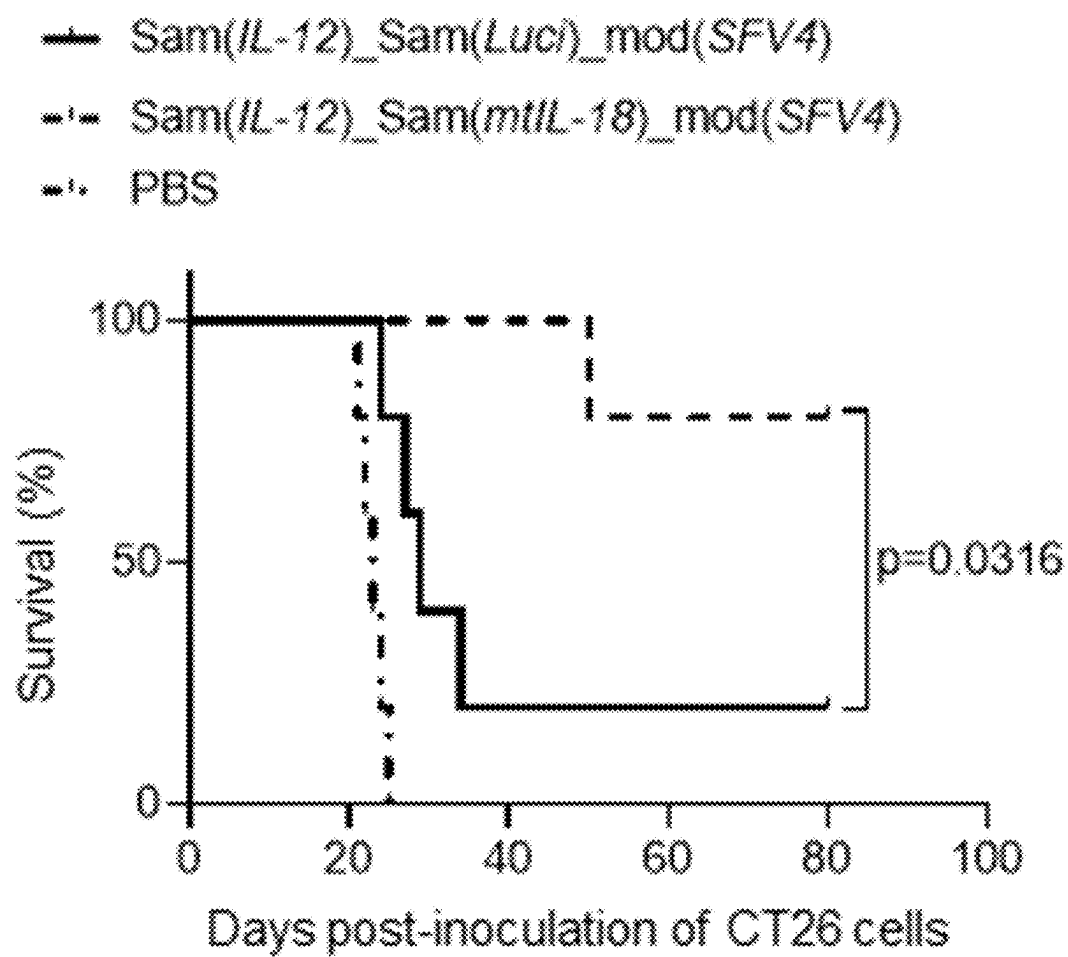


Fig. 12B

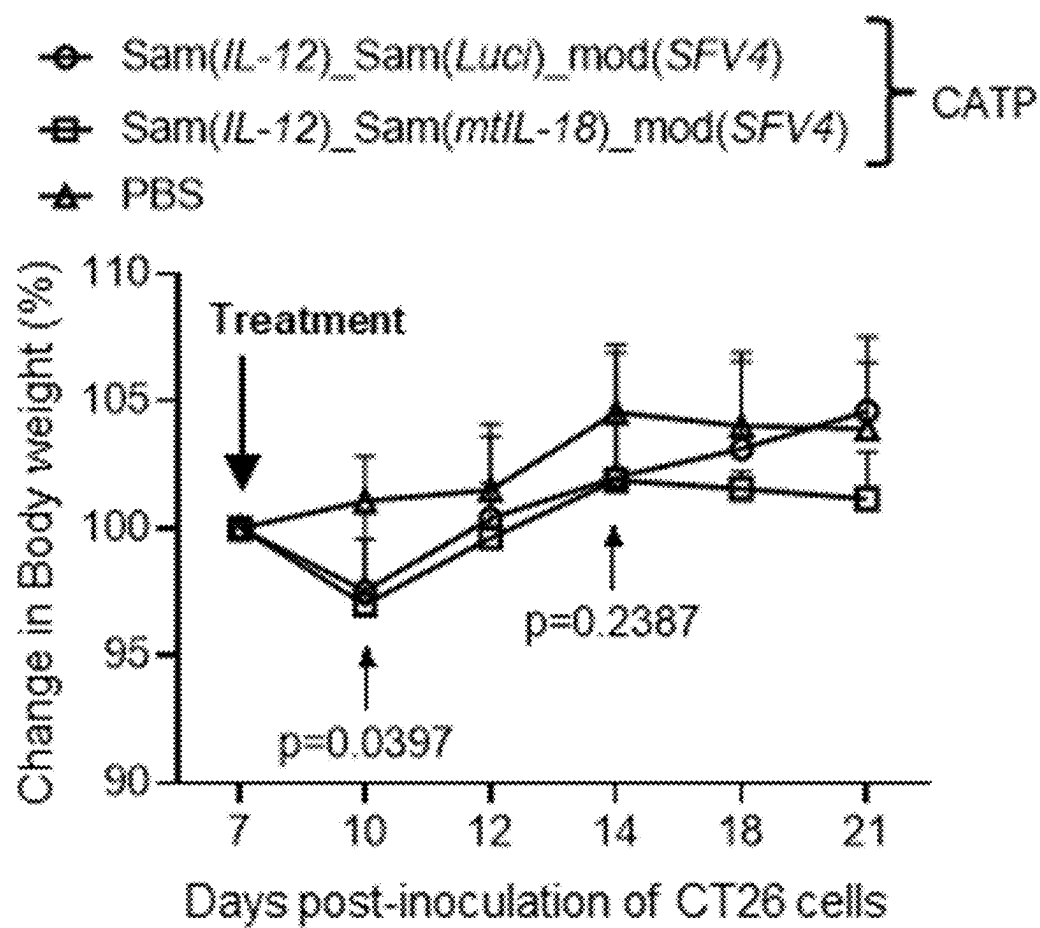


Fig. 12C

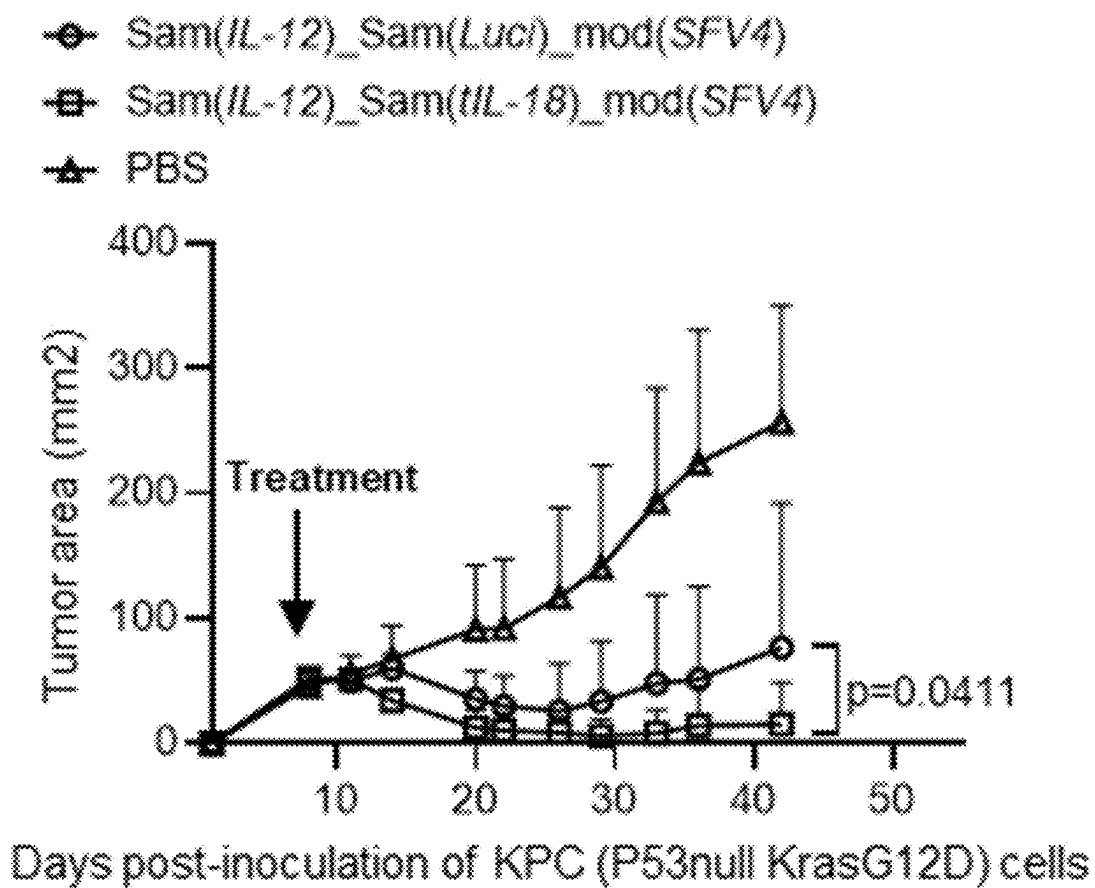


Fig. 12D

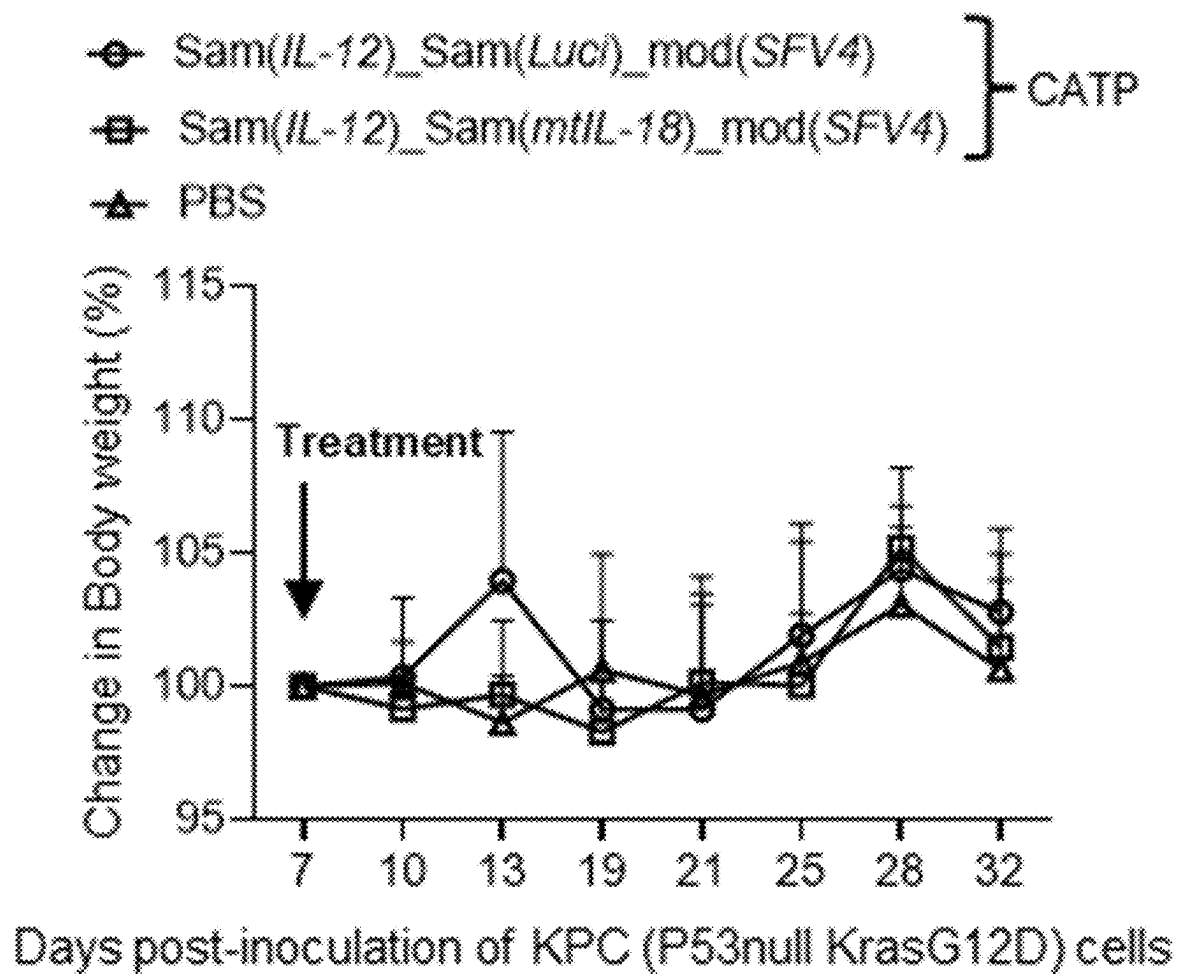


Fig. 12E

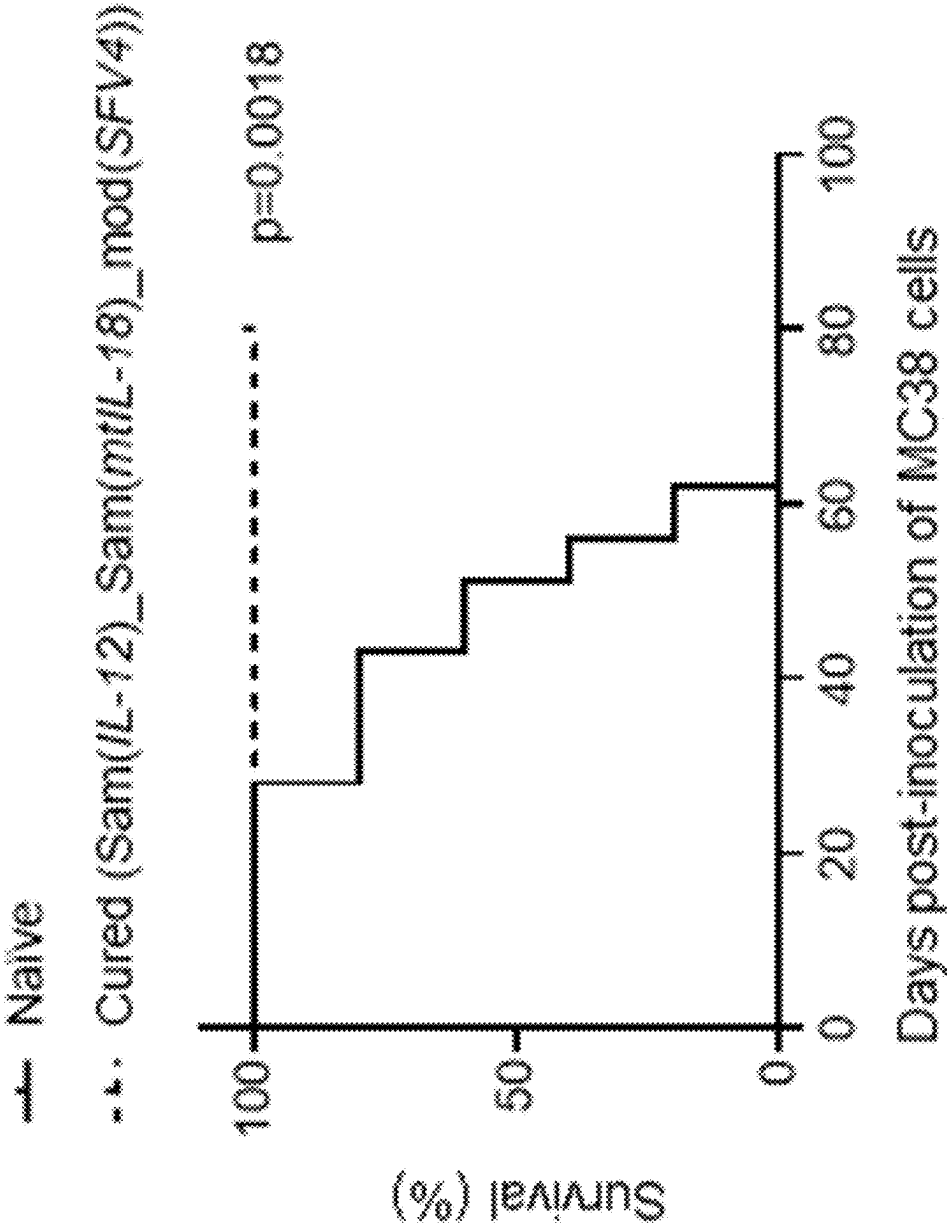


Fig. 13A

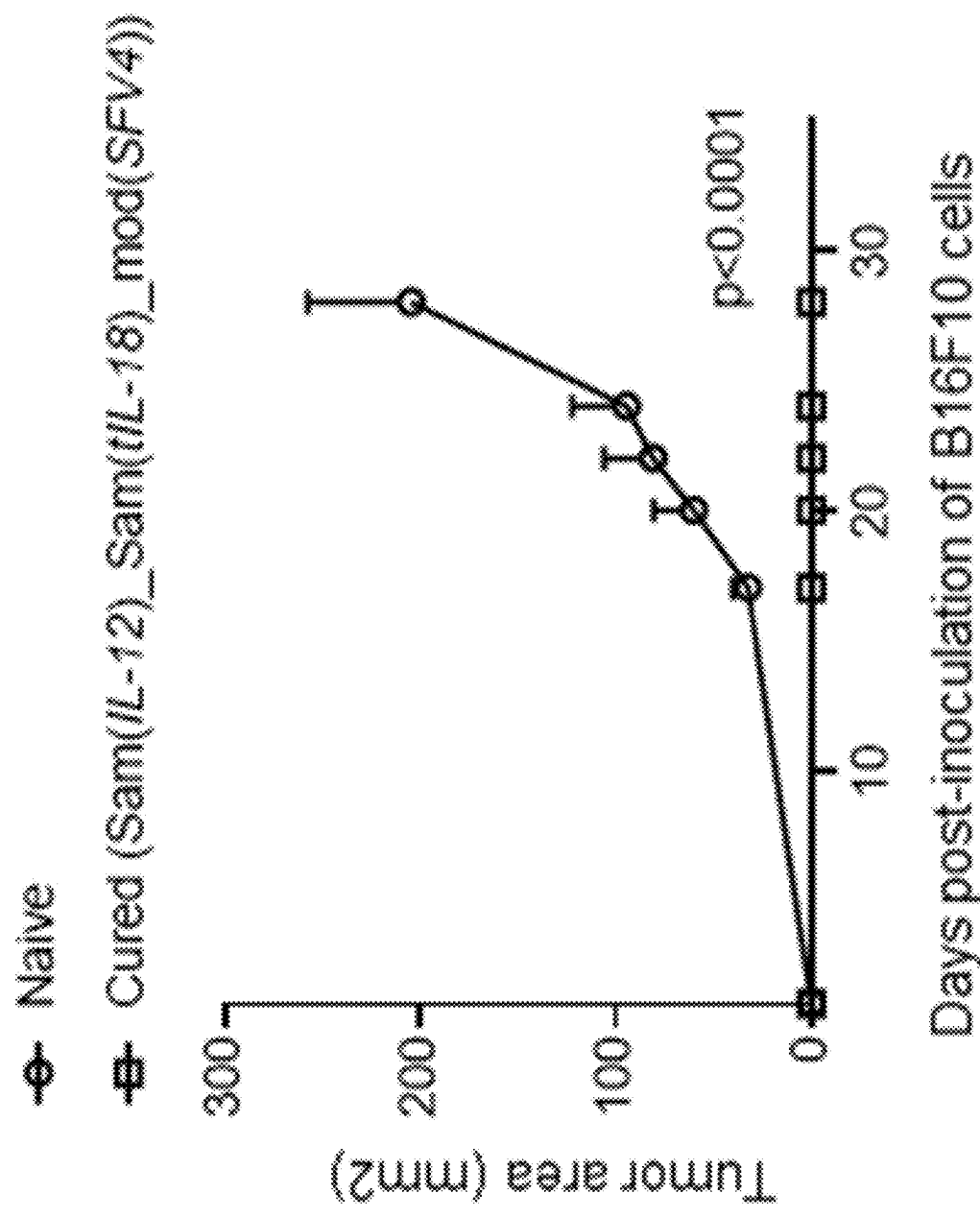


Fig. 13B

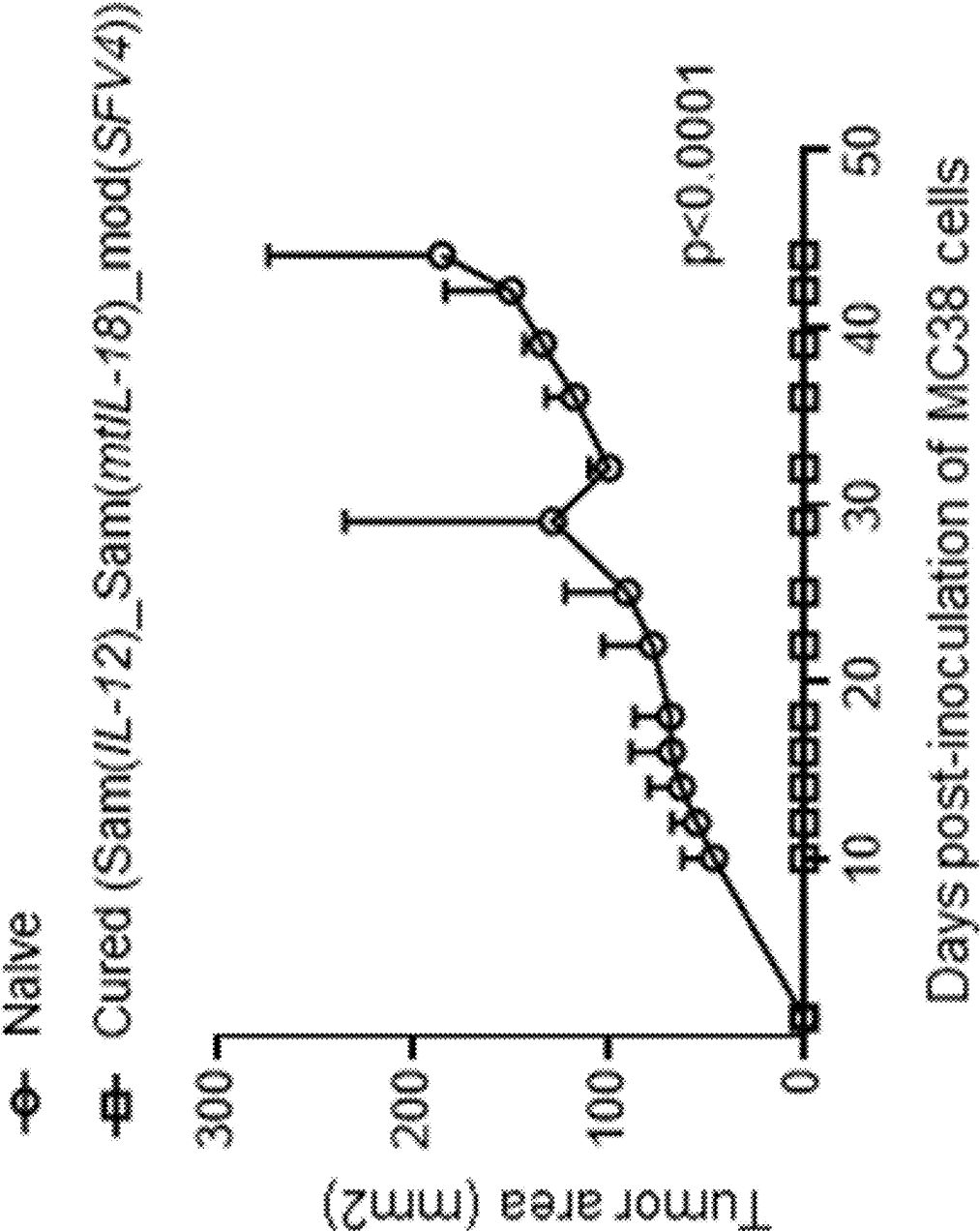


Fig. 13C

METHODS FOR CASCADE AMPLIFICATIONS OF THERAPEUTIC PAYLOADS (CATP) & COMPOSITIONS FOR CANCER IMMUNOTHERAPIES AND GENE THERAPY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority benefit of U.S. Provisional Appl. No. 63/563,062 filed on 8 Mar. 2024 and U.S. Provisional Appl. No. 63/738,692 filed on 24 Dec. 2024, the disclosures of each of which are incorporated herein by reference in their entireties.

SEQUENCE LISTING STATEMENT

[0002] The instant application contains a Sequence Listing in electronic format which has been submitted electronically. Said Sequence Listing, created on 25 Feb. 2025, is named "5292-114PCT-ST26.xml" and is 444,100 bytes in size. The information in the electronic format of the Sequence Listing is part of the present application and is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0003] The present disclosure provides oncolytic defective virus compositions and methods of synthesis thereof, for use in cancer immunotherapies.

BACKGROUND

[0004] It is a challenge in the field of therapeutics to design, synthesize and deliver a therapeutic agent, e.g., a ribonucleic acid (RNA) for example, a messenger RNA (mRNA) or self-amplifying mRNA (sa-mRNA) inside a cell in vivo, such as to effect physiologic outcomes which are beneficial to the cell, tissue or organ and ultimately to an organism.

[0005] In particular, current cancer treatments, including existing viral vectors, chemotherapy, radiation, and surgery, lack the specificity to selectively treat cancerous cells, while maintaining the health and viability of normal, non-cancerous cells, and can produce undesirable off-target effects. As such, there is a need in the art for cancer therapies that are broadly efficacious in multiple cancers, are capable of selectively eliminating cancerous cells and produce less undesirable off-target effects.

[0006] While the use of oncolytic viruses has been contemplated, a principal danger of the use of viruses in cancer treatment is the possibility of dissemination of a wild-type virus reconstituted in the patient's healthy cells or tissues. Oncolytic viruses are capable of selective replication in dividing cells (e.g., cancer cells) while leaving non-dividing cells (e.g., healthy cells) unharmed. As the infected dividing cells are destroyed by lysis, they release new infectious virus particles to infect the surrounding dividing cells. Cancer cells are ideal hosts for many viruses because they have the antiviral interferon pathway inactivated or have mutated tumor suppressor genes that enable viral replication to proceed unhindered. Use of oncolytic viruses carries the risk of non-specific viral infection of healthy cells, leading to the death of non-cancerous cells and tissues.

[0007] Moreover, achieving sufficient production of oncolytic virus therapies in vitro remains difficult because (1) in vitro manufacturing processes need to be established for

each OV individually; (2) different virus features-particle size, presence/absence of an envelope, and host species-require specific requirements to ensure sterility, for handling, and for toxicity testing; and (2) optimization of serum-free culture conditions, increasing virus yields, development of scalable purification strategies, and formulations guaranteeing long-term stability require further optimization and characterization (Ungerechts et al., 2006, Mol Ther Methods Clin Dev 3: 16018).

[0008] Nucleic acid therapies, such as mRNA therapies, are able to deliver one or more gene or genes of interest that are encoded by a nucleic acid to a subject in need thereof. However, achieving adequate gene expression is a medical challenge for nucleic acid therapies because the number of RNA transcripts available in vivo is proportional to the number of nucleic acid molecules successfully delivered during administration, thus existing nucleic acid therapies may require large doses or repeated administrations. Large doses and repeated administrations of nucleic acid therapies can elicit undesirable immune responses and repeated administration can render subsequent administration of the same therapeutic less effective.

SUMMARY OF THE INVENTION

[0009] The present disclosure includes a composition comprising: a first nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) encoding at least one gene of interest (GOI) or a plurality of GOIs; a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein; and at least one payload delivery system, wherein the at least one payload delivery system is a non-viral payload delivery system and wherein the payload is at least one nucleic acid construct.

[0010] In one aspect, the composition comprises a first nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) comprising SEQ ID NO: 45 and at least one of SEQ ID NOs: 54, 101 or 102; a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein from SFV4; and at least one payload delivery system, wherein the non-viral payload delivery system is an LNP, wherein the LNP comprises the compound (IVe), wherein the payload is at least one nucleic acid construct.

[0011] In some aspects, the at least one virus structural protein encodes a viral capsid protein, a viral envelope protein, or a combination thereof. In some aspects, the first nucleic acid construct is or encodes a sa-mRNA molecule. In some aspects, the second nucleic acid construct is or encodes a mRNA molecule. In some aspects, the non-viral payload delivery system is an LNP. In some aspects, the LNP comprises an ionizable lipid, a phospholipid, a steroid, a polyethylene glycol (PEG) lipid, a modular lipid, or a combination thereof.

[0012] In one aspect, the first nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[0013] [5' UTR]-[nsP 1-4]-[SGP]-[GOI]-[3' UTR]-[Poly A]

[0014] wherein, 5' UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, GOI is one gene of interest or a plurality of genes of interest, 3' UTR is a 3' untranslated region, and poly A is a poly A tail. In some aspects, the first nucleic construct com-

prise more than one GOI operably linked to one or more SGP. In some aspects, each GOI is linked to a different SGP.

[0015] In some aspects, the at least one virus structural protein encoded by the second nucleic acid construct is from an alphavirus, adenovirus, a HSV, an echovirus, a polio virus, a vaccinia virus, a measles virus, a vesicular stomatitis, an orthomyxovirus, a parvovirus, a maraba virus, a coxsackievirus, an autonomous parvovirus, a myxoma virus, a Newcastle disease virus, a reovirus, a seneca valley virus morbillivirus virus, a retrovirus, an influenza virus, a sindbis virus, semliki forest virus, venezuelan equine encephalitis virus, or a poxvirus.

[0016] In one aspect, the present disclosure includes a pharmaceutical composition comprising: composition of the disclosure and a pharmaceutically acceptable carrier. In some aspects, the first nucleic acid construct and the second nucleic acid construct are present in the pharmaceutical composition in a 1:100 to 100:1 ratio. In some aspects, the first nucleic acid construct and the second nucleic acid construct are present in the pharmaceutical composition in a 50:1 to 1:50, 40:1 to 1:40, 30:1 to 1:30, 25:1 to 1:25, 20:1 to 1:20, 15:1 to 1:15, 10:1 to 1:10, 9:1 to 1:9, 8:1 to 1:8, 7:1 to 1:7, 6:1 to 1:6, 5:1 to 1:5, 4:1 to 1:4, 3:1 to 1:3, 2:1 to 1:1, or 1:1 ratio.

[0017] In one aspect, the present disclosure provides a method of treating a subject having a tumor or cancer, comprising administering to a subject the pharmaceutical composition of the disclosure. In some aspects, the method induces apoptosis of a cancerous cell or tumor, the method comprising contacting the cancerous cell in vivo with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and at least one virus structural protein replicate within said cancerous cell, to express the GOI, wherein the first nucleic acid replication, expression of the GOI and the at least one virus structural protein replication within the cancerous cell results in cell death.

[0018] In one aspect, the present disclosure provides method of increasing expression of a polypeptide encoded by a GOI in a cell comprising contacting the cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said cell, and the first nucleic acid construct expresses the GOI within said cell, wherein the at least one virus structural protein form a pseudoviral particle that encapsulate one or more of the sa-mRNA, and wherein expression of the polypeptide is increased by 50-fold, 100-fold, 200-fold, 300-fold, 400-fold, 500-fold compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

[0019] In one aspect, the present disclosure provides a method of increasing central memory CD8 T cell in a tumor draining lymph node or spleen comprising contacting a tumor with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said tumor to produce the at least one virus structural protein and the sa-mRNA, and the first nucleic acid construct expresses the GOI within said tumor, wherein the at least one virus structural protein form a pseudoviral particle, which encapsulate one or more of the sa-mRNA, and wherein central memory CD8 T cell count is increased by 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, or

more compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

[0020] A method of treating is disclosed herein for a subject having a tumor or cancer, comprising administering to a subject a composition comprising a nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) comprising SEQ ID NO: 45 and at least one of SEQ ID NOs: 54, 101 or 102.

[0021] Each of the aspects of the present disclosure can encompass various elements of the present disclosure. It is, therefore, anticipated that each of the aspects of the present disclosure involving any one element or combinations of elements can be included in each aspect of the present disclosure. This disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following detailed description or illustrated in the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1A shows a diagram of introducing one embodiment of the composition of the present disclosure to a host cell, the result of the expression of the payload of the composition in the host cell, and the effect of the expression of the payload of the composition on the host cell and adjacent cells. FIG. 1B shows a schematic of the cascade amplification of therapeutic payload process. The production of defective viruses enables the infections of adjacent cells, further amplifying the therapeutic payloads and enhancing overall efficacy.

[0023] FIG. 2A shows the oncolytic effects of the composition of the present disclosure in HEK293 cells in vitro. FIG. 2B shows the same effects in APRE-19 cells in vitro.

[0024] FIGS. 3A-3B show the oncolytic effects of the composition of the present disclosure in C57B6L mice with B16F10 melanoma in vivo. FIG. 3A shows a diagram of the treatment plan.

[0025] FIG. 3B shows mIL12 expression in blood serum and tumor of the C57B6L mice with B16F10 melanoma.

[0026] FIGS. 4A-4C show the effects of the composition of the present disclosure in the central memory CD8 T cells, in the tumor draining lymph node (TDLN) and the effector CD8 T cells in the TDLN and spleen. The statistical analysis were performed by 2-way ANNOVA. “****” means p-value smaller than 0.001. FIG. 4A shows a diagram of the treatment plan. Six to eight weeks old C57B6L mice (5 replicates in each group) were subcutaneously inoculated with 1 million B16F10 melanoma cells. FIG. 4B shows the number of central memory (CD62L+ CD122+) and effector (CD62L- KLRG1+) CD8 T cells in the TDLN. FIG. 4C shows the number of central memory (CD62L+ CD122+) and effector (CD62L- KLRG1+) CD8 T cells in the spleen.

[0027] FIGS. 5A-5D show the oncolytic effects of the composition of the present disclosure in C57B6L mice with B16F10 melanoma in vivo. The statistical analysis were performed by 2-way ANNOVA. “****” means p-value smaller than 0.001. FIG. 5A shows a diagram of the treatment plan. FIG. 5B shows tumor area of the B16F10 melanoma (Y-Axis) versus days post intratumorally injection (X-Axis) in the treated mice. FIG. 5C shows the survival curve of the treated mice. FIG. 5D shows body weight changes of the treated mice (Y-Axis) versus days post intratumoral injection (X-Axis).

[0028] FIG. 6 shows the oncolytic effects of the composition of the present disclosure in C57B6L mice with MC38 in vivo. The statistical analysis were performed by 2-way ANNOVA.

[0029] FIG. 7 shows a schematic representation of a linearized SAM002 that is used as a template for production of sa-mRNA. The definitions of the abbreviations in the nucleotide sequence map are as follows: 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, Puromycin R is the puromycin resistance gene, 3' UTR is a 3' untranslated region and contigs are subgenomic intervals generated as vectors to facilitate sequencing and numbered for the identification of mutations after directed evolution.

[0030] FIG. 8 shows survival rates (Y-axis) versus days post-B16F10 cell inoculation (X-axis) treated by 10 μ g SamRNA encoding with mIL-12 and 1 μ g modified mRNA encoding with VEE, SFV4 or SIN groups. The P-Values were determined by a Comparison of Survival Curves (Kaper-myer) test.

[0031] FIGS. 9A-9D show the optimization of therapeutic payloads of CATP and mechanism of long-term immune memory by CATP. FIG. 9A shows the tumor area (Y-axis) versus days post-B16F10 cell inoculation (X-axis) and survival rate (Y-axis) versus days post-B16F10 cell. FIG. 9B shows body weight changes (Y-axis) versus days post-B16F10 cell inoculation (X-axis). FIG. 9C shows inoculation (X-axis) and survival rates (Y-axis) versus days post-B16F10 cell. FIG. 9D shows re-challenges of tumor free mice in the group LNP encapsulated 5 μ g of SamRNA encoding with mouse mutant IL-18 plus 5 μ g of SamRNA encoding mouse IL-12 and 1 μ g of modified mRNA encoding with SFV4 capsids/envelop proteins. The cured mice (n=10) and naïve mice (n=7) matched with age and sex and were challenged with 0.1 million B16F10 cells. Results are shown as survival rate (Y-axis) versus days post-B16F10 cell inoculation (X-axis).

[0032] FIGS. 10A-C show schemes of CATP by different viral envelopes and capsids. FIG. 10A shows illustrations of constructs for CATP. FIG. 10B-10C show a comparison of therapeutic efficacies by CATP with VEE, SIN, and SFV4 capsids/envelop. Shown are tumor area (Y-axis) versus days post cancer cell inoculation (X-axis). FIG. 10B shows body weight changes (Y-axis) versus days post cancer cell inoculation (X-axis). FIG. 10C shows p-values labeled, which was determined by two-way ANOVA test.

[0033] FIGS. 11A-C show the optimization of therapeutic payloads of CATP in MC38 colon cancer model. The P-Values labeled was determined by a two-way ANOVA test or Comparison of Survival Curves (Kaper-myer) test. FIG. 11A shows the tumor area (Y-axis) versus days post MC38 cancer cell inoculation (X-axis). FIG. 11B shows the survival rate (Y-axis) versus days post MC38 cancer cell inoculation (X-axis). FIG. 11C shows the body weight changes (Y-axis) versus days post MC38 cancer cell inoculation (X-axis).

[0034] FIGS. 12A-12E show the therapeutic efficacy of CATP with optimized therapeutic payloads and SFV4 capsids in CT26 colon cancer and KPC (P53^{mut} KRas^{G12D}) pancreatic duct cancer model. The P-Values labeled was determined by a two-way ANOVA test or Comparison of Survival Curves (Kaper-myer) test. FIGS. 12A-12C used six- to eight-week-old Balb/c mice. FIGS. 12D-12E used C57BL/6 mice (n=5 per group). FIG. 12A and FIG. 12D

show tumor area (Y-axis) versus days post cancer cell inoculation (X-axis). FIG. 12B show survival rate (Y-axis) versus days post cancer cell inoculation (X-axis). FIGS. 12C and 12E show body weight changes (Y-axis) versus days post cancer cell inoculation (X-axis).

[0035] FIGS. 13A-13C show CATP with mouse IL-12 and mutant IL-18 forms long-term memory against tumor recurrences indicated by re-challenges of tumor free mice. The P-Values labeled was determined by a two-way ANOVA test or Comparison of Survival Curves (Kaper-myer) test. FIG. 13A shows the survival rate (Y-axis) versus days post rechallenges of MC38 cancer cell (X-axis). FIG. 13B shows the tumor area (Y-axis) versus days post-inoculation of tumor cells B16F10 cells (X-axis). FIG. 13C shows the tumor area (Y-axis) versus days post-inoculation of tumor cells MC38 cells (X-axis).

[0036] Note that any one of more of the illustrative components of the molecules and methods are optional and the present disclosure includes aspects that contain fewer than all of the illustrated elements.

DETAILED DESCRIPTION

[0037] The disclosure relates to novel compositions, and methods for the preparation, manufacture and therapeutic use thereof. In one aspect, the present disclosure provides a composition comprising: a first nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) encoding at least one gene of interest (GOI) or a plurality of GOIs; a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein; and at least one payload delivery system, wherein the at least one payload delivery system is a non-viral payload delivery system and wherein the payload is at least one nucleic acid construct. In some aspects, the at least one virus structural protein encodes a viral capsid protein, viral envelope protein, or a combination thereof. In some aspects, the present disclosure provides pharmaceutical compositions comprising the composition of the disclosure and methods of treating a proliferative disease.

[0038] It is important to note that while many of the approaches described in this specification and the examples given are focused on cancer treatment, they are equally applicable to other uses, such as for vaccine development, gene therapy or gene regulation.

[0039] Although the present disclosure is described in detail below, it is to be understood that this disclosure is not limited to the particular methodologies, protocols and reagents described herein as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the present disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

[0040] In one aspect, the present disclosure provides constructs of the sa-mRNAs and variations thereof as shown and described, and methods of making and using the constructs. The present disclosure includes noncytopathic and cytopathic versions of the sa-mRNAs and variations thereof. The present disclosure includes sequences and engineering of conjugations between elements of the constructs as shown and described. The present disclosure includes self-amplifying mRNAs that reduce the transcription numbers of sa-mRNA (e.g., nsP3) and subgenome (e.g., eGFP) to make

less-cytopathic versions of the sa-mRNA. The present disclosure includes methods and constructs including structure-based engineering to control replication rate and interferon responses of sa-mRNAs.

[0041] In one aspect, the present disclosure provides constructs of the oncolytic viruses and pseudoviral variations thereof as shown and described, and methods of making and using the constructs. The present disclosure includes pseudoviral variations of oncolytic viruses that are replication-defective or replication-impaired.

[0042] The present disclosure includes a composition comprising a payload delivery system, and nucleic acid constructs encoding at least an oncolytic virus and sa-mRNA encoding one or more GOI for use for the treatment of proliferative diseases such as cancer.

[0043] The present invention relates to a method for treating a disease or disorder comprising administering a composition comprising a payload delivery system, one or more nucleic acid construct(s) encoding at least one structural protein an oncolytic virus and one or more GOI encoded by an sa-mRNA. In particular, the composition as described herein is for use for treating a proliferative disease and, especially, for treating a cancer in a subject having or at risk of having a cancer.

[0044] Although the present disclosure is described in detail below, it is to be understood that this disclosure is not limited to the particular methodologies, protocols and reagents described herein as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the present disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

[0045] Additionally, several documents are cited throughout the text of this specification. Each of the documents cited herein (including all patents, patent applications, scientific publications, manufacturer's specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety.

Definitions

[0046] As used herein, the terms "approximately" and "about," as applied to one or more values of interest, refer to a value that is $\pm 10\%$ of the recited value.

[0047] As used herein, the terms "obtained from", "derived from", "originating" and "originate" are used to identify the original source of a component (e.g. polypeptide, nucleic acid molecule) but is not meant to limit the method by which the component is made which can be, for example, by chemical synthesis or recombinant means.

[0048] As used herein, the term "oncolytic virus" refers to a virus capable of selectively replicating in dividing cells (e.g. a proliferative cell such as a cancer cell) with the aim of slowing the growth and/or lysing said dividing cell, either in vitro or in vivo, while showing no or minimal replication in non-dividing cells. Typically, an oncolytic virus contains a viral genome packaged into a viral particle (or virion) and is infectious (i.e. capable of infecting and entering into a host cell or subject). A oncolytic virus may be a pseudovirus.

[0049] As used herein, the term "pseudoviral particle", "pseudovirus particle" or "pseudovirus" refers to a recombinant virus which lack certain gene sequences of the wild-type (WT) virus(es) from which it was derived. For

example, a pseudovirus may be composed of core or backbone and surface proteins derived from different viruses and/or are able to infect susceptible host cells but can only replicate intracellularly for a single round. For example, a pseudoviral particle may be composed of at least one virus structural protein derived from the same or different wild-type viruses.

[0050] As used herein, the term "pseudo-viral particle sequence", "defective viral genome", or "DVG" means that the nucleic acid sequence/viral genome contains all the genes necessary for the expression of a pseudo-retroviral particle with the exception of the envelope (env) or capsid (cap) gene and possibly comprising one or more mutations to attenuate infectiveness of the pseudo-retroviral system, modify the tropism of the pseudo-retroviral system compared to the virus from which it was derived, or deliver therapeutic genes.

[0051] As used herein the term "virus structural protein" or "structural protein" refer to a viral capsid protein, a viral envelope protein, a fragment thereof or a complex thereof. Thus, virus structural proteins used for the present invention may consist of or comprise a capsid protein and/or an envelope protein and/or a fragment thereof. Some viruses are naturally enveloped. Such viruses include, but are not limited to, the Retroviridae (e.g. HIV, Moloney Murine Leukaemia Virus, Feline Leukaemia Virus, Rous Sarcoma Virus), the Coronaviridae, the Herpesviridae, the Hepadnaviridae, and the Orthomyxoviridae (e.g. Influenza Virus). However, naturally non-enveloped viruses may form envelopes and these are also encompassed by the invention. Naturally non-enveloped viruses include the Picornaviridae, the Reoviridae, the Adenoviridae, the Papillomaviridae and the Parvoviridae.

[0052] As used herein the term "viral capsid", "capsid" or "cap" refer to a viral capsid protein. As used herein the term "viral envelope", "envelope" or "env" refer to a viral envelope protein.

[0053] As used herein, the terms "gene of interest," "genes of interest," "gene or genes of interest," or "GOI" refers to the nucleotide sequence which encode the therapeutic polypeptide, prophylactic polypeptide, diagnostic polypeptide, antigen, non-coding gene that encodes regulatory structure, or reporter as a result of expression of the payload, such as a nucleic acid molecule, including a mRNA or a sa-mRNA. For example, a GOI, for the purposes of this disclosure, includes, but is not limited to, polynucleotides encoding immunomodulators (such as IL12 and IL21).

[0054] As used herein, the term "modified nucleotide" refers to a nucleotide that contains one or more chemical modifications (e.g. substitutions) in or on the nitrogenous base of the nucleoside (e.g., cytosine (C), thymine (T) or uracil (U)), adenine (A) or guanine (G)). A nucleotide analog can contain further chemical modifications in or on the sugar moiety of the nucleoside (e.g., ribose, deoxyribose, modified ribose, modified deoxyribose, six-membered sugar analog, or open-chain sugar analog), or the phosphate. These modified nucleobases can be engineered to provide polynucleotide molecules having enhanced properties, e.g., increased stability such as resistance to nucleases. There are more than 96 naturally occurring modified nucleosides found on mammalian RNA. See, e.g., Limbach et al, *Nucleic Acids Research*, 22(12):2183-2196 (1994). Modified nucleosides may include a compound selected from the following non-limiting group: pyridin-4-one ribonucleoside, 5-aza-uridine,

2-thio-5-aza-uridine, 2-thiouridine, 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyl-uridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine, 1-taurinomethyl-4-thio-uridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxyuridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, 5-aza-cytidine, pseudoisocytidine, 3-methyl-cytidine, N4-acetylcytidine, 5-formylcytidine, N4-methylcytidine, 5-hydroxymethylcytidine, 1-methyl-pseudoisocytidine, pyrrolo-cytidine, pyrrolo-pseudoisocytidine, 2-thio-cytidine, 2-thio-5-methyl-cytidine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-cytidine, 2-methoxy-5-methyl-cytidine, 4-methoxy-pseudoisocytidine, 4-methoxy-1-methyl-pseudoisocytidine, 2-aminopurine, 2, 6-diaminopurine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-aminopurine, 7-deaza-8-aza-2-aminopurine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diaminopurine, 1-methyladenosine, N6-methyladenosine, N6-isopentenyladenosine, N6-(cis-hydroxyisopentenyl)adenosine, 2-methylthio-N6-(cis-hydroxyisopentenyl)adenosine, N6-glycinylicarbamoyladenine, N6-threonylicarbamoyladenine, 2-methylthio-N6-threonylicarbamoyladenine, N6,N6-dimethyladenosine, 7-methyladenine, 2-methylthio-adenine, 2-methoxy-adenine, inosine, 1-methyl-inosine, wyosine, wybutosine, 7-deaza-guanosine, 7-deaza-8-aza-guanosine, 6-thio-guanosine, 6-thio-7-deaza-guanosine, 6-thio-7-deaza-8-aza-guanosine, 7-methyl-guanosine, 6-thio-7-methyl-guanosine, 7-methylinosine, 6-methoxy-guanosine, 1-methyl-guanosine, N2-methyl-guanosine, N2,N2-dimethyl-guanosine, 8-oxo-guanosine, 7-methyl-8-oxo-guanosine, 1-methyl-6-thio-guanosine, N2-methyl-6-thio-guanosine, N1-Methylpseudouridine-5'-Triphosphate, and N2,N2-dimethyl-6-thio-guanosine. The preparation of nucleotides and modified nucleotides and nucleosides are well-known in the art, e.g. from U.S. Pat. Nos. 4,373,071, 4,458,066, 4,500,707, 4,668,777, 4,973,679, 5,047,524, 5,132,418, 5,153,319, 5,262,530, 5,700,642 all of which are incorporated by reference in their entirety herein, and many modified nucleosides and modified nucleotides are commercially available.

[0055] As used herein, the term “regulatory element” refers to a nucleotide sequence that controls, at least in part, the transcription of a gene or genes of interest. Regulatory elements may include promoters, enhancers, and other nucleic acid sequences (e.g., polyadenylation signals) that control or help to control nucleic acid transcription or translation. Examples of transcription regulatory elements are described, for example, in Goeddel, *Gene Expression Technology: Methods in Enzymology* 185 (Academic Press, San Diego, Calif, 1990).

[0056] As used herein, the term “open reading frame”, “ORF”, “expression unit” or “coding region” refer to a sequence of nucleotides that is capable of transcribing or translating into one or more gene products. A gene product

may comprise, but is not limited to, a polypeptide, a protein, a regulatory structure, or a combination thereof.

[0057] As used herein, the term “operably linked” refers to a first molecule joined to a second molecule, wherein the molecules are so arranged that the first molecule affects the function of the second molecule. The two molecules may or may not be part of a single contiguous molecule and may or may not be adjacent. For example, a promoter is operably linked to a GOI if the promoter modulates transcription of said GOI in a cell. Additionally, two portions of a transcription regulatory element are operably linked to one another if they are joined such that the transcription-activating functionality of one portion is not adversely affected by the presence of the other portion. Two transcription regulatory elements may be operably linked to one another by way of a linker nucleic acid (e.g., an intervening non-coding nucleic acid) or may be operably linked to one another with no intervening nucleotides present.

[0058] As used herein, the term “linker” refers to a moiety connecting two moieties, for example, a nucleotide sequence added between two nucleotide sequences to connect said two nucleotide sequences. There is no particular limitation regarding the linker sequence.

[0059] As used herein, the term “subgenomic promoter”, is a promoter that can be used to transcribe the subgenome of alphaviruses encoding structural proteins by RNA dependent RNA polymerase encoded by nsP. When two or more subgenomic promoters are present in a nucleic acid comprising multiple expression units, the subgenomic promoters can be the same or different. In certain aspects, subgenomic promoters can be modified using techniques known in the art in order to increase or reduce viral transcription of the proteins, see e.g., U.S. Pat. No. 6,592,874, which is incorporated by reference in its entirety herein.

[0060] As used herein, “expression” of a nucleic acid sequence refers to transcription of DNA into RNA (such as a regulatory element or mRNA, including sa-mRNA) and/or translation of an mRNA into a peptide, polypeptide, or protein, assembly of multiple polypeptides (e.g., heavy chain or light chain of antibody) into an intact protein (e.g., antibody) and/or post-translational modification of a polypeptide or fully assembled protein (e.g., antibody). In this application, the terms “expression” and “production,” and grammatical equivalents, are used interchangeably.

[0061] As used herein, the term “reporter” relates to a molecule, typically a peptide or protein, which is encoded by a reporter gene and measured in a reporter assay. Existing systems usually employ an enzymatic reporter (e.g. GFP or Luciferase) and measure the activity of said reporter.

[0062] As used herein, the phrase “biologically active” refers to a characteristic of any substance that has activity in a biological system and/or organism. The term “biologically active agent” refers to any agent that, when administered to a subject, has a therapeutic, diagnostic, and/or prophylactic effect and/or elicits a desired biological and/or pharmacological effect. Such agents include, but are not limited to, cytotoxins, radioactive ions, chemotherapeutic agents, small molecule drugs, proteins, and nucleic acids, such as therapeutic RNA. As used herein, the term “therapeutic RNA” refers to a RNA molecule that has activity and/or elicits a desired biological and/or pharmacological effect in a biological system and/or organism.

[0063] As used herein, the term “payload” refers to a moiety whose biological activity is desired to be delivered

(in)to and/or localize at a cell or tissue. Payloads include, but are not limited to biologically active agents and the like. In some aspects, the payload may be a nucleic acid that encodes a protein or polypeptide. In some aspects, the payload may include or encode a cytokine, a chemokine, an antibody or antibody fragment, a receptor or receptor fragment, an enzyme, an enzyme inhibitor, a hormone, a lymphokine, a plasminogen activator, a natural or modified immunoglobulin or a fragment thereof, an antigen, a chimeric antibody receptor, variable or hypervariable regions of light and/or heavy chains of an antibody (V_L , V_H), variable fragments (Fv), Fab' fragments, F(ab')₂ fragments, Fab fragments, single chain antibodies (scAb), single chain variable regions (scFv), complementarity determining regions (CDR), domain antibodies (dAbs), single domain heavy chain immunoglobulins of the BHH or BNAR type, single domain light chain immunoglobulins, or other polypeptides known in the art containing an AB capable of binding target proteins or epitopes on target proteins, or any other desired biological macromolecule. In some aspects, the payload may include or encode a regulatory element. In some aspects, the payload may include or encode a small interfering RNA (siRNA), a micro-RNA (miRNA), an asymmetrical interfering RNA (aiRNA), a Dicer-substrate RNA (dsRNA), a small hairpin RNA (shRNA), small activating RNA (saRNA), transfer RNA (tRNA), or long intergenic non-coding (lincRNA).

[0064] As used herein, the term “transcription” comprises “in vivo transcription” and “in vitro transcription” wherein the term “in vitro transcription” relates to a method in which RNA, in particular sa-mRNA, is synthesized in vitro in a cell-free manner.

[0065] As used herein, two nucleic acids are substantially homologous when the nucleotide sequences have at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. Sequence homology for nucleic acids, which can also be referred to as percent sequence identity, is typically measured using sequence analysis software. See, e.g., the Sequence Analysis Software Package of the Genetics Computer Group (GCG), University of Wisconsin Biotechnology Center, 910 University Avenue, Madison, Wis. 53705. A typical algorithm used comparing a molecule sequence to a database containing a large number of sequences from different organisms is the computer program BLAST (Altschul, 1990; Gish, 1993; Madden, 1996; Altschul, 1997; Zhang, 1997), especially blastp or tblastn (Altschul, 1997).

[0066] As used herein, “encapsulation efficiency” refers to the amount of a biological agent that becomes part of a nanoparticle composition, relative to the initial total amount of biologically active agent used in the preparation of a nanoparticle composition. For example, if 97 mg of biologically active agent are encapsulated in a nanoparticle composition out of a total 100 mg of biologically active agent initially provided to the composition, the encapsulation efficiency may be given as 97%. As used herein, “encapsulation” may refer to complete, substantial, or partial enclosure, confinement, surrounding, or encasement.

[0067] As used herein, a “payload delivery system” or “delivery system” are systems to deliver biologically active

agents/payloads into target host cells. Such delivery systems include, for example lipid nanoparticle based delivery (Debs and Zhu (1993) WO 93/24640; Mannino and Gould-Fogerite (1988) BioTechniques 6(7): 682-691; Rose U.S. Pat. No. 5,279,833; Brigham (1991) WO 91/06309; and Feigner et al (1987) Proc. Natl. Acad. Sci. USA 84: 7413-7414), as well as use of viral vectors {e.g., adenoviral (see, e.g., Berns et al (1995) Ann. NY Acad. Sci. 772: 95-104; Ali et al (1994) Gene Ther. 1: 367-384; and Haddada et al. (1995) Curr. Top. Microbiol. Immunol. 199 (Pt 3): 297-306 for review), papillomaviral, retroviral (see, e.g., Buchscher et al. (1992) J. Virol. 66(5) 2731-2739; Johann et al. (1992) J. Virol. 66 (5): 1635-1640 (1992); Sommerfelt et al. (1990) Virol. 176:58-59; Wilson et al. (1989) J. Virol. 63:2374-2378; Miller et al, J. Virol. 65:2220-2224 (1991); Wong-Staal et al, PCT/US94/05700, and Rosenberg and Fauci (1993) in Fundamental Immunology, Third Edition Paul (ed) Raven Press, Ltd., New York and the references therein, and Yu et al, Gene Therapy (1994) supra.), and adeno-associated viral vectors (see, West et al (1987) Virology 160:38-47; Carter et al (1989) U.S. Pat. No. 4,797,368; Carter et al WO 93/24641 (1993); Kotin (1994) Human Gene Therapy 5:793-801; Muzyczka (1994) J. Clin. Invest. 94:1351 and Samulski (supra) for an overview of AAV vectors; see also, Lebkowski, U.S. Pat. No. 5,173,414; Tratschin et al (1985) Mol. Cell. Biol. 5(11):3251-3260; Tratschin, et al (1984) Mol. Cell. Biol. 4:2072-2081; Hermonat and Muzyczka (1984) Proc. Natl. Acad. Sci. USA, 81:6466-6470; McLaughlin et al (1988) and Samulski et al (1989) J. Virol., 63:03822-3828), and the like.

[0068] The nucleic acid construct(s) encoding the pseudoviral particle sequence/defective viral genome (DVG) and the GOI may be carried together, either by the same plasmid or payload delivery system; or separately by separate nucleic acid molecules and payload delivery systems. Three particularly useful delivery systems are (i) LNPs (ii) non-toxic and biodegradable polymer microparticles (iii) cationic submicron oil-in-water emulsions. In one aspect, the payload of the present disclosure is delivered using LNPs.

[0069] As used herein, a “nanoparticle composition” or “LNP formulation” refer to a type of non-viral payload delivery system comprising one or more lipids. Nanoparticle compositions are typically sized on the order of micrometers or smaller and may include a lipid bilayer.

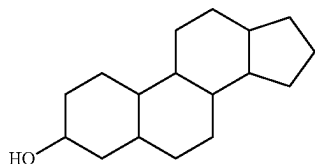
[0070] For example, the lipid component of a nanoparticle composition may include one or more cationic/ionizable, PEGylated, structural, or other lipids, such as phospholipids; or a LNP and lipid components of a nanoparticle composition according to the LNPs and lipid components disclosed in PCT Patent Application Nos. PCT/US2023/085919 and PCT/US2023/017777, which are fully incorporated herein by reference.

[0071] As used herein, a “lipid component” is that component of a nanoparticle composition that includes one or more lipids. For example, the lipid component may include one or more cationic/ionizable, PEGylated, structural, or other lipids, such as phospholipids.

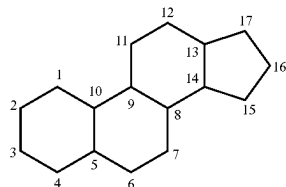
[0072] As used herein “pharmaceutically acceptable” refers to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues, organs, and/or bodily fluids of human beings and animals without

excessive toxicity, irritation, allergic response, or other problems or complications commensurate with a reasonable benefit/risk ratio.

[0073] As used herein the term “sterol” or “sterol derivative” refer to a class of compounds with a four ring 17 carbon cyclic structure which can further comprises one or more substitutions. A sterol derivative comprises any molecule having the 4-member ring structure characteristic of steroids and a hydroxyl (—OH) or ester (—OR) substitution at the 3-carbon position, for example, having the exemplary structure below:



The skilled artisan will understand that a sterol derivative can be further substituted at one or more of the other ring carbons, for example, with one or more non-alkyl substitutions, including alkyl groups, alkoxy groups, hydroxy groups, oxo groups, acyl groups, or a double bond between two or more carbon atoms:



Sterols useful in the compositions and methods of the present disclosure may be selected from the following non-limiting group: cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, tomatine, ursolic acid, alpha-tocopherol, cholesteryl chloroformate, sitosterol, ergosterol, lanosterol, desmosterol, or a derivative thereof.

[0074] As used herein, the terms “PEG lipid” or “PEGylated lipid” refer to a lipid comprising a polyethylene glycol component. For example, a PEG lipid may be selected from the following non-limiting group: PEG-modified phosphatidylethanolamines, PEG-modified phosphatidic acids, PEG-modified ceramides, PEG-modified dialkylamines, PEG-modified diacylglycerols, PEG-modified dialkylglycerols, and mixtures thereof. For example, a PEG lipid may be PEG-c-DMG, PEG-DMG, PEG-DLPE, PEG-DMIPE, PEG-DPPC, or a PEG-DSPE lipid.

[0075] As used herein, the terms “phospholipid” or “helper lipid” refer to a lipid that includes a phosphate moiety and one or more carbon chains, such as unsaturated fatty acid chains. A phospholipid may include one or more multiple (e.g., double or triple) bonds (e.g., one or more unsaturations). Particular phospholipids may facilitate fusion to a membrane. For example, a cationic phospholipid may interact with one or more negatively charged phospholipids of a membrane (e.g., a cellular or intracellular mem-

brane). Fusion of a phospholipid to a membrane may allow one or more elements of a lipid-containing composition to pass through the membrane permitting, e.g., delivery of the one or more elements to a cell. In general, phospholipids may include a phospholipid moiety and one or more fatty acid moieties.

[0076] Phospholipids useful in the compositions and methods of the disclosure may be selected from the non-limiting group consisting of 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-dilinoleoyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dimyristoyl-sn-glycero-phosphocholine (DMPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-diundecanoyl-sn-glycero-phosphocholine (DUPC), 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC), 1,2-di-O-octadecenyl-sn-glycero-3-phosphocholine (18:0 Diether PC), 1-oleoyl-2-cholesterylhemisuccinoyl-sn-glycero-3-phosphocholine (OChemPC), 1-hexadecyl-sn-glycero-3-phosphocholine (C16 Lyso PC), 1,2-dilinolenoyl-sn-glycero-3-phosphocholine, 1,2-diarachidonoyl-sn-glycero-3-phosphocholine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphocholine, 1,2-diphytanoyl-sn-glycero-3-phosphoethanolamine (ME 16.0 PE), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinoleoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilino-lenoyl-sn-glycero-3-phosphoethanolamine, 1,2-diarachidonoyl-sn-glycero-3-phosphoethanolamine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphoethanolamine, 1,2-dioleoyl-sn-glycero-3-phospho-rac-(1-glycerol) sodium salt (DOPG), and sphingomyelin. In some aspects, a nanoparticle composition includes DSPC. In certain aspects, a nanoparticle composition includes DOPE. In some aspects, a nanoparticle composition includes both DSPC and DOPE.

[0077] As used herein, the term “substantially” refers to the qualitative condition of exhibiting total or near-total extent or degree of a characteristic or property of interest.

[0078] As used herein, “size” or “mean size” in the context of nanoparticle composition, such as an LNP, refers to the mean diameter of the LNP.

[0079] The present disclosure may include uses or methods of administration that may include intravenous, intranasal, intratracheal, intracerebral, intratumoral, intraperitoneal, intramuscular, intradermal, intravitreal, subretinal, subcutaneous, or other methods of delivering a composition to a subject. A method of administration may be selected to target delivery (e.g., to specifically deliver) to a specific region, cell, tumor, organ, or system of a body.

[0080] As used herein, the terms “analog,” “mutant,” “modified” and “variant” refer to a molecule (polypeptide or nucleic acid) exhibiting one or more sequence modification(s) with respect to the native or wildtype counterpart. Any modification(s) can be envisaged, including substitution, insertion and/or deletion of one or more nucleotide/amino acid residue(s). Preferred are analogs that retain a degree of sequence identity of at least 80%, preferably at least 85%, more preferably at least 90%, and even more preferably at least 98% identity with the sequence of the native counterpart.

[0081] As used herein, the term “host cell” should be understood broadly without any limitation concerning particular organization in tissue, organ, or isolated cells. Such cells may be of a unique type of cells or a group of different types of cells such as cultured cell lines, primary cells and

dividing cells. In the context of the invention, the term “host cells” include prokaryotic cells, lower eukaryotic cells such as yeast, and other eukaryotic cells such as insect cells, plant and mammalian (e.g. human or non-human) cells as well as cells capable of producing the oncolytic virus and/or therapeutic RNA(s) for use in the invention. This term also includes cells which can be or has been the recipient of the biologically active agents described herein as well as progeny of such cells.

[0082] The expression “target cell” means the cell which it is desired to treat by introduction of a biologically active agent.

[0083] As used herein, the term “proliferative disease” encompasses any disease or condition resulting from uncontrolled cell growth and spread including cancers as well as diseases associated to an increased osteoclast activity (e.g. rheumatoid arthritis, osteoporosis, etc.) and cardiovascular diseases (restenosis that results from the proliferation of the smooth muscle cells of the blood vessel wall, etc.). The term “cancer” may be used interchangeably with any of the terms “tumor”, “malignancy”, “neoplasm”, etc. These terms are meant to include any type of tissue, organ or cell, any stage of malignancy (e.g. from a prelesion to stage IV).

Oncolytic Virus

[0084] The oncolytic virus for use in the present invention can be obtained from any member of virus identified at present time provided that it is oncolytic by its propensity to selectively replicate and kill dividing cells as compared to non-dividing cells. It may be a native virus that is naturally oncolytic or may be engineered by modifying one or more viral genes so as to increase tumor selectivity and/or preferential replication in dividing cells, such as those involved in DNA replication, nucleic acid metabolism, host tropism, surface attachment, virulence, lysis and spread (see for example Kirm et al., 2001, *Nat. Med.* 7: 781; Wong et al., 2010, *Viruses* 2: 78-106). One may also envisage placing one or more viral gene(s) under the control of event or tissue-specific regulatory elements (e.g. promoter).

[0085] Exemplary oncolytic viruses include, without limitation, Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus (EVEV), Mucambo virus (MUCV), Pixuna virus (PIXV), Western Equine Encephalitis virus (WEE), Sindbis virus (SINV), Semliki Forest virus (SFV), Middelburg virus (MIDV), Chikungunya virus (CHIKV), O'nyong-nyong virus (ONNV), Ross River virus (RRV), Barmah Forest virus (BFV), Getah virus (GETV), Sagiya virus (SAG), Bebaru virus (BEBV), Mayaro virus (MAYV), Una virus (UNAV), Aura virus (AURAV), Whataroa virus (WHAV), Babanki virus (BBKV), Kyzylagach virus (KYZV), Highlands J virus (HJV), Fort Morgan virus (FMV), Ndumu virus (NDUV), Buggy Creek virus (BCRV), reovirus, Seneca Valley virus (SVV), vesicular stomatitis virus (VSV), Newcastle disease virus (NDV), herpes simplex virus (HSV), morbillivirus virus, retrovirus, influenza virus, Sinbis virus, poxvirus, adenovirus, or the like.

[0086] In one aspect, the at least one virus structural protein encoded by the second nucleic acid construct is from an alphavirus, adenovirus, a HSV, an echovirus, a polio virus, a vaccinia virus, a measles virus, a vesicular stomatitis, an orthomyxovirus, a parvovirus, a maraba virus, a coxsackievirus, an autonomous parvovirus, a myxoma virus, a Newcastle disease virus, a reovirus, a seneca valley virus

morbillivirus virus, a retrovirus, an influenza virus, a sindbis virus, semliki forest virus, venezuelan equine encephalitis virus, or a poxvirus. In some aspects, the at least one virus structural protein is a H101 protein, a T-VEC protein, a ECHO-7 protein, a teserpaturev protein, a nadofaragene firadenovec protein or a HSV1 protein. In some aspects, the at least one virus structural protein is SEQ ID NO: 26, SEQ ID NO: 56, or SEQ ID NO: 57.

[0087] In one aspect, the oncolytic virus for use in the present invention is obtained from a VEE. Representative examples for use in the invention are described in the literature (e.g. Lundstrom, 2022, *Front. Mol. Biosci.* 9:864781).

[0088] In one aspect, the oncolytic virus for use in the present invention is a generated chimeric virus that is derived from Venezuelan equine encephalitis virus (VEE), the same technical principles as discussed herein may be applied to construct chimeric viruses derived from other viruses, such as alphaviruses (including western equine encephalitis virus, eastern equine encephalitis virus or other related viruses). The second nucleic acid construct encoding an mRNA encode structural proteins of VEE, which may include viral envelope glycoproteins, viral capsid proteins or both. A representative example of the strains of VEE from where the nucleic acid sequence is derived from may include but is not limited to the TC-83 strain.

Non-Viral Payload Delivery System

[0089] In some aspects, the non-viral payload delivery system of the composition of the disclosure is a lipidoid, a polymer, a core-shell nanoparticle, a nanoparticle mimic, a lipid nanoparticle (LNP), a polymeric nanoparticle, a micelle, liposome, a virus-like particle (VLP), a lipoplex, a polyplex or a lipopolyplex. In some aspects, the non-viral payload delivery system is a LNP non-viral payload delivery system. In some aspects, the LNP comprises an ionizable lipid, a phospholipid, a steroid, a polyethylene glycol (PEG) lipid, a modular lipid, or a combination thereof.

[0090] In some aspects, the diameter of the LNP is 1 m or shorter (e.g., 1 m, 900 nm, 800 nm, 700 nm, 600 nm, 500 nm, 400 nm, 300 nm, 200 nm, 175 nm, 150 nm, 125 nm, 100 nm, 75 nm, 50 nm, or shorter), e.g., when measured by dynamic light scattering (DLS), transmission electron microscopy, scanning electron microscopy, or another method. Nanoparticle compositions include, for example, lipid nanoparticles (LNPs), liposomes, lipid vesicles, and lipoplexes. In some aspects, nanoparticle compositions, such as LNPs, are vesicles including one or more lipid bilayers. In certain aspects, a nanoparticle composition includes two or more concentric bilayers separated by aqueous compartments. Lipid bilayers may be functionalized and/or cross-linked to one another. Lipid bilayers may optionally further include one or more ligands, proteins, or channels.

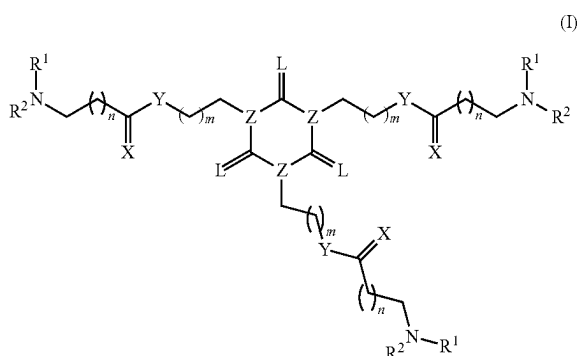
[0091] Typically the LNPs used as the delivery system in the research and development of new drugs, including FDA approved mRNA vaccines such as the mRNA COVID vaccines, and FDA approved siRNA therapies, such as the siRNA therapy for the treatment of polyneuropathy in people with hereditary transthyretin-mediated amyloidosis, use a four component LNP delivery system. In a four component LNP delivery system, phospholipids function to increase transfection efficacy of nucleic acids; cationic/ionizable lipids function to stabilize nucleic acids within the lipid nanoparticle; stabilizing lipids serve as the “lipid raft,”

which stabilizes the integrity of the LNP; and PEG-lipids inhibit aggregation and prevent clearance by macrophages, monocytes, or other phagocytic cells in vivo. The LNPs of the present disclosure may be a two-, three-, four-, five- or more component LNP.

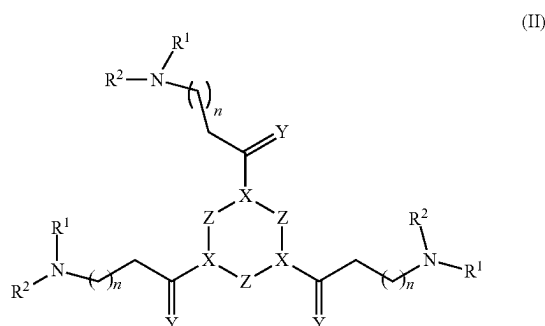
Cationic/Ionizable Lipids

[0092] As used herein, the terms “ionizable lipid” or “cationic lipid” are lipids that may have a positive or partial positive charge at physiological pH. A nanoparticle composition may include one or more ionizable lipids in addition to a lipid according to lipids disclosed in PCT Patent Application No. PCT/US2023/017777, which is fully incorporated herein. A nanoparticle composition may include one or more ionizable lipids in addition to a lipid according to Formula (I)-(VII).

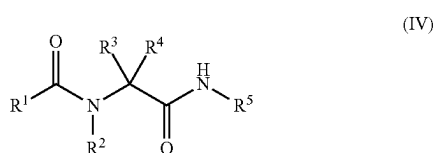
[0093] In one aspect, the present disclosure provides an LNP comprising an ionizable lipid compound selected from Formulae I, II, IV, V, IV, and VII, wherein Formula I is



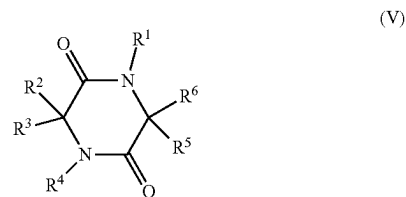
or a salt or isomer thereof; wherein Formula II is



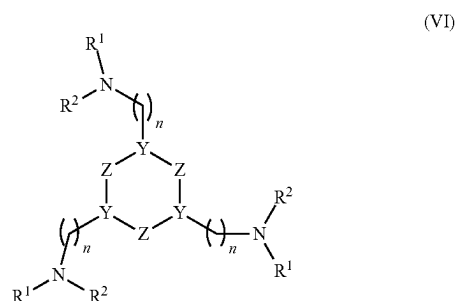
or a salt or isomer thereof; wherein Formula IV is



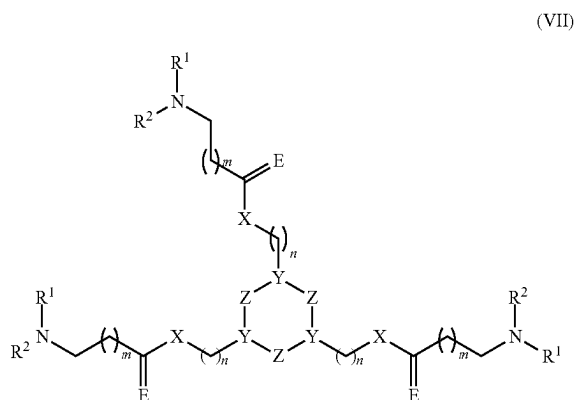
or a salt or isomer thereof; wherein Formula V is



or a salt or isomer thereof; wherein Formula VI is

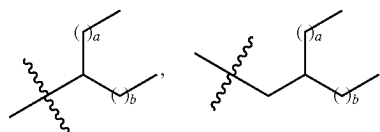


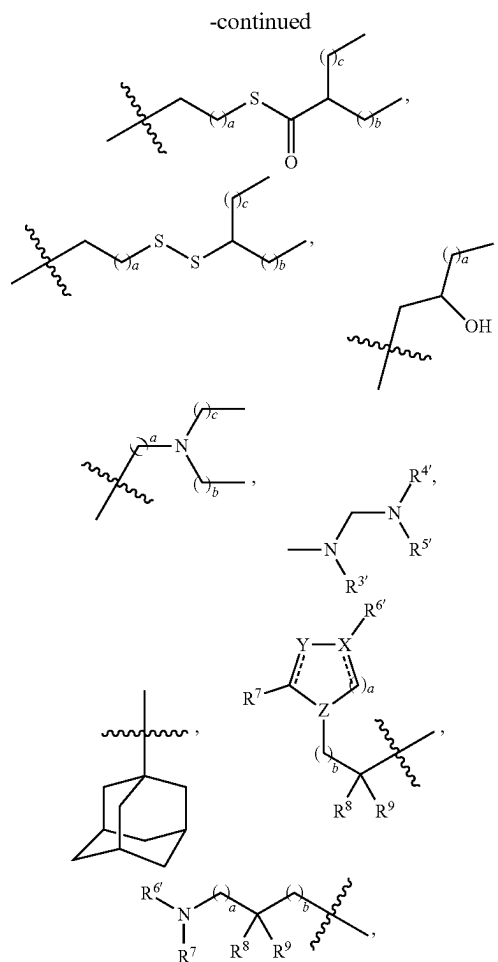
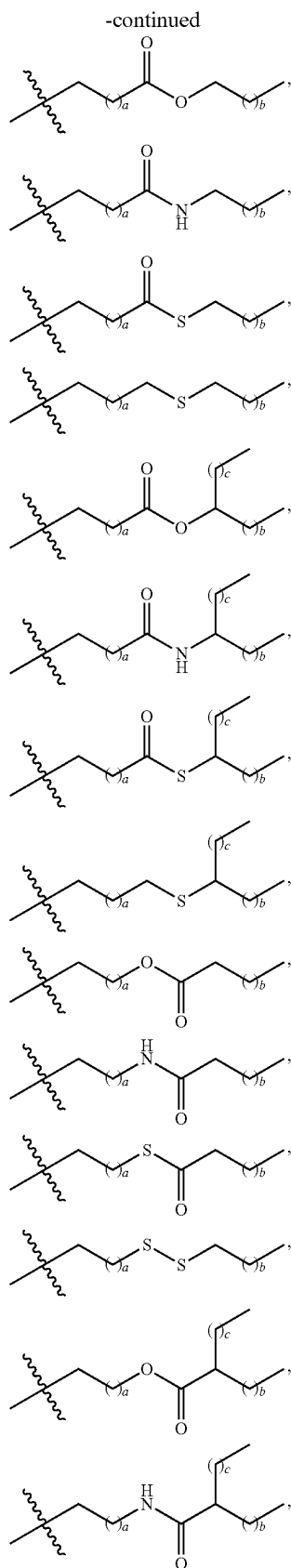
or a salt or isomer thereof; wherein Formula VII is



or a salt or isomer thereof; wherein, each m and n are independently an integer from 0-10;

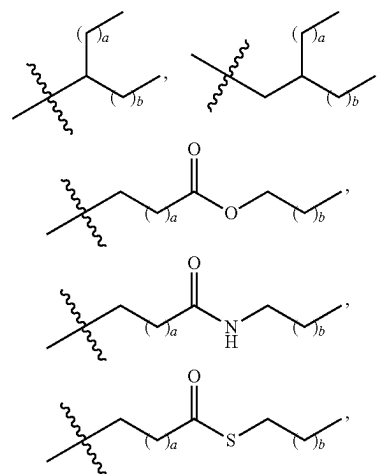
[0094] each R¹, R², R³, R⁴, R⁵ and R⁶ are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



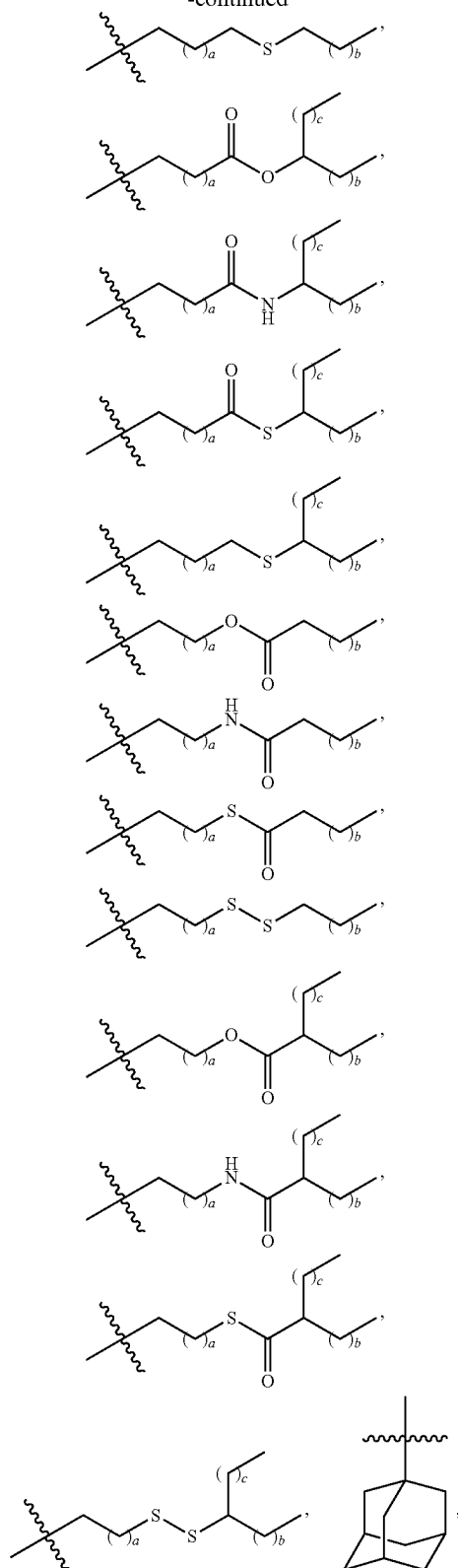


[0095] a, b and c are each independently an integer from 0-24;

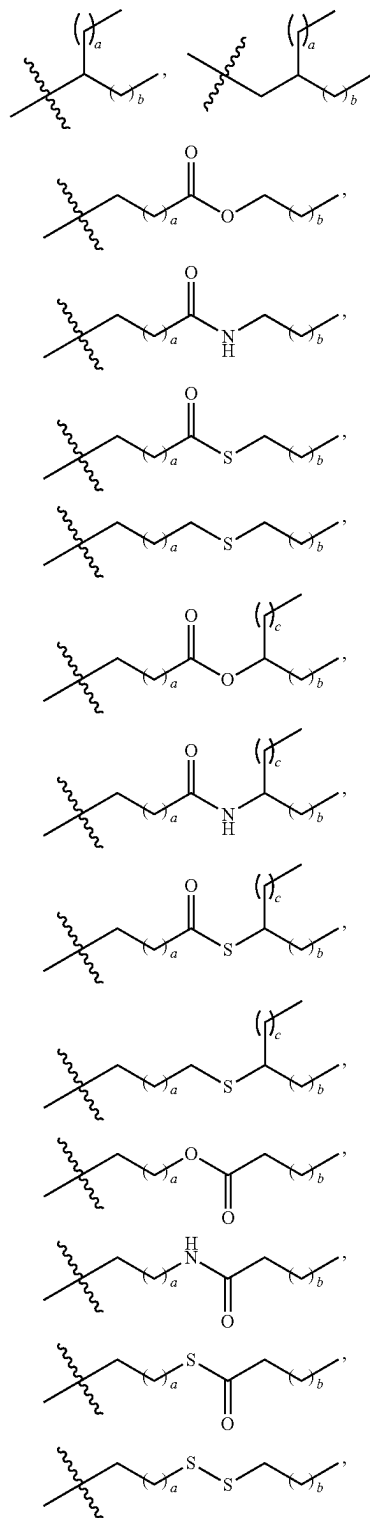
[0096] each R^{6'}, R⁷, R⁸ and R⁹ are independently selected from H, C1-C24 alkyl, C1-C24 alkenyl, C1-C24 alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl,



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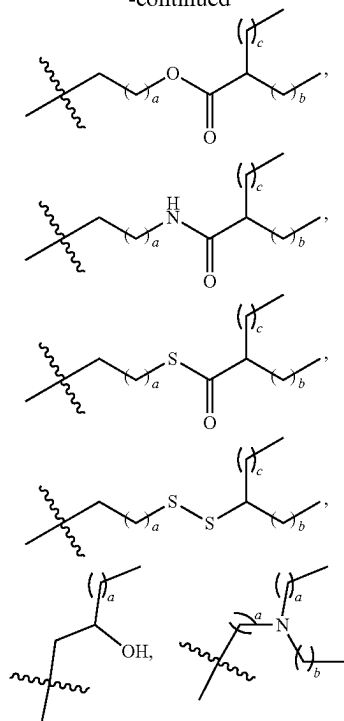


substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



[0097] each $R^{3'}$, $R^{4'}$ and $R^{5'}$ are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl,

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[0098] a, b and c are each independently an integer from 0-24;

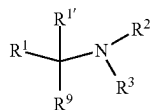
[0099] each X is independently selected from CH₂, NH, O, or S;

[0100] each Y is independently selected from CH₂, NH, O, or S;

[0101] each Z is independently selected from CH, N, O, or S; and

[0102] each E is independently selected from CH₂, NH, O, or S.

[0103] In some aspects, the present disclosure provides an LNP comprising an ionizable lipid compound selected from Formula III, wherein Formula III is

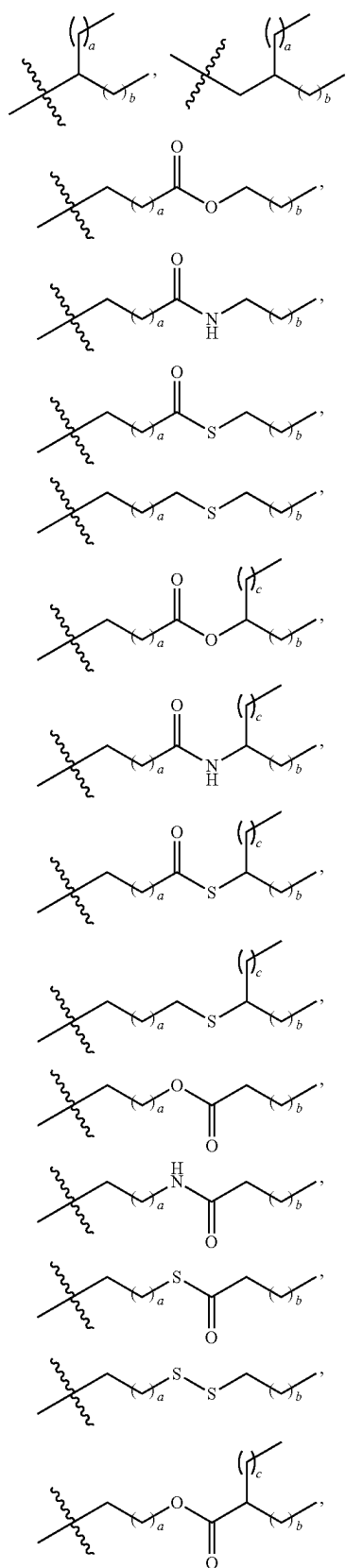


(III)

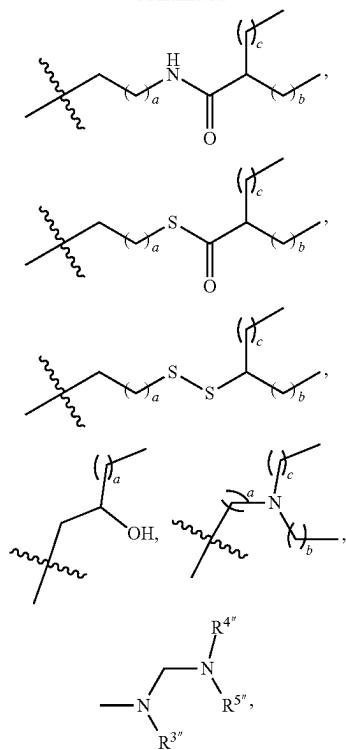
or a salt or isomer thereof, wherein

[0104] each n is independently an integer from 0-10;

[0105] each R¹, R^{1'}, R² and R³ are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,

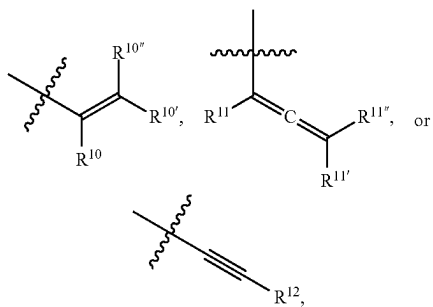


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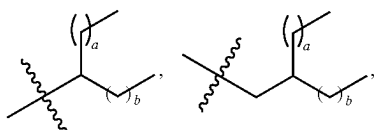


[0106] a, b and c are each independently an integer from 0-24;

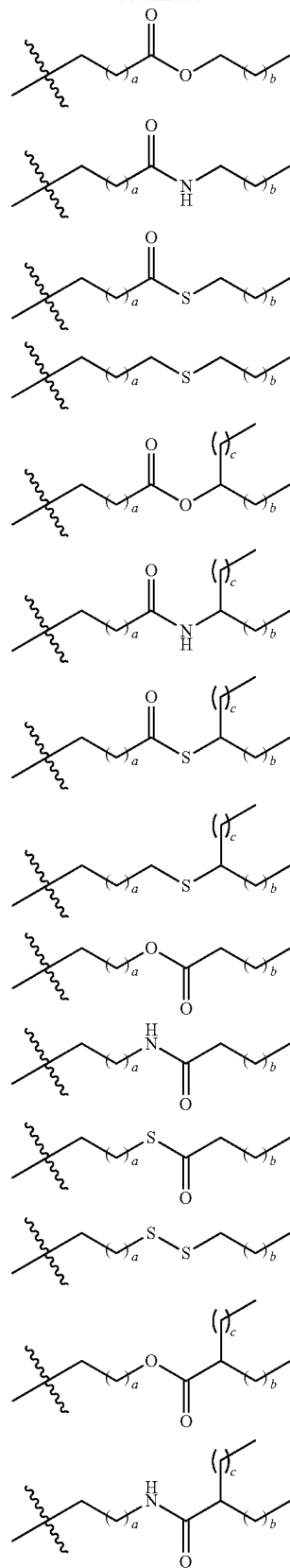
[0107] each R⁹ is independently selected from



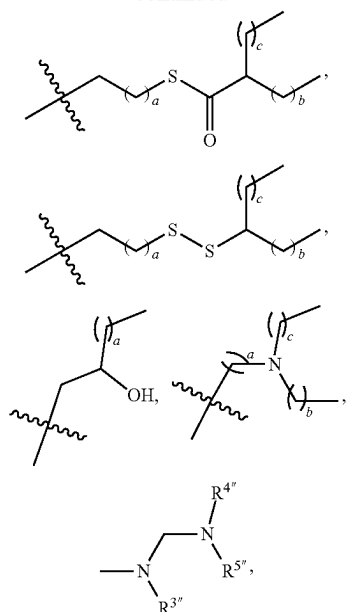
wherein each R¹⁰, R^{10'}, R^{10''}, R¹¹, R^{11'}, R^{11''}, and R¹² are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



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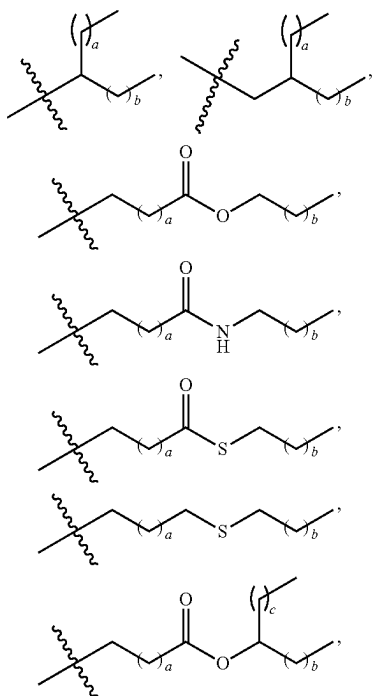


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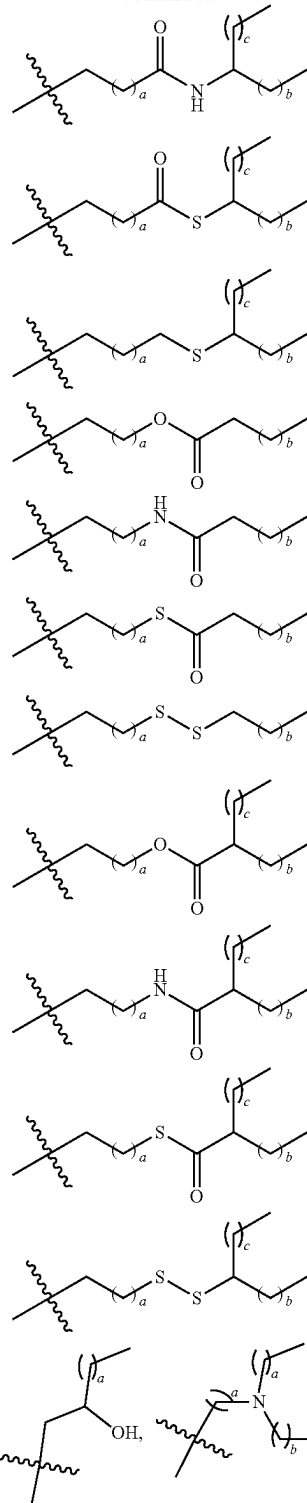


[0108] a, b and c are each independently an integer from 0-24;

[0109] R^{3''}, R^{4''}, and R^{5''} are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,

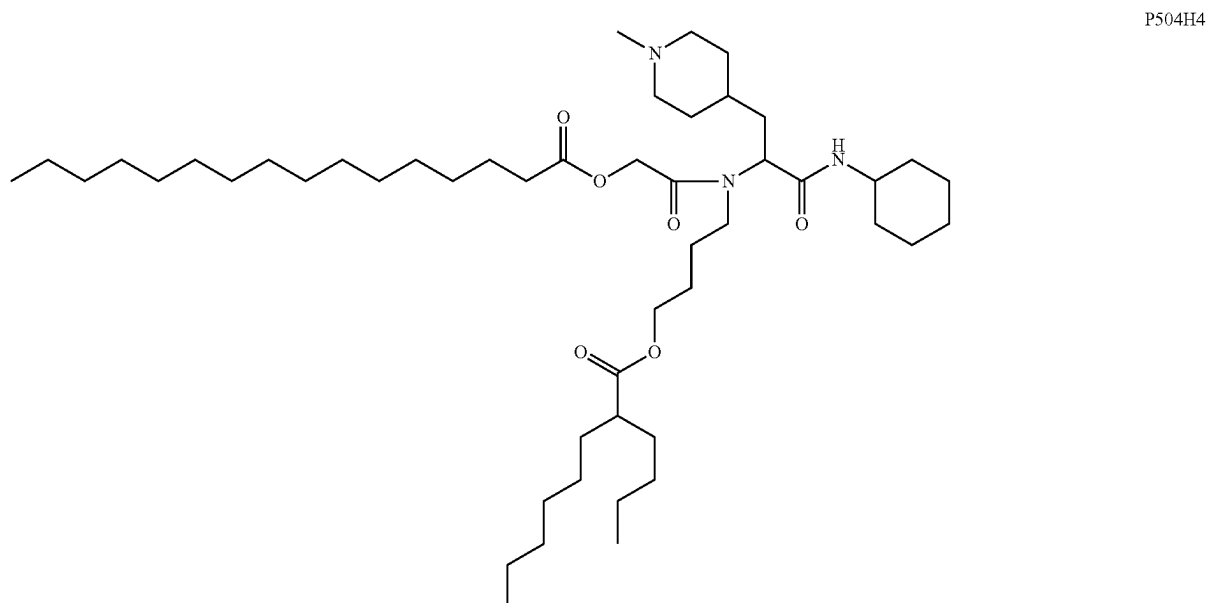
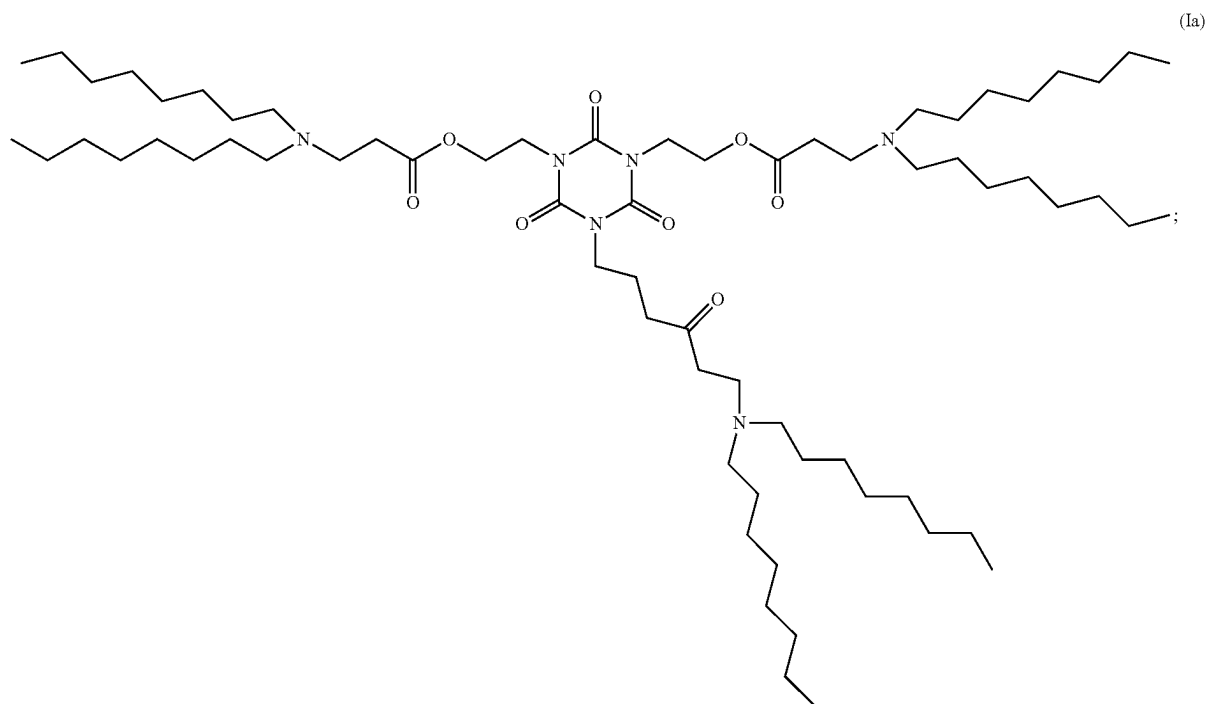


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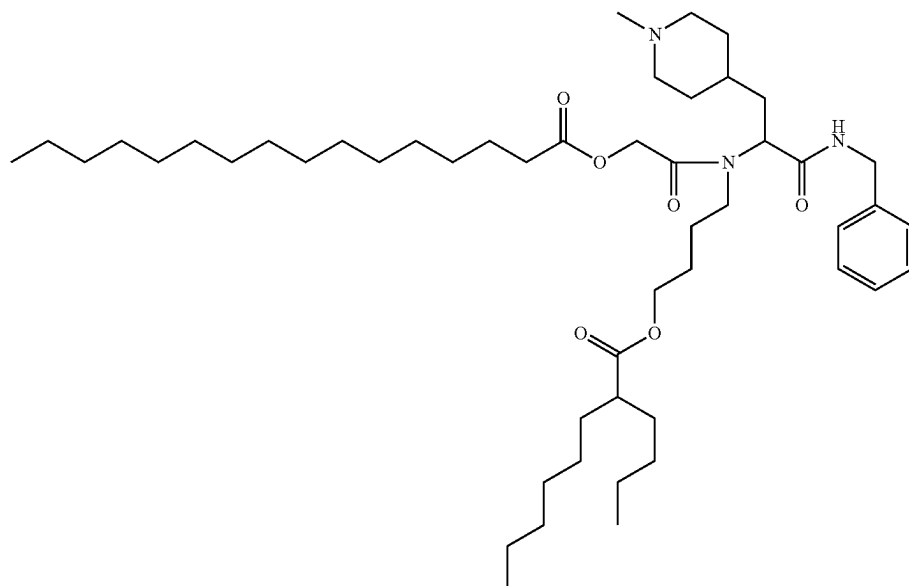
[0110] a, b and c are each independently an integer from 0-24

[0111] In some aspects, the LNP comprises an ionizable lipid compound, wherein the ionizable lipid is.

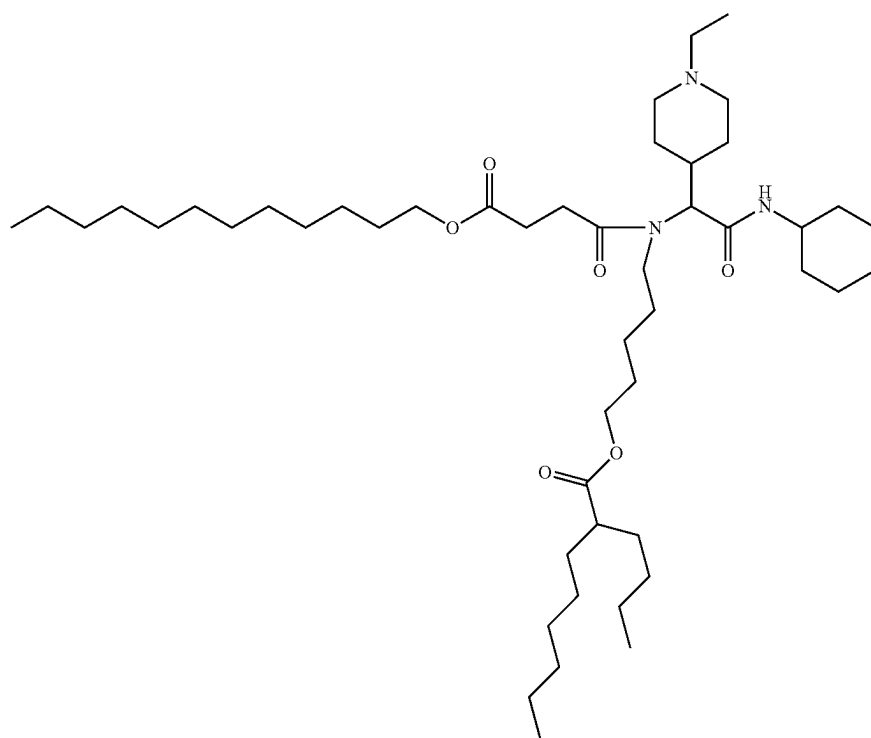


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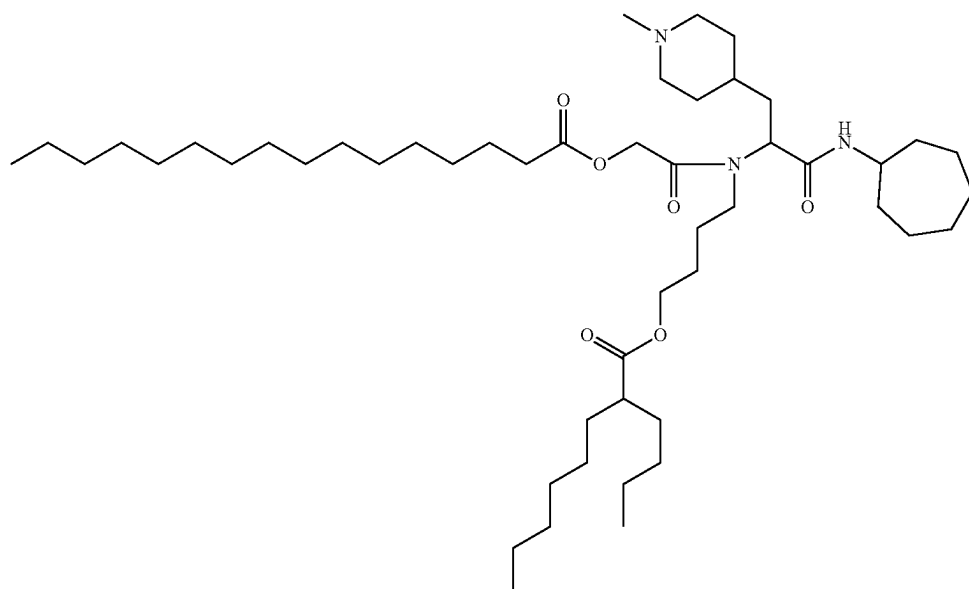


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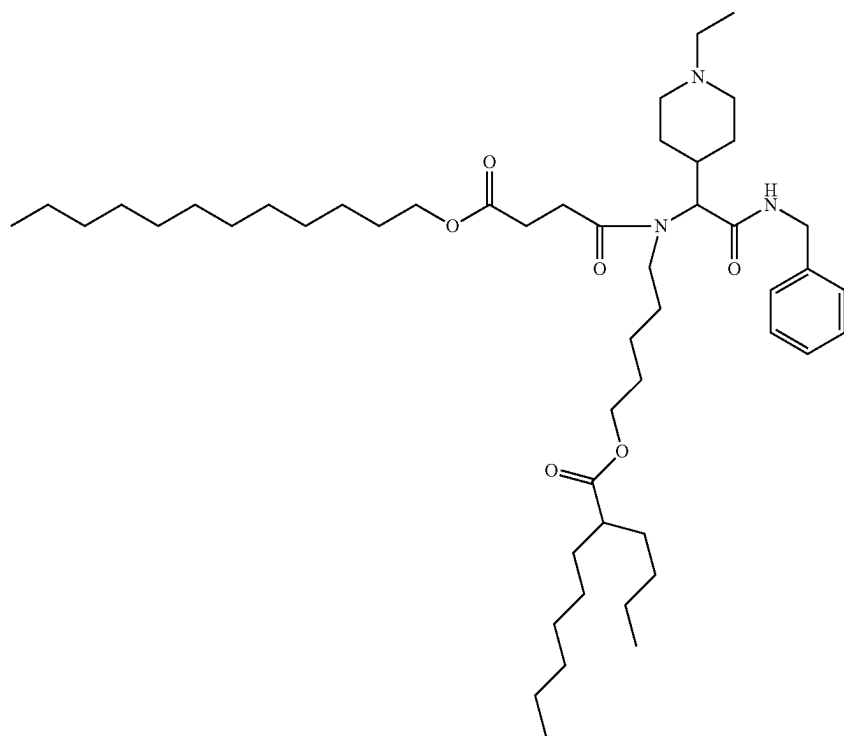


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P504H9

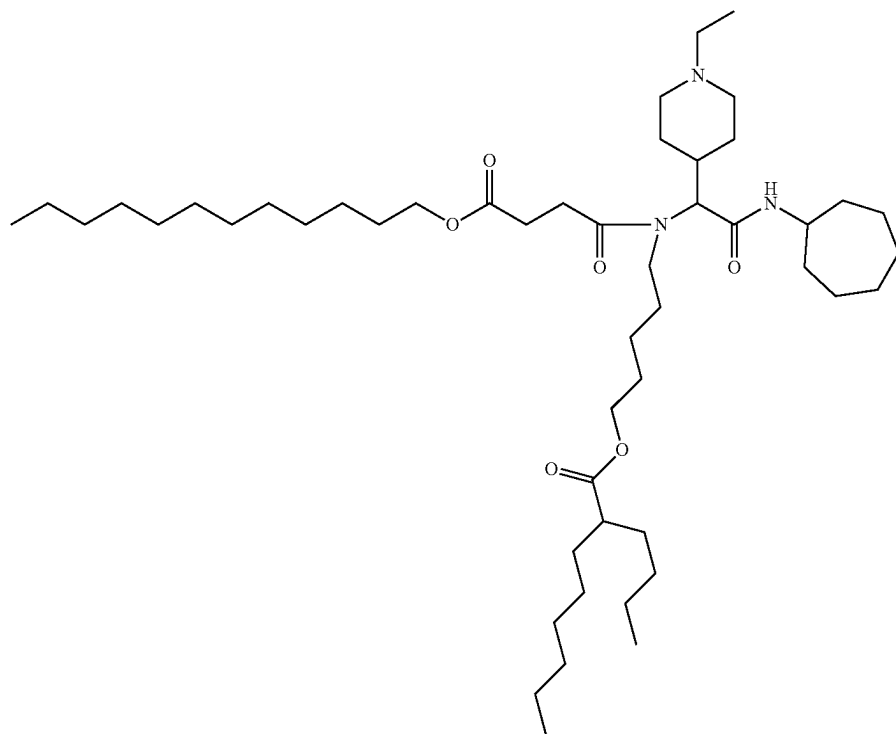


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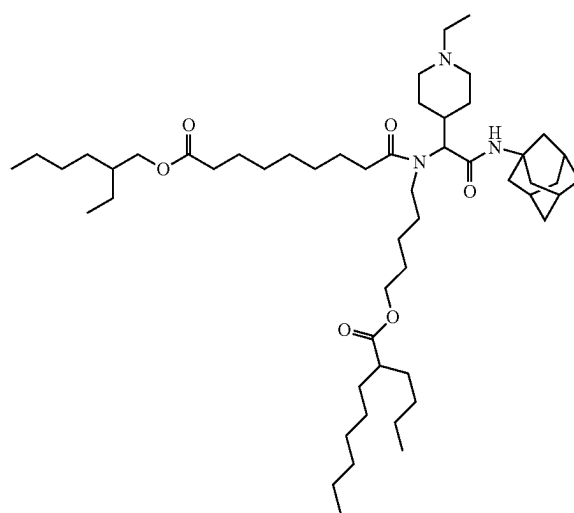
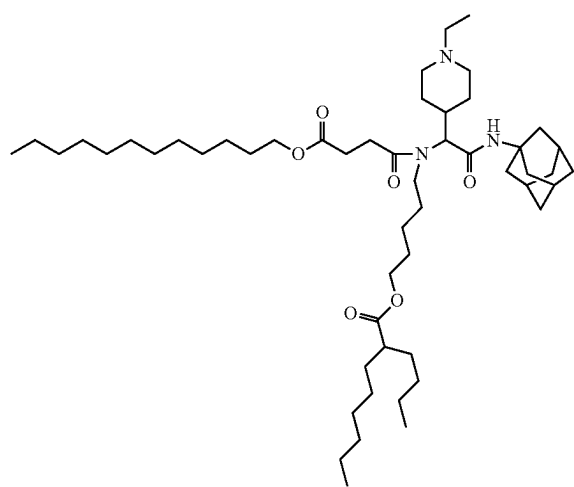
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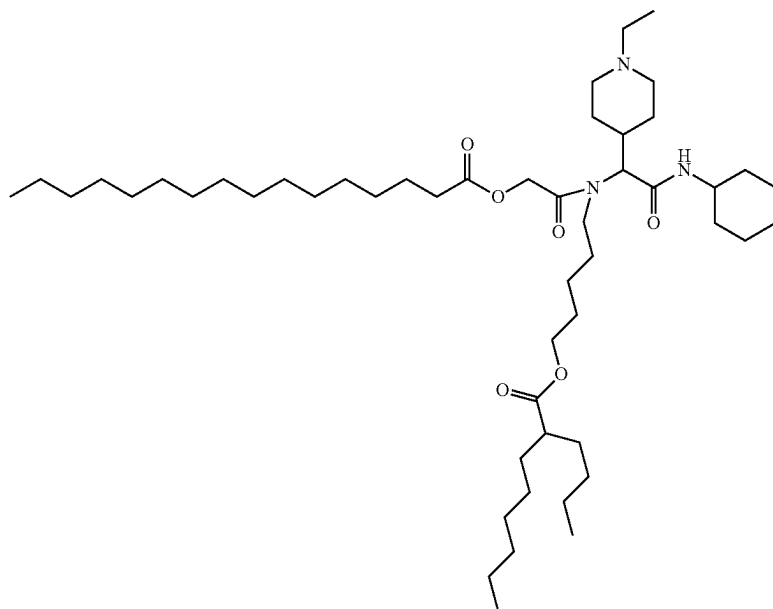
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P505C10

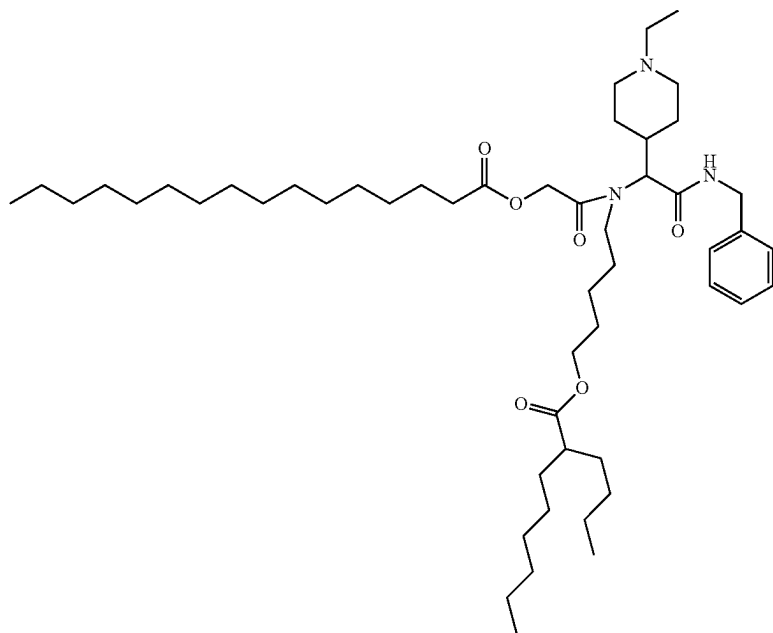


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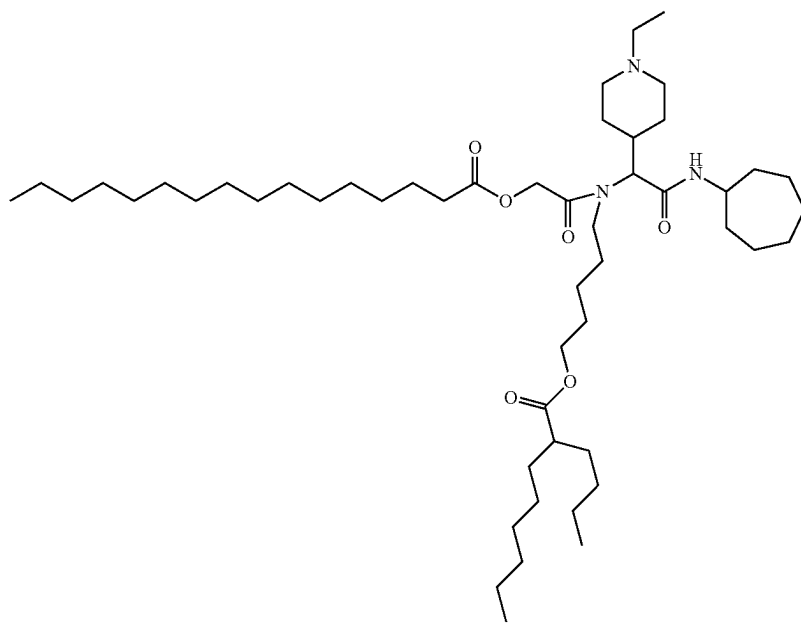


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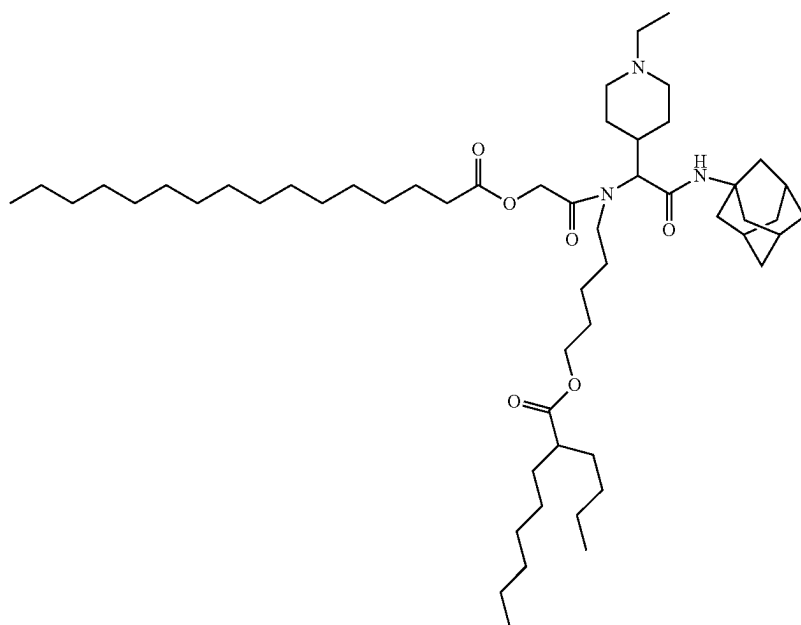


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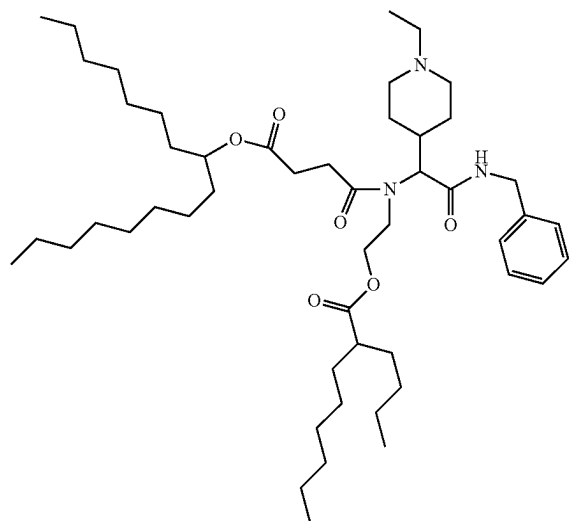


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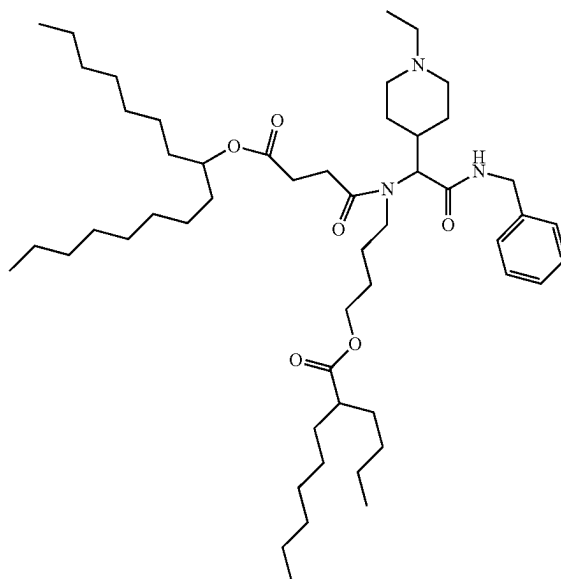


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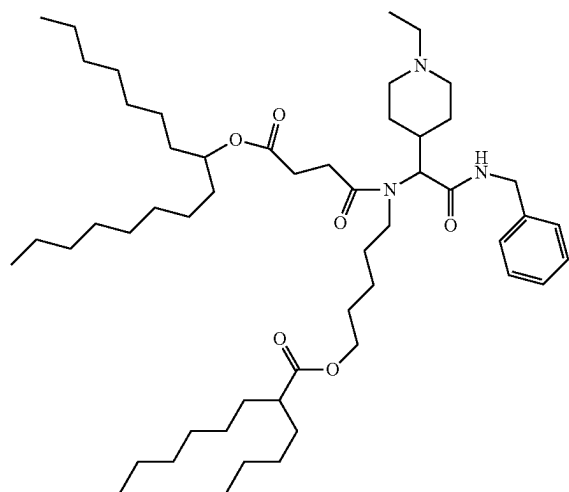
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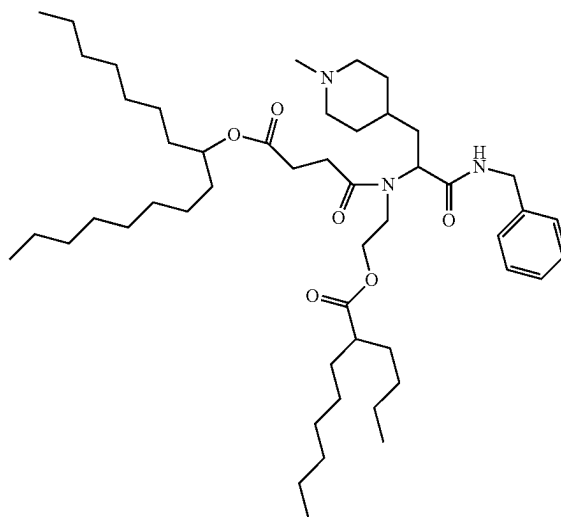
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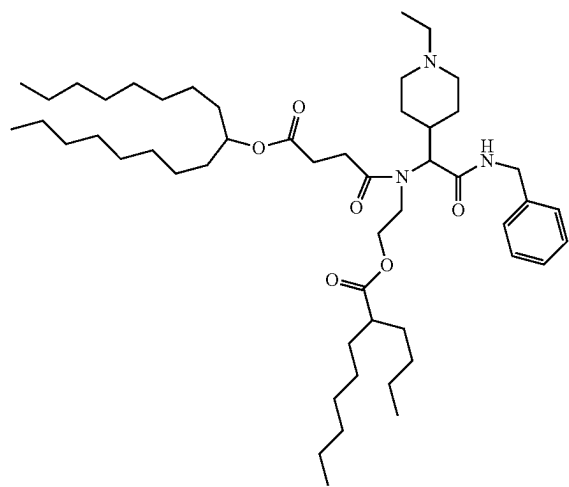
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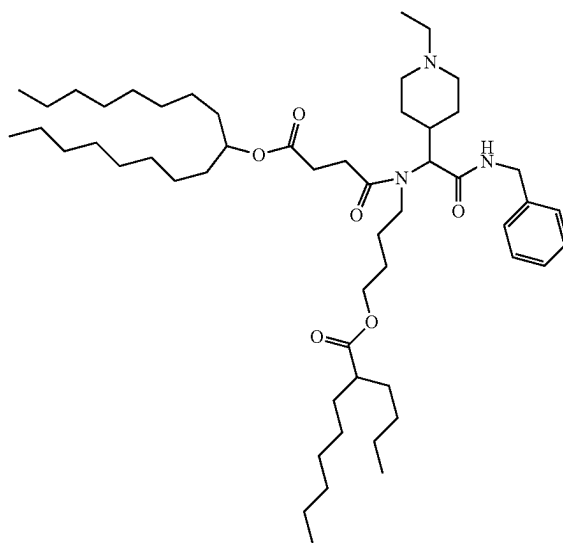
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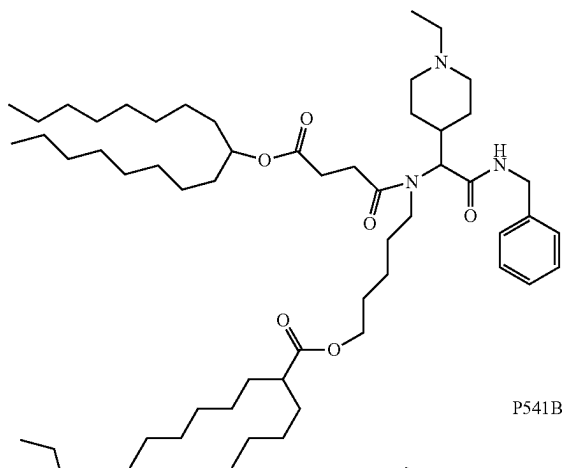
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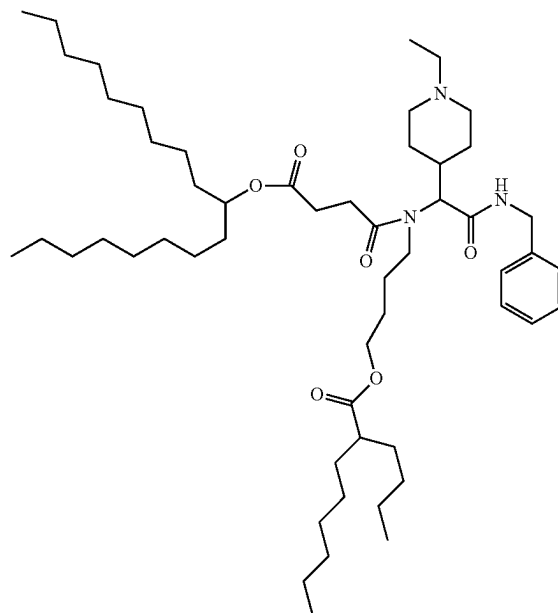


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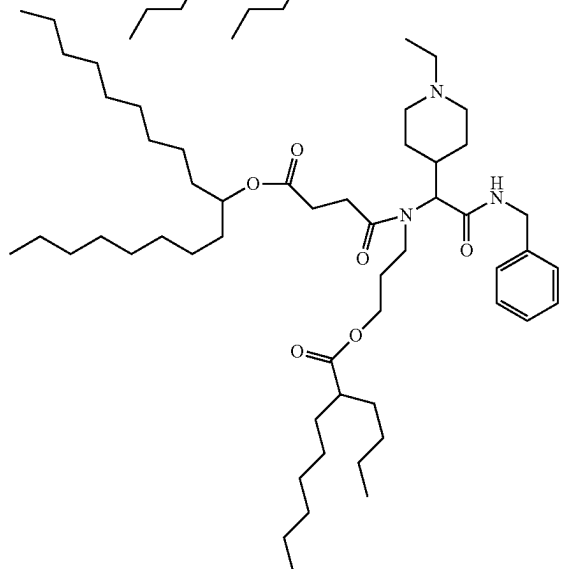


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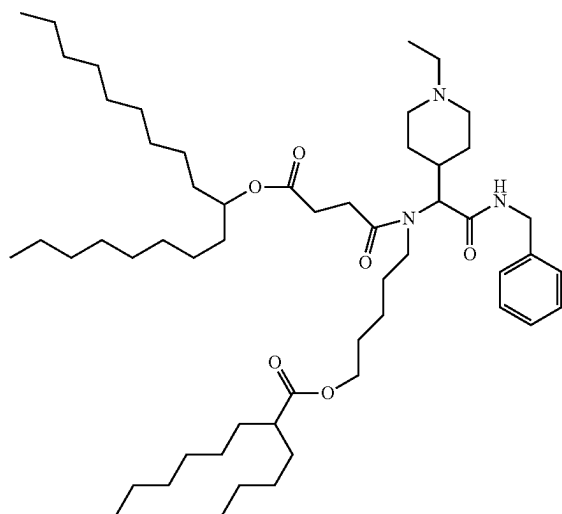


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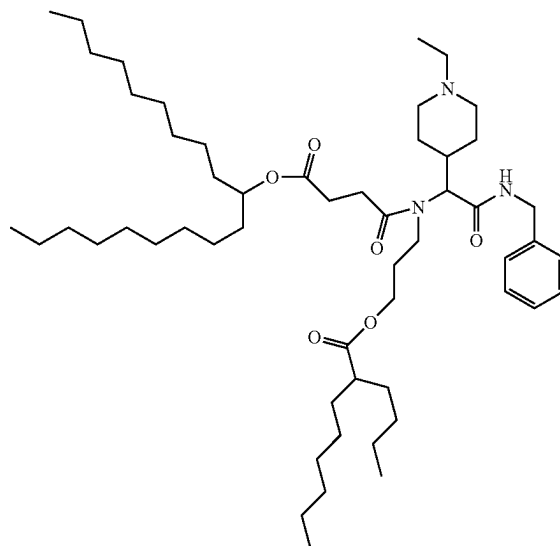
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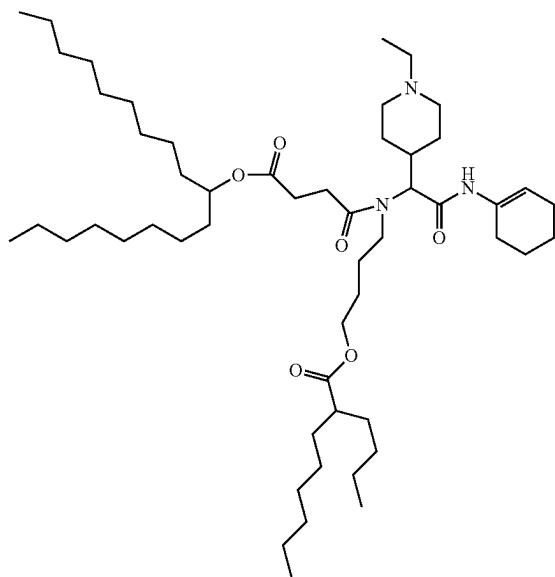


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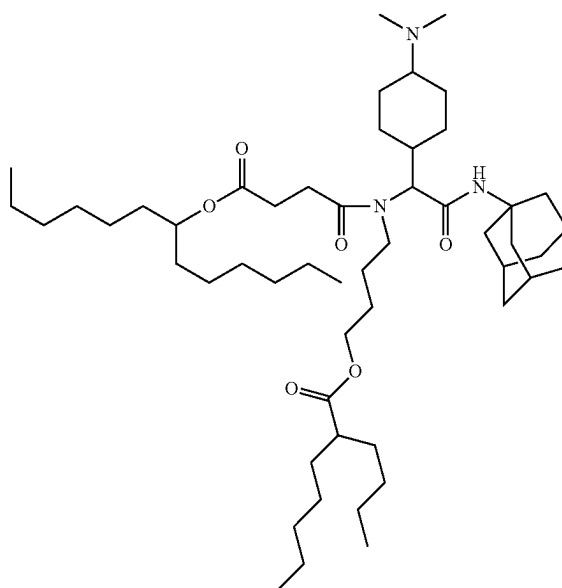
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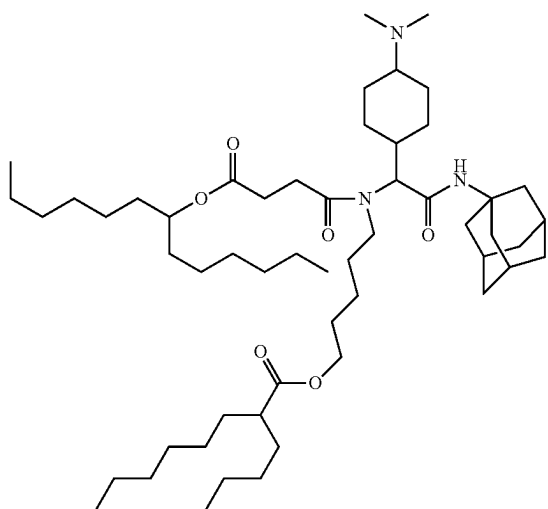


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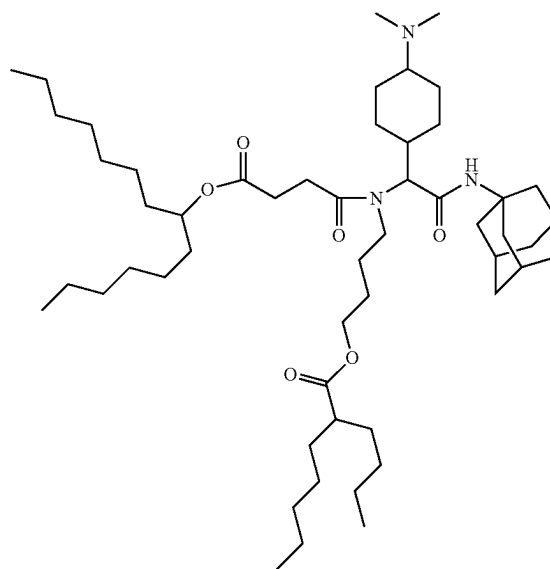
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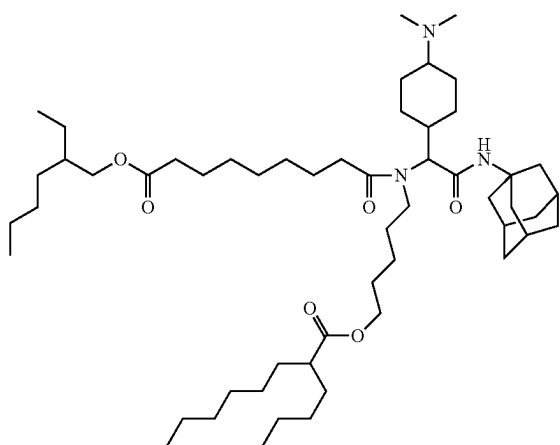


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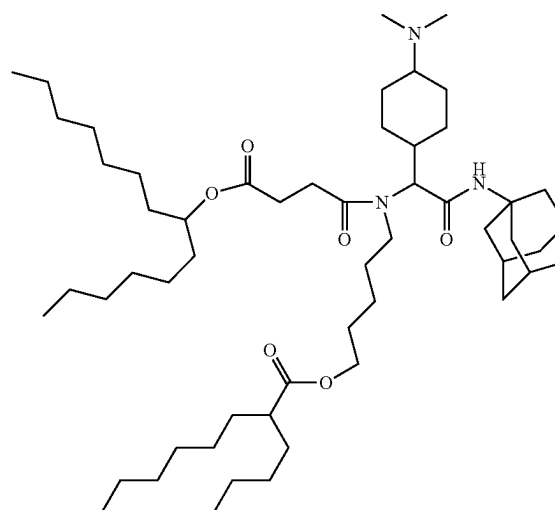
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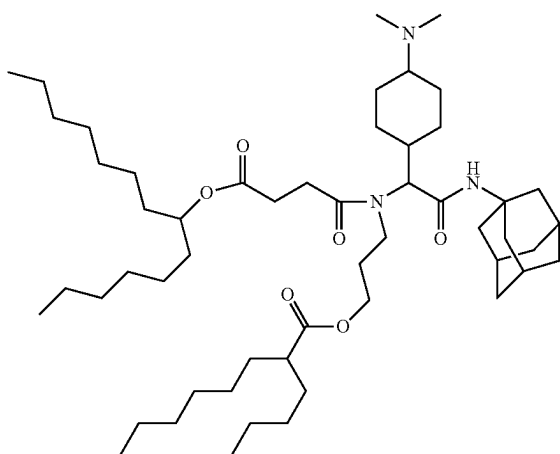
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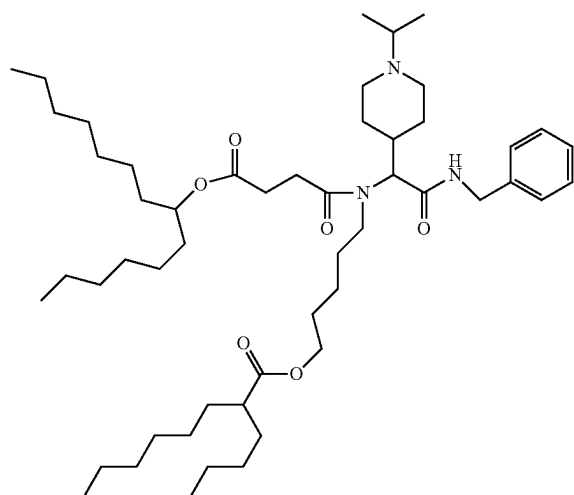
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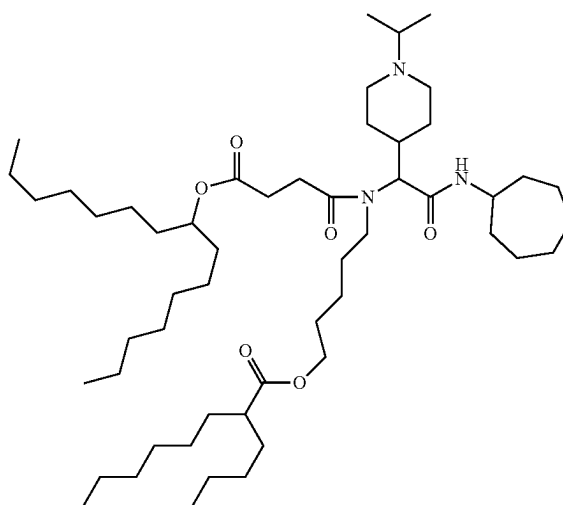
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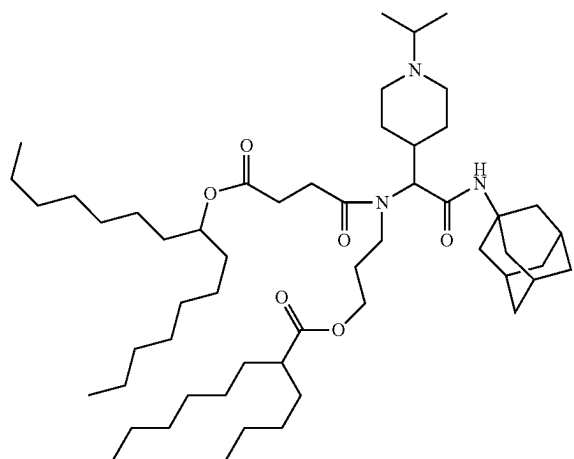
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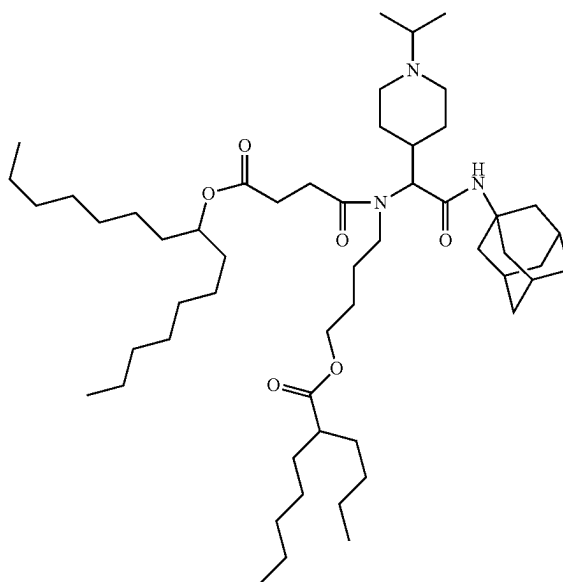
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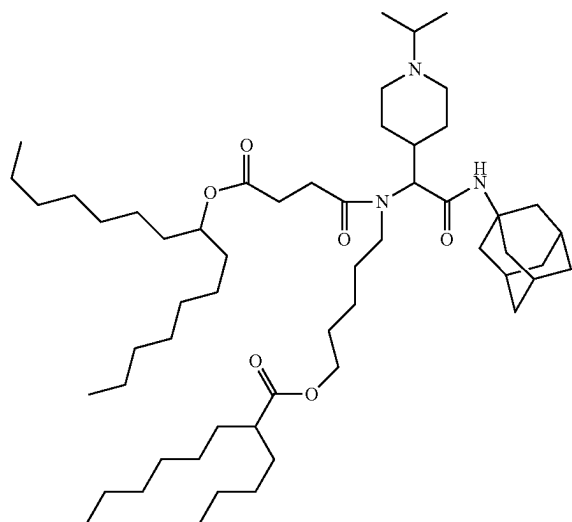


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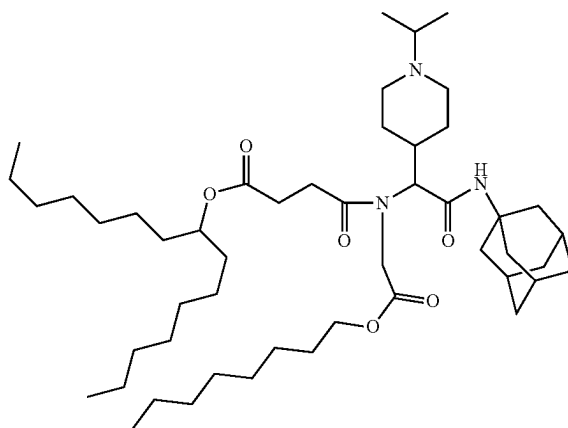


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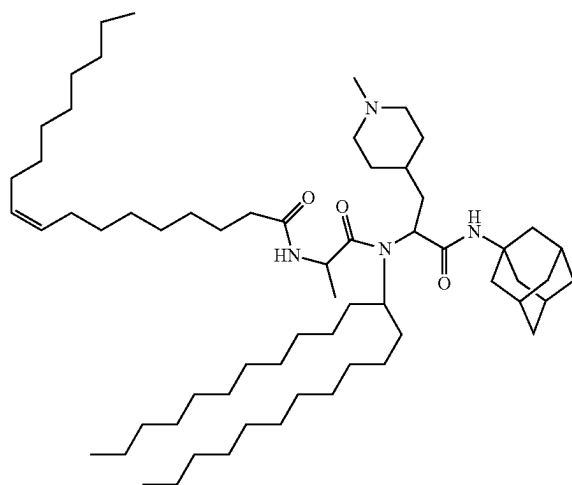
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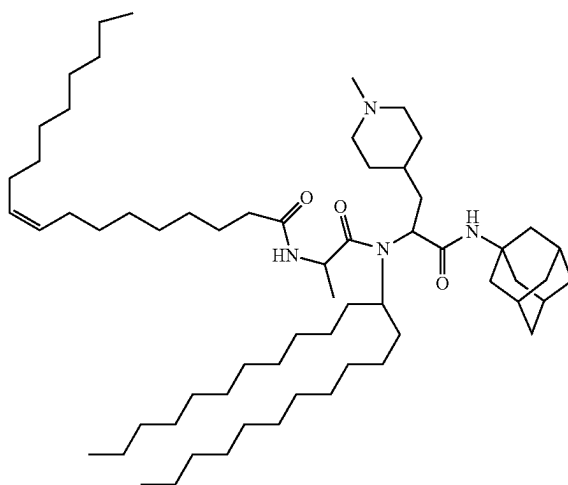
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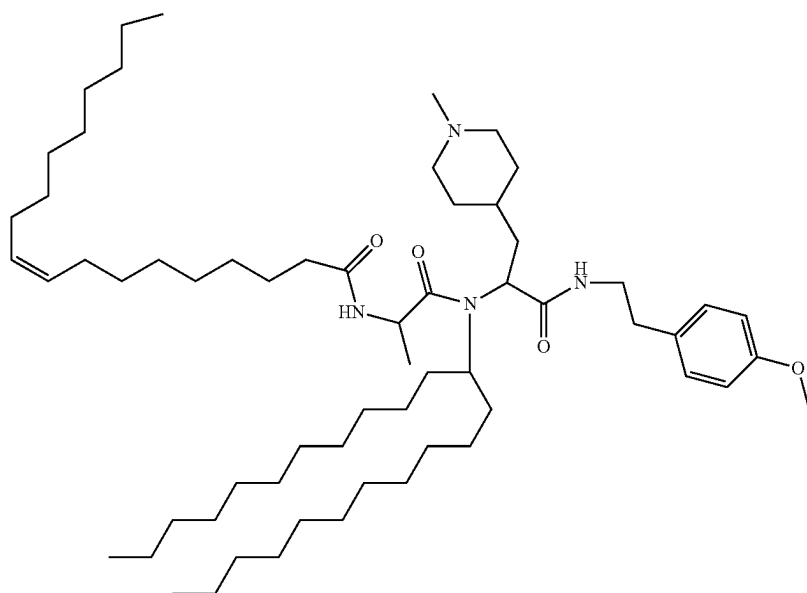


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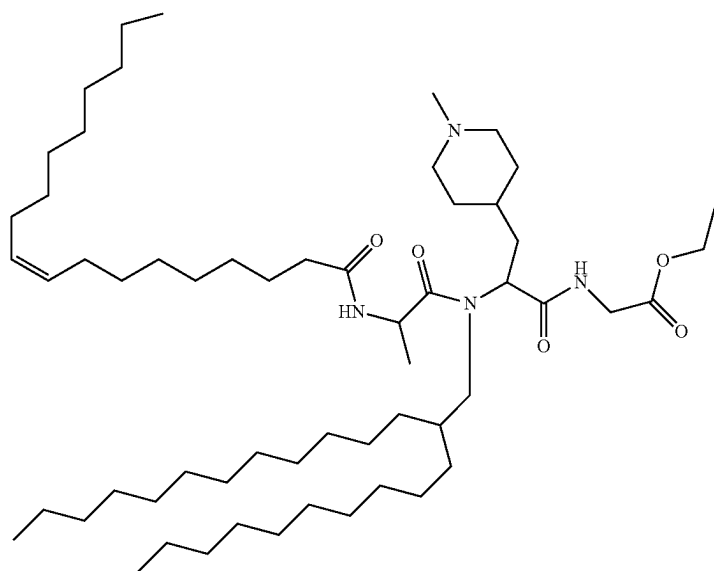


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HP20F10

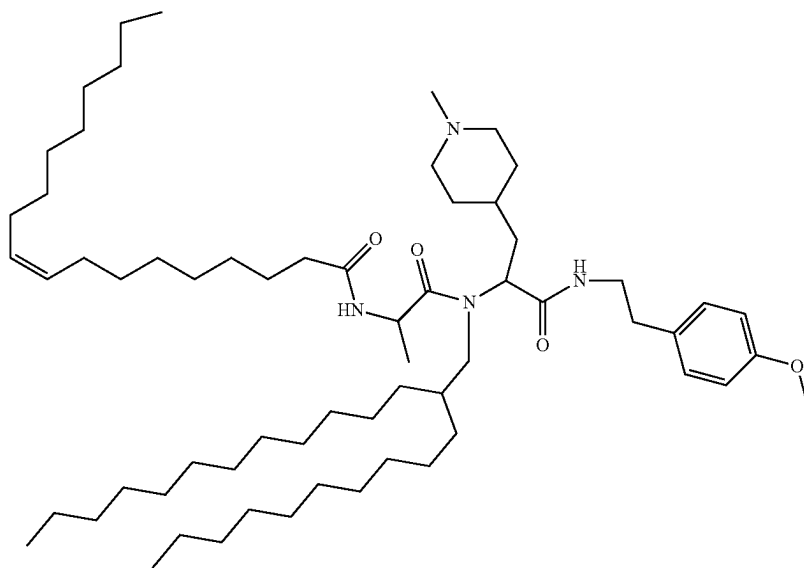


HP20G3



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HP20G10



or

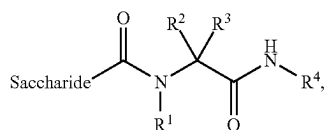
[0112] a salt or isomer thereof.

Modular Lipids

[0113] As used herein, the term “modular lipid” refer to lipids comprising two or more functional groups and at least one linker between at least two functional groups. In some aspects, one of the two or more functional groups is a lipid group, a lipid raft group, a cationic ionizable group, a steric group, a sterol group, a saccharide group, a folate group, a GalNAc group, an oligo peptide group, or an oligo nucleotide group. In some aspects, the linker is covalently linked to two or more functional groups. In some aspects, the two or more functional groups comprise a lipid raft group and a cationic ionizable group. In one aspect, the two or more functional groups comprise a sterol group and a cationic ionizable group. In one aspect, the two or more functional groups comprise a saccharide group and at least one of a sterol group and a PEG group.

[0114] In one aspect, a modular lipid is derived from viral envelope lipids with saccharide modifications. In one aspect, the modular lipid is modified with monosaccharide, disaccharide, oligosaccharide or polysaccharide modifications.

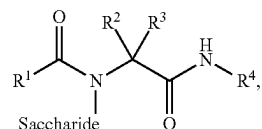
[0115] In one aspect, the LNP comprises a modular lipid compound selected from Formulae VIII-IX, wherein Formula VIII is



(VIII)

or a salt or isomer thereof;

[0116] wherein Formula IX is

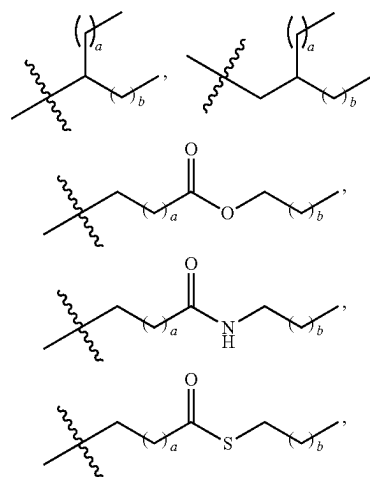


(IX)

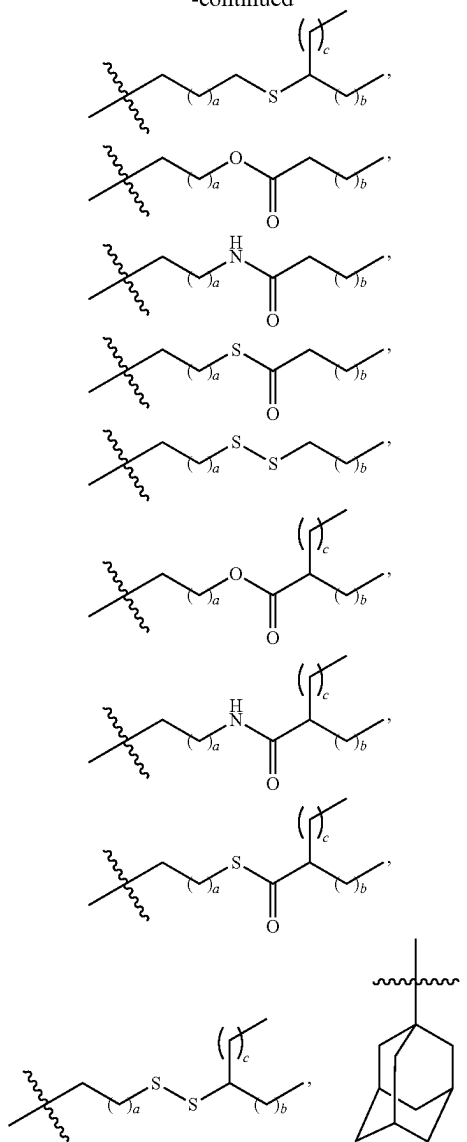
or a salt or isomer thereof;

[0117] wherein

[0118] each of R^1 , R^2 , R^3 and R^4 is independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG),



-continued



[0121] a, b and c are each independently an integer from 0-24;

[0122] each X is independently selected from CH or N;

[0123] each Y is independently selected from CH₂, NH, O, or S;

[0124] each Z is independently selected from CH or N; and

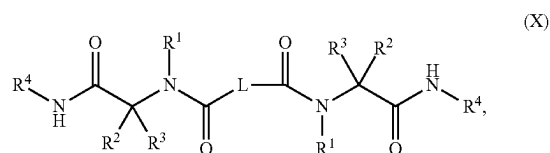
[0125] each saccharide is independently selected from monosaccharides, disaccharides, oligosaccharides, and polysaccharides.

[0126] Monosaccharides useful in the composition of the disclosure include trioses (ketotriose, aldotriose), tetroses (ketotetrose, aldotetroses), pentoses (ribulose, xylulose, ribose, arabinose, xylose, lyxose, deoxyribose), hexoses (psicose, fructose, sorbose, tagatose, allose, altrose, glucose, mannose, fucose, fuculose, rhamnose, heptose, octose, nonose, gulose, idose, galactose, talose), sedoheptulose. Disaccharides useful in the composition of the disclosure include sucrose, lactose, maltose, trehalose, turanose, cellobiose.

Oligosaccharides useful in the composition of the disclosure include raffinose, melezitose, maltotriose, acarbose, stachyose, fructooligosaccharide, galactooligosaccharides, mannanoligosaccharides. Polysaccharides useful in the composition of the disclosure include ployglycitol, n-acetylglucosamine, chitin.

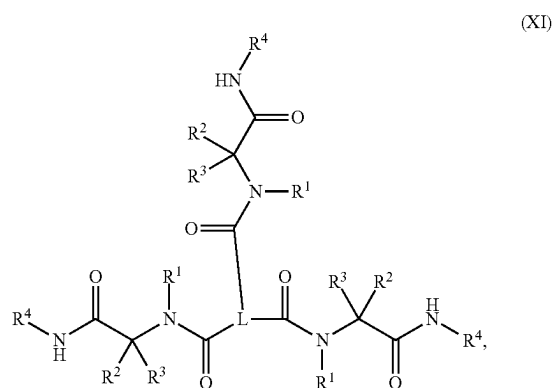
[0127] In one aspect, the LNP comprises a modular lipid compound selected from Formulae X-XX,

[0128] wherein Formula X is



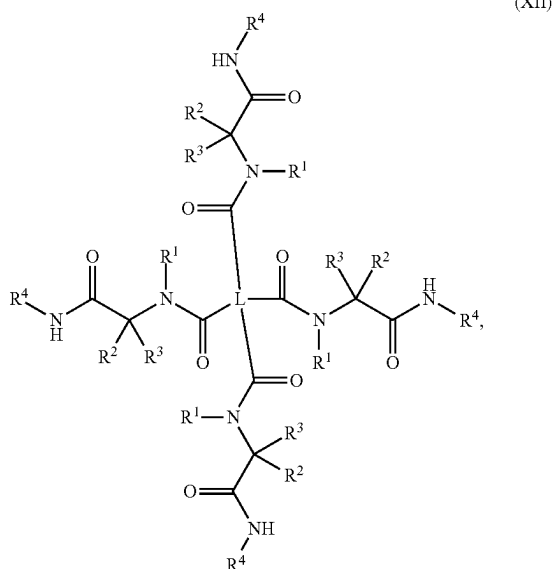
or a salt or isomer thereof,

[0129] wherein Formula XI is



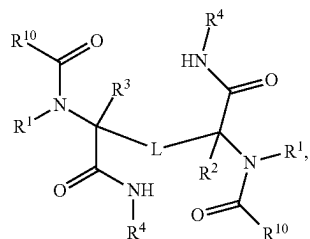
or salt or isomer thereof,

[0130] wherein Formula XII is



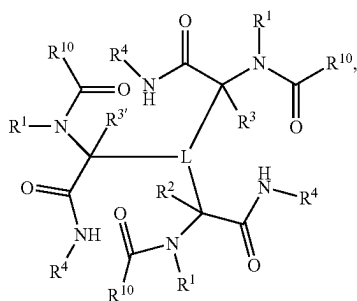
or salt or isomer thereof.

[0131] wherein Formula XIII is



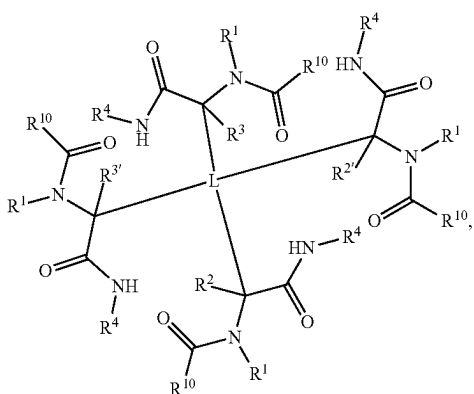
or salt or isomer thereof;

[0132] wherein Formula IXX is



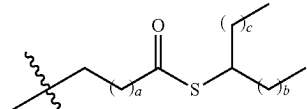
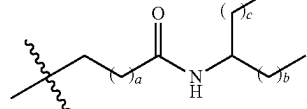
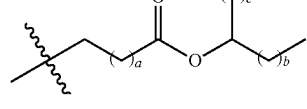
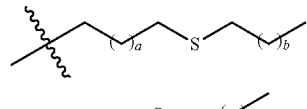
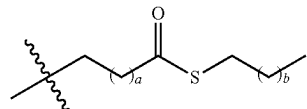
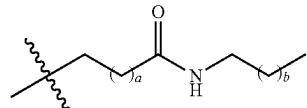
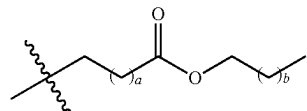
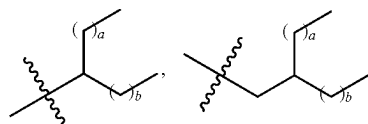
or a salt or isomer thereof,

[0133] wherein Formula XX is



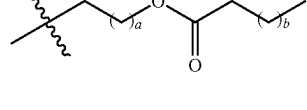
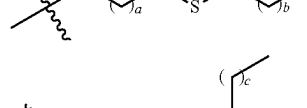
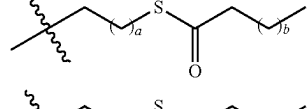
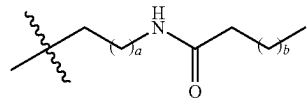
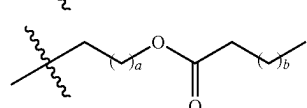
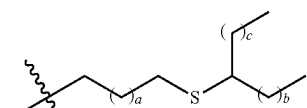
or a salt or isomer thereof, wherein each R^1 , R^4 , and R^{10} , is independently selected from C_2 - C_{24} alkyl, C_2 - C_{24} alkenyl, C_2 - C_{24} alkynyl, substituted C_2 - C_{24} alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly (ethylene glycol) (PEG),

(XIII)

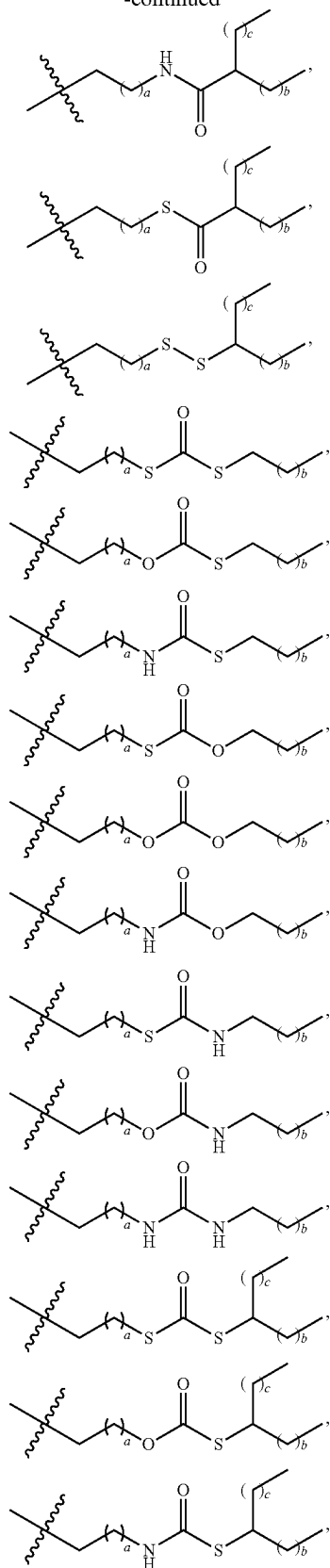


(IXX)

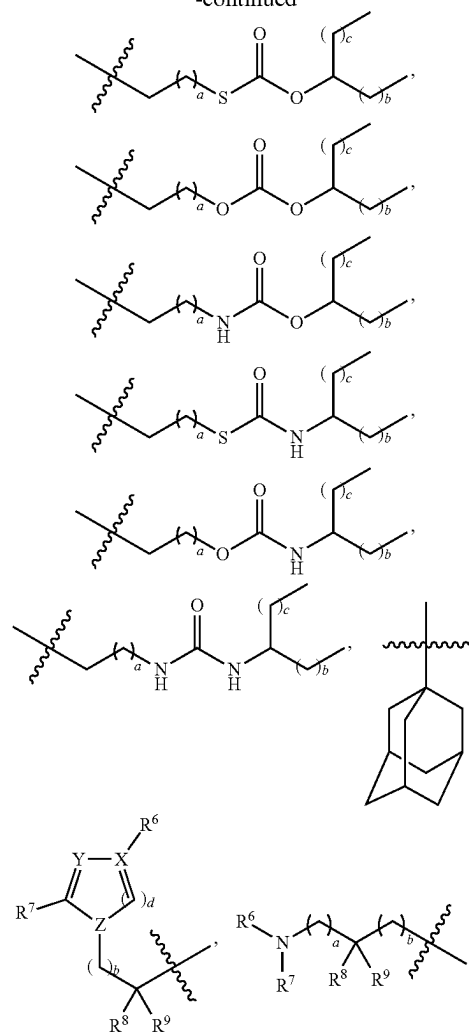
(XX)



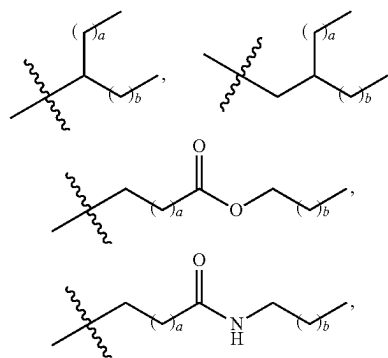
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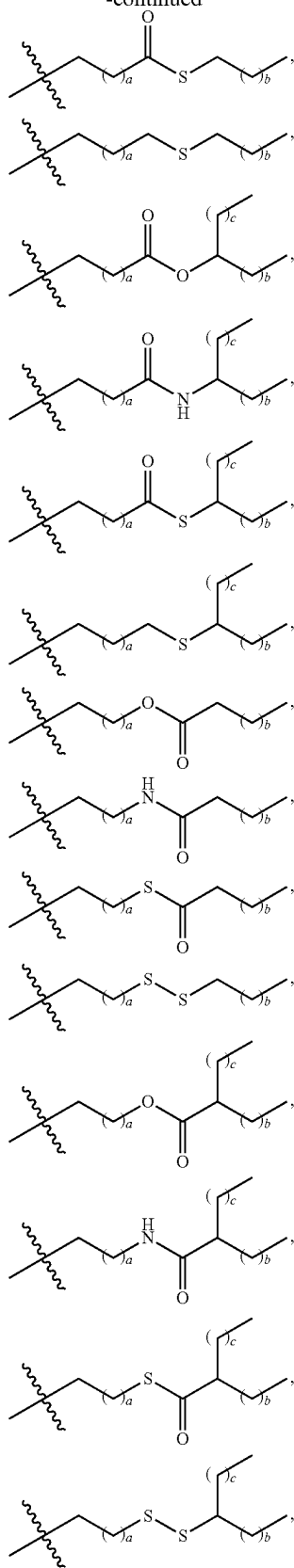
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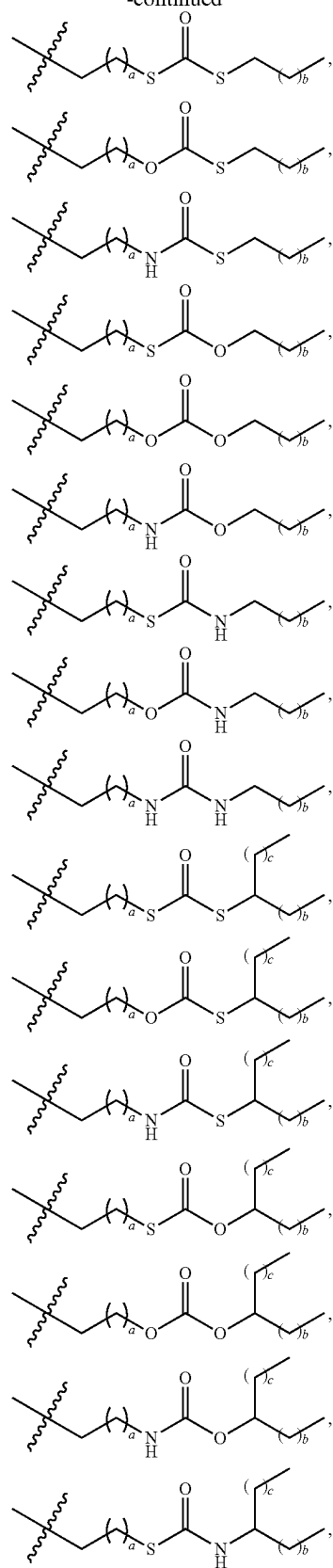
[0134] wherein each R^2 , $R^{2'}$, R^3 and $R^{3'}$ is independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG) and



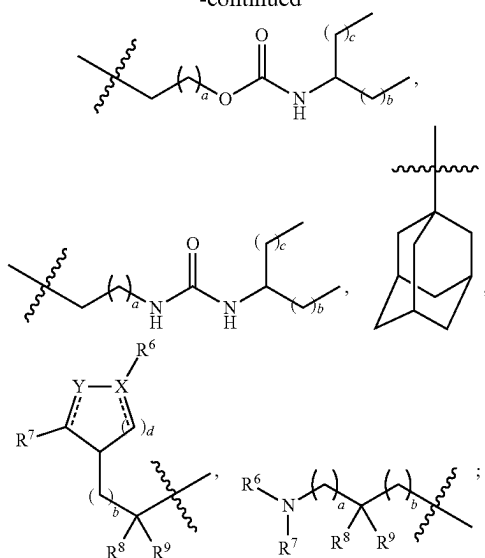
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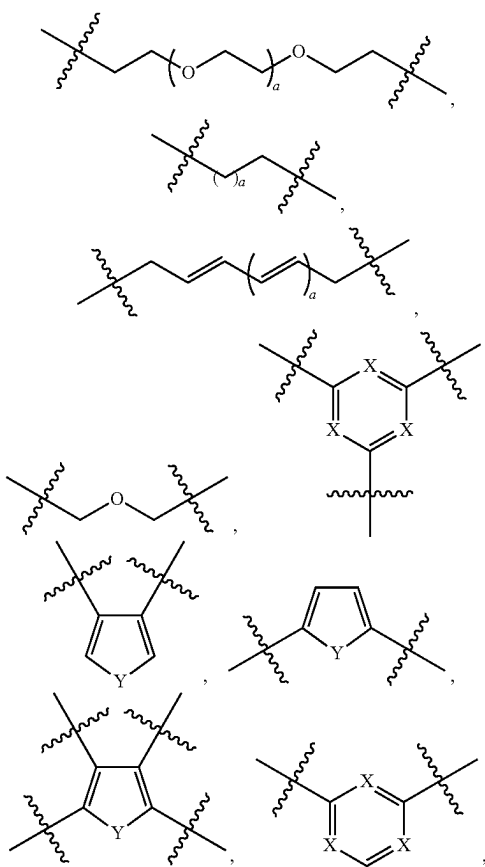
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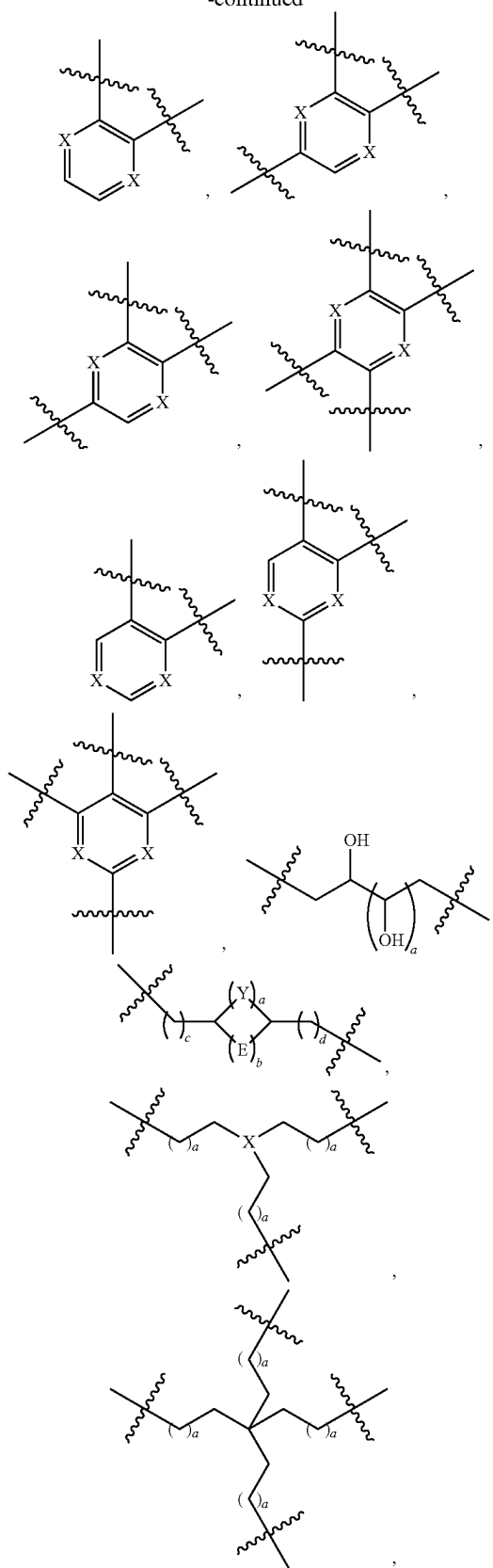
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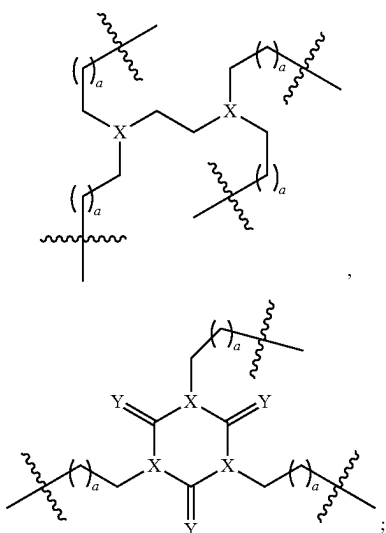
[0135] wherein each L is independently selected from alkyl, alkenyl, alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG) and



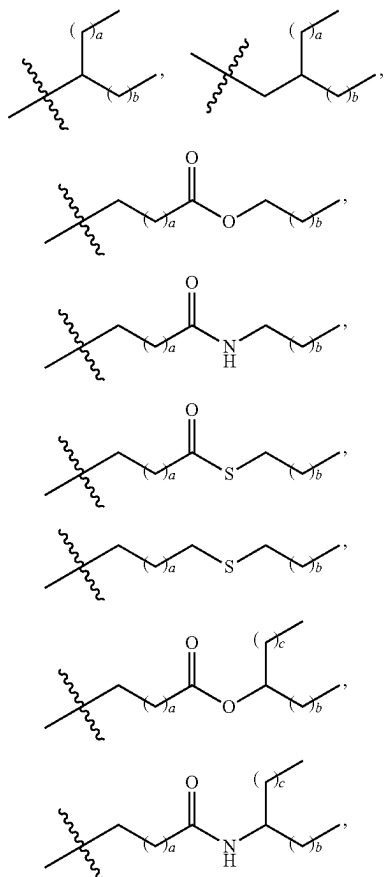
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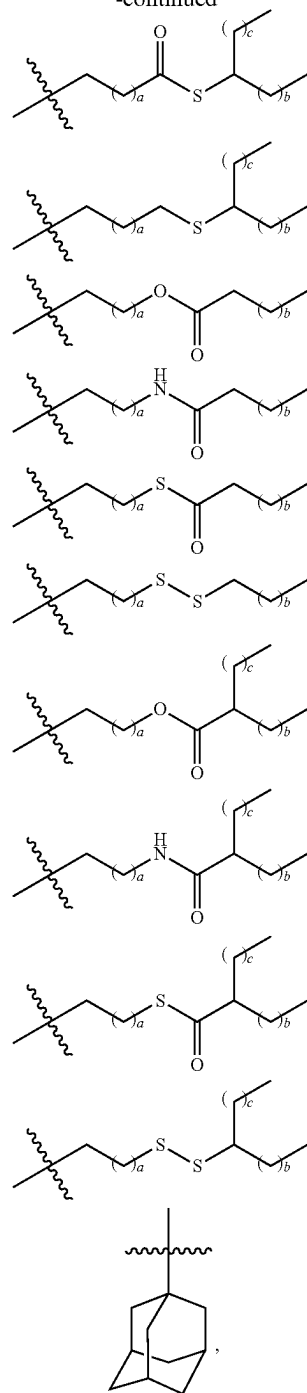
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[0136] wherein R^6 , R^7 , R^8 and R^9 are each independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl,



-continued



[0137] a, b, c and d are each independently an integer from 0-24;

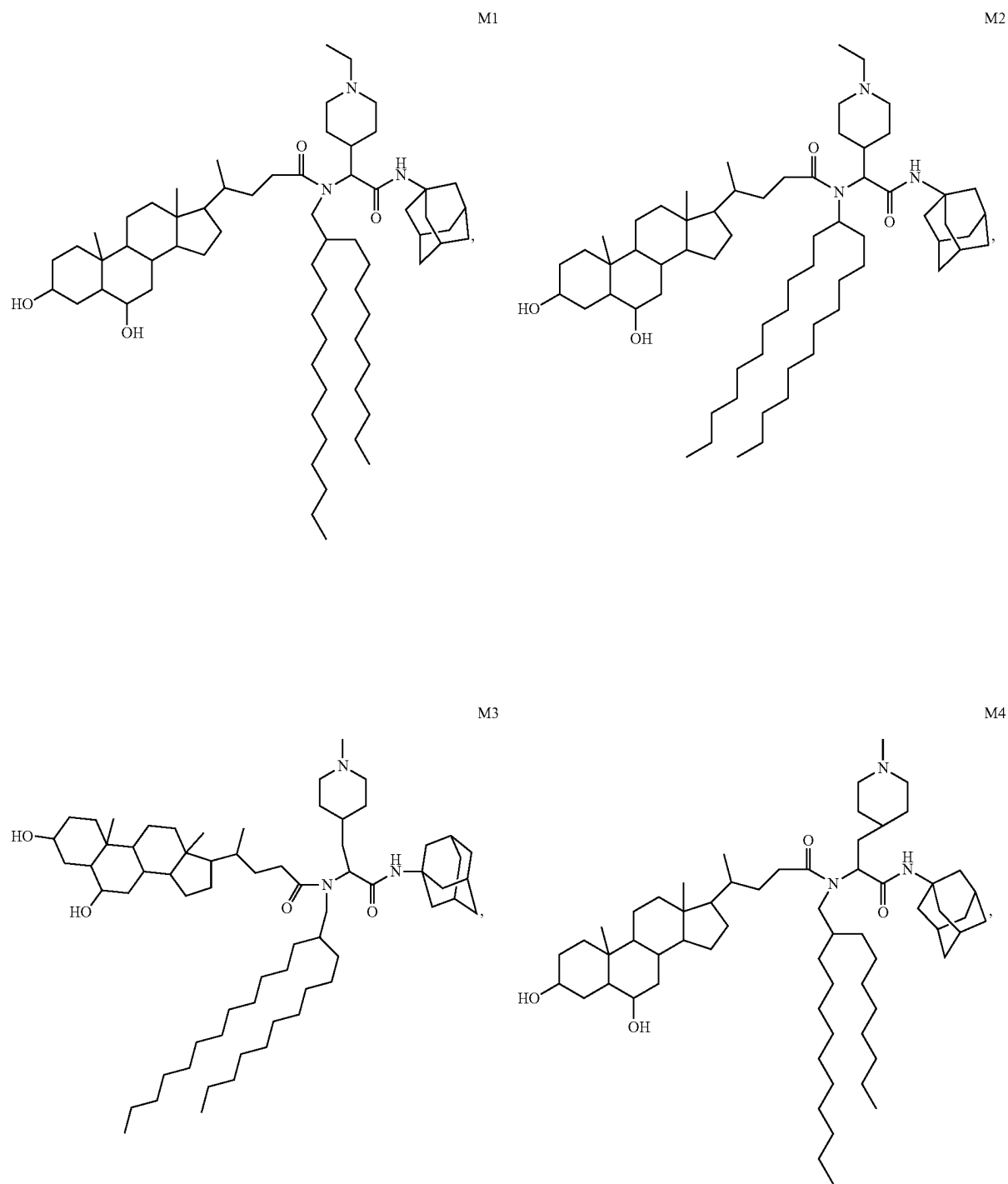
[0138] each E is independently selected from CH_2 , NH, O, or S;

[0139] each X is independently selected from CH or N;

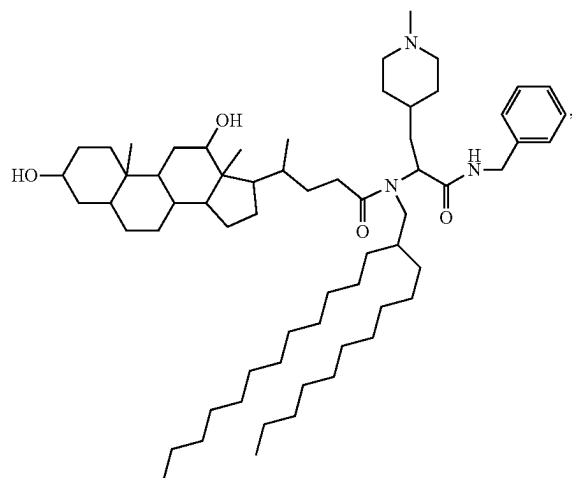
[0140] each Y is independently selected from CH_2 , NH, O, or S; and

[0141] each Z is independently selected from CH or N.

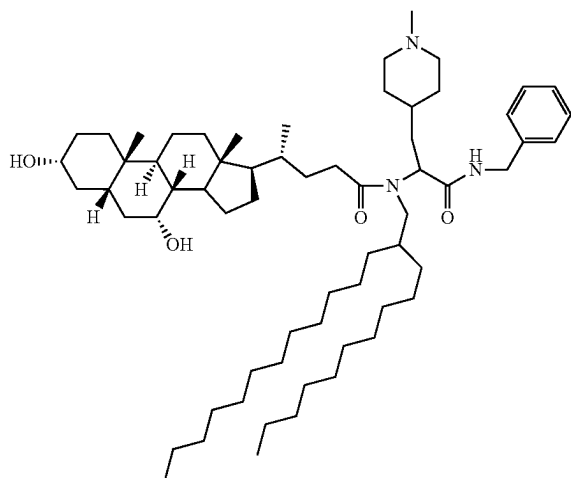
[0142] In some aspects, the modular lipid is:



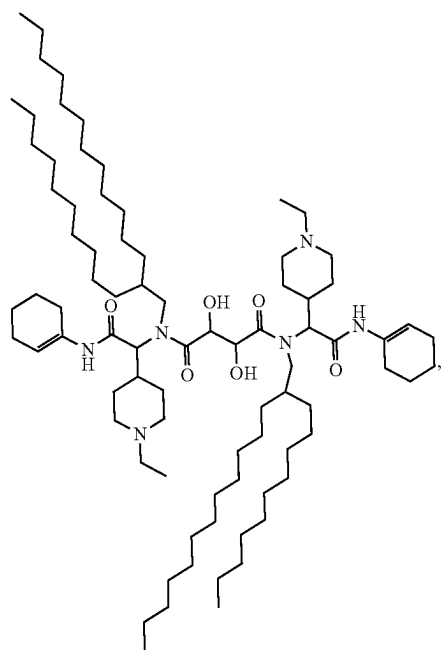
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M4



M5

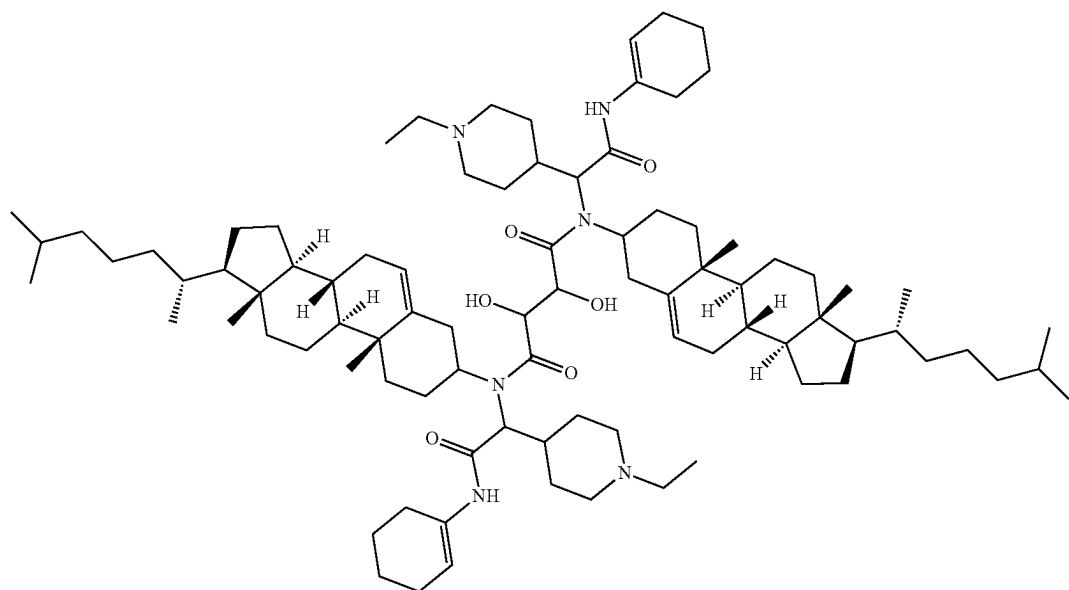


M6



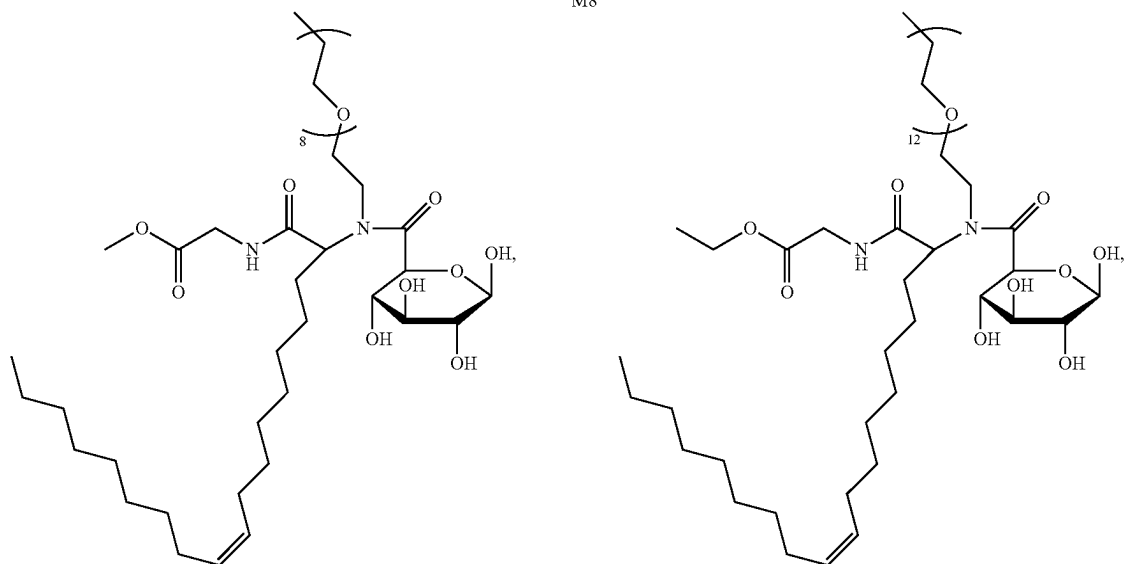
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M7

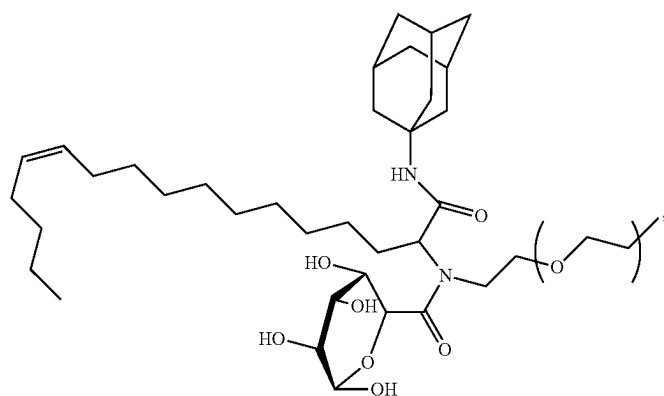


M8

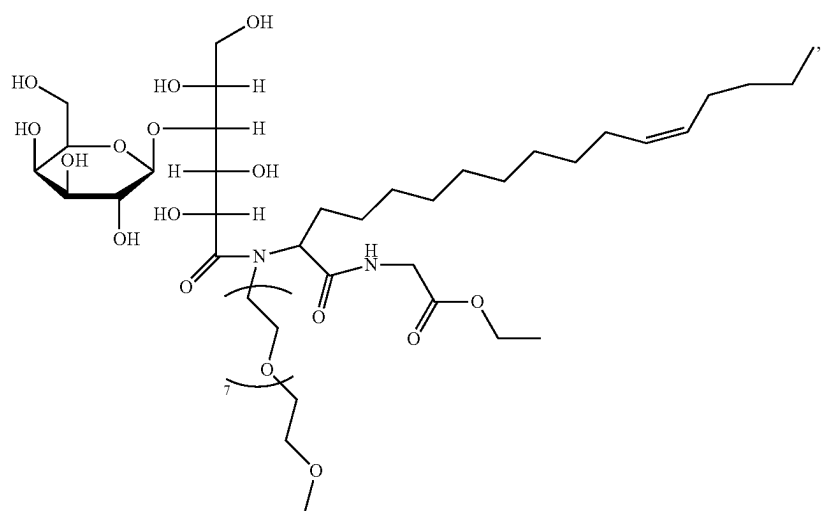
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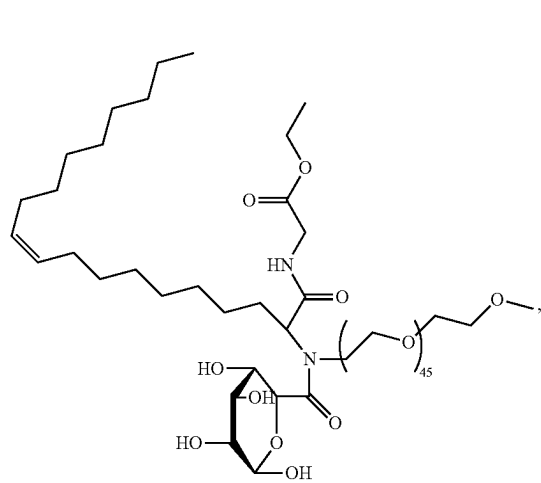
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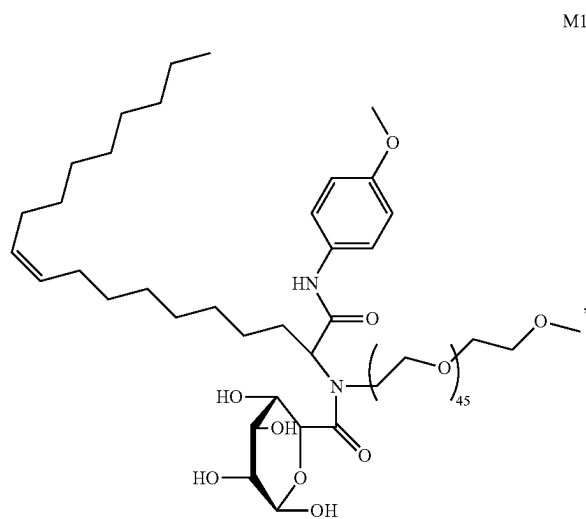
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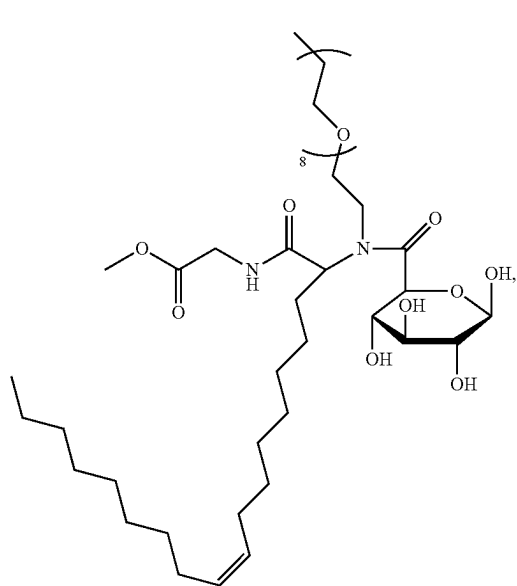
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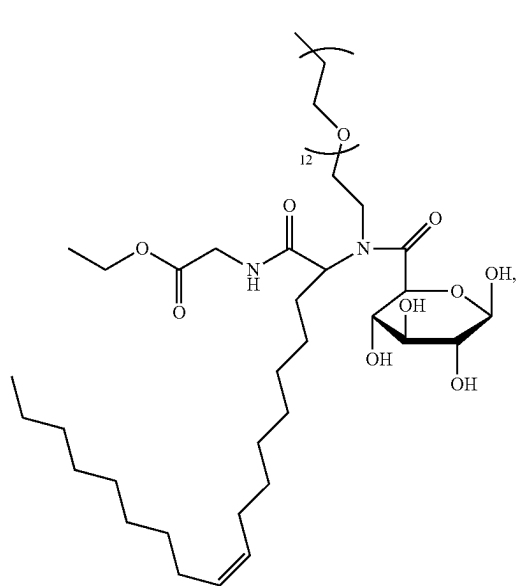
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M12



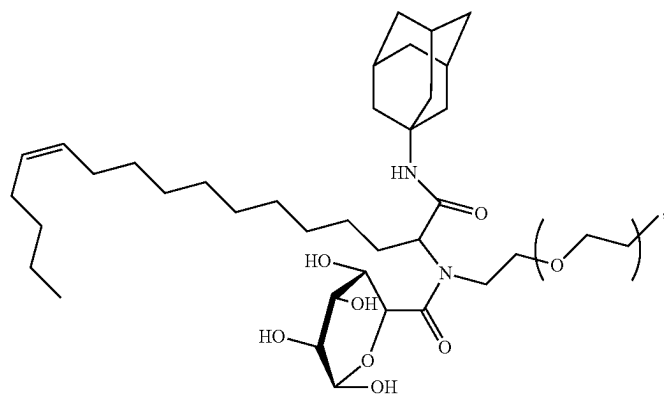
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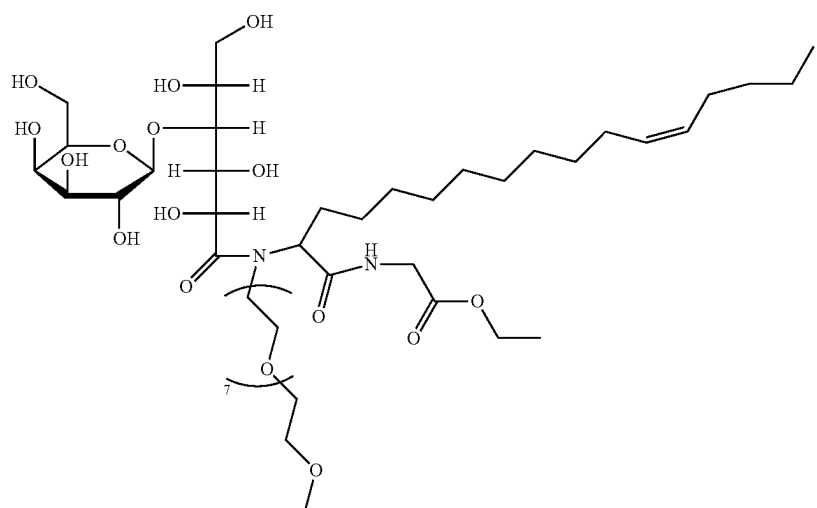
M14

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M15

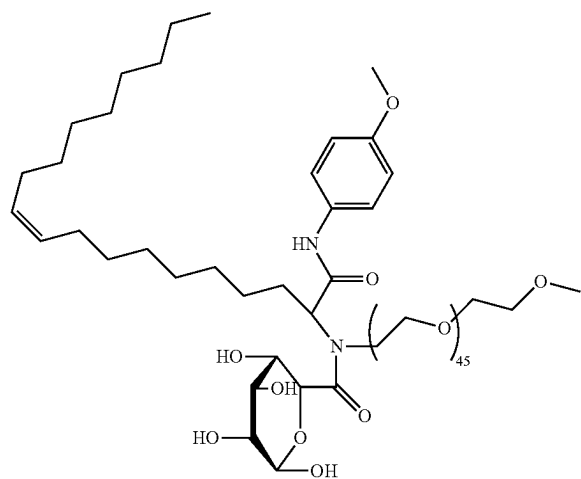
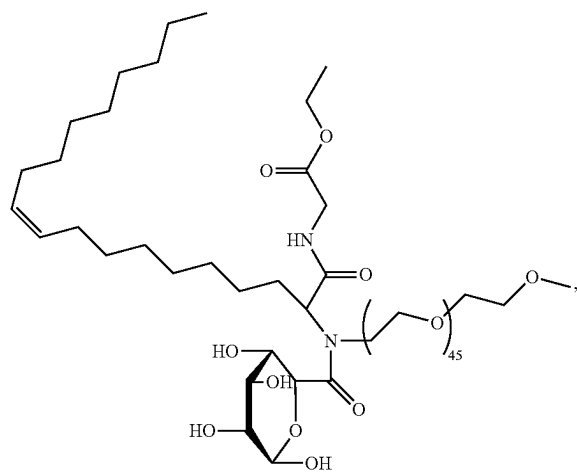


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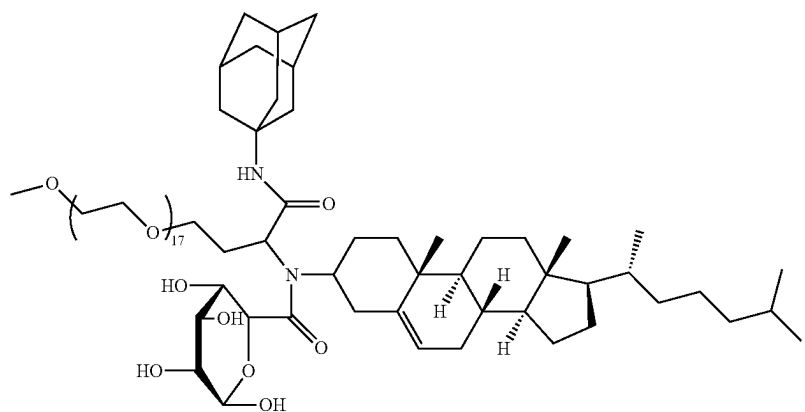
M17

M17

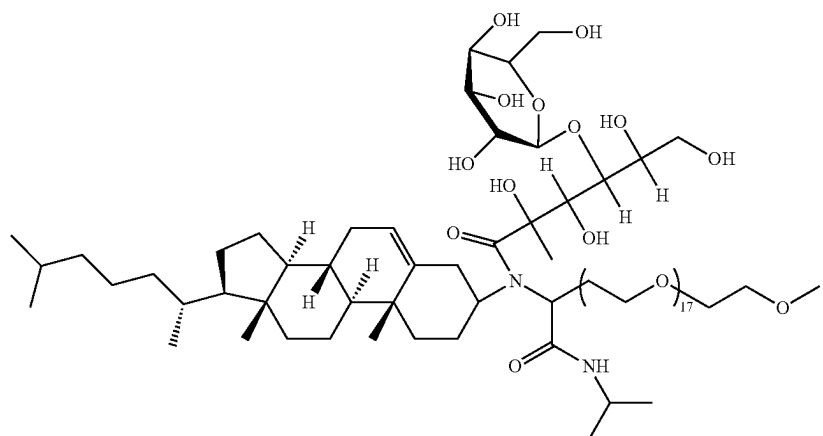


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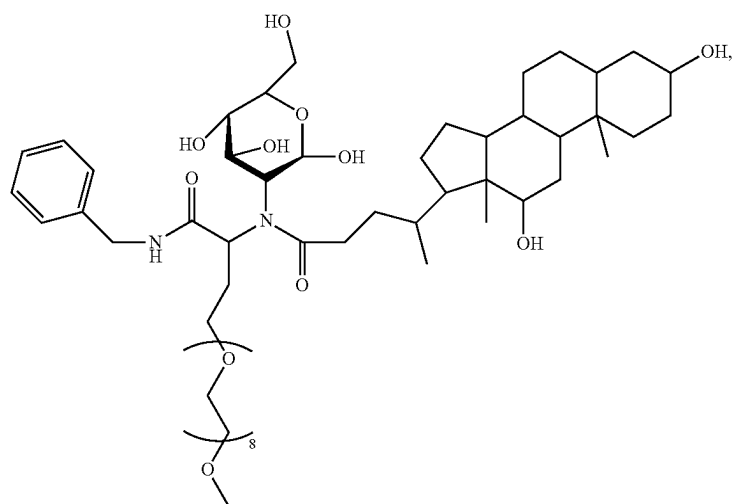
M18



M19

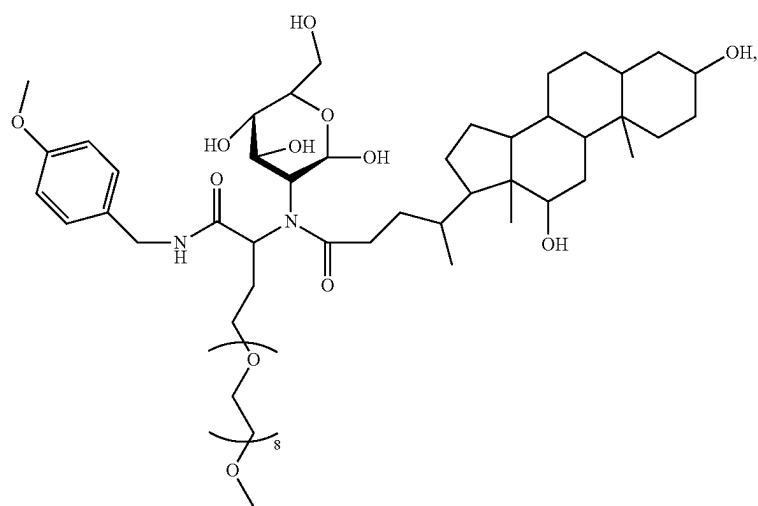


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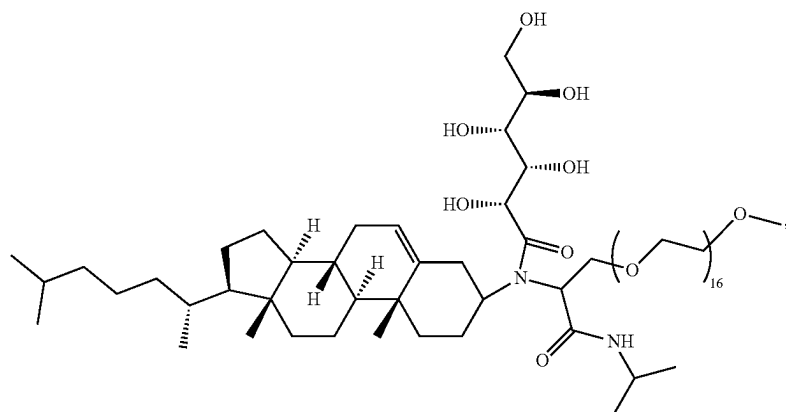


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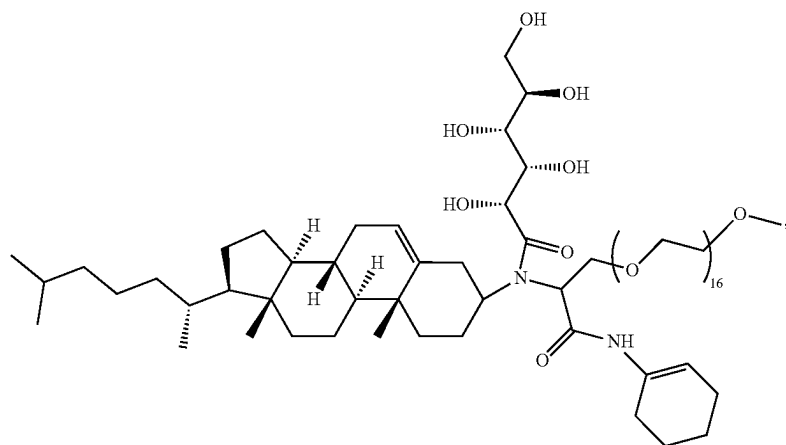
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M22

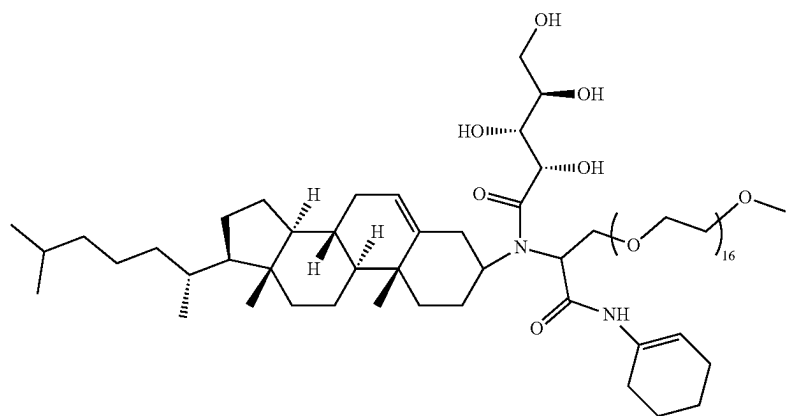


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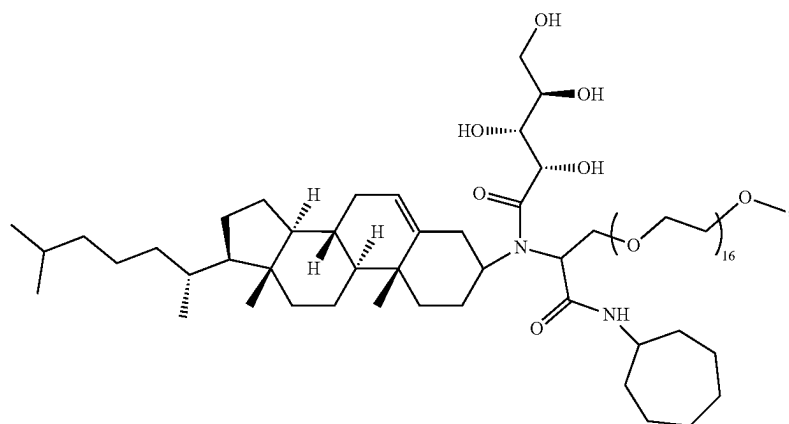


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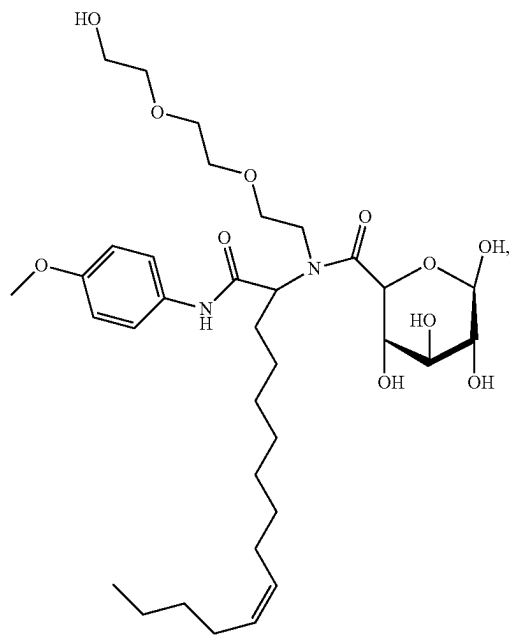
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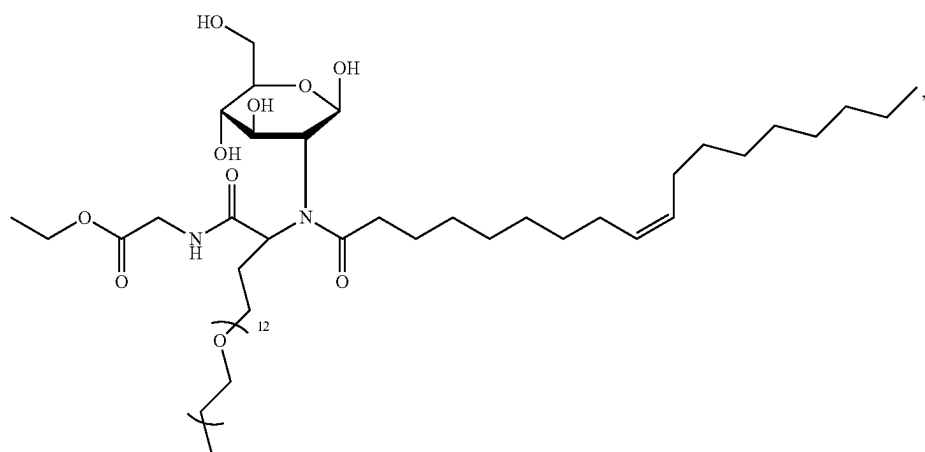
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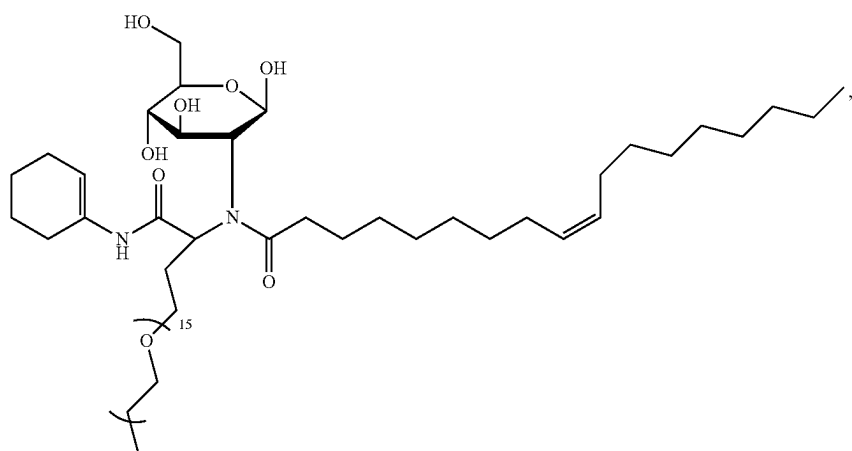
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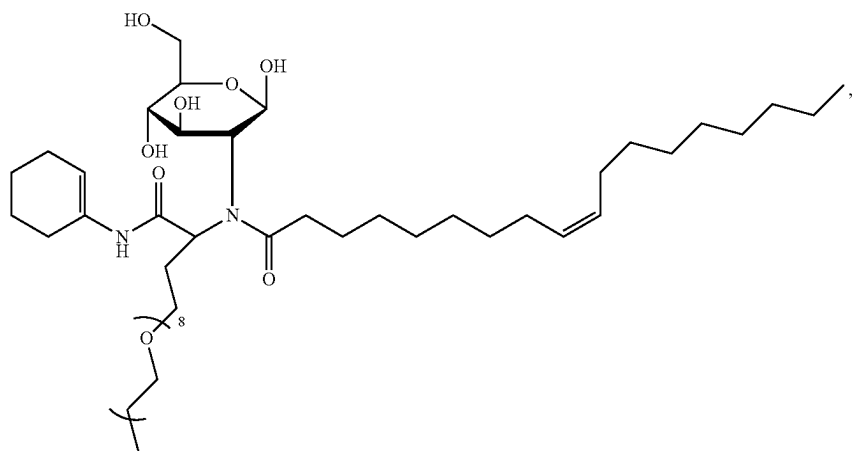
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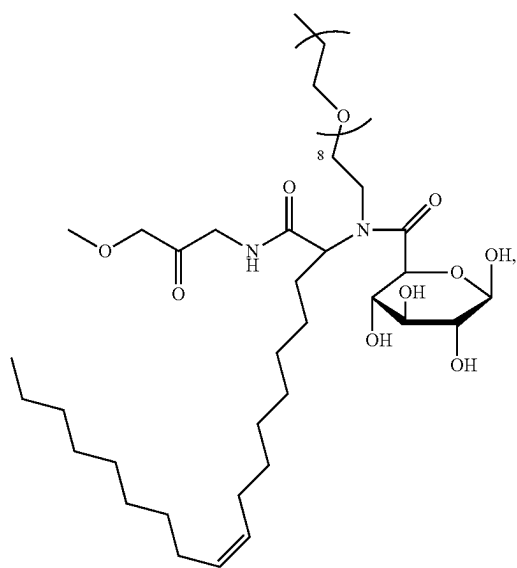
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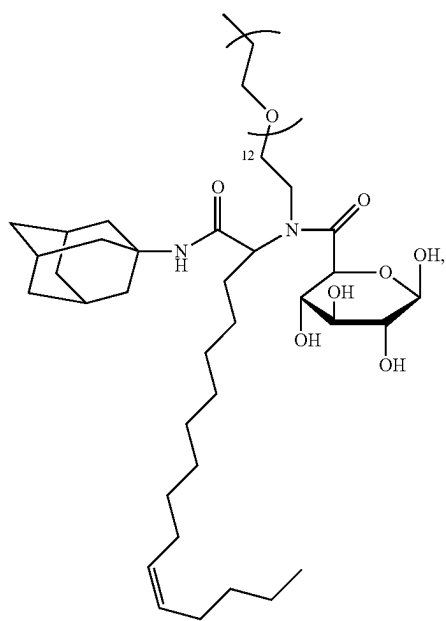
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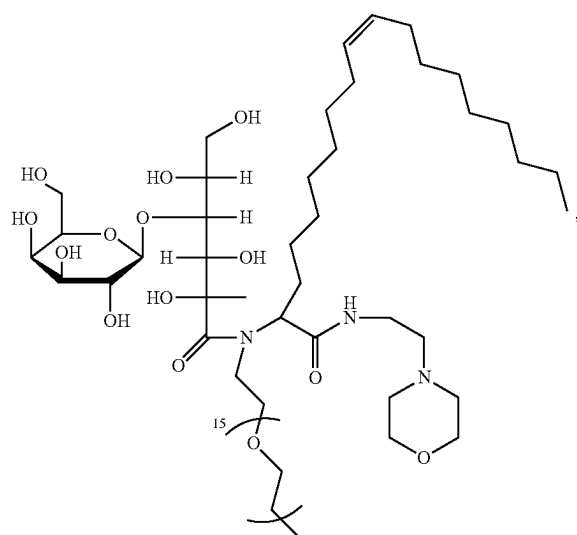
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P1_E2



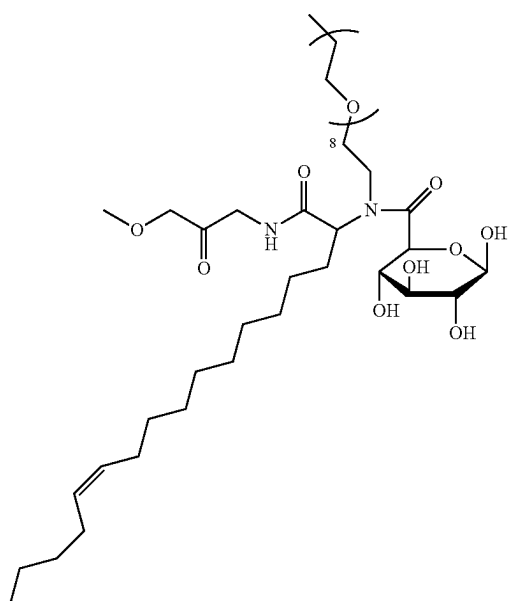
P1_F11

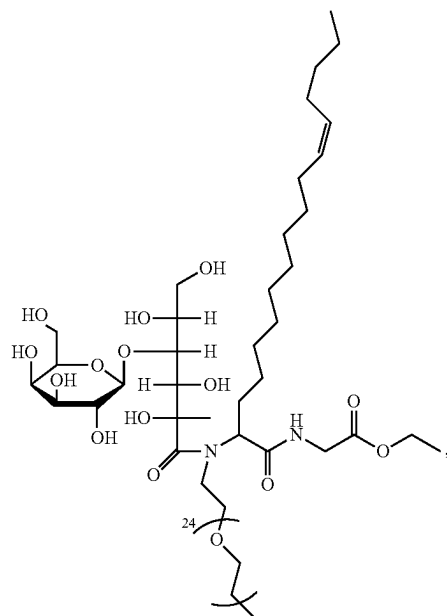


P1_C6

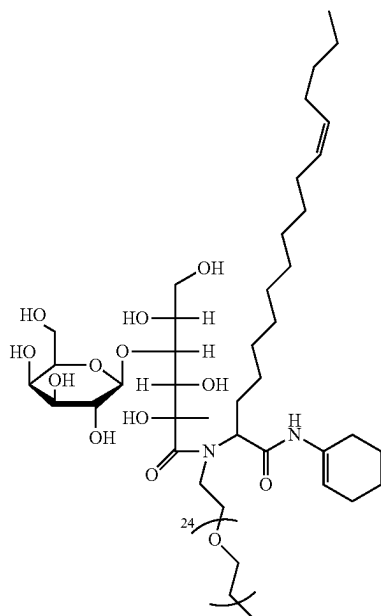


P1_E8



-continued
P1_D8

P1_D9



or a salt or isomer thereof.

Adjuvants

[0143] In some aspects, a nanoparticle composition that includes one or more lipids described herein may further include one or more adjuvants, e.g., Glucopyranosyl Lipid Adjuvant (GLA), CpG oligodeoxynucleotides (e.g., Class A or B), poly(I.C), aluminum hydroxide, and Pam3CSK4.

Biologically Active Agents

[0144] Payload delivery systems encapsulate a payload. The payload may comprise one or more biologically active agents, which may be a therapeutic, diagnostic or prophylactic. The disclosure features methods of delivering a biologically active agent to a cell or organ and treating a disease or disorder in a subject in need thereof comprising administering to a subject and/or contacting a cell with a payload delivery system including a biologically active agent.

[0145] A biologically active agent may be a substance that, once delivered to a cell or organ, brings about a desirable therapeutic change in the cell, organ, or other bodily tissue or system. Such species may be useful in the treatment of one or more diseases, disorders, or conditions.

[0146] In some aspects, the payload delivery system of the disclosure encapsulate more than one biologically active agent as its payload. In some aspects, at least one of the biologically active agents of the payload is a small molecule drug useful in the treatment of a particular disease, disorder, or condition. Examples of drugs useful in the nanoparticle compositions include, but are not limited to, antineoplastic agents (e.g., vincristine, doxorubicin, mitoxantrone, camptothecin, cisplatin, bleomycin, cyclophosphamide, methotrexate, and streptozotocin), antitumor agents (e.g., actinomycin D, vincristine, vinblastine, cytosine arabinoside, anthracyclines, alkylating agents, platinum compounds,

antimetabolites, and nucleoside analogs, such as methotrexate and purine and pyrimidine analogs), anti-infective agents, local anesthetics (e.g., dibucaine and chlorpromazine), beta-adrenergic blockers (e.g., propranolol, timolol, and labetalol), antihypertensive agents (e.g., clonidine and hydralazine), anti-depressants (e.g., imipramine, amitriptyline, and doxepin), anti-convulsants (e.g., phenytoin), antihistamines (e.g., diphenhydramine, chlorpheniramine, and promethazine), antibiotic/antibacterial agents (e.g., gentamycin, ciprofloxacin, and cefoxitin), antifungal agents (e.g., miconazole, terconazole, econazole, isoconazole, butaconazole, clotrimazole, itraconazole, nystatin, naftifine, and amphotericin B), antiparasitic agents, hormones, hormone antagonists, immunomodulators, neurotransmitter antagonists, antiglaucoma agents, vitamins, narcotics, and imaging agents.

[0147] In some aspects, at least one of the biologically active agents of the payload is a cytotoxin, a radioactive ion, a chemotherapeutic, a vaccine, a compound that elicits an immune response, and/or another therapeutic and/or prophylactic. A cytotoxin or cytotoxic agent includes any agent that may be detrimental to cells. Examples include, but are not limited to, taxol, cytochalasin B, gramicidin D, ethidium bromide, emetine, mitomycin, etoposide, teniposide, vincristine, vinblastine, colchicine, doxorubicin, daunorubicin, dihydroxyanthracenedione, mitoxantrone, mithramycin, actinomycin D, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, maytansinoids, e.g., maytansinol, rachelmycin (CC-1065), and analogs or homologs thereof. Radioactive ions include, but are not limited to iodine (e.g., iodine 125 or iodine 131), strontium 89, phosphorous, palladium, cesium, iridium, phosphate, cobalt, yttrium 90, samarium 153, and praseodymium. Vaccines include compounds and preparations that are capable of providing immunity against one or more conditions related to infectious diseases such as influenza, measles, human papillomavirus (HPV), rabies, meningitis,

whooping cough, tetanus, plague, hepatitis, and tuberculosis and can include mRNAs encoding infectious disease derived antigens and/or epitopes. Vaccines also include compounds and preparations that direct an immune response against cancer cells and can include mRNAs encoding tumor cell derived antigens, epitopes, and/or neoepitopes. Compounds eliciting immune responses may include vaccines, corticosteroids (e.g., dexamethasone), and other species.

[0148] In some aspects, at least one of the biologically active agents of the payload is a polypeptide or protein. Therapeutic proteins useful in the nanoparticles in the disclosure include, but are not limited to, gentamycin, amikacin, insulin, erythropoietin (EPO), granulocyte-colony stimulating factor (G-CSF), granulocyte-macrophage colony stimulating factor (GM-CSF), Factor VIR, luteinizing hormone-releasing hormone (LHRH) analogs, interferons, heparin, Hepatitis B surface antigen, typhoid vaccine, and cholera vaccine.

[0149] In some aspects, at least one of the biologically active agents of the payload is a polynucleotide or nucleic acid (e.g., ribonucleic acid or deoxyribonucleic acid). The term “polynucleotide,” in its broadest sense, includes any compound and/or substance that is or can be incorporated into an oligonucleotide chain. Exemplary polynucleotides for use in accordance with the present disclosure include, but are not limited to, one or more of deoxyribonucleic acid (DNA), ribonucleic acid (RNA) including messenger mRNA (mRNA), hybrids thereof, RNAi-inducing agents, RNAi agents, gRNA, shRNA, siRNAs, shRNAs, miRNAs, tRNA, antisense RNAs, ribozymes, catalytic DNA, RNAs that induce triple helix formation, aptamers, vectors, etc. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a RNA or DNA molecule.

[0150] In some aspects, the payload comprise to nucleic acid constructs. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a DNA construct. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a double stranded DNA (dsDNA) construct. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a single stranded DNA (ssDNA) construct. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a linear nucleic acid molecule. In some aspects, at least one of the first nucleic acid construct or the second nucleic acid construct is a circular nucleic acid molecule. RNAs useful in the compositions and methods described herein can be selected from the group consisting of, but are not limited to, shortmers, antagomirs, antisense, ribozymes, small interfering RNA (siRNA), asymmetrical interfering RNA (aiRNA), microRNA (miRNA), Dicer-substrate RNA (dsRNA), small hairpin RNA (shRNA), transfer RNA (tRNA), messenger RNA (mRNA), and mixtures thereof. In some aspects, the first nucleic acid construct is a sa-mRNA molecule. In some aspects, the second nucleic acid construct is a mRNA molecule.

Polynucleotides and Nucleic Acids

[0151] In some aspects, the payload delivered in a payload delivery system of the disclosure is a first nucleic acid construct encoding a sa-mRNA encoding at least one gene of

interest (GOI) or a plurality of GOIs; and a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein.

[0152] The amount of a biologically active agent in a payload delivery system may depend on the size, composition, desired target and/or application, or other properties of the payload delivery system as well as on the properties of the biologically active agent. For example, the amount of a nucleic acid useful in a payload delivery system may depend on the size, sequence, and other characteristics of the nucleic acid. The relative amounts of a biologically active agent and other elements (e.g., lipids) in a payload delivery system may also vary. In some aspects, the wt/wt ratio of the lipid component to a biologically active agent in a payload delivery system may be from about 1:1 to about 60:1, such as 1:1, 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 11:1, 12:1, 13:1, 14:1, 15:1, 16:1, 17:1, 18:1, 19:1, 20:1, 25:1, 30:1, 35:1, 40:1, 45:1, 50:1, and 60:1. For example, the wt/wt ratio of the lipid component to a biologically active agent may be from about 1:1 to about 4:1. In certain aspects, the wt/wt ratio is about 20:1. In certain aspects, the wt/wt ratio is about 60:1. The amount of a biologically active agent in a payload delivery system may, for example, be measured using absorption spectroscopy (e.g., ultraviolet-visible spectroscopy).

Self-Amplifying mRNA

[0153] Sa-mRNAs of the disclosure have the ability to self-replicate in cells and, thus, can be used to induce expression of encoded gene products, such as proteins (e.g., antigens) and regulatory structures (e.g. siRNA, miRNA, saRNA, tRNA, and lincRNA) encoded by the sa-mRNA. In addition, sa-mRNAs are generally based on the genome of an RNA virus (e.g. a Group IV positive single strand RNA virus), and therefore are foreign nucleic acids that can stimulate the innate immune system (e.g. induce an interferon response). This can lead to undesired consequences and safety concerns, such as rapid inactivation and clearance of the RNA, injection site irritation and/or inflammation and/or pain.

[0154] The sa-mRNAs of the present disclosure contain modified structures and have reduced capacity to stimulate the innate immune system, which will lead to rapid decay of the sa-mRNA and its associated gene products. Rapid decay of the sa-mRNA and its associated gene products will lead to increased frequency of administration, which is associated with safety concerns and reduced therapeutic efficacy. Thus one aspect of the invention is sa-mRNAs that have reduced cytotoxic effects on the host cell or subject. This provides for enhanced safety of the sa-mRNAs of the present disclosure and provides additional advantages. For example, an advantage of a sa-mRNA with low cytotoxicity allows for administration of a large dose of the sa-mRNAs to produce high expression levels of the encoded gene product with reduced risk of undesired effects, such as injection site irritation and or pain. Because CATP can increase therapeutic payloads dramatically, one can reduce the therapeutic dosage to reduce the toxicity. In certain embodiments, toxicity is reduced at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 75%, at least about 80%, or at least about 90%, for example about 20% to about 90%, about 20% to about 75%, about 30% to about 50%, or about 45% to about 55%.

[0155] One suitable system for producing a sa-mRNA of the present disclosure is to use an alphavirus-based RNA replicon. Alphavirus-based replicons are positive (+)-single stranded replicons that can be translated after delivery to a cell to give a replicase (or replicase-transcriptase). The replicase is translated as a polyprotein which auto-cleaves to provide a replication complex, comprising plurality of non-structural replicase domain sequences, which creates genomic (−)-strand copies of the (+)-strand delivered RNA. These (−)-strand transcripts can themselves be transcribed to give further copies of the (+)-stranded parent RNA and also to give an mRNA transcript which encodes the desired gene product. Translation of the subgenomic transcript thus leads to in situ expression of the desired gene product by the infected cell.

[0156] Whereas natural alphavirus genomes encode structural proteins in addition to the non-structural replicase, in one aspect, an alphavirus based sa-mRNA does not encode alphavirus structural proteins. Thus the sa-mRNA can lead to the production of RNA copies of itself in a cell, but not to the production of RNA-containing alphavirus virions. The inability to produce these virions means that, unlike a wild-type alphavirus, the sa-mRNA cannot perpetuate itself in infectious form. The alphavirus structural proteins which are necessary for perpetuation in wild-type viruses are absent from self-amplifying mRNAs and their place is taken by the GOI, such that the sa-mRNA transcript encodes the desired gene product rather than the structural alphavirus virion proteins.

[0157] The sa-mRNAs described herein may be engineered to express multiple GOI, from two or more coding regions, thereby allowing co-expression of proteins and/or regulatory structures, such as a two or more antigens together with cytokines or other immunomodulators, which can enhance the generation of an immune response. Such a sa-mRNA might be particularly useful, for example, in the production of various gene products (e.g., proteins) at the same time.

[0158] In one aspect, the payload delivered in a payload delivery system of the disclosure include a first nucleic acid construct encoding a sa-mRNA encoding at least one gene of interest (GOI) or a plurality of GOIs; and a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein.

[0159] In one aspect, the first nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[0160] [5' UTR]-[nsP]-[SGP]-[GOI]-[3' UTR]-[Poly A]

[0161] wherein, 5' UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, GOI is one gene of interest or a plurality of genes of interest, 3' UTR is a 3' untranslated region, and Poly A is a poly A tail. In some aspects, the first nucleic acid construct comprise more than one GOI operably linked to one or more SGP. In some aspects, each GOI is linked to a different SGP.

[0162] In some aspects, the first nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[0163] [5' UTR]-[nsP]-[ORF]-[3' UTR]-[Poly A]

[0164] wherein 5' UTR is a 5' untranslated region, nsP is at least one non-structural replicase domain sequence, ORF is two or more open reading frames, 3'

UTR is a 3' untranslated region, and Poly-A is a 3' poly-A tail, wherein the ORF comprises at least two SGP operably linked to at least one GOI.

[0165] In some aspects, the operably linked nucleic acid sequence further comprises one or more linkers. In some aspects, the one or more linkers are independently selected from a sequence comprising any one of SEQ ID NOs: 27-37.

[0166] In one aspect, the first nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[0167] [5' UTR]-[nsP]-[SGP]-[L]-[GOI]-[L]-[3' UTR]-[Poly A]

[0168] wherein, 5' UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, L is a linker independently selected from a sequence comprising any one of SEQ ID NOs: 27-37, SGP is a subgenomic promoter, GOI is one or more genes of interest, 3' UTR is a 3' untranslated region, and poly-A is a poly-A tail.

[0169] In some aspects, the plurality of non-structural replicase domain sequences are obtained from a Group IV positive single strand RNA virus selected from Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, or Astroviridae. In some aspects, the plurality of non-structural replicase domain sequences are obtained from an alphavirus selected from Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiyama virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylgach virus, Highlands J virus, Fort Morgan virus, Ndumu virus, or Buggy Creek virus. In some aspects, the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4). In some aspects, the plurality of non-structural replicase domain sequences are obtained from the TC-83 strain of VEE. In some aspects, the nsP is SEQ ID NO: 19.

[0170] In some aspects, the 5' UTR is selected from a sequence comprising any one of SEQ ID NO: 15, 16, 17, 18, 71, or 72. In some aspects, the 3' UTR comprises SEQ ID NO: 21, 22, 69 or 70.

[0171] In some aspects, the GOI encodes a therapeutic polypeptide, a prophylactic polypeptide, a diagnostic polypeptide, an antigen, a non-coding gene that encodes regulatory structure, or a reporter. In some aspects, the regulatory structure is selected from small interfering RNA (siRNA), micro-RNA (miRNA), an asymmetrical interfering RNA (aiRNA), a Dicer-substrate RNA (dsRNA), a small hairpin RNA (shRNA), small activating RNA (saRNA), transfer RNA (tRNA), or long intergenic non-coding (lincRNA).

[0172] In some aspects, the GOI encodes an infectious pathogen antigen, an allergen antigen, a tumor antigen, or a combination thereof. In some aspects, the GOI encodes an immunomodulator. In some aspects, the immunomodulator comprises a cytokine, a chemokine, anti-neoplastic fusion protein, single domain antibody, bispecific antibody, multi-specific antibody, a stem cell growth factor, a lymphotoxin, a stem cell growth factor, an erythropoietin, a thrombopoietin, an interleukin, an interferon, a fusion protein, a colony stimulating factor, or a combination thereof. In some aspects, the interleukin comprises IL-1, IL-1a, IL-10,

IL-1Ra, IL-2, IL-3, IL-4, IL-6, IL-7, IL-9, IL-10, IL-12, IL-13, IL-15, IL-3, IL-5, IL-6, IL-11, IL-12, IL-10, IL-20, IL-14, IL-16, IL-17, IL-18 or IL-21. In some aspects, the interleukin is selected from SEQ ID NOs: 4-9 or 45-55. In some aspects, the interferon is an interferon- α , - β and - γ . In some aspects, the colony stimulating factor is a granulocyte-colony stimulating factor (G-CSF), or a macrophage colony stimulating factor (M-CSF), or a granulocyte macrophage-colony stimulating factor (GM-CSF), a transforming growth factor beta, a transforming growth factor alpha, a bone morphogenetic protein, a fibroblast growth factor, an insulin-like growth factor, a neurotrophic factor, thrombopoietin.

[0173] In some aspects, the GOI comprises SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47, SEQ ID NO: 48, SEQ ID NO: 49, SEQ ID NO: 50, SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54, SEQ ID NO: 101, or SEQ ID NO: 55.

[0174] In some aspects, the SGP is a viral promoter that is recognized by viral RNA dependent RNA polymerase. In some aspects, the SGP is SEQ ID NO: 1, 2 or 3.

[0175] In some aspects, the first nucleic acid construct comprises SEQ ID NO: 41, 42, 73, 74, 75, 76, 77, 78, 79, 80, or 81.

[0176] In one aspect, the first nucleic acid construct is a sa-mRNA, wherein the sa-mRNA is SEQ ID NO: 73 comprising at least one substitution selected from:

- [0177]** a substitution of C with A at position 2623;
- [0178]** a substitution of T with C at position 3309;
- [0179]** a substitution of A with C at position 3493;
- [0180]** a substitution of A with G at position 3867;
- [0181]** a substitution of A with G at position 4674; or
- [0182]** a substitution of A with G at position 5897.

[0183] In some aspects, the first nucleic acid construct is a sa-mRNA. The sa-mRNAs of the present disclosure may be produced from a nucleic acid template in the form of recombinant DNA expression vectors, RNA replicons or plasmids. The nucleic acid template of the present disclosure encodes two expression units comprising: i) an origin of replication sequence (Ori); ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter; and iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter, wherein the first expression unit encodes a selectable marker and the second expression unit encodes a sa-mRNA.

[0184] In some aspects, the second promoter is a promoter that drives transcription of the encoded self-amplifying mRNA using the second expression unit as a template for in vitro transcription of nucleic acid, e.g. mRNA. Suitable promoters include, for example, T7 promoter, T3 promoter, SV40 promoter, SP6 promoter, T5 promoter, β -lactamase promoter, *E. coli* galactose promoter, arabinose promoter, alkaline phosphatase promoter, tryptophan (trp) promoter, lactose operon (lac) promoter, lacUV5 promoter, trc promoter, tac promoter, Klebsiella phage promoter (K11 RNAP) and the like, or mutants of these promoters. In some aspects, the SP6 promoter comprises SEQ ID NO: 82 (ATTTAGGTGACAC). In some aspects, the K11 RNAP promoter comprises SEQ ID NO: 83 (AATTAGGGCACAC). A sa-mRNA can be prepared by transcribing (e.g., in vitro transcription) a DNA that encodes the

sa-mRNA using a suitable DNA-dependent RNA polymerase, such as: T7 phage RNA polymerase, SP6 phage RNA polymerase, T3 phage RNA polymerase, T5 phage RNA polymerase, RNA polymerase III, RNA polymerase II, Taq polymerase, Vent polymerase, and the like, or mutants of these polymerases. The transcription reaction will contain nucleotides, including modified nucleotides in some aspects, and other components that support the activity of the selected polymerase, such as a suitable buffer, and suitable salts. In some aspects, nucleotide analogs will be incorporated into a sa-mRNA to, for example, alter the stability of such RNA molecules, to increase resistance against RNases, to establish replication after introduction into appropriate host cells ("infectivity" of the RNA), and/or to induce or reduce innate and adaptive immune responses.

[0185] In one aspect, the second promoter is a mutant T7 promoter. In some aspects, a modified T7 promoter comprises at least one insertion at position at the 5' end of the wildtype T7 promoter nucleotide sequence. The modification may be, for example, insertion of a single guanine (G) at the 5' end of the wildtype T7 promoter. In some aspects, the mutant T7 promoter comprises SEQ ID NO: 13 (TAATACGACTCACTATAGGATAGG). In some aspects, the wildtype T7 promoter comprises SEQ ID NO: 14 (TAATACGACTCACTATAGG).

[0186] In one aspect, the present disclosure relates to a method of obtaining RNA, including mRNA or sa-mRNA, comprising: a) performing an in vitro transcription reaction using an initial amount of a nucleic acid molecule comprising a promoter and a nucleic acid sequence encoding a RNA, and b) producing the RNA by in vitro transcription, using the nucleic acid molecule as a template and a RNA polymerase (e.g., T7 polymerase). In some aspects, the promoter is a mutant T7 promoter or a wildtype T7 promoter. In some aspects, an increased copy number of the RNA is produced when the promoter is a mutant T7 promoter compared to when the promoter is a wildtype T7 promoter.

Pharmaceutical Compositions

[0187] Compositions of the disclosure may be formulated in whole or in part as pharmaceutical compositions. For example, a pharmaceutical composition may include a pharmaceutical composition comprising the composition of the disclosure and a pharmaceutically acceptable carrier. Pharmaceutical compositions may further include one or more pharmaceutically acceptable excipients or accessory ingredients such as those described herein. General guidelines for the formulation and manufacture of pharmaceutical compositions and agents are known in the art, for example, Remington's *The Science and Practice of Pharmacy*, 21st Edition, A. R. Gennaro; Lippincott, Williams & Wilkins, Baltimore, Md., 2006. Conventional excipients and accessory ingredients may be used in any pharmaceutical composition, except insofar as any conventional excipient or accessory ingredient may be incompatible with one or more components of a payload delivery system. An excipient or accessory ingredient may be incompatible with a payload delivery system if its combination with the component may result in any undesirable biological effect or otherwise deleterious effect.

[0188] In some aspects, the first nucleic acid construct and the second nucleic acid construct are present in the pharmaceutical composition in a 1:100 to 100:1 ratio. In some aspects, the first nucleic acid construct and the second

nucleic acid construct are present in the pharmaceutical composition in a 50:1 to 1:50, 40:1 to 1:40, 30:1 to 1:30, 25:1 to 1:25, 20:1 to 1:20, 15:1 to 1:15, 10:1 to 1:10, 9:1 to 1:9, 8:1 to 1:8, 7:1 to 1:7, 7:1 to 1:7, 6:1 to 1:6, 5:1 to 1:5, 4:1 to 1:4, 3:1 to 1:3, 2:1 to 1:1, or 1:1 ratio. In some aspects, the first nucleic acid construct and the second nucleic acid construct are present in the pharmaceutical composition in a 1:1 ratio.

[0189] In some aspects, one or more excipients or accessory ingredients may make up greater than 50% of the total mass or volume of a pharmaceutical composition including a composition of the disclosure. For example, the one or more excipients or accessory ingredients may make up 50%, 60%, 70%, 80%, 90%, or more of a pharmaceutical convention. In some aspects, a pharmaceutically acceptable excipient is at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% pure. In some aspects, an excipient is approved for use in humans and for veterinary use. In some aspects, an excipient is approved by United States Food and Drug Administration. In some aspects, an excipient is pharmaceutical grade. In some aspects, an excipient meets the standards of the United States Pharmacopoeia (USP), the European Pharmacopoeia (EP), the British Pharmacopoeia, and/or the International Pharmacopoeia.

[0190] Relative amounts of the one or more payload delivery systems encapsulating biologically active agents, the one or more pharmaceutically acceptable excipients, and/or any additional ingredients in a pharmaceutical composition in accordance with the present disclosure will vary, depending upon the identity, size, and/or condition of the subject treated and further depending upon the route by which the composition is to be administered.

[0191] In certain aspects, the compositions of the disclosure and/or pharmaceutical compositions of the disclosure are refrigerated or frozen for storage and/or shipment (e.g., being stored at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C. For example, the pharmaceutical composition comprising a composition of the disclosure is a solution that is refrigerated for storage and/or shipment at, for example, about -20° C., -30° C., -40° C., -50° C., -60° C., -70° C., or -80° C. In certain aspects, the disclosure also relates to a method of increasing stability of the compositions of the disclosure by storing the compositions at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C., e.g., about -5° C., -10° C., -15° C., -20° C., -25° C., -30° C., -40° C., -50° C., -60° C., -70° C., -80° C., -90° C., -130° C. or -150° C.). For example, the compositions of the disclosure and/or pharmaceutical compositions disclosed herein are stable for about at least 1 week, at least 2 weeks, at least 3 weeks, at least 4 weeks, at least 5 weeks, at least 6 weeks, at least 1 month, at least 2 months, at least 4 months, at least 6 months, at least 8 months, at least 10 months, at least 12 months, at least 14 months, at least 16 months, at least 18 months, at least 20 months, at least 22 months, or at least 24 months, e.g., at a temperature of 4° C. or lower (e.g., between about 4° C. and -20° C.). In one embodiment, the formulation is stabilized for at least 4 weeks at about 4° C. In certain aspects, the pharmaceutical composition of the disclosure comprises a composition of the disclosure disclosed herein and a pharmaceutically acceptable carrier selected from one or more of Tris, an acetate (e.g., sodium acetate), an citrate

(e.g., sodium citrate), saline, PBS, and sucrose. In certain embodiments, the carrier may be at a concentration of 1-100 mM (e.g., including but not limited to any numerical value or range within the range of 1-100 mM such as 1, 2, 3, 4, . . . 97, 98, 99, 100, 10-90 mM, 20-80 mM, 30-70 mM and so on).

[0192] In certain aspects, the pharmaceutical composition of the disclosure has a pH value between about 5 and 8 (e.g., 5, 5.5, 6, 6.5, 6.8 6.9, 7.0, 7.1, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9 or 8.0, or between 7.5 and 8 or between 7 and 7.8). For example, a pharmaceutical composition of the disclosure comprises a composition of the disclosure disclosed herein, Tris, saline and sucrose, and has a pH of about 7.5-8, which is suitable for storage and/or shipment at, for example, about -20° C. For example, a pharmaceutical composition of the disclosure comprises one or more compositions of the disclosure disclosed herein and PBS and has a pH of about 7-7.8, suitable for storage and/or shipment at, for example, about 4° C. or lower. "Stability," "stabilized," and "stable" in the context of the present disclosure refers to the resistance of compositions of the disclosure and/or pharmaceutical compositions disclosed herein to chemical or physical changes (e.g., degradation, particle size change, aggregation, change in encapsulation, etc.) under given manufacturing, preparation, transportation, storage and/or in-use conditions, e.g., when stress is applied such as shear force, freeze/thaw stress, etc.

[0193] In certain embodiments, the pharmaceutical composition of the disclosure contain the biologically active agent(s) at a ratio of 0.01 to 25 mg/ml, 0.1 to 20 mg/ml, 0.2 to 18 mg/ml, 0.5 to 15 mg/ml, 0.7 to 12 mg/ml, 0.9 to 10 mg/ml, 1 to 8 mg/ml, 1.5 to 6 mg/ml, 2 to 5 mg/ml, 2.5 to 4 mg/ml, 0.5 to 3.0 mg/ml, 0.2 to 4.0 mg/ml, 0.4 to 2.0 mg/ml, and any numerical value or range within the range of 0.01 to 25 mg/ml.

[0194] Compositions and/or pharmaceutical compositions of the present disclosure of the disclosure may be administered to any patient or subject, including those patients or subjects that may benefit from a therapeutic effect provided by the delivery of a biologically active agent to one or more particular cells, tissues, organs, or systems or groups thereof. Although the descriptions provided herein of the compositions of the disclosure and pharmaceutical compositions including compositions of the disclosure are principally directed to compositions which are suitable for administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to any other mammal. Modification of compositions suitable for administration to humans in order to render the compositions suitable for administration to various animals is well understood, and the ordinarily skilled veterinary pharmacologist can design and/or perform such modification with merely ordinary, if any, experimentation. Subjects to which administration of the compositions is contemplated include, but are not limited to, humans, other primates, and other mammals, including commercially relevant mammals such as cattle, pigs, horses, sheep, cats, dogs, mice, and/or rats.

[0195] A pharmaceutical composition including the composition of the disclosure may be prepared by any method known or hereafter developed in the art of pharmacology. In general, such preparatory methods include bringing the active ingredient into association with an excipient and/or one or more other accessory ingredients, and then, if desir-

able or necessary, dividing, shaping, and/or packaging the product into a desired single- or multi-dose unit.

[0196] Injectable preparations, for example, sterile injectable aqueous or oleaginous suspensions may be formulated according to the known art using suitable dispersing agents, wetting agents, and/or suspending agents. Sterile injectable preparations may be sterile injectable solutions, suspensions, and/or emulsions in nontoxic parenterally acceptable diluents and/or solvents, for example, as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are water, Ringer's solution, U.S.P., and isotonic sodium chloride solution. Sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose any bland fixed oil can be employed including synthetic mono- or diglycerides. Fatty acids such as oleic acid can be used in the preparation of injectables.

[0197] Injectable formulations can be sterilized, for example, by filtration through a bacterial-retaining filter, and/or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved or dispersed in sterile water or other sterile injectable medium prior to use.

Methods of Delivering Therapeutic Agents to Cells and Organs

[0198] The present disclosure provides methods of delivering a biologically active agent to a cell or organ.

[0199] In some aspects, a composition of the disclosure may target a particular type or class of cells (e.g., cells of a particular organ or system thereof or cancerous cells). For example, a composition of the disclosure including a biologically active agent of interest may be specifically delivered to a liver, kidney, spleen, femur, or lung. Specific delivery to a particular class of cells, an organ, or a system or group thereof implies that a higher proportion of the composition of the disclosure including a biologically active agent are delivered to the destination (e.g., tissue) of interest relative to other destinations, e.g., upon administration of a composition of the disclosure to a mammal. In some aspects, specific delivery may result in a greater than 2 fold, 5 fold, 10 fold, 15 fold, or 20 fold increase in the amount of biologically active agent per 1 g of tissue of the targeted destination (e.g., tissue of interest, such as a liver) as compared to another destination (e.g., the spleen). In some aspects, the tissue of interest is selected from the group consisting of a liver, kidney, a lung, a spleen, a femur, an ocular tissue (e.g., via intraocular, subretinal, or intravitreal injection), vascular endothelium in vessels (e.g., intra-coronary or intra-femoral) or kidney, and tumor tissue (e.g., via intratumoral injection).

[0200] As another example of targeted or specific delivery, an mRNA that encodes a protein-binding partner (e.g., an antibody or functional fragment thereof, a scaffold protein, or a peptide) or a receptor on a cell surface may be included in a composition of the disclosure. An mRNA may additionally or instead be used to direct the synthesis and extracellular localization of lipids, carbohydrates, or other biological moieties. Alternatively, other biologically active agents or elements (e.g., lipids or ligands) of a the composition of the disclosure may be selected based on their affinity for particular receptors (e.g., low density lipoprotein receptors) such that the payload delivery system of the composition of the disclosure may more readily interact with

a target cell population including the receptors. For example, ligands may include, but are not limited to, members of a specific binding pair, antibodies, monoclonal antibodies, Fv fragments, single chain Fv (scFv) fragments, Fab' fragments, F(ab')₂ fragments, single domain antibodies, camelized antibodies and fragments thereof, humanized antibodies and fragments thereof, and multivalent versions thereof; multivalent binding reagents including mono- or bi-specific antibodies such as disulfide stabilized Fv fragments, scFv tandems, diabodies, tribodies, or tetrabodies; and aptamers, receptors, and fusion proteins.

[0201] In some aspects, a ligand may be a surface-bound antibody, which can permit tuning of cell targeting specificity. This is especially useful since highly specific antibodies can be raised against an epitope of interest for the desired targeting site. In one embodiment, multiple antibodies are expressed on the surface of a cell, and each antibody can have a different specificity for a desired target. Such approaches can increase the avidity and specificity of targeting interactions.

[0202] In certain aspects, compositions in accordance with the present disclosure may be administered at dosage levels sufficient to deliver from about 0.0001 mg/kg to about 10 mg/kg, from about 0.001 mg/kg to about 10 mg/kg, from about 0.005 mg/kg to about 10 mg/kg, from about 0.01 mg/kg to about 10 mg/kg, from about 0.05 mg/kg to about 10 mg/kg, from about 0.1 mg/kg to about 10 mg/kg, from about 1 mg/kg to about 10 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 5 mg/kg to about 10 mg/kg, from about 0.0001 mg/kg to about 5 mg/kg, from about 0.001 mg/kg to about 5 mg/kg, from about 0.005 mg/kg to about 5 mg/kg, from about 0.01 mg/kg to about 5 mg/kg, from about 0.05 mg/kg to about 5 mg/kg, from about 0.1 mg/kg to about 5 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 5 mg/kg, from about 0.0001 mg/kg to about 2.5 mg/kg, from about 0.001 mg/kg to about 2.5 mg/kg, from about 0.005 mg/kg to about 2.5 mg/kg, from about 0.01 mg/kg to about 2.5 mg/kg, from about 0.05 mg/kg to about 2.5 mg/kg, from about 0.1 mg/kg to about 2.5 mg/kg, from about 1 mg/kg to about 2.5 mg/kg, from about 2 mg/kg to about 2.5 mg/kg, from about 0.0001 mg/kg to about 1 mg/kg, from about 0.001 mg/kg to about 1 mg/kg, from about 0.005 mg/kg to about 1 mg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 1 mg/kg, from about 0.1 mg/kg to about 1 mg/kg, from about 0.0001 mg/kg to about 0.25 mg/kg, from about 0.001 mg/kg to about 0.25 mg/kg, from about 0.005 mg/kg to about 0.25 mg/kg, from about 0.01 mg/kg to about 0.25 mg/kg, from about 0.05 mg/kg to about 0.25 mg/kg, or from about 0.1 mg/kg to about 0.25 mg/kg of a biologically active agent (e.g., an mRNA) in a given dose, where a dose of 1 mg/kg (mpk) provides 1 mg of a biologically active agent per 1 kg of subject body weight. In some aspects, a dose of about 0.001 mg/kg to about 10 mg/kg of a biologically active agent (e.g., mRNA) of the composition of the present disclosure may be administered. In other aspects, a dose of about 0.005 mg/kg to about 2.5 mg/kg of a biologically active agent may be administered. In certain aspects, a dose of about 0.1 mg/kg to about 1 mg/kg may be administered. In other aspects, a dose of about 0.05 mg/kg to about 0.25 mg/kg may be administered. A dose may be administered one time or multiple times in the same or a different amount, to obtain a desired level of sa-mRNA and mRNA expression and/or therapeutic, diagnostic, or prophylactic effect. In

some aspects, a single dose may be administered, for example, prior to or after a surgical procedure or in the instance of an acute disease, disorder, or condition.

[0203] The composition of the disclosure may be used in combination with one or more other therapeutic, prophylactic, diagnostic, or imaging agents. By “in combination with,” it is not intended to imply that the agents must be administered at the same time and/or formulated for delivery together, although these methods of delivery are within the scope of the present disclosure. For example, one or more compositions of the disclosure may be administered in combination. Compositions can be administered concurrently with, prior to, or subsequent to, one or more other desired therapeutics or medical procedures. In general, each agent will be administered at a dose and/or on a time schedule determined for that agent. In some aspects, the present disclosure encompasses the delivery of compositions, or imaging, diagnostic, or prophylactic compositions thereof in combination with agents that improve their bioavailability, reduce and/or modify their metabolism, inhibit their excretion, and/or modify their distribution within the body.

[0204] It will further be appreciated that biologically active, or imaging active agents utilized in combination may be administered together in a single composition or administered separately in different compositions. In general, it is expected that agents utilized in combination will be utilized at levels that do not exceed the levels at which they are utilized individually. In some aspects, the levels utilized in combination may be lower than those utilized individually.

[0205] The particular combination of therapies (therapeutics or procedures) to employ in a combination regimen will take into account compatibility of the desired therapeutics and/or procedures and the desired therapeutic effect to be achieved. It will also be appreciated that the therapies employed may achieve a desired effect for the same disorder (for example, a composition useful for treating cancer may be administered concurrently with a chemotherapeutic agent), or they may achieve different effects (e.g., control of any adverse effects, such as infusion related reactions).

Methods of Treating Diseases

[0206] The present disclosure provides methods of delivering a biologically active agent to a cell or organ. Compositions of the disclosure may be useful for treating a disease, disorder, or conditions characterized by dysfunctional or aberrant protein or polypeptide activity. Diseases, disorders, and/or conditions characterized by dysfunctional or aberrant protein or polypeptide activity for which a composition of the disclosure may be administered include, but are not limited to, rare diseases, infectious diseases (as both vaccines and therapeutics), cancer and proliferative diseases, genetic diseases (e.g., cystic fibrosis), autoimmune diseases, diabetes, neurodegenerative diseases, cardio- and reno-vascular diseases, lysosomal storage disorders and metabolic diseases. In particular, such compositions may be useful in treating cancer. The present disclosure provides a method for treating such diseases, disorders, and/or conditions in a subject by administering a composition of the disclosure wherein the payload induces apoptosis of cancer cells. In one aspect, the compositions of the present disclosure may be useful for treating adenocarcinoma, anal cancer, basal cell carcinoma, brain cancer, bladder cancer, bone cancer, breast cancer, cervical cancer, cholangiocarcinoma, colorectal can-

cer, endometrial cancer, esophageal cancer, gall bladder cancer, gastric cancer, glioblastoma, head and neck cancer, liver cancer, lung cancer, ovarian cancer, pancreatic cancer, prostate cancer, renal cancer, sarcoma, small bowel carcinoma, skin cancer, testicular cancer, uterine cancer, colon cancer, rectal cancer, intrahepatic bile duct cancer, thyroid cancer, eye cancer, squamous cell carcinoma and melanoma. In some aspects, the disorder is lymphoma, e.g. Epstein-Barr virus associated lymphoma, B-cell lymphoma, T-cell lymphoma Hodgkins and non-Hodgkins lymphoma. In some aspects, the cancer is a squamous cell cancer. In some aspects, the cancer is a head and neck squamous cell cancer. In some aspects, the cancer is a skin squamous cell carcinoma. In some aspects, the cancer is an esophageal squamous cell carcinoma. In some aspects, the cancer is a head and neck squamous cell carcinoma. In some aspects, the cancer is a lung squamous cell carcinoma. In some aspects, the compositions of the present disclosure may be useful for treating MC38 colon cancer, 4T1 breast cancer, Lewis lung carcinoma (LLC1), CT26 colon cancer, P53null KRasG12D pancreatic duct cancer or B16F10 melanoma.

[0207] A composition of the disclosure may be administered by any route. In some embodiments, compositions, including prophylactic, diagnostic, or imaging compositions including one or more nanoparticle compositions described herein, are administered by one or more of a variety of routes, including oral, intravenous, intramuscular, intra-arterial, intramedullary, intrathecal, intraparenchymal, subcutaneous, intraventricular, trans- or intradermal, interdermal, rectal, intravaginal, intraperitoneal, intraocular, subretinal, intravitreal, topical (e.g. by powders, ointments, creams, gels, lotions, and/or drops), mucosal, nasal, buccal, enteral, vitreal, intratumoral, sublingual, intranasal; by intratracheal instillation, bronchial instillation, and/or inhalation; as an oral spray and/or powder, nasal spray, and/or aerosol, and/or through a portal vein catheter. In some embodiments, a composition may be administered intravenously, intramuscularly, intradermally, intra-arterially, intratumorally, subcutaneously, intraocularly, subretinally, intravitreally, intraparenchymally or by any other parenteral route of administration or by inhalation. However, the present disclosure encompasses the delivery or administration of compositions described herein by any appropriate route taking into consideration likely advances in the sciences of drug delivery. In general, the most appropriate route of administration will depend upon a variety of factors including the nature of the composition including one or more biologically active agents (e.g., its stability in various bodily environments such as the bloodstream and gastrointestinal tract), the condition of the patient (e.g., whether the patient is able to tolerate particular routes of administration), etc.

[0208] In one aspect, the present disclosure provides a method of treating a subject having a tumor or cancer, comprising administering to a subject the pharmaceutical composition of the disclosure. In some aspects, the subject is a mouse, a rat, a rabbit, a cat, a dog, a horse, a donkey, a non-human primate, or a human. In one aspect, the pharmaceutical composition is administered intravenously, subcutaneously, intratumorally, intramuscularly, intratracheal, intraperitoneal, or intranasally.

[0209] In one aspect, the method comprises performing a first administering step, comprising administering the pharmaceutical composition of the disclosure wherein the at least one virus structural protein is a first viral capsid, a second

envelope or a combination thereof, and performing a second administering step, comprising administering the pharmaceutical composition of the disclosure wherein the at least one virus structural protein is a second viral capsid that differs from the first viral capsid, a second viral envelope that differs from the first viral envelope or a combination thereof.

[0210] In one aspect, the method induces apoptosis of a cancerous cell or tumor, the method comprising contacting the cancerous cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and at least one virus structural protein replicate within said cancerous cell, to express the GOI, wherein the first nucleic acid replication, expression of the GOI and the at least one virus structural protein replication within the cancerous cell results in cell death. In some aspects, the contacting is *in vivo*.

[0211] In one aspect, the present disclosure provides a method of delivering a GOI encoding a therapeutic gene product to cancerous cells *in vivo* comprising contacting the cancerous cells with the pharmaceutical composition of the disclosure.

[0212] In one aspect, the present disclosure provides a method of effecting *in vivo* synthesis of an oncolytic defective viral particle comprising contacting a cancerous cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the at least one virus structural protein replicate within said cancerous cell and express the GOI.

[0213] In one aspect, the present disclosure provides a method of inducing production of 1) a therapeutic payload 2) a pseudoviral particle and 3) sa-mRNA in a cell comprising contacting a cancerous cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein, the sa-mRNA, and the GOI within said cancerous cell, and wherein the at least one virus structural protein form the pseudoviral particle, which encapsulate one or more of the sa-mRNA.

[0214] In one aspect, the present disclosure provides a method of increasing expression of a polypeptide encoded by a GOI in a cell comprising contacting the cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said cell, and the first nucleic acid construct expresses the GOI within said cell, wherein the at least one virus structural protein form a pseudoviral particle that encapsulate one or more of the sa-mRNA, and wherein expression of the polypeptide is increased by at least 100-fold compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

[0215] In one aspect, the present disclosure provides a method of increasing expression of a polypeptide encoded by a GOI in a cell comprising contacting the cell with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said cell, and the first nucleic acid construct expresses the GOI within said cell, wherein the at least one virus structural protein form a pseudoviral particle, which encapsulate one or more of the sa-mRNA, and wherein expression of the polypeptide is increased by

50-fold, 100-fold, 200-fold, 300-fold, 400-fold, 500-fold, or more compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

[0216] In one aspect, the present disclosure provides a method of increasing central memory CD8 T cell in a tumor draining lymph node or spleen comprising contacting a tumor with the pharmaceutical composition of the disclosure such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said tumor to produce the at least one virus structural protein and the sa-mRNA, and the first nucleic acid construct expresses the GOI within said tumor, wherein the at least one virus structural protein form a pseudoviral particle, which encapsulate one or more of the sa-mRNA, and wherein central memory CD8 T cell count is increased by 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, or more compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

[0217] In one aspect, the present disclosure provides a method of treating cancer, comprising administering to a subject a composition comprising an IL-12 and an IL-18. In some aspects, the IL-12 is a wildtype human IL-12. In some aspects, the IL-12 is a variant IL-12 having at least 90% amino acid sequence identity to a wildtype human IL-12. In some aspects, the IL-12 is a wildtype human IL-18. In some aspects, the IL-18 is a variant IL-18 having at least 90% amino acid sequence identity to a wildtype human IL-18. In some embodiments, the variant IL-18 is encoded by SEQ ID NO. 101.

[0218] In one aspect, the present disclosure provides a method of treating a subject with a payload delivery system encapsulating a mRNA encoding at least one gene of interest (GOI) or a plurality of GOIs and minimizing immunogenicity of the payload delivery system comprising administering the pharmaceutical composition of the disclosure to the subject such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within a cell of the subject, and the first nucleic acid construct expresses the GOI within said cell, wherein the at least one virus structural protein form a pseudoviral particle, which encapsulate one or more of the sa-mRNA, and wherein said non-viral payload delivery system has reduced immunogenicity compared to viral payload delivery systems.

[0219] Compounds of the present disclosure can be prepared in a variety of ways using commercially available starting materials, compounds known in the literature, or from readily prepared intermediates, by employing standard synthetic methods and procedures either known to those skilled in the art, or which will be apparent to the skilled artisan in light of the teachings herein. Standard synthetic methods and procedures for the preparation of organic molecules and functional group transformations and manipulations can be obtained from the relevant scientific literature or from standard textbooks in the field. Although not limited to any one or several sources, classic texts such as Smith, M. B., March, J., March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, 5th edition, John Wiley & Sons: New York, 2001; Greene, T. W., Wuts, P. G. M., Protective Groups in Organic Synthesis, 3rd edition, John Wiley & Sons: New York, 1999; R. Larock, Comprehensive Organic Transformations, VCH Publishers

(1989); L. Fieser and M. Fieser, *Fieser and Fieser's Reagents for Organic Synthesis*, John Wiley and Sons (1994); and L. Paquette, ed., *Encyclopedia of Reagents for Organic Synthesis*, John Wiley and Sons (1995), incorporated by reference herein by reference, are useful and recognized reference textbooks of organic synthesis known to those in the art. The following descriptions of synthetic methods are designed to illustrate, but not to limit, general procedures for the preparation of compounds of the present disclosure.

EXAMPLES

[0220] While the standard viral genome encodes all the viral proteins required for viral replication, some viral variants contain mutations that result from either small mutations or drastic modifications of the viral genome, which may render the virus unable to complete a full replication cycle, called defective viral genome (DVG) (Vignuzzi, 2019, *Nat Microbiol* 4: 1075-1087). DVGs play complicated roles on interactions between the host cell and the virus (e.g. Rezeli, et al. 2021, *Nat Commun* 12: 2290). The oncolytic defective virus used in the experiments described in the following sections was a VEE (TC-83 strain) with a DVG engineered using with the structural proteins deleted for use as a cancer immunotherapy.

[0221] The sa-mRNA used in the experiments described in the following sections was derived from the viral genome of Venezuelan equine encephalitis virus (TC-83) and was engineered by replacing the subgenome that encodes structural proteins for capsids and envelopes with therapeutic payloads, as described in PCT/US2023/066903, which is fully incorporated herein by reference.

[0222] Typically, the therapeutic oncolytic defective virus is engineered in vitro, which is very time consuming and expensive (see, e.g., Ungerechts, et al., 2016, *Mol. Ther. Methods Clin. Dev.* 3: 16018). In one aspect, the present invention is a method of producing pseudoviral particles encapsulating sa-mRNA encoding a therapeutic GOI, such as IL12, and a therapeutic in vivo using the composition of the disclosure, which comprises a payload delivery system encapsulating a first nucleic acid construct encoding a sa-mRNA encoding the therapeutic and a second nucleic acid construct encoding an mRNA encoding at least one virus structural protein.

[0223] Notably, the composition used in the studies below, composed of LNPs encapsulating sa-mRNA encoding IL12 (SamRNA-IL12; SEQ ID NO: 41) in combination with a modified mRNA encoding the structural proteins from TC-83 strain of VEE (SEQ ID NO: 26), showed a over 500x expression of IL12 in the intratumorally injection site compared to mice treated with a negative control (FIG. 3B). Furthermore, the composition of the disclosure dramatically enhanced central memory CD8 T cell presence in the tumor draining lymph node, showing 10x higher numbers of effective CD8 T cells in both tumor-draining lymph node (TDLN) and spleen (FIGS. 4B-4C). Consequently, tumor growth was significantly suppressed in mice treated with the composition of the disclosure compared to the negative control (FIG. 5B), with 40-60% of the treated mice being tumor free after a single treatment with the composition of the disclosure. Importantly, the treatment showed reduced side effects such as body weight changes in the treated mice compared to mice treated with the negative control.

Example 1: In Vitro Testing

[0224] In one aspect, after the composition of the present disclosure is introduced to a host cell, the payload of the composition in the host cell will be expressed, and the expression of the payload of the composition will cause cell death of the host cell and adjacent cells as shown in FIG. 1. On the left side of FIG. 1, a LNP delivery system encapsulating a sa-mRNA encoding a therapeutic GOI and a mRNA encoding the capsid and envelope proteins of an oncolytic virus is introduced into the host cell. The payloads are released from the LNP inside the host cell and produces a therapeutic, new copies of the sa-mRNA and the capsid and envelope proteins of an oncolytic virus. The capsid and envelope proteins of an oncolytic virus form a pseudoviral particle encapsulating the new copies of the sa-mRNA, which is released to infect adjacent cells. The host cell will release the therapeutic out of the cell, which will affect adjacent cells. In the middle of the diagram, an adjacent cell is infected with the pseudoviral particle encapsulating the copies of the sa-mRNA and produces the therapeutic. On the right side of the diagram, the oncolytic effects of the pseudoviral particle and increased adaptive immune responses triggered by the therapeutic cause immunogenic cell death in affected cells, including the host cell and the adjacent cell. By co-delivering SamRNA from VEE encoding therapeutic payloads and modified mRNA encoding optimized viral capsids/envelop using lipid nanoparticles, the inherent amplification of self-amplifying mRNA is leveraged, which is referred to as cascade amplification of therapeutic payloads (CATP) in this work. Additionally, as shown in FIG. 10A, the production of defective viruses enables the infections of adjacent cells, further amplifying the therapeutic payloads and enhancing overall efficacy.

[0225] As shown in FIG. 10A, HEK293 cells were treated by LNP encapsulated SamRNA mRNA encoding with eGFP and modified mRNA encoding with envelop and capsids from VEE or LNP encapsulated SamRNA mRNA encoding with eGFP and modified mRNA encoding firefly Luciferase. The GFP percentages were then determined by flow cytometer at day 1, 2, 3, 4 post transfection.

[0226] VEE SamRNA plasmid DNA was prepared based on the constructs previously developed in the PCT/US2023/066903, "Synthetic SamRNA Molecules with Secretion Antigen and Immunomodulator". Modified mRNA plasmid DNA was prepared based on the constructs previously developed in the PCT/US2023/085919, "Compositions and Methods for Delivering Molecules" the contents of each of which are herein incorporated by reference.

[0227] eGFP, firefly luciferase, mouse IL-12, and mouse IL-18 were cloned into XbaI and ClaI after the subgenomic promoter of VEE SamRNA plasmid DNA. Capsids/envelops from VEE, Sindbis virus, and Semliki Forest virus (SFV) 4 were cloned after T7 promoter using seamless cloning technology.

[0228] SamRNA were in vitro transcribed (IVT) from the templates of linearized VEE DNA constructs using the NEB RNA synthesis kit (Catalog No. E2050S). The synthesized mRNAs were then capped and methylated by Cellscript kits (Catalog No. C-SCCS1710). Modified mRNAs were in vitro transcribed (IVT) from the templates of linearized VEE DNA constructs using the NEB RNA synthesis kit (Catalog No. E2040S) plus pseudouridine (Catalog No. N-1019) and Cleancap (Catalog No. N-7413) from Trilink. The quantity

and purity of SamRNA and modified mRNA were assessed by Nanodrop and gel electrophoresis.

[0229] The size, polydispersity index and zeta potentials of LNPs were measured using dynamic light scattering (Z-100-Z2(MTS), HORIBA, Ltd.). Diameters are reported as the intensity mean peak average. To calculate the nucleic acid encapsulation efficiency, a modified Quant-iT RiboGreen RNA assay (Invitrogen) was used. Results of encapsulation efficiency can be found in Table A below.

TABLE A				
Encapsulation efficiency and particle properties of LNPs				
	LNP-Sam (IL12) _{mod} (Lucif)	LNP-Sam (IL12) _{mod} (VEE)	LNP-Sam (IL12) _{mod} (SFV4)	LNP-Sam (IL12) _{mod} (SfM)
Size (nm)	119.3 ± 0.6	123.2 ± 2.1	122.2 ± 4.0	120.0 ± 1.8
PDI	0.17 ± 0.02	0.17 ± 0.04	0.17 ± 0.02	0.21 ± 0.01
Zeta Potential	-3.2 ± 3.6	-6.8 ± 1.9	-7.1 ± 4.3	-4.3 ± 1.0
EE(%)	90.5 ± 1.4	93.4 ± 0.6	93.5 ± 1.1	92.8 ± 1.5

[0230] For in vitro transfection, the cells were plated at 30% confluence at day 0. The formulated nanoparticles containing 100 ng mRNA were added to 100 μ l of medium in a well of a 96-well plate at day 1. Then the transfected cells were followed with analysis accordingly.

[0231] Cell lines HEK293 (ATCC CRL-1573), B16F10 (ATCC CRL-6475), CT26 (ATCC CRL-2638), were cultured following vendor instructions. MC38 and KPC cells were maintained in the Li lab at the University of Michigan. Female C57BL/6 (Charles River Lab), Balb/C (Charles River Lab) mice at 6-8 weeks of age were purchased and maintained in the animal facility at the Mispro Biotech Service Corporation, Alewife. All animal studies and procedures were carried out following federal, state and local guidelines under an institutional animal care and use committee-approved animal protocol by the Committee of Animal Care at Mispro.

[0232] One million B16F10, MC38, CT26 or KPC cells in 50 μ l of sterile PBS were s.c. injected into the flank of mice. At 7 days later when tumors reached ~50 mm² in size, animals were injected intratumorally with PBS (control) or LNPs in 50 μ l of PBS as indicated. For the rechallenge, 0.1 million B16F10 or 0.2 million MC38 cancer cells were s.c. injected into another side flank of tumor free mice or naïve mice.

[0233] Antibodies against mouse CD4 (Catalog No. 100412), CD8 (Catalog No. 100766), CD3e (Catalog No. 155612), CD62L (Catalog No. 104406), CD122 (Catalog No. 123216), KLRG1 (Catalog No. 138418), 7-AAD (Catalog No. 420404), and Zombie Aqua (Catalog No. 423102) were bought from Biolegend. All the antibodies were diluted 1:50. The live/dead dye Zombie Aqua was diluted 1:300. The single-cell suspensions were filtered by 70- μ m nylon strainers and stained as described 20. Stained samples were analyzed using a Symphony A5 FACS analyzer from BD Biosciences. All flow cytometry data were analyzed using FlowJo software (Flowjo LLC).

[0234] Tumors were collected and ground in tissue protein extraction reagent (T-PERTM, Thermo Fisher Scientific, cat. no. 78510) in the presence of 1% proteinase and phosphatase inhibitors (Thermo Fisher Scientific, cat. no. 78442). The lysates were incubated at 4° C. for 30 min with slow rotation then centrifuged to remove debris. The supernatants were

transferred to a clean tube for ELISA or Luminex analysis. Mouse IL-12 in tumor tissue supernatants or in serum were measured by ELISA kits from R&D (Catalog No. DY419) following the manufacturer's instructions.

[0235] Data were statistically analyzed by one-way or two-way ANOVA or by Student's t-test using GraphPad PRISM as indicated. Animals were randomized to treatment groups once the mean tumor size around 50 mm2 was reached by the tumor-inoculated cohort. No data were excluded from the analyses. The investigators were not blinded to allocation during experiments and outcome assessments. The samples sizes for in vitro analysis were three (triplicates) and for in vivo analysis are as annotated in figure legends. The details of statistical analysis for figures and Extended Data Figures are included in the Source Data files.

Example 2: In Vitro Testing

[0236] To study the effects of delivering the composition of the disclosure to a host cell and production of the sa-mRNA, GOI and pseudoviral particle by the host cell in vitro. Sa-mRNA encoding GFP (SamRNA-GFP; SEQ ID NO: 42) and modified mRNA encoding luciferase (mRNA-Luc) (SEQ ID NO: 43) or modified mRNA encoding Cap/Env were encapsulated in an LNP comprising lipid components SV1 and P6 and transfected into HEK293 cells. The SamRNA-GFP with modified mRNA encoding Cap/Env showed a more gradual increase in GFP expression compared to mRNA-Luc post transfection, likely due to higher immunogenicity of Cap/Env compared to Luc, which triggered innate immunity and resulted in feedback inhibitions. Interestingly, as shown in FIG. 2A, while GFP expression showed a more gradual increase in days 2 to 3, GFP expression increased at day 4.

[0237] To validate the platform in vitro, HEK293 cells were co-transfected with SamRNA encoding eGFP and modified mRNA encoding VEE capsids/envelop (CATP: Sam(eGFP)_mod(VEE)) and were compared to cells transfected with SamRNA encoding eGFP (Sam(eGFP)_mod(Luci)). Transfection efficiency was determined by flow cytometry and the percentages of GFP cells were normalized by transfection efficiencies in the group of encapsulated self-amplifying mRNA (Sam002) encoding eGFP and modified mRNA encoding luciferase at day 1 post treatments. Statistic significance was analyzed by two-way ANNOVA. As anticipated, the percentages of eGFP-positive cells were initially lower in CATP group than control group within the first three days post transfection. However, the CATP group markedly expanded on day 3 and surpassed the control group after day 4, while the control group started to decay after day 2, as shown in FIG. 2A. This observation underscored the key feature of CATP in terms of amplifying reporter genes in vitro.

[0238] Having validated this platform in HEK293 cells, a similar experiment was conducted in ARPE-19 cells (human retinal pigment epithelial cells). APRE-19 cells were co-transfected with SamRNA (Sam002) encoding eGFP and modified mRNA encoding: VEE capsids/envelop (CATP: Sam(eGFP)_mod(VEE)); SFV4 capsids/envelop (CATP: Sam(eGFP)_mod(SFV4)); or SINV capsids/envelop (CATP: Sam(eGFP)_mod(SINV)). The capsid/envelope-transfected APRE-19 were compared to cells transfected with Sam002

RNA encoding eGFP (Sam(eGFP)_and modified RNA encoding luciferase (mod(Luci)). The results are shown in FIG. 2B.

[0239] P6 ionizable lipid was designed and synthesized in the patent (PCT/US2023/017777). An ethanol phase was prepared by dissolving P6, (DOPE (870341, Avanti Research), cholesterol (7001, Avanti Research), and DMG-PEG2000 (880151, Avanti Research) at a predetermined molar ratio of 30:15:50:1.5. The aqueous phase was prepared in 50 mM citrate buffer (pH 4.5, AAJ60024AK, Fisher Scientific) containing SamRNA and/or mRNA. All mRNA samples were stored at -80°C . and thawed on ice prior to use. The aqueous and ethanol phases were combined at a 3:1 ratio with a lipid-to-RNA (N/P) ratio of 4.2, using a microfluidic chip device (iNano™ L system, Micro & Nano) at a flow rate of 12 mL/min. The resulting lipid nanoparticles (LNPs) were dialyzed against 1×PBS (MT21040CMX, Fisher Scientific) using a Slide-A-Lyzer™ MINI Dialysis Devices, 20K MWCO (Catalog No. 88405 Fisher Scientific) at 25°C . for 80 min and stored at 4°C . prior to injection.

Example 3: In Vivo Testing

[0240] To study the effects of delivering the composition of the disclosure to a host cell and production of the sa-mRNA, GOI and pseudoviral particle by the host cell in vivo. The amplification capacity of CATP was performed in a highly immune resistant mouse B16F10 melanoma model via a single intratumoral injection of lipid nanoparticles encapsulating CATP encoding mouse cytokine IL-12. Delivery of recombinant IL-12 via different technologies has been shown to exhibit potent anti-tumor immunity by stimulating CD4, CD8 T cells, NK cells, and B cells, while its prolonged systemic exposure is associated with host toxicity. As shown in FIG. 3A, sa-mRNA encoding mIL12 and mRNA-Luc or modified mRNA encoding Cap/Env were encapsulated in an LNP comprising lipid components SV1 and P6 and transfected into C57BL6 mice seven days after the mice were inoculated with 1 million B16F10 melanoma cells. Seven days post-inoculation, mice were intratumorally treated with PBS, or LNP encapsulating 10 g SamRNA encoding with mouse IL-12 (mIL-12) and 1 g modified mRNA encoding with firefly Luciferase or LNP encapsulated 10 g of SamRNA encoding mIL-12 and 1 g modified mRNA encoding capsids/envelops from VEE, SFV4 and SIN.

[0241] Serum was collected at day 1, 3, and 7 post injection, and tumor tissues were collected at day 7 post injection. At day 1 post injection, the average level of mouse IL12 within each group of 5 replicates was 5× higher in the mice treated with the composition including SamRNA-mIL12 and mRNA encoding Cap/Env compared to the composition including SamRNA-mIL12 and mRNA-Luc. At day 3 and 7 post injection, the amount of mIL12 in the serum showed a significant decrease in all three groups. Surprisingly, the level of mIL12 in the tumor of the mice treated with SamRNA-IL12 and modified mRNA encoding Cap/Env is 500× higher than the mice treated with SamRNA-IL12 and mRNA-Luc. Compared to the control group containing SamRNA encoding mouse IL-12 (Contr: Sam(IL-12)_mod(Luci)), the CATP group containing SamRNA-IL12 and VEE capsids/envelop (CATP: Sam(IL-12)_mod(VEE)) displayed five-fold increase in mouse serum IL-12 levels on day 1, remained detectable on day 3, and became undetectable by day 7. Notably, the mouse IL-12 levels in tumor sites increased by 525 times in CATP

group relative to that of control group, as shown in FIGS. 3A-3B. These results demonstrated that the CATP system outperformed the conventional SamRNA platforms in the expression of transgenes both in vitro and in vivo. To further evaluate whether the increased therapeutic levels achieved through a single dose of CATP-mIL-12 improves therapeutic efficacies in the B16F10 melanoma model that's a poorly immunogenic and highly aggressive model for cancer immunotherapy. On day 7 post-inoculation of cancer cells when tumor area reached approximately 50 mm, mice were intratumorally treated with a single dose of CATP or corresponding controls as described in FIG. 3A. Consistent with the increased therapeutic levels observed in FIG. 3B, tumors in the CATP-IL-12 group showed improved regression, as shown in FIG. 5B, resulting in 40% of tumor-free mice ($p=0.0357$, as shown in FIG. 5C). Notably, body weight changes remained below 4%, as shown in FIG. 5D, aligning with the undetectable serum IL-12 levels on day 7 post injections of LNP-mRNA, as shown in FIG. 3B. In summary, the CATP not only increases the expression of therapeutic payloads but also enhances therapeutic efficacy while maintaining the same dosage and safety profiles, demonstrating its potential as a safe and effective cancer immunotherapy strategy.

Example 4: In Vivo Immunogenicity Testing

[0242] To better understand the immune responses of the in vivo testing of Example 3, three groups of five C57BL6 mice were subcutaneously inoculated 1 million B16F10 cells. Seven days later, the tumors, which were approximately 50 mm² in size, were intratumorally treated according to the treatment plan described in Example 3 and shown in FIG. 4A. Then the spleen (FIG. 4C) and the TDLN (FIG. 4B), here the inguinal lymph node, were isolated and prepared into single cell suspensions for fluorophore activated cell sorting. The live CD8 T cells were gated and further analyzed by central markers (e.g. (CD8+ CD62L+ CD122+) CD62L and CD122, or by effective makers (CD8+ CD62L-KLRG1+) CD62L and KLRG1 (e.g. Liu, et al., 2015, Front Immunol 6:494; Herndler-Brandstetter, et al., 2018, Immunity 48:716-729 e718)). In contrast to the tumors treated with PBS, both groups treated with a composition including SamRNA-mIL12 showed 10× higher effector CD8 T cell levels. Interestingly, the central memory CD8 T levels were significantly higher in the group treated with the composition including SamRNA-IL12 and mRNA encoding Cap/Env than the group treated with the composition including SamRNA-IL12 and mRNA-Luc.

Example 5: In Vivo Therapeutic Efficacy Testing

[0243] To better understand the therapeutic efficacy of the in vivo testing of Example 3, three groups of five C57BL6 mice were subcutaneously inoculated 1 million B16F10 cells. Seven days later, the tumors, which were approximately 50 mm in size, were intratumorally treated according to the treatment plan described in Example 3 and shown in FIG. 5A. Then the tumor size, survival rate, and body weight were measured, as shown in FIGS. 5B-5D respectively. Tumor growth was significantly delayed in the group treated with the composition including SamRNA-mIL12 and mRNA encoding Cap/Env. 2 of the 5 mice were tumor free in the group treated with the composition including SamRNA-mIL12 and mRNA encoding Cap/Env, which is

consistent with the increase of mIL12 and central memory CD8 T cells in vivo seen in Example 3 and 4 (FIGS. 3B and 4B). To determine side effects of the treatment, the body weight of the tumor bearing mice were measured and showed only a slight, less than 5%, decrease 2 days post injection, which gradually recovered to pre treatment levels after day 2.

[0244] To understand the therapeutic efficacy of the composition of the disclosure on a different cancer type, a similar study was conducted using MC38 colon cancer cells in vivo. Six to eight weeks old C57B6L mice (5 replicates in each group) were subcutaneously inoculated with 0.5 million MC38 colon cancer cells. Seven days post inoculation, the mice were intratumorally injected with PBS as a negative control; an LNP comprising lipid components SV1 and P6 encapsulating SamRNA-mIL12 and a modified mRNA encoding capsid and envelop proteins from VEE virus; an LNP comprising lipid components SV1 and P6 encapsulating a sa-mRNA encoding GOI mIL18 (SEQ ID NO: 53) and a modified mRNA encoding capsid and envelop proteins from VEE virus; or an LNP comprising lipid components SV1 and P6 encapsulating a sa-mRNA encoding mIL18 mutant (SEQ ID NO: 54) and a modified mRNA encoding capsid and envelop proteins from VEE virus. FIG. 6 shows tumor area of the MC38 colon cancer tumor (Y-Axis) versus days post intratumorally injection (X-Axis) in the treated mice. The statistical analysis were performed by 2-way ANNOVA.

[0245] To investigate the roles of different capsids/envelops in determining the amplification efficiency for CATP, an evaluation of the capsids/envelops derived from three commonly used alphaviruses, including VEE, Sindbis virus (SIN), and Semliki Forest virus (SFV4) was conducted. Interestingly, the SFV4-derived capsids/envelops exhibited the strongest regression of tumor growth, as shown in FIG. 10B, resulting in a 60% tumor free rate with minimal weight changes, as shown in FIG. 8 and FIG. 10C. These findings suggest that CATP leveraging SFV4 capsids/envelops may generate oncolytic virus-like particles, in coincidence with other researchers' observations of oncolytic effects associated with SFV4 viruses.

[0246] To further enhance therapeutic efficacy, the therapeutic payloads were optimized by combining mouse IL-12 with mouse IL-18, as IL-18 enhances polarizations towards Type 1 inflammation in the presence of IL-12. However, IL-18 binding protein (IL-18BP) has been reported to be upregulated in solid tumors thereby inhibiting IL-18 activity. Therefore, wildtype mouse IL-18 was compared to a mutant form, which was designed to disrupt interactions with IL-18BP.

[0247] Following a single dosing in tumors with a starting size of ~50 mm, the combination of mIL-12 with mutant mIL-18 in the CATP system significantly inhibited the tumor growth and increased the tumor-free rates to 80% in the B16F10 melanoma model. This outcome markedly surpassed the 40% tumor-free rates achieved by mIL-12 plus wild-type mIL-18, or mIL-12 alone, as shown in FIG. 6 and FIG. 9A. While the CATP with mouse mutant IL-18 alone effectively inhibited tumor growth, it failed to cure any tumor-bearing mouse, as shown in FIG. 6 and FIG. 9A, indicating that the anti-tumor effects of mutant IL-18 were dependent on mouse IL-12. Although the combination of mIL-12 with mutant mIL-18 was initially associated with body weight loss of >6% after CATP administration, they

returned to normal body weights similar to the control group (PBS) one week after the dosing (14 days post tumor inoculation, $p=0.8421$), as shown in FIG. 9B. These findings suggest that the CATP system combining mIL-12 and mutant IL-18 displayed superior efficacy without compromising the safety following a single administration in tumors.

[0248] To generalize the CATP system to other syngeneic mouse cancer models, the MC38 colorectal cancer model was evaluated, where the CATP therapy combining mIL-12 and mutant mIL-18 eradicated all the tumors, as shown in FIGS. 11A-11C. In the CT26 colon cancer model, the CATP therapy with mIL-12 and mutant mIL18 achieved 80% tumor-free rates, outperforming 20% tumor free rates with CATP mIL-12 alone. There is no significant difference between the body weights of CATP therapy combining mIL-12 and mutant mIL-18 group and the control group (PBS) one week after the dosing, as shown in FIGS. 12A-12C. To assess the efficacy of this approach in genetically driven cancers, mice were subcutaneously inoculated with P3^{mut} Kras^{G12D} pancreatic duct cancer¹⁶ cells and treated intratumorally. Consistent with previous findings, the CATP combination therapy with mIL-12 and mutant mIL-18 resulted in better inhibition of tumor growth and 80% of mice being tumor free, compared to 60% with IL-12 alone, as shown in FIG. 9C and FIGS. 12D-12E. In summary, CATP combining IL-12 and mutant IL-18 demonstrates superior efficacy across multiple cancer models. Moreover, no overt body weight differences were detected between the combination and the vehicle control group (PBS) one week after dosing, although the combination treatment was associated with an initial decrease in body weight within 2-3 days after dosing, as shown in FIGS. 11C, 12C, and 12E.

[0249] To determine whether tumor-free mice treated with the CATP combination therapy developed long-term immune memory in B16F10 and MC38 models, the cured mice were rechallenged with original tumor cells. Of note, all tumor-free mice were able to reject the tumor cells, in contrast to the age- and sex-matched naïve mice that developed tumors, as shown in FIG. 9D and FIGS. 13A-13C. These results suggested that the CATP system induces robust long-term immune memory. To investigate the mechanisms underlying this immune memory, B16F10 melanoma was intratumorally treated with the CATP therapy using mouse IL-12 alone, as shown in FIG. 4A. Flow cytometry analysis of tumor draining lymph nodes (TDLN) and spleen showed that CATP group significantly ($p=0.0006$) increased the number of memory precursor cells characterized by CD62L⁺ CD122⁺ CD8 T cells in TDLN, rather than in spleen, compared to the control treatment group, as shown in FIGS. 4A-4B. However, no significant differences were observed in CD62L⁺ KLRG1⁺ cytotoxic CD8 T¹⁹ cells either in TDLN or in spleen, as shown in FIG. 4C. These findings suggested that the CATP therapy likely promotes the increase of the memory precursor CD8 T cells, contributing to long term protection against tumor recurrence. A novel cascade amplification of therapeutic payloads (CATP) strategy was developed to overcome the challenges of achieving therapeutic payload thresholds in cancer therapy. The CATP system demonstrated a remarkable 525-fold increase in mouse IL-12 expression in the B16F10 melanoma model compared to conventional samRNA delivery methods. This innovative approach significantly enhances therapeutic

potency, presenting a viable solution for gene therapies requiring high payloads without escalating dosage levels.

[0250] It was also shown that the CATP system co-expressing mouse IL-12 and a mutant form of IL-18 was effective against multiple cancer types, including B16F10 melanoma, MC38 and CT26 colon cancers, and P53^{null} K^{RasG12D} pancreatic duct cancer. Importantly, the system does not require personalized tumor-associated antigens, enabling its broad applicability across diverse malignancies. These findings position the CATP strategy as a promising platform for developing broad-spectrum cancer immuno-therapies with robust therapeutic potential.

Example 6: Production of sa-mRNA

[0251] A study was conducted to produce sa-mRNA constructs. In one aspect, the first nucleic construct is an sa-mRNA designed using the method described herein. Directed evolution of SAM001 and SAM002 (SEQ ID NOs: 58 and 59 respectively), encoding GFP (SEQ ID NO: 24) and puromycin resistance gene (puromycin) (SEQ ID NO: 68) according to Tables 1-2 below, was performed at different concentrations (1 ug/ml or 10 ug/ml) of puromycin in C2C12 cells for 60 days.

TABLE 1

Puromycin (ug/ml)	Contigs	Nucleotide	Amino Acids	Region
1	5	G5795A	Glu to Lys	nsP4
10	2	C2623A	Asp to Glu	nsP2
10	3	T3309C	Met to Thr	nsP2
10	3	A3493C	Glu to Asp	nsP2
10	3	A3867G	Lys to Arg	nsP2
10	4	A4674G	Lys to Arg	nsP3
10	5	A5897G	Arg to Gly	nsP4

TABLE 2

Contigs	2	3	4	5
	WT	WT	WT	WT
	a1(C2623A)	b1(T3309C) b2(A3493C) b3(A3867G)	c1(A4674G)	d1(A5897G)
Allele No	2	4	2	2

[0252] RNA dependent RNA polymerases are known to have a high error rate, which will cause mutants of SAM002 to appear over time. The SAM002 construct as shown in FIG. 7 was divided into 8 contigs, marked 1-8 on FIG. 7, comprising SEQ ID NOs. 60-67. The 8 overlapping contigs facilitate cloning to a vector form suitable for Sanger DNA sequencing. As shown in Table 1, only 1 mutation was found at nsP4 in 1 ug/ml puromycin. In contrast, there were 6 mutations found in nsP2, nsP3, and nsP4 at 10 ug/ml puromycin. The 6 mutations were in the 2nd, 3rd, 4th, 5th contigs. The 6 mutations of SAM002 is numbered as alleles 2, 4, 2, 2 respectively, which could form 32 variants. Tables 3-13 below show 66 variations of nucleic acid templates capable of producing 66 sa-mRNA variations produced using the method described above using SAM001 and SAM002 as a starting point, wherein the position of the mutations are counted from the first nucleotide of SEQ ID NOs: 58 (SAM001) or 59 (SAM002):

TABLE 3

No.	% of GFP normalized at day 3		
1	a1BCD-Sam002	93.60967	85.55008
2	a1BCD-Sam001	26.3245	26.94064
3	Ab1CD-Sam002	84.06015	83.50669
4	Ab1CD-Sam001	20.36517	21.08844
5	Ab2CD-Sam002	88.83792	89.1369
6	Ab2CD-Sam001	35.46667	36.71875
7	Ab3CD-Sam002	42.59259	41.76991
8	Ab3CD-Sam001	7.362637	10.01764
9	Abc1D-Sam002	75.67568	74.12224
10	Abc1D-Sam001	25.06739	19.02174
11	ABCD1-Sam002	88.4273	84.55657
12	ABCD1-Sam001	34.73282	31.71355
13	a1b1CD-Sam002	82.53968	75.84803
14	a1b1CD-Sam001	24.43325	22.82472
15	a1b2CD-Sam002	90.82707	85.01441
16	a1b2CD-Sam001	43.10345	43
17	a1b3CD-Sam002	47.98808	51.11441
18	a1b3CD-Sam001	13.7415	11.36182
19	a1Be1D-Sam002	68.26265	70.17544
20	a1Be1D-Sam001	24.86842	26.52005
21	a1BCd1-Sam002	81.64557	85.66929
22	a1BCd1-Sam001	23.58621	28.09587

TABLE 4

No.	% of MFI normalized at day 3		
1	a1BCD-Sam002	26.70408	24.86716
2	a1BCD-Sam001	92.66064	71.8318
3	Ab1CD-Sam002	29.60338	27.95987
4	Ab1CD-Sam001	54.84388	72.12059
5	Ab2CD-Sam002	25.2417	22.28458
6	Ab2CD-Sam001	79.93834	75.08098
7	Ab3CD-Sam002	62.02842	59.2609
8	Ab3CD-Sam001	72.97279	46.89674
9	Abc1D-Sam002	44.37571	45.86502
10	Abc1D-Sam001	80.15279	73.20619
11	ABCD1-Sam002	19.66246	19.21909
12	ABCD1-Sam001	64.61213	70.72639
13	a1b1CD-Sam002	26.06985	28.32976
14	a1b1CD-Sam001	63.195	56.67116
15	a1b2CD-Sam002	21.32604	21.28533
16	a1b2CD-Sam001	69.66121	71.22326
17	a1b3CD-Sam002	60.85935	58.26164
18	a1b3CD-Sam001	61.42157	45.29323
19	a1Be1D-Sam002	49.57803	46.91416
20	a1Be1D-Sam001	68.03935	78.72531
21	a1BCd1-Sam002	18.37446	19.01418
22	a1BCd1-Sam001	61.15648	70.99497

TABLE 5

No.	nsP3 transcripts normalized with Actin		
1	a1BCD-Sam002	9.447941	12.38052
2	a1BCD-Sam001	51.62507	58.08123
3	Ab1CD-Sam002	14.42001	24.59
4	Ab1CD-Sam001	84.44851	119.4282
5	Ab2CD-Sam002	14.92853	16.91229
6	Ab2CD-Sam001	63.11889	64.89341
7	Ab3CD-Sam002	36.75835	41.35529
8	Ab3CD-Sam001	177.294	199.4661
9	Abc1D-Sam002	19.42712	17.5087
10	Abc1D-Sam001	66.71781	57.68003
11	ABCD1-Sam002	11.71269	10.48315
12	ABCD1-Sam001	60.96883	58.89201
13	a1b1CD-Sam002	17.75311	17.02992
14	a1b1CD-Sam001	148.0561	121.9377
15	a1b2CD-Sam002	8.75435	8.15241
16	a1b2CD-Sam001	7.568461	6.453134

TABLE 5-continued

No.	nsP3 transcripts normalized with Actin		
17	a1b3CD-Sam002	56.88593	47.83518
18	a1b3CD-Sam001	182.2784	178.5272
19	a1Bc1D-Sam002	36.50444	37.27147
20	a1Bc1D-Sam001	120.2589	124.4998
21	a1BCd1-Sam002	10.77787	11.47164
22	a1BCd1-Sam001	27.09585	22.3159

TABLE 6

No.	eGFP transcripts normalized with Actin		
1	a1BCD-Sam002	5.133704	4.958831
2	a1BCD-Sam001	37.79177	40.50421
3	Ab1CD-Sam002	7.110741	10.05611
4	Ab1CD-Sam001	62.6829	72.50457
5	Ab2CD-Sam002	6.062866	5.61778
6	Ab2CD-Sam001	41.64294	49.86653
7	Ab3CD-Sam002	21.55574	22.47112
8	Ab3CD-Sam001	90.50967	92.41147
9	Abc1D-Sam002	8.514961	6.868523
10	Abc1D-Sam001	67.64915	56.49299
11	ABCD1-Sam002	3.41054	3.24901
12	ABCD1-Sam001	24.93327	26.72281
13	a1b1CD-Sam002	6.543216	6.147501
14	a1b1CD-Sam001	86.82268	76.10926
15	a1b2CD-Sam002	3.116658	2.8481
16	a1b2CD-Sam001	2.789487	2.584706
17	a1b3CD-Sam002	25.81254	23.91759
18	a1b3CD-Sam001	51.62507	33.3589
19	a1Bc1D-Sam002	12.81712	11.15795
20	a1Bc1D-Sam001	50.21338	50.91434
21	a1BCd1-Sam002	2.694467	3.052518
22	a1BCd1-Sam001	3.97237	3.917681

TABLE 7

No.	% of GFP normalized at day 3		
23	Ab1c1D-Sam002	57.94271	55.04711
24	Ab1c1D-Sam001	14.20749	14.68531
25	Ab2c1D-Sam002	70.09736	68.3844
26	Ab2c1D-Sam001	20.82067	23.27965
27	Ab3c1D-Sam002	34.7651	33.24397
28	Ab3c1D-Sam001	13.96648	20.9589
29	Ab1Cd1-Sam002	74.92308	76.68232
30	Ab1Cd1-Sam001	15.24476	20.0569
31	Ab2Cd1-Sam002	84.28571	82.7957
32	Ab2Cd1-Sam001	33.70474	30.24251
33	Ab3Cd1-Sam002	44.31138	39.58333
34	Ab3Cd1-Sam001	19.94498	10.28037
35	Abc1d1-Sam002	65.26868	67.29798
36	Abc1d1-Sam001	22.30483	21.73397
37	a1b1c1D-Sam002	51.07731	52.57069
38	a1b1c1D-Sam001	18.37769	16.13723
39	a1b2c1D-Sam002	63.92496	56.13772
40	a1b2c1D-Sam001	18.06167	17.48148
41	a1b3c1D-Sam002	18.5	19.86755
42	a1b3c1D-Sam001	7.973761	10.42735
43	a1b1Cd1-Sam002	73.34315	73.23308
44	a1b1Cd1-Sam001	14.30536	13.49501

TABLE 8

No.	% of MFI normalized at day 3		
23	Ab1c1D-Sam002	50.962	48.41584
24	Ab1c1D-Sam001	45.2422	52.21293
25	Ab2c1D-Sam002	40.1987	36.34074

TABLE 8-continued

No.	% of MFI normalized at day 3		
26	Ab2c1D-Sam001	77.60194	64.59525
27	Ab3c1D-Sam002	75.13296	69.43731
28	Ab3c1D-Sam001	82.07372	41.10864
29	Ab1Cd1-Sam002	21.75325	22.44224
30	Ab1Cd1-Sam001	56.84459	57.05329
31	Ab2Cd1-Sam002	16.99651	16.89642
32	Ab2Cd1-Sam001	54.45878	52.79225
33	Ab3Cd1-Sam002	47.26541	48.89045
34	Ab3Cd1-Sam001	57.46982	80.52686
35	Abc1d1-Sam002	38.89447	36.95033
36	Abc1d1-Sam001	73.61894	61.88906
37	a1b1c1D-Sam002	48.99976	52.12974
38	a1b1c1D-Sam001	36.22922	41.67375
39	a1b2c1D-Sam002	40.58023	43.91632
40	a1b2c1D-Sam001	77.31859	68.19837
41	a1b3c1D-Sam002	83.91999	67.79338
42	a1b3c1D-Sam001	47.01905	34.42404
43	a1b1Cd1-Sam002	25.29739	23.40211
44	a1b1Cd1-Sam001	83.93086	70.15867

TABLE 9

No.	nsP3 transcripts normalized with Actin		
23	Ab1c1D-Sam002	50.21338	43.71329
24	Ab1c1D-Sam001	79.89316	81.57188
25	Ab2c1D-Sam002	36.50444	35.0174
26	Ab2c1D-Sam001	115.3601	106.8913
27	Ab3c1D-Sam002	37.27147	41.06963
28	Ab3c1D-Sam001	69.55103	77.1717
29	Ab1Cd1-Sam002	27.85762	29.65082
30	Ab1Cd1-Sam001	163.1438	172.4459
31	Ab2Cd1-Sam002	34.29675	32.89964
32	Ab2Cd1-Sam001	166.5718	155.4169
33	Ab3Cd1-Sam002	0.752623	0.779165
34	Ab3Cd1-Sam001	1.536875	1.231144
35	Abc1d1-Sam002	2.770219	2.496661
36	Abc1d1-Sam001	147.0334	132.5139
37	a1b1c1D-Sam002	91.13921	96.33579
38	a1b1c1D-Sam001	191.3407	207.9366
39	a1b2c1D-Sam002	15.45498	14.6213
40	a1b2c1D-Sam001	101.1253	101.8287
41	a1b3c1D-Sam002	0.447513	0.397768
42	a1b3c1D-Sam001	116.1625	124.4998
43	a1b1Cd1-Sam002	22.7848	22.62742
44	a1b1Cd1-Sam001	210.8393	210.8393

TABLE 10

No.	eGFP transcripts normalized with Actin		
23	Ab1c1D-Sam002	16.44982	16
24	Ab1c1D-Sam001	38.05463	41.64294
25	Ab2c1D-Sam002	11.47164	12.90627
26	Ab2c1D-Sam001	49.86653	48.50293
27	Ab3c1D-Sam002	9.063071	6.19026
28	Ab3c1D-Sam001	21.55574	20.39297
29	Ab1Cd1-Sam002	3.41054	3.5801
30	Ab1Cd1-Sam001	26.72281	32.89964
31	Ab2Cd1-Sam002	3.340352	3.530812
32	Ab2Cd1-Sam001	21.40684	17.75311
33	Ab3Cd1-Sam002	0.005486	0.005839
34	Ab3Cd1-Sam001	0.00982	0.008729
35	Abc1d1-Sam002	0.011438	0.010167
36	Abc1d1-Sam001	10.41073	7.727491
37	a1b1c1D-Sam002	24.42015	20.39297
38	a1b1c1D-Sam001	86.22295	89.88447
39	a1b2c1D-Sam002	0.406126	0.539614
40	a1b2c1D-Sam001	68.11969	76.10926
41	a1b3c1D-Sam002	0.004944	0.004072

TABLE 10-continued

No.	eGFP transcripts normalized with Actin
42	a1b3c1d-Sam001 134.3637 140.0696 138.141214
43	a1b1Cd1-Sam002 4.890561 4.563055 5.16941132
44	a1b1Cd1-Sam001 50.21338 49.18001 46.5271206

TABLE 11

No.	% of GFP normalized at day 3
45	a1b2Cd1-Sam002 86.72087 85.7337 85.2367688
46	a1b2Cd1-Sam001 29.46429 27.00922 31.53846154
47	a1b3Cd1-Sam002 37.07025 36.80124 37.57668712
48	a1b3Cd1-Sam001 5.844828 6.109091 6.763110307
49	Ab1c1d1-Sam002 42.32633 37.85124 41.06870229
50	Ab1c1d1-Sam001 9.556314 5.738832 7.123966942
51	Ab2c1d1-Sam002 60.93294 60 60.34732272
52	Ab2c1d1-Sam001 22.2964 16.52893 20.51282051
53	Ab3c1d1-Sam002 30.11583 30.46272 22.75132275
54	Ab3c1d1-Sam001 9.871959 9.257362 9.638242894
55	a1Bc1d1-Sam002 53.46535 49.91334 58.52842809
56	a1Bc1d1-Sam001 11.69697 11.47692 13.87442573
57	a1b1c1d1-Sam002 45.08929 40.68554 38.14102564
58	a1b1c1d1-Sam001 12.20979 12.54795 21.51724138
59	a1b2c1d1-Sam002 68.81579 70.96774 67.578125
60	a1b2c1d1-Sam001 21.31783 21.50259 25.89641434
61	a1b3c1d1-Sam002 26.46675 24.3807 29.17214192
62	a1b3c1d1-Sam001 9.371795 10.38119 17.31984829
63	ABCD-Sam001 42.87516 41.27182 47.11779449
64	ABCD-Sam002 88.08777 85.36585 87.59231905
65	ABCD2-Sam002 76.58897 82.50433 90.74117236
66	ABCD2-Sam001 25.07269 23.83486 27.81926965

TABLE 12

No.	% of MFI normalized at day 3
45	a1b2Cd1-Sam002 21.59581 21.38896 21.2904946
46	a1b2Cd1-Sam001 80.67199 62.76361 60.79245886
47	a1b3Cd1-Sam002 59.51047 50.75597 49.29017604
48	a1b3Cd1-Sam001 37.66094 100.0263 53.2477737
49	Ab1c1d1-Sam002 46.83513 47.13787 50.37907506
50	Ab1c1d1-Sam001 59.35291 51.54369 51.29244805
51	Ab2c1d1-Sam002 49.5077 41.19688 43.92891221
52	Ab2c1d1-Sam001 54.3385 71.99053 72.05443699
53	Ab3c1d1-Sam002 68.90104 77.1418 86.83062969
54	Ab3c1d1-Sam001 97.9473 55.35554 49.14956012
55	a1Bc1d1-Sam002 47.22042 38.97181 40.27795818
56	a1Bc1d1-Sam001 55.91852 77.0978 55.41929666
57	a1b1c1d1-Sam002 45.95797 40.88739 55.26597644
58	a1b1c1d1-Sam001 39.12734 47.70419 30.85771948
59	a1b2c1d1-Sam002 36.97674 36.20338 35.07727652
60	a1b2c1d1-Sam001 61.17269 58.38926 53.09734513
61	a1b3c1d1-Sam002 77.3002 69.05956 44.25943546
62	a1b3c1d1-Sam001 54.34123 134.8006 109.557945
63	ABCD-Sam001 75.08005 63.53236 58.12901796
64	ABCD-Sam002 25.37244 25.65748 24.13470845
65	ABCD2-Sam002 22.24231 25.40835 20.73913043
66	ABCD2-Sam001 87.19298 79.77941 87.31617647

TABLE 13

No.	nsP3 transcripts normalized with Actin
45	a1b2Cd1-Sam002 41.06963 36.75835 35.2609637
46	a1b2Cd1-Sam001 128 137.187 121.937664
47	a1b3Cd1-Sam002 49.18001 51.98415 48.8402947
48	a1b3Cd1-Sam001 136.2394 146.0178 165.421162
49	Ab1c1d1-Sam002 60.54769 63.55792 70.5219274
50	Ab1c1d1-Sam001 257.7806 243.8753 266.871235

TABLE 13-continued

No.	nsP3 transcripts normalized with Actin
51	Ab2c1d1-Sam002 38.58585 43.71329 42.2242531
52	Ab2c1d1-Sam001 62.24992 69.55103 71.5063768
53	Ab3c1d1-Sam002 56.49299 60.96883 60.5476894
54	Ab3c1d1-Sam001 139.1021 141.0439 141.043855
55	a1Bc1d1-Sam002 26.90869 31.55945 31.3414495
56	a1Bc1d1-Sam001 58.48521 64 62.682899
57	a1b1c1d1-Sam002 29.04061 34.29675 39.3966212
58	a1b1c1d1-Sam001 103.2501 109.1373 135.298309
59	a1b2c1d1-Sam002 27.47409 26.53823 28.0513831
60	a1b2c1d1-Sam001 60.12946 74.02804 66.2569551
61	a1b3c1d1-Sam002 51.26847 55.71524 58.4852128
62	a1b3c1d1-Sam001 128.8903 129.7868 132.51391
63	ABCD-Sam001 69.07061 61.3929 63.1188931
64	ABCD-Sam002 13.92881 13.8326 9.64646262
65	ABCD2-Sam002 47.17662 47.50475 44.0173382
66	ABCD2-Sam001 13.08643 13.64216 15.1369223

TABLE 14

No.	eGFP transcripts normalized with Actin
45	a1b2Cd1-Sam002 1.931873 1.292353 1.86606598
46	a1b2Cd1-Sam001 30.90996 28.44297 23.7523771
47	a1b3Cd1-Sam002 16.56424 17.14838 13.3614067
48	a1b3Cd1-Sam001 48.50293 56.49299 61.8199251
49	Ab1c1d1-Sam002 14.3204 15.24221 15.5624792
50	Ab1c1d1-Sam001 80.44886 81.57188 75.5835303
51	Ab2c1d1-Sam002 9.189587 10.05611 8.9382971
52	Ab2c1d1-Sam001 32 35.50622 35.0173984
53	Ab3c1d1-Sam002 28.05138 26.17287 28.6408023
54	Ab3c1d1-Sam001 49.18001 36.50444 48.1678959
55	a1Bc1d1-Sam002 14.22148 12.04197 11.7941537
56	a1Bc1d1-Sam001 37.01402 36.25228 30.6964518
57	a1b1c1d1-Sam002 13.36141 14.52031 18.6357374
58	a1b1c1d1-Sam001 61.3929 64 75.0614368
59	a1b2c1d1-Sam002 9.646463 8.815241 8.6938789
60	a1b2c1d1-Sam001 32.89964 34.77552 33.8245773
61	a1b3c1d1-Sam002 22.16175 23.75238 25.281322
62	a1b3c1d1-Sam001 71.01245 64.89341 71.0124462
63	ABCD-Sam001 35.50622 22.94328 27.857618
64	ABCD-Sam002 14.22148 5.897077 6.45313407
65	ABCD2-Sam002 44.6318 40.78594 30.4844159
66	ABCD2-Sam001 6.868523 7.310652 8.45614432

[0253] Tables 3-14 show characterizations of 66 sa-mRNA mutants. To generate these mutants, C2C12 cells were transfected with SAM002 encoding puromycin resistance gene and GFP encapsulated by an LNP comprising the ionizable lipid P6 (P6-LNP). The transfected cells were cultured for 2 months at 1 or 10 μ g/ml puromycin. At 2 months post transfection, the total RNA of selected cells was extracted and reverse transcribed. The specific primers covering contigs from 1 to 6 were for amplicons and sub-cloning. For each contig, 8 clones were cultured and isolated using a Mini-Prep procedure to isolate small plasmid DNA from bacteria while limiting contaminating proteins and genomic DNA for Sanger Sequencing. The contig sequences comprise SEQ ID NOS. 60-67, which correspond to contigs 1-8, respectively. The identified mutations could make 66 combinations and were further engineered into SAM001 or SAM002 at the specified location for each mutation. Thus, this study identified 67 constructs of cytopathic and noncytopathic sa-mRNA, including the mutation at 1 μ g/ml puromycin and the 66 mutants shown in Tables 3-14.

[0254] A characterization study was conducted to study the expression level of the 66 sa-mRNA variants identified using the method described above. The 66 sa-mRNA vari-

ants were transcribed in vitro and transfected to C2C12 cells using a LNP comprising an ionizable lipid P6 as defined in PCT Patent Application No. PCT/US2023/017777 (P6-LNP), which is fully incorporated herein by reference. The transfected cells were performed by fluorescence-activated cell sorting (FACS) at day 1 and 3 post transfection. The percentages of GFP and mean fluorescent intensities (MFI), representing gene product expression of each variant, were analyzed. The percentage and MFI of GFP at day 3 were normalized compared to the data from day 1. Total RNA of the transfected cells was extracted and reverse transcribed as complementary DNA for quantification polymerase chain reaction (qPCR) by specific probes nsP3 and eGFP.

[0255] To characterize the identified variants overtime, the variants were transfected using P6-LNP into mouse myoblast C2C12 cells, analyzed by flow cytometer at day 1 and 3 post transfection, and quantified the transcript number of each sa-mRNA construct using nsP3 specific probes. The subgenomic transcripts were quantified using GFP specific

probes. The decrease of GFP cells and intensity of GFP ranged broadly between the tested variants.

[0256] In one aspect, the first nucleic construct is an sa-mRNA comprising one or more mutations compared to a sa-mRNA construct encoding a GOI (SEQ ID NO: 41, 42, 73, 74, 75, or 76) according to Table 15 below, wherein the position of the mutations are counted from the first nucleotide of each sequence:

TABLE 15

Nucleotide Change Compared to SEQ ID NO: 41, 42, 73, 74, 75, or 76	
Mutation 1	C2623A
Mutation 2	T3309C
Mutation 3	A3493C
Mutation 4	A3867G
Mutation 5	A4674G
Mutation 6	A5897G

TABLE 16

SEQUENCES:		
Description	SEQ ID NO.	Sequence
SGP	1	taacctgaatggactacgacatagtctagtcgcgccaag
	2	gaacttccatcatagttatggccatgactactctagctagcagtgtaaatacttcagctacctgagaggggcccctataaactctctacggc
	3	atggactacgacatagtctagtcgcgccaag
Murine interleukin-12 comprising heavy chain p40 (mIL12 P40)	4	atgtgggagcttgaaaaagacgtctatgtagtagaagtggactggacacctgatgctcctggcgagacagttaacctcacatgcgataccctgaggaagatgatatcacctggacttctgaccagagacacggggtgattgggagcgggcaaacctgacgatcactgtgaaggagtttctggacgcggccagataacctgtcacagggggggagacctgagtcacagccacctgtgctgcacaagaagagagaatggcatctgggtctacagagatcctgaagaactttaagaacaagaccttctgaagtgtgaagcaccaaaactacagtggtcgctttacctgcagctggtctgggtccaaagaacatggacctgaaatttaataataaagagtagctcttcgagtcctgatccagggccgtgacgtgcggcatggcaagcctttcagccgaaaaagtcacgctggatcagcgagactatgagaagtacagcgttagctgtcaggagcgttaacttgcccgaactgcggaggaactctgccatagagctcgctctggaggccaggcagacagaacaaatagagaattacagcactagttctttattagagacatcatcaaacccgacccaccaagaatctgcagatgaagccgctgaagaatagtcaggtcgaggtttcctgggaatatccagattcoatggtccactccgcattcttatttttctctaaaattcttgttaggattcagcggaaaagaagaaagatgaaagagacggaggaagggtgcaaccagaagggggccttcctgggtggaaaagacaagcactgaggtccaatgtaagggtgggaacgtttgctgcaggctcaggatcgctactacaacagcagtgctctaaagtgggcctgcgtacctgtgcgcgtcaggagt
Murine interleukin-12 comprising light chain p35 (mIL12-P35)	5	aggggtgatcccagtgctctgggcccggcccggttgcttgctcaatccagaaacctcctcaagaccactgacgatatggtaaagactgcccagagagaagctaaaaactactctgtacagctgaagatatagaccatgaggatataacacgggacagacctctactctgaaaaacctgtctgcctcttgagctgcacaagaacgagtctgtctggctaccccgcaaacctcaagcacaccagaggtagttgctgccccacaaaagacatcgcttatgatgaccttggtctgggatctatttatgaggacctgaagatgtaccaaaactgagttccaggcaataaatgctgctctcagaatcacaaatcatcaacaaatcatccttgataaggggatgctggtcgcaatcgacgagctcatgcaatcgctgaaccacaatggggaaacctcaggcagaaaccacgggtgggagagggccgacctaccgtgttaaaatgaagtgtgtgtattcttttgatgcattctctacaagagtcgttaccatcaatcgctcatggggtacctgtcatcagc
Murine interleukin-21 (mIL21)	6	gggtacctgtcatcagcggcggttagtggtgggtgggagccacaagtcctcccccagggtccggatcggtcttgatcagactgagacatctgattgatattgtcgagcagttgaagatctatgagaatgacctcgatcctgagttactgagtgccccacaggacgttaagggcactgtgaacacgcgcgatttgctgttttcagaaggccaagctgaaaccttctaataccgggaataacaaaactttcattatcgatctcgtcgcgagctgaggcgcgacttctgcacggcgggggggg

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		aaaaagcaaaagcacatcgcaaagtgtccctcatgacgactcttaccgagaaa cgtacccttaaggagttccttgaagactcaaatggctgctgcaaaagatg atccaccagcatctcagc
Human interleukin-12 comprising heavy chain p40 (hIL12 P40)	7	atatgggagctgaagaaggacgtgtatgtcgtggagctggactggtagccca gatgctcctggcgaaatgggtggttttaacatgtgatacccccgaggaggac ggcatcacatggactctggaccagagttctgagggtgctgggggtccggcaag actctgacaatccaggttaaggagttcggcgacgcaggacagtacacttgt cacaaggagggtgagggtgttctcacagcctgttctgctcctataagaag gaagacggtatttgggtcaaccgacatcctcaaggaccagaaggagccaaa aacaagacctttctgagatgtgaggccaagaattacagcggtagattcact tgttgggtggctcaccaccatattccacagacttgaccttcagtgtcaaaagt tcacgagggagctcagatcctcaaggcggtacctgtggcgacgcagcgtg tccgcagaaagagtcaggggagacaacaaggaatacagagtactctgtcag tgccaggaggattccgcctgtccggccgcagaggagttcttacctattgag gtgatggctgatgccgtgcacaagcttaagtacgagaattacacatcaagt ttttctatccgcgacatcattaaacctgatccaccaagaacctgcagctc aagcctctgaagaatagcaggcaggtcgaggtaagctgggagtagctcctgat acctgggtccacccccacagttatttcagcctcacctctcgcgtccaaagt cagggaagagcaagagagagaagaaggatagggtgttcacagataagact tcagctactgtgatctgcagaaaagaatgcgtctatctctgtgcgagcaca gacaggtactacagttctagctggagcgagtgggcatcagtcacctgcagt
Human interleukin-12 comprising light chain p35 (hIL12-P35)	8	aggaacctcccagttgtctacacctgacccgggaatgtttccatgcctccac cattcccagaatctcctccgagccgtgtccaatattgctgcaaaaggctcgg cagaccttggagttttacccttgccactcagaagaatcgatcatgaggat atcacaaggataaagacgagcactgttgaggcatgctgccccctggagcta actaagaatgagttctgcctgaacagcaggagagactccttcattaccaac ggtagctgtcttgcagcaggaagacatctttatgatggccctgtgtcta tctagcatatatgaagacctgaagatgtaccaggtggaattcaaaaccatg aatgctaaagcttctcatggatcccaaggagcaaatctcctggaccagaat atgcttgcctcatagatgaactgatgcaggcgttgaattttaacagcgag acggtgcctcaaaaaagctcactggaagaacctgatttttataaaaacgaag atcaagctgtgtattttactacagcctttagaatccgcgctgttaccatc gacagagtaatgtcctacctaaatgcttca
Human interleukin-21 (hIL21)	9	caggacaggcatatgatccggatgcggcagctgatcgatattgtagaccag ttgaagaattatgtgaacgacttagtgccggaattcctccccgccccgag gacgtggagactaatgtgagtggtctgcattctcatgcttccaaaaagca cagctgaagagtgccaataccggcaataacgaaaggatcatcaatgtaagt ataaagaagttaaaacgcaaacgccccagtagccaacgctggacgcaggcaa aaacacaggtgacatgccctcgtgtgatctgtacgaaaaaaaacctcca aaggaattcctggaaagggtcaagtcctattacagaaaatgattcaccag cacctgagtagtaggacccacggatccgaagactcc
ORI	10	tttccataggtccgccccctgacgagcatcacaaaaatcgacgctcaag tcagaggtggcgaaacccgcagcagactataaagataccaggcggtttcccc tggaaagctcctcgtgcgctctcctgttccgacctgcgccttaccggata cctgtccgcctttctccttcgggaagcgtggcgctttctcatagctcagc ctgtaggatctcagttcgggtgtaggtcgttcgctccaagctgggctgtgt gcacgaacccccctgcagcccgacccgtgcgccttatccggtaactatcg tcttgagttccaacccggttaagacacgacttatcgccactggcagcagccac tggtaacaggattagcagagcaggtatgtaggcgggtgtacagagttctt gaagtgggtggcctaactacggctacactagaagaacagtagtttggtagctg cgctctgctgaagccagttaccttcggaagaaagagttggtagctcttgatc cggcaaaccaaacaccgctggtagcgggtgttttttggttgcaagcagca gattacgcgcagaaaaaaaggatctcaa
First Expression Unit (Kana/Neo Cassette)	11	atgagtaacctgaggctatggcagggcctgccgccccgacgttggtgcga gccctgggcttccaccgaacttgggggggtgggggtggggaaaaggagaaa gcggggcgataggcccaatgggggtctcgggtgggggtatcgacagagtgcc agccctgggaccgaaccccgctttatgaacaaacgaccaacacctgcg ttttattctgtcttttattgcgctcatagcgcgggttctctccggtattg tctctcctgtgtttcagttagcctccccctagggtggggcgaagaactcca gcatgagatccccgcgctggaggatcatccagccggcgtcccggaaaacga ttccgaagcccaacctttcatagaaggcggcggtggaatcgaaatctcgtg atggcaggttggggcgtcgttgggtcggtcatttcgaaccccgaggtccgc tcagaagaactcgtcaagaaggcgatagaaggcgatgcgctgcgaatcggg agcggcgataaccgtaaagcacgaggaagcggtagccatttcgcgcgaag ctcttcagcaatatacgggttagccaacgctatgtcctgatagcgatccgc cacaccagccggccacagtcgtagaatccagaaaagcggccattttccac

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		catgataattcggcaagcaggcatcgccatgggtcacgacgagatcctcgccgtcgggcatgctcgccctgagcctggcgaacagttcggctggcgcgagcccctgatgctcttcgtccagatcactcctgatcgacaagaccggcttccatccgagtacgtgctcgctcgatcgatggtttcgcttgggtggaatgggcaggtagccggatcaagcgtatgcagccgcccattgcatcagccatgatggatacttctcggcaggagcaaggtgagatgacaggagatcctgccccggcacttcgccaatagcagccagtccttcccgttcagtgacaacgtcgagcacagctgcgaaggaaaccccgtcgtggccagccagatagccgcgtgcctcgctcttgagttcatcagggcacccgacaggtcggctctgacaaaaagaaccggcgccccctgcgtgacagccggaacacggcggcatcagagcagccgattgtctgttgtgcccagtcatagcgaatagcctctccaccaagcggccggagacactgctgcaatccatctgttcaatcatgcgaaacgatcctcatcctgtctcttgatcgatctttgcaaaagcctaggcctccaaaaagcctcctcactacttctggaatagctcagaggccgagcggcctcgccctctgcataaataaaaaaattagtcagccatggggcggagaatgggcggaactgggcggagttaggggcgggagttaggggcgggactatgggtgctgactaattgagatgcatgctttgcatacttctgcctgctggggagcctggggactttccacacctggttgcataaattgagatgcatgctttgcatacttctgcctgctggggagcctggggactttccacacccctaaactgacacacattccacagctgggtccttccgcctcaggactcttcccttttcaatattatgaagcatttatcagggtattgtctcatgagcggatcacatatttgaatgtatttagaaaaataacaaatagggttccgcg
First Expression Unit (Amp Cassette)	12	atgagtaaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatcttctggtcctcatccatagttgcctgactcccgcgtgtgagataaactacgatacgggagggttacatctggcccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaaccagccagccggaaggccgagcgcagaaagtggtcctgcaactttatccgcctccatccagctctaatattgttgccgggaagctagagtaagtagttcgcagttaatagtttgccgaacggtgtgcccattgctacaggcatcgtggtgtcacgctcgctgttggtatgggttcattcagctccgggttcccaacgatcaaggcgagttacatgatccccatgttggtgcaaaaagcggtagctccttcgggtctcccgatcgttgtagaagtaagttggccgcagtggttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgctttctgtgactgggtgagtagtcaaccaagtcattctgagaatagtgtagcggcgaccgagttgctcttgcggcggtcaatcgggataataccgcgccacatagcagaactttaaagtgctcatcattggaaaacggttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatctttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaagggaataaggcgacacgggaatgttgaatactatactcttcccttttcaatattatgaagcatttatcagggtattgtctcatgagcggatcacatatttgaatgtatttagaaaaataacaaatagggttccgcg
Modified T7 Promoter	13	taatacgactcactataggatagg
Wildtype T7 Promoter	14	taatacgactcactatagg
SP6 promoter	82	atthaggtgacac
Kil RNAP promoter	83	aattagggcacac
5' UTR	15	cggcgcatgagagaagcccagaccaattacctaaccctaaa
5' UTR	16	ttcgaaggcgcgcctctagagccacc
5' UTR	17	gcgcgcatgagagaagcccagaccaattacctaaccctaaa
5' UTR	18	acatttgcttctgacacaactgtgttcactagcaacctcaaacagacacc
5' UTR (human actin)	71	accgcccagaccgcgtccgccccgcgagcacagagcctcgctttgcccgatccgcccgcgtccacacccgcgcagctcaccatggat
5' UTR (human Albumin)	72	ctagcttttctcttctgtcaacccacacgcctttggcaca

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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Signal Peptide	20	atggactggacctggcgaatactgttcttgggtgccgccgctacagggaact cacgca
3' UTR 1	21	catcgatgatatacgggcccatacagcagcaattggcaagctgcttacat agaactcggcggttggtgcatgccgccttaaaatttttattttattttct ttcttttccgaatcggttttggtttttaatat
3' UTR 2	22	gctcgtttcttgcgtgtccaatttctattaaaggttcctttgttccctaag tccaaactactaaactggggatattatgaaggccttgagcatctggattc tgccataaaaaaacatttattttcattgcaaagccacaccctggagctag c
3' UTR (human beta actin)	69	gctgactgtgacttagttgcattacaccctttcttgacaaaacctaaacttg cacagaaaacacgatgagattggcatggctttattgtttttgtttttgtt tggtttgtttgttttggcttgactcaggatttaaaactggaacgggtgaagg tgacagcagttggttggagcgagcatccccaaagtctcgaatgtggccg agaactttgatgtacattgttcttttttaattgggtcattccaaatcgt gagatgcattgttacaggaagtccttgccctctaaaagccacccacttc tctctaaggagaatggccagtcctctcccgagtcacacacgggaggttaa tagcattgcttttgggttaaattatataatgcaaaaatcttttaacttttgc cttaacttttttatttttttaatttgaatgatcagcctttgtgactcc cctttttgtcaccacacttgagatgtatgaaggcttttcatctccctcgg agtggtggaggcgagccagggttaactgtacactaacttgagaccagttg aataaaagtgcatacca
3' UTR (human Albumin for regular mRNA)	70	catcacatttaaaagcatctcagcctaccatgagaataagagaaagaaaat gaagatcaaaagcttatcctctgtttttcttttctgttggtgtaaaagcca acaccctgtctaaaaaacataaatttctttaactcattttgcctcttttctc tgtgcttcaattataaaaaatggaaagaatcctaagagtggtacagcac tgttatttttcaagatgtgttgctatcctgaaaattctgtagggtctgtg gaagttccagtggttctctctattccacttcggttagaggatttctagtctc ttgtgggtcaattataataaatcatttaactcttctaagttatggattata aacattcaaaataatattttgacattatgataattctgaataaaagaacaa aaacca
IL12	23	atgtgggagcttgaaaaagacgtctatgtagtagaagtggactggacacct gatgctcctggcgagacagttaacctcacatgcgataccctgaggaagat gatataccctggactcttgaccagagacacggggtgattgggagcgga acccagcagatcactgtgaaggagtttctggacgcccggcagtagacctgt cacaagggggggagaccctgagtcataagccacctgttgctgcacaagaag gagaatggcatctggctacagagatcctgaagaactttaagaacaagacc ttcctgaagtgtgaagcaccaactacagtggtcgctttacctgcagctgg ctggtccaaagaaacatggacctgaaatttaataaagagtagttcctcg agtctgatccagggcctgacgtgcccagatggcaagccttccagccgaa aaagtacgctggatcagcgagactatgagaagtacagcgttagctgtcag gaggacgtaacttgcccagctgccgaggagactctgcccatagagctcgt ctggaggccaggcgagcagaacaaatagagaattacagcactagttcttt attagagacatcatcaaccgcgacccccaagaatctgcagatgaagccg ctgaagaatagtcaggtcgaggtttcctgggaatatccagattcatggtcc actccgcatcttatttttcttaaaattctttgttaggatccagcggaaa aaagaaaagatgaagagacggaggaagggtgcaaccagaagggggcttc ctggtggaagacaagcactgaggtccaatgtgaagggtgggaacgtttgc gtgcaggctcaggatcgctactacaacagcagtgctctaaagtgggctgc gtacctgtcgctcaggagtgagggggggtcaggggggtggctcaggcgcc ggcagtgggggcagcagggtgatcccagtgctctggccggcccggttcttg tctcaatccagaaacctcctcaagaccactgacgatattgtaagactgcc cgagagaagctaaaacactactctgtacagctgaagatatagacctgag gatataacacgggaccagacctctactctgaaaacctgtctgctcttgag ctgcacaagaacgagtcctgtctggctacccgcgaaacctcaagcacaacc agaggtagttgcctgccccacaaaagacatcgcttatgatgaccttggtg ctgggatctattatgaggacctgaagatgtacaaaactgagttccaggca ataaatgctgctctccagaatcacaatcatcaacaatcatccttgataag gggatgctggtcgcaatcgacgagctcatgcaatcgctgaaccacaatggg gaaacctcaggcagaaccacgggtgggagaggccgaccctaccgtgtt

TABLE 16-continued

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Luciferase	25	atggaggacgctaaaaatatcaagaagggtcccgtcctttctatccactc gaggacggtaccgctggtgagcaactgcacaagggtatgaaacgggtatgct cttgtgccaggcaccatagctttcacgcagcccacatagaggtggacatt acttacgcagagtactttgagatgtctgtccggctggcagaggcaatgaaa agatatggcctcaacacaaacctataggatagtggtttgcagcgagaattct cttcagtttttatgccgtcctcggggtccttttcatcgccgtggctgtc gcccccccaacgacatatacaacgagcagagctgctgaacagcatggga atcagtcagcctacagtggtttttgtatcaaaagaagggttgcaaaagatc cttaacgtacagaaaaagctccctattatcagaagatcatataatggat tcaaaagaccgactatcagggtatccagtcattgtatactttcgtcacttcc caccttctcctgggtttaatgaatacagacttctgtcccgaaggttctgat cgggacaaaactattgcaactcatcatgaactcctcgtgggtccactgggtg cctaaagggtgtcgcaacttcccacaggactgcattgtgaaggttctctcac gcacgagatcccatattcggtaaccaaactcattcctgataccgctatctc tctgtagtgcctttccaccatggattcgccatgttccaccctctgggttac ttgatgtgggttcgcgtggtccttatgtacagatttgaaggaggactc tttttgaggagcttgaggactacaaaatccaaagtgtctcctcgttcccc acattgttctcattcttcgccaactcactctcatcgacaagtacgatttg agcaacttgcaacgaaattgcttctggggggggtccactctccaaagaagt ggcgaagccgtgtgtaaaagggtccacttgccaggcattcgccagggtgat ggccttactgagacaaacctcagctattctcattaccctgagggtgacgat aagcctgggtgccgtcggtgaaggtagtagcatttttcgaggcaaaagggtg gacttggtatccggaaaaaacctcgggtgtgaatcagagaggcgaattgtgt gtcaggggtccaatgatcatgagcgggtacgtcaataatccgaggccact aacgccttgattgataaagatgggtggcttcattcaggtgatatagcatac tgggatgaggatgaacatttctcattgtcgataggctcaaaagcctgatc aagtataagggttaccagtggtccctgcggaactcgaatctatactgctg caacatccaaacatttttgacgctggcgtggccggactgcccgatgatgat gctggtgaactcctgcccgcgtgtgctgctgggaacatggtaagaccatg acagaaaaggagattgtagattacgtggcttcacaagtgactacagctaac aattgccgggggtgtcgtttctgtggacgaggtaccgaaggccctcact gggaagctcgacgccagaaaaatccgagaaactcgtattaaagcaaaagaag ggcgaagattgcagtgtag
VEE Structural Proteins	26	atgttcccgttccagccaatgtatccgatgcagccaatgccctatcgcaac ccgttcgcggccccgcgcaggccctggttccccagaaccgacctttctg gcgatgcaggtgcaggaataaacccgctcgatggctaacctgacgttcaag caacgcgggacgcgccacctgaggggcatccgctaagaaaccgaagaag gagccctcgcaaaaacagaaggggaggccaaagggaagaagaagaagaac caagggaagaagaaggctaagacaggccgcctaatccgaaggccacagaat ggaaacaagaagaagaccaacaagaaccaggcaagagacagcgcgtggtc atgaaattggaatctgacaagacgttcccaatcatgttgaagggaagata aacggctacgcttgtgtggtcgagggaagttattcaggccgatgcattgtg gaaggcaagatcgacaacgacgttctggccgcgcttaagacgaagaagca tccaaatcagatcttgatgtgagatgtgcccacagaacatcgggccgat acattcaaatacaccatgagaaaccccaaggctattacagctggcatcat ggagcagtcacatataaaaatggcggtttcacgggtgccgaaggagtggg gccaagggaagacagcggacgaccttctggataaccagggaagggtgggtc gctattgtgctgggaggtgtgaatgaaggatctaggacagccctttcagtc gtcatgtggaacgagaaggaggtaccgtgaagtatactccggagaactgc gagcaatgggtcactagtgaaccacatgtgtcgtcgcgaatgtgacgttc ccatgtgctcaaccaccaatttgctacgacagaaaaccagcagagactttg

TABLE 16-continued

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Linker	27	cgcgtagataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggcca
		ggaaccgtaaaaaggccgcttgctggcgctt
	28	cacatttcccgaaggaagtgccacctgagctc
	29	ttcgaaggcgccctctagagccacc
	30	catcgatgatatcgccggccgcatacagcagc
	31	ggaggggggtcaggggggtggctcaggcggcggcagtgggggcagc
	32	ggcggtagtggtggtgggagc
	33	ggtggcggaagcgagggggcagcggagggtgggagcggaggggagc
	34	ggagggtcaggaggaggatcc
	35	gaagatcctttgatctttctacgggggtctgacgctcagtggaacgaaaaac
		tcacgttaagggaatttgggtcatgagattatcaaaaggatcttccactag
		atccttttaaatataaaatgaagttttaaatcaatctaaagtatat

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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	37	cgcgatcgatgatatcgcgccgcatacagcagcaattggcaagctgctt acatagaactcgcgccgattggcatgccccttaaaattttttattttattt ttctttttcttttcgaatcggaattttgtttttaatatttcaaaaaaaaaa aaaaaaaaaaaaaaaaacgcgtgataacgcaggaaagaacatgtgagcaaaag gccagcaaaaggccaggaaacgtaaaaaggccgcgttgtctggcggt
	38	cttgatgttacaagagcctggggccggctcagtgagacacctcgtggctt gataaagggttaccagctacgatggcgaggacaagatcggtctctacgctgt gctttctccgcaggctgtactcaagagtgaataattatcttgcatccacc tctcgtgaacaagtcatagtataacacactctggccgaaaaggcggtta tgccgtggaaccataccatggtaagtagtggtgccagagggacatgcaat accggtccaggactttcaagctctgagtgaagtgccaccattgtgtacaa cgaaagtgagttcgtaaacaggtacctgcaccatattgccacacatggagg agcgctgaacactgatgaagaattattcaaaactgtcaagcccagcgagca cgacggcggaatacctgtacgacatcgacaggaaacagtgcgtaagaaaga actagtcactgggctagggtcacaggcgagctgggtggtatcctccctcca tgaattcgctacgagagcttgagaacacgaccagccgctccttaccagt accaaccataggggtgtatggcggtgccaggatcaggcaagtctggcatcat taaaagcgagtcacaaaaaagatctagtggtagcgccaaagaaagaaa ctgtgcagaattataagggacgtcaagaaaaatgaaagggtggagcgtcaa tgccagaactgtggactcagtgctcttgaatggatgcaaacaccccgtaga gacctgtatattgacgaagcttttgcttgtcatgcaggtactctcagagc gctcatagccattataagacctaaaaaggcagtgctctcgcggggatcccaa acagtgccggttttttaacatgatgtgcctgaaagtgcattttaaccacga gatttgacacaaagcttctccacaaaagcatctctcgccgttgactaaatc tgtgacttcggctcgtctcaaccttggttttacgcaaaaaaatgagaacgac gaatccgaaagagactaagattgtgattgacactaccggcagtagcaaac taagcaggacgatctcattctcacttggttcagaggggtgggtgaagcagtt gcaaatagattacaaaggcaacgaaataatgacggcagctgcctctcaagg gctgacccgtaaaagggtgtgtatgccgttcggtacaaggtgaatgaaatcc tctgtacgcaaccacctcagaacatgtgaaagctcctactgacccgcacgga ggaccgcatcgtgtggaaaaacactagccggcgacccatggataaaaacact gactgccaagtaccctgggaatttcactgccacgatagaggagtggaagc agagcatgatgccatcatgaggcacatcttgagagacccggaccctaccga cgtcttccagaataaaggcaaacgtgtgttgggccaaggctttagtgcgggt gctgaagaccgctggcatagacatgacactgaacaatggaacactgtgga ttattttgaacggaacaaagctcactcagcagagatagtattgaaccaact atgcgtgaggttctttggactcagatctggactccggtctatttctgcacc cactgttcggttatccattaggaataatcactgggataaactcccgctcgcc taacatgtacgggctgaataaagaagtggtccgtcagctctctcgaggtta cccacaactgcctcgggcagttgccactggaagagttctatgacatgaacac tggtacactgcgcaattatgatccgcccataaacctagtacctgtaaacag aagactgcctcatgctttagtctccaccataatgaaccccacagagtgaa cttttcttcatctgtcagcaaatgaagggcagaactgtcctgggtggcg ggaaaagtgtccgtcccaggcaaaatgggtgactggttgtcagacggcc tgaggctaccttcagagctcggctggatttaggcatcccaggtgatgtgcc caaatatgacataatatttgtaagtgtgaggaccccatataaataccatca ctatcagcagtggtgaagaccatgccattaaagcttagcatgtgaccaagaa agcttgtctgcatctgaatcccggcggaacctgtgtcagcataggttatgg ttacgctgacaggggccagcgaaagcatcatttggtgctatagcgcgagctt caagttttccgggtatgcaaacggaatcctcacttgaagagacggaagt tctgtttgtatctattgggtacgatcgcaaggcccgtagcacaattctta caagctttcatcaaccttgaccaacattttatacaggttccagactccacga agccggatgtgcacctcatatcatgtggtgagggggaatttgccacggc caccgaaggagtgatataaatgctgctaacagcaaaaggacaaactggcg aggggtgtgaggagcgctgtataagaaatcccggaaagcttcgatttaca ccgatcgaagtaggaaaagcgagctgggtcaaaagtgagctaaacatata cattcatgcccgtaggacaaacttcaacaaagtttcggagggtgaaggtga caaacagttggcagaggttataggtccatcgctaaagattgtcaacgataa caattacaagtcagtagcgattccactgtgtgccacggcatctttccgg gaacaaagatcgactaacccaatcattgaaccatttctgtacagctttaga caccactgatgcagatgtagccatatactgcagggaacaaatgggaaat gactctcaaggaaagcagtggtaggagagaagcagtgaggagatagcat atccgacgactcttcagtgacagaacctgatgcagagctgggtgaggggtgca tccgaagagttctttggctggaaggagggtacagcacaaagcgatggcaa aactttctcatatttggaagggaacaaagtttccacggcgggcgaaggtat agcagaattaatgccatgtggcccggttgcaacggaggccaatgagcaggt atgcatgtatatcctcggagaaagcatgagcagttattaggtcgaaatgcc cgtcgaaagagtcggaagcctccacaccactagcacgctgcttctgtgtg catccatgccatgactccagaaaggtacagcgccctaaaagcctcacgtcc

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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Description	SEQ ID NO.	Sequence
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Second nucleic acid construct, mRNA encoding luciferase	43	acatttctcttgacacaactgtgttcactagcaacctcaaacagacacca tggaggacgctaaaaatatcaagaagggtcccgctcctttctatccactcg aggacggtagccgctggtgagcaactgcacaaggctatgaaacggtagtctc ttgtgccaggcaccatagctttcaccgacgcccacatagaggtggacattta ctacgcagagtactttgagatgtctgtccggtggcagaggcaatgaaaa gatattggcctcaacacaaacataggatagtggtttgacgcgagaattctc ttcagttttttatgcccgctcctcggggtccttttcatcgcggtggctgtcg ccccgcacaacgacatatacaacgagcgagagctgctgaacagcatgggaa tcagtcagcctacagtgggttttgtatcaagaagggttgcaaaagatcc ttaacgtacagaaaaagctccctattatcagaagatcattataatggatt caaagaccgactatcagggattccagtcgaatgtatccttctgcacttccc accttctcctgggtttaatgaatcgaacttcgttcccgaagtttcgatc gggcaaaaaactattgcactcatcatgaactcctctgggtccactgggtgctg ctaaagggtgtcgacttcccacaggactgcagtggtgaaggttctctcacg cacgagatcccatattcggtaaccaaatcattcctgataccgctatcctct ctgtagtgcctttccaccatggattcggcatgttcaccactctgggttact tgatattgtgggttcgcggtggctccttatgtacagattgaagaggagctct ttttgaggagcttgacggactacaaaatccaaagtgcctcctcgtcccca cattgtctcattcttcgcaaatctactctcatcgacaagtacgatttga gcaacttgcacgaattgcttctgggggggtccactctccaaagaagtgcg gcgaagccggttgctaaaagggtccacttgccaggcattcgccaggggtatg gccttactgagacaacctcagctattctcattaccctcgaggtgacgata agcctggtgcccgtcggtaaggtagtaccattttcgaggcaaaagtggtcg acttggataccggaaaaaacctcgggtgtgaatcagagaggcgaattgtgtg tcaggggtccaatgatcatgagcgggtacgtcaatatacccgaggccacta acgccttgattgataaagatggttggttcatcagggtgatatagcatact gggatgaggatgaacatttcttcattgtcgataggctcaaaagcctgatca agtataagggttaccaggtggccctgcgaactcgaatctatctgctgc aacatccaaacattttgacgctggcgtggccggactgccgatgatgatg ctggtgaactccctgcccgggtgtcgtcctggaacatggtgaagaccatga cagaaaaaggagattgtagattacgtggcttcacaagtgcactacagctaaga aattgcgcgggggtgtcggtttcgtggacgaggtaccgaaggccctcactg ggaaagctcgacgccagaaaaatccgagaaatcctgatataagcaaaagagg gcggaaagattgcagtgtaggctcgtttcttctgctgtccaatttctattaa aggctcctttgttccctaagtcctaactactaaactgggggatattatgaag ggccttgagcatctggattctgcctaataaaaaaacatttatttctattgca aagccacaccctggagctagcaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa aaaaaaaaaaaaaaaaaaaaa
GOI, sa-mRNA encoding human IL12	44	atggactggacctggcgaatactgttcttggttgcgcccgtacagggact cacgcaatatgggagctgaagaaggacgtgtatgtcgtggagctggactgg taccagatgctcctggcgaatgggtggttttaacatgtatacccccgag gaggacggcatcacatggactctggaccagagtcttgaggtgctgggggtcc ggcaagactctgacaatccaggttaaggagttcggcgacgcaggacagatc acttgtcacaaaggaggtgaggtgcttctcacagcctgttgcgtgctccat aagaaggaagacggtatttgggtcaaccgacatcctcaaggaccagaaggag ccccaaaacaagaccttctgagatgtgaggccaagaattacagcgtgaga ttcacttgttgggtggctcaccaccatataccacagactgaccttcagtgctc aaaagttcacgagggagctcagatcctcaaggcgttacctgtggcgacgag acgctgtccgcagaaagagtacggggagacaaaggaatacagtagtactct gtcgagtgccaggaggattccgcctgtccggccgcagaggagtctttacct attgaggtgatggtcgatgccgtgcacaagcttaagtagcagaattacaca tcaagttttttcatccgcgacatcattaaacctgatccaccaagaacctg cagctcaagcctctgaagaatagcaggcaggtcgaggtgaagctgggagtat cctgataacctggtccacccccacagttatttcagcctcacctctcgcgtc caagtcagggaagagcaagagagagaagaaggatagggtgttcacagat aagacttcagctactgtgatctgcagaaagaatgcgtctatctctgtgcga

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		gcacaagacaggtactacagttctagctggagcagtgaggcatcagtcctc tgagtggtggcggaagcggagggggagcggaggtgggagcggaggagc aggaaacctccagttgctacacctgacccgggaatgtttccatgcctccac cattcccagaatctcctccgagccgtgtccaatagctgcaaaaggctcgg cagacctggagttttacccttgacctcagaagaaatcgatcatgaggat atcacaaggataagacgagcactgttgaggcatgcctgcccctggagcta actaagaatgagttctgcctgaacagcaggagagacttccctcattaccaac ggtagctgtcttgccagcaggaagacatctttatgatggccctgtgtcta tctagcatatgaagacctgaagatgtaccaggtggaattcaaaaccatg aatgtctaatcttctcatggatcccaagggcaaatcttccctggaccagaat atgcttgctgtcatagatgaactgatgcaggcgttgaaatttaacagcgag acggtgcctcaaaaaagctcactggaagaacctgatttttataaaaacgaag atcaagctgtgtattttactacacgcctttagaatccgcgctgttaccatc gacagagtaatgtcctacctaataatgcttcatag
GOI, human IL2	SEQ ID NO. 45	atgtaccgcagtcagttgctgtcttgattgcttagcctcgcttggtc acaacagtgccccgacctccagttctaccaagaagacgcaactccagctt gagcatcttctcctggacttgagatgatattgaatggatcaacaattac aagaatcccaaaactcaccaggatgcttacttttaagttttatatgcctaag aaagcgacggaactgaaacaccttcaatgcttgagggaagaactgaaccg cttgaggaggtactgaatttggcacagtcaaaaaacttccacctgaggcca agggacttgatcagtaacataaacgtaatcggtttggagcttaaggggagc gaaaccacctttatgtgcgagtacgcagacgagacggccacgattgtagaa tttttgatcgctggattactttttgccaagatcatcagtacactcacc tag
GOI, mouse IL2	SEQ ID NO. 46	atgtattcaatgcagcttgccagttgtgtcacactcactcttgctcctgtg gttaattccgccccaaacttcaagccccaccagctccccacctcaagttca accgctgaagcacagcaacagcaacaacacagcaacacttggaacaaatc ctcatggatctccaggaactgctttctcgaatggagaactaccggaatctg aaacttcccagaatgctgaccttcaagttctacctccctaacgaggtact gaacttaagatttgcaatgtcttgaagatgaactgggcccacttcgacac gtatggatctgacccagtcaaaatccttcaattggaggacgcagagaac ttcataagcaacatcagggtcaccgtggtgaagctcaaggggtctgacaat acatttgatgtcaatttgatgatgagtcagccaccgtcgtcgatttttg cggagatggatcgcatctgcccagacataatttctaccagccctcagtag
GOI, human IL15	SEQ ID NO. 47	atgtatcgaatgcaattgctgtcctgcattgcgctctctctggcgcttgct accaattcaaaactgggttaacgtaatatctgatttgaaaaaatcgaagac ctcatcacagagtatgcacatagatgctacgtgtacacagaaagcgagta cacccgagctgtaaggtcactgccatgaagtgtcttctctcgaactccaa gttatcagctctcgaatccggggatgctcaatccatgacactgttgagaat ctcattattctcgcaacaactcactcagctcaaatgggaacgtcactgaa tccggctgcaagggaatgcgaggaattggaagagaagaatataaagagttc ctgcagagtttcgttcatattgttcaaatgtttatcaacacaagttag
GOI, mouse IL15	SEQ ID NO. 48	atgtattcaatgcagcttgccagttgtgtcacactcactcttgctcctgtg gttaattccaattggatcgatgtccggtatgatttggaagatcgaatct ctcatccaaagcatcacattgacaccactttgtatactgatagcgatttt caccctcttgtaaggtgaccccatgaattgctttctgttggaacttcaa gttatacttcacagatttcaaacatgacactgaatgaaactgtccgaaac gtactctaccttgctaaactccaccttctcatcaataaagaacgtagctgag agcggatgtaaggagtgtaggagctcgaagagaagacatttactgaattt cttcagagttttatccgcatagtgcaaatgtttattaacacctcctag
GOI, human IL15/IL15Rα	SEQ ID NO. 49	atgtatcgaatgcaattgctgtcctgcattgcgctctctctggcgcttgct accaattcaaaactgggttaacgtaatatctgatttgaaaaaatcgaagac ctcatcacagagtatgcacatagatgctacgtgtacacagaaagcgagta cacccgagctgtaaggtcactgccatgaagtgtcttctctcgaactccaa gttatcagctctcgaatccggggatgctcaatccatgacactgttgagaat ctcattattctcgcaacaactcactcagctcaaatgggaacgtcactgaa tccggctgcaagggaatgcgaggaattggaagagaagaatataaagagttc ctgcagagtttcgttcatattgttcaaatgtttatcaacacaagtgaggg gggtcaggggggtggctcaggcggcgagtgaggggcagcataaactgtcca cctccgatgagtgtggaacatgctgacatttgggttaagagttactcactc tattcacgagagcgatacatctgtaaatagtggttttaaaagaaaagccggt acatcctctcttactgaatgcgtactcaataaggcgaccaacgtcgcgcac tggaccactccttcttgaaatgcatccgcgacccgctctcgtttag

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
GOI, human IL15R α /IL15	SEQ ID NO. 50	atgtaccgcatgcaacttttgtcttgcaattgcacttagcttggcactcgtc actaatccattacatgccctccgctatgtctgtggagcacgcagacatc tgggtaaaagtcttactccctctactcaagagaacgatataatctgtaattca gggtttaagcgaaaagcaggaacctcttcactgacagaatgctgtctta aagcgacgaacgttgctcattggactacacctagtttgaagtgtattcgg gatccggcactcgttaggggggtcaggggggtggctcaggcgccggcagt gggggcagcaattgggtaaacgtaattagcgatctgaagaagatcgaggac ttgatacaaaagtatgcatatcgatgctactcttataccgagctcgatgta caccctcttgtaaaagtaacagcgatgaagtgtctttgttggagtgcga gtgataagcctggaaagtggggatgctccatccacgataccgctcgagaac ttgataaactggtggcaacaacagctctctagtaacggtaacgtaaccgag agcgggtgtaaagaatgcgaagaattggaagagaaaaatataaagaattc cttcagtcatttgtgcatatagtgcagatgtttatcaacacatcctag
GOI, mouse IL15/IL15R α	SEQ ID NO. 51	atgtattcaatgcagcttgccagttgtgtcacactcactcttgctctgtg gttaattccaattggatcgatgtccggtatgatttggagaagatcgaatct ctcatccaaagcatcacacttgacacacctttgtatactgatagcgatttt caccatcttgtaaggtgaccgccatgaattgctttctgttggaaactcaa gttatacttcacgagttatcaaacatgacactgaatgaaactgtccgaaac gtactctaccttgctaaactccacctttcatctaaagaacgtagctgag agcggatgtaaggagtgtagggagctcgaagagaagacatttactgaattt cttcagagttttatccgcatagtgcgaatgtttatcaacacctccggaggg gggtcagggggtggctcaggcgccggcagtgggggcagcggaaactacatgt ccacctccgctcagtatcgaaacatgcagatcagagttaaaaattatca gtgaacagcagggagcgctacgtgtgttaatagtggttttaaaaggaaagcc ggcactctacattgattgaatgcgttatcaataagaataactaacgtagct cactggaccacacctctctgaagtgtatccgagaccttcacttgccctag
GOI, mouse IL15R α /IL15	SEQ ID NO. 52	atgtattcaatgcagcttgccagttgtgtcacactcactcttgctctgtg gttaattccggaactacttgcaccaccacctcagtatcgagcacgctgat ataagagttaaaaactattctgtaaattctagggaagatagtatgtaac agcggtttcaaaagggaaggcagggacatcaaccttatcgagtgcgtcatt aacaataacaaactcgcccatggaccactccatcactcaaatgtata cgagacctagcttggcaggaggggggtcaggggggtggctcagggcgccgc agtggggcgagcaattggatagatgtccggtacgacctcgagaagatcgag tcacttattcaatctatacatattgacactacctgtatactgacagcgac tttcatccaagtgttaagtcaactgctatgaattgctttttgctcgagttg caggtcatctgcatgagttatcaatagacttgaatgaaacgctcaga aatgtcttgatcttgctaacagtaaccttagctccaacaagaatgttgcc gaatcagggttgtaagaatgcgaggagttggaagagaaaaccttaccgaa tttcttcaatcattcatcagaattgtacaatgttcatcaatacttcatag
GOI, mouse IL18-WT	SEQ ID NO. 53	atgacctcccgcttgtgagggtaactggctgctgctatgctggtggctgct gctgtgagtggtggcaactttgggagactccactgtaccactgcccgttata cggaataataacgatcaggtctctttagtagacaagagacagccagtttcc gaggacatgactgatatagatcagagtgccctcgagcctcaaacctcggtc ataatatacatgtataaggatagcgaggttcggggacttgccgtgactttg tcctgtaaagactcaaaaatgagcacacctgtcatgcaaaaataagataatt tccttcgaggaaatggacctcccgagaacattgatgatataccagtcggac ttgatattttccagaaacgggttcccggtcatacaaaaatggagtttgag tcctcattgtacgaggggtcatttttggcatgccagaaggaggatgatgcc ttcaaacgtatttgaagaaaaaagacgagaacggagacaagctgtcatg tttacacttaccaaccttcataatcctag
GOI, mouse IL18-mutant	SEQ ID NO. 54	atgacctcccgcttgtgagggtaactggctgctgctatgctggtggctgct gctgtgagtggtggcacacttcggacgactccattgtactacagccgttata cgaaataataacgatcaggtgtgtttgttagacaaaaggcagcccggtgtt gaagacatgactgatattgatcagagtgccctcgagcccgactagactg attatatatgcttatggcgactctagagctagagggaagcagtaactctg tcagtgaagacagcaagatgtctacctgtcttgcaagaataaaaataatc tcatttgaggagatggacccccctgaaaacatagatgacatccaaagtgac ctgatattcttccagaagcgggtaccgggtcatacaaaaatggagttcgag tcaagcctctatgaggggtcactttctcgctgcccagaaggaagatgacgcc ttcaaatgtatctgaaaaaaaagacgagaatgggtgataagctgtgtcatg ttcacactcactaatttgcatcagtcatag

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
GOI, mouse mCXCL10-9-11-4 f	SEQ ID NO. 55	atgaatccatccgcagctgtaattttctgccttatcctcctggggtgagt ggaaccaaggaatccccctcgctcgaaacagtacgctgcaactgtatccac atagacgatggcccggttcgaatgagggtatcggaagctcgaaataata cccgctctctctcatgcccccgctagaaataatagcaactatgaagaag aacgacgagcaacgatgcttgaaccagaaagcaagactatcaagaacttg atgaaagcttttagccaaaagcggagttaacgagccccggggaggcggc tccggcggaggaggagtggtggcgggtggatccggaggcgggtgggagtacc cttgtgattaggaatgcacgatgcagctgtataagcacctccccgggacact attcactataagagccttaagatctcaagcagtttgcaccatctccaaat tgtaacaaaactgagattattgctacactgaagaacggggaccacacctgc ctggaccagatagtgcctaatgttaaaaagcttatgaaggaaatgggagaaa aaaatcagtcaaaagaaaaagcaaaaacgcggtaaaaaacaccagaaaaac atgaagaatcggaaccaaaaacccacagttctaggagaaggagtcgaaaa acaacagggggaggcggaaagtggcggcgggggagcgggtggcggggatca ggcggcgggggagcttcccttatgttcaacaaggcgggtgctgtgtata gggccaggtatgaagcagtgaaagtggctgaaatcgaaaaagcaagtgtg atctaccctagtatggctgtgataaagtgaaagtattgtgactatgaaa gctcacaagagacagcgggtgttggatccatagatccaaacaagcccgactc atcatgcaagccatagaaaaagaagacttcctcggaaggcagaatatgggt ggaggtggaagcgggtggcgggggtagcggcgggggaggtagtgggtggggg ggtagcgttaactagtctggaccagaggagtcagatggggatctctcttgt gtctgcgttaagaccatcatccgggattcaccttaagcacattaccagc ctcgaaagtgtcaaaagctggtaggcatttgtgcctgcccagctcatagcc acatgaagaatgggaggaaaaattgccttgatagacaggcaccatttgtat aagaagtcattaaaaaattcttgaatcatag
GOI, structural proteins from semliki forest virus SVF4	56	atgaattacatccctacgcaaacgttttacgggcgcagatggcgacctcga cccgccgctcgccctggcctttgcaagctacgccgtagcccccggtggtc ccagactttcaggcacagcaaatgcagcagcaactcatcagcgccgtaaaat gcgctgacaatgagacagaaacgcaattgctcctgctaggcctcccaaacca aagaagaagaagacaacaaaccaaagccgaaaacgcagcccaagaagatc aacggaaaaacgcagcagcaaaaagaagaagacaagcaagccgacaagaag aagaagaaacccggaaaaagagaaagaaatgtgcatgaagatgaaaaatgac tgtatcttcgaagtcaaacacgaaggaaaggtcactgggtacgcctgcctg gtggggcacaaggtcatgaaacctgcccacgtgaaaggagtcacgcacaac cgggacctggcacaagctagcttcaagaaatcgagcaagtatgaccttgag tgtgccagataccagttccatgaggtcggatgcctcaaagtacacgcat gagaagcccaggggacactataactggcaccacggggctgttcagtagacg ggaggtagggttcaactataccgacaggagcgggcaaacggggagacagtggc cggccctctttgacaacaaggggagggtagtgcgtatcgtcctgggcggg gccaacgagggtcagcgcacgactgtcggtggtcacctggaaacaaagat atggtgactagagtgaaccccaggggtccgaagagtggtccgcccgcgtg atactgccatgtgtgtccttgccaatgctacctcccgtgcttccagccc ccgtgtgtaccttgcctgatgaaaaacaacgcagagggccacactacggatg ctgaggataaacgtggataggccagggtactacgacctcctcaggcagcc ttgacgtgccgaaacggaacaagacacccggcgcagcgtgtcgcaacacttc aacgtgtataaaggctacacgcccttacatcgcgtagctgcgcgactcgga gcagggcactcgtgtcatagccccgtagcaattgaagcgggtcagggtccgaa gctaccgacgggatgctgaagattcagttctcggcacaatgtggcatagat aagagtgaacatcatgactacacgaagataaggtacgcagacgggcacgcc attgagaatgccgtccgggtcatcttgaaggtagccacctccggagactgt ttcgctccatggcacaatgggacatttcatactggcaagtgcccacccgggt gaattcctgcaggtctcgatccaggacaccagaaacgcgggtccgtgctgc agaatacaatatcatcatgacctcaaccgggtgggtagagaaaaattaca attagaccacactatgaaaagagatcccttgaccacttatcaacagacc acagcgaagaccgtggaggaaatcgacatgcatatgccgccagatacgccg gacaggacgttgctatcacagcaatctggcaatgtaaagatcacagtcgga ggaaagaaggtgaaatacaactgcacctgtggaaccggaaacgttggcact actaatccggacatgacgatcaacacgtgtctaataagagcagtgccacgtc tcagtgacggaccataaagaatggcagtcgaactcaccttctgcctccgaga gccgacgaaccggctagaaaaggcaagtcctatccattcccggttggaac aacatcacatgcagagttccaatggcgcgcggaaccaaccgctcatccacggc aaaagagaagtgacactgcaccttcaccagatcatcccagctcttttcc taccgcacactgggtgaggaccccgagttatcacgagggaatgggtgacagcg gcggtggaacggaccataccggtaccagtggaagggtggagtaccactgg ggaaacaacgaccagtgaggcttgggtctcaactcaccactgaagggaaa ccgacggctggccgcatcagatcgtacagtaactactatgggctttaccgg gccgctacagtatccgcggtcgtcgggatgagcttactggcgttgatatcg atcttcgctcgtgctacatgctggttgcgcccgagtaagtgttgacc ccttatgctttaacaccaggagctgcagttccgtggacgctggggatactc

TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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[illegible]

TABLE 16-continued

[illegible]

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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Contig 4	63	gaataatcactgggataactcccgtcgccctaacatgtacgggtgaataa agaagtgggtccgtcagctctctcgcaaggtagccacaaactgcctcgggcagt tgccactggaagagtctatgacatgaacactggtagactcgcaattatga tccgcgcataaacctagtacctgtaaacagaagactgcctcatgctttagt cctccaccataatgaacaccacagagtactttcttcatctgctcagcaa attgaagggcagaactgtcctgggtggtcggggaaaagtgtccgtcccagg caaaatgggtgactgggtgtcagaccggcctgaggctacctcagagctcg gctggatttaggcatcccagggtgatgtgcccaaatatgacataaatattgt taatgtgaggaccccatataaataccatcactatcagcagtgtagaacca tgccattaaagcttagcatgttgaccaagaagcttgtctgcatctgaatcc cgccggaacctgtgtcagcataggttatgggttacgctgacagggccagcga aagcatcattgggtgctatagcgcggcagttcaagtttcccggtgatgcaa accgaaatcctcacttgaagagacggaagtctctgtttgtatcattgggtga cgatcgcaaggcccgtagcaccaattcttacaagcttcatcaaccttgac caacatttatacaggttccagactccacgaagccggatgtgcacctcata tcatgtggtgcgaggggatattgccacggccaccgaaggagtgattataa tgctgctaacagcaaggacaaacctggcgagggtgtgcggagcgctgta taagaaattcccgaaagcttcgatttacagccgatcgaagtaggaaaagc gcgactggtcaaaaggtgcagctaaacatcatctcgcgtaggacccaaa cttcaacaaagtttcggaggttgaaggtgacaaacagttggcagaggctta tgagtcctatcgctaagattgtcaacgatacaattacaagtcagtagcgat tccactgtgtccaccggcatctttccgggaacaaagatcgactaaccca atcattgaaccatttgctgacagctttagacaccactgatgcagatg
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Contig 6	65	agaaccagcctagtttccaccccgccaggcgtgaatagggtgatcactaga gaggagctcgaggcgcttaccggctcacgcactcctagcaggtcggtctcg agaaccagcctggtctccaacccgccaggcgtaaatagggtgattacaaga gaggagtttggagcgttcgtagcacaacaacaatgacgggttgatgcccgt gcatacatcttttctcgcacaccggtcaagggtatttacaacaaaaatca gtaaggcaaacgggtgctatccgaagtgggtgtggagaggaccgaattggag

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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Contig 7	66	gctagcaacagcgtatctgtgcggaatccaccgagagctggttaggagatt aaatgcggtcctgcttccgaacattcatcactgtttgatattgctcggtga agactttgacgctattatagccgagcacttccagcctggggtattgtgttct ggaaactgacatcgcgtcgtttgataaaagtgaggacgacgccatggctct gaccgcgttaaatgattctggaagacttaggtgtggacgcagagctgttgac gctgattgaggcggttctcgccgaaatttcatcaatacatttgcacctaa aactaaatttaaatccggagccatgatgaaatctggaatgttccctcacact gtttgtgaacacagtcattaacattgtaatcgcaagcagagtggttgagaga acgctaacgggatcccatgtgcagcattcatggagatgacaatcgt gaaaggagtcaaatcggacaaattaatggcagacaggtgcgccacctgggt gaatatggaagtcaagattatagatgctgtggtggcgagaaaagcgctta tttctgtggagggtttattttgtgtgactccgtgaccggcacagcgtgcgc tgtggcagacccccctaaaaaggctgtttaagcttggcaaacctctggcagc agacgatgaacatgatgatgacaggagaagggcatgtcatgaagagtcaac acgctggaaaccgagtggtattctttcagagctgtgcaaggcagtagaatc aaggtatgaaaccgtaggaaacttccatcatagttatggccatgactactct agctagcagtggttaaatcatcagctacctgagagggggccctataactct ctacggctaacctgaatggactacgacatagctctagtcgcccaagtccgaa ggccgcctctagagccaccatgaccgagtacaagcccacgggtgcgcctcg ccacccgcgacgactccccaggccgtacgcacctcgcgcgcgcgttcg ccgactaccccgccacgcgcacacccgtcgatccggaccgccacatcgagc gggtcaccgagctgcaagaactcttcct
Contig 8	67	cacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggaacgacggcgc cgcggtggcggtctggaccacgcccggagagcgtcgaaagcggggcggtgtt cgccgagatcggcccgcgcatggccgagttgagcgggttcccggtcggccgc cgacaaacagatggaaggcctcctggcgcgcacccggcccaaggagccgcg gtggttcctggccaccgtcggcgtctcgcccgaccacaggggcaagggtct gggcagcgcgctgctgtccccggagtgaggcgccgagcgcgcgggggt gccgccttctggagacctcgcgcgcccgcaacctcccttctacgagcg gctcggcttcaccgtcaccgcccagctcgaggtgccggaaggaccgcgcac ctggtgcatgaccgcgaagccgggtgctgacatcgatgatcgcggccg catacgcagcaattggcaagctgcttacatagaactcgcggcgattggca tgccgccttaaaatttttattttttttttttttttttttccgaatcggatt ttgtttttaatatttcaaaaaaaaaaaaaaaaaa
Selective marker, puromycin resistance gene	68	atgagtaaaacttggtctgacagttaccaatgcttaatcagtgaggcaccta tctcagcgatctgtctatttcgttcatccatagttgacctgactccccgtcg ttagataaactacgatacgggagggttaccatctggccccagtgctgcga tgataccgcgagaccacgctcaccggctccagatttatcagcaataaac agccagccgaaggcgagcgcagaagtggctcctgcaactttatccgcct ccatccagctctattaattgttgcgggaagctagagtaagtgttcgccag ttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcac gctcgtcgtttggtatgggttcatcagctccgggtcccaacgatcaaggc gagtacatgatccccatgttgtgcaaaaaagcgggtagctccttcgggtc ctccgatcgttgcagaagttaagtggccgcagtggttatcactcatggtta tggcagcactgcataattctcttactgtcatgccatccgtaagatgctttt ctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggc gaccgagttgctcttgcggcgctcaatcgggataataccgcgcacata gcagaactttaaaagtgtcatcattggaaaaagcttcttcggggcgaaaac tctcaaggatcttaccgctgttgagatccagttcgatgaacccactcgtg caccacactgatcttcagcatctttactttcaccagcgttctcgggtgag

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
First nucleic acid construct, sa-mRNA encoding mouse IL15Ra (Sam002-mIL15Ra)	78	gcggcgcgatgagagaagcccagaccaattacctacccaaaatgggaaaagt tcacgttgacatcgaggaagacagccattcctcagagctttgcagcggag cttcccgcagtttgaggtagaagccaagcaggctcactgataatgaccatgc taatgccagagcgttttcgcatctggcttcaaaactgatcgaaacggaggt ggacccatccgacacgatccttgacattggaagtgcgcccgcccgagaaat gtattctaagcacaagtatcattgtatctgtccgatgagatgtgcggaaga tccggacagattgtataagtatgcaactaagctgaagaaaaactgtaagga aataactgataaggaattggacaagaaaaatgaaggagctggccgcgtcat gagcgacccctgacctggaaactgagactatgtgcctccacgacgacgagtc gtgtcgctacgaagggaagtcgctgtttaccaggatgtatcgcggttga cggaccgacaagtctctatcaccaagccaataagggaggttagagtcgccta ctggataggctttgacaccacccctttatgtttaagaacttggtcggagc atatccatcatactctaccaactgggcccagcaaacccgtgttaacggctcg taacataggcctatgcagctctgacgttatggagcggctcacgtagagggat gtccattcttagaaagaagtattgaaaccatccaacaatgttctattctc tgttggctcgacctctaccacgagaagaggacttactgaggagctggca cctgcgctctgtatttcaactacgtggcgaagcaaaatcacacatgtcgtg tgagactatagttagttgcgacgggtacgtcgttaaaagaatagctatcag tccaggcctgtatgggaagcctcaggtatgtctgctacgatgcaccgca gggattcttctgtctgcaaaagtgcagacacattgaacggggagaggggtctc tttcccggtgtgcacgtatgtgccagctacattgtgtgaccaaatgactgg catactggcaacagatgtcagtcgggacgacgcgcaaaaactgctgggttg gctcaaccagcgtatagtcgtcaacggctgcacccagagaaaacaccaatac catgaaaaattacctttgcccgtagtggcccaggcatttgcaggtgggc aaaggaatataaggaagatcaagaagatgaaaggccactaggactacgaga tagacagttagtcatggggtgtgttgggcttttagaaggcacaagataac atctattataagcgcgcggataacccaaaccatcatcaaagtgaacagcga tttccactcatctcgtgctgccaggataggcagtaacacattgagatcgg gctgagaacaagaatcagggaaatgttagaggagcacaaggagccgtcacc tctcattaccgcccaggagctacaagaagctaagtgcgcagccgatgagcg taaggaggtgcgtgaagccgaggaggtgcgcgcagctctaccacctttggc agctgatgttgaggagccactctggaagccgatgtcgacttgatgttaca agaggtctggggccggctcagtgagagacacctcgtggcttgataaagggtac cagctacgatggcgaggacaagatcggctcttacgctgtgctttctccgca ggctgtactcaagagtgaaaaattatcttgcatccaccctctcgctgaaca agtcatagtgataaacacactctggccgaaaaggcgcttatgcccgtggaacc atccatggtaaaagtgtgtgcccagaggacatgcaataccctgccagga ctttcaagctctgagtgaaagtgccaccatgtgtacaacgaacgtgagtt cgtaaaacaggtacctgcaccatattgccacacatggaggagcgtgaaac tgatgaagaatattacaaaactgtcaagcccagcgagcagcagggcgaata cctgtacgacatcgacaggaacagtgctgcaagaaagaactagtcactgg gctagggtcacaggcgagctgggtggatcctcccttccatgaattcgctca cgagagctcgagaacacgaccagccgctccttaccaaagtaaccaaccatagg gggtgatggcggtgccaggatcaggcaagctcggcatcattaaaagcgcagt caccaaaaaagatctagtggtgagcgccaaagaaagaaaactgtgcagaaat tataagggacgtcaagaaaatgaaagggtggacgtcaatgccagaactgt ggactcagtgctcttgaaatggatgcaaacaccccgtagagaccctgtatat tgacgaagcttttgccttgcatgcaggtactctcagagcgtctatagccat tataagacctaaaaggcagtgctctgcggggatcccaaacagtgcggttt ttttaacatgatgtgctgaaagtgcattttaaccacgagatttgcacaca agtcctcccaaaaagcatctcgcgcttgcaactaaatctgtgacttcggt cgtctcaacctgttttgcgacaaaaaatgagaacgacgaatccgaaaga gactaagatttgtgatgacactaccggcagtaaccaacctaaagcaggacga tctcatctcacttggttcagagggtgggtgaagcagttgcaaatagattaca aaaaggcaacgaaataatgacggcagctgcctctcaagggtgacccgtaa agggtgtgatgcgcttcggtaacagggtgaatgaaaatcctctgtacgcacc cacctcagaacatgtgaacgtcctactgacccgcacggaggacgcgatcgt gtggaaaacactagccggcgacccatggataaaaacactgactgccaaagta ccctgggaatttcactgccacgatagaggagtggaagcagagcatgatgc catcatgaggcacatctggagagacggacccataccgacgtcttcagaa taaggcaaacgtgtgttgggccaaggctttagtgccggtgctgaagaccgc tgccatagacatgaccactgaacaatggaacactgtggattattttgaaac ggacaaaagtcactcagcagagatagtattgaaccaactatgcgtgagggt ctttggactcgatctggactccggtctattttctgcaccaactgttccgtt atccattaggaataatcactgggataaactccccgtgcctaacatgtacgg gctgaataaagaagtggccgctcagctctctcgcaggtaccacaactgcc tcgggcagttgccactggaagagtctatgacatgaacactggtacactgcg caattatgatccgcgcataaacctagtacctgtaaacagaagactgcctca tgctttagtcctccaccataatgaaccccacagagtgaactttctctcat cgtcagcaaatgaagggcagaactgtcctgggtggcgggaaaagtgtgc

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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Nucleic Acid Template, sa- mRNA encoding GFP (SVP-018) - the selectable marker (Amp Cassette, SEQ ID NO: 12) is bolded and italicized; the coding sequences of structural proteins of of SFV4 are italicized and underlined; and viral structural proteins from SFV4 is bolded. The selectable marker and GOI can be substituted with one or more different selectable markers and GOIs.	84	aaaaaaaaaacgctgataacgcaggaagaacatgtgagcaaaaggccag caaaaggccaggaaacgtaaaaggccgcgttgctggcggtttttccatagg ctccgccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtgg cgaaaccgacaggactataaagataaccaggcggtttccccctggaaagctcc ctctgctgcgtctctctgttccgaccctgccgcttaccggataacctgtccgcc tttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggat ctcagttcggtgtaggctcgttcgctccaagctgggctgtgtgcacgaaccc ccggttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtc aaccggtaagacacgacttatcgccactggcagcagccactggtaacagg attagcagagcaggtatgtaggcggtgctacagagttcttgaaagtggg cctaactacggctacactagaagaacagttatgggtatctgcgctctgctg aagccagttaccttcggaaaaagagttggtagctcttgatccggcaaaaca accccgctggtagcggttttttggttgcaagcagcagattacgcgcaga aaaaaaggatctcaagaagatcctttgatctttctacggggtctgacgct cagtggaacgaaaaactcacgttaagggattttgggtcatgagattataaaa aggatcttcacctagatccttttaaataaaaatgaagttttaaatcaatc taaagtatatatgagtaaaacttggctctgacagttaccaatgcttaactcagt gaggcacctatctcagcgatctgtctatttctgttcacatcattgtgctga ctccccctgctgtgtagataaactacgatacgggagggttaccatctggcccc agtgctgcaatgataccgcgagaccacgcctcaccggctccagatttatca gcaataaaaccagccagccgaaggggcgagcgcagaagtggctctgcaact ttatccgctccatccagctctatttaattgttgccgggaagctagagtaagt agttcgccagtttaattgttgccgaacgttgggtgcccattgctacaggcatc gtggtgtcacgctcgtcgtttgggtatgggttcattcagctccgggtcccaa cgatcaaggcgagttacatgatccccatgttggtgcaaaaaagcggttagc tccttcgggtctccgatcgtgtgcagaagttaagtggcccgagtggtatca ctcatgggtatggcagcactgcataattctcttactgtcatgccatccgta agatgcttttctgtgactgggtgagtagtcaaccaagtcattctgagaaatag tgtagtgcggcgaccgagttgctcttgcccggtcgaatacgggataatacc ggcaccacatagcagaactttaaaagtgcctcatcattggaaaaagcttcttcg gggcgaaaaactctcaaggatcttaccgctgttgagatccagttcagatgtaa cccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtt tctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagg gcgacacggaaatgtgaataactatactcttcttttcaatattatga agcatttatcagggttattgtctcatgagcggatataatttgaatgtatt

TABLE 16-continued

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Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

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Nucleic Acid Template, modified mRNA encoding viral structural proteins from SFV4 (SVP- 431) -the selectable marker (Amp Cassette, SEQ ID NO: 12) is bolded and italicized and the GOI (GFP, SEQ ID NO: 24) is bolded. The selectable marker and GOI can be substituted with one or more different selectable markers and GOIs.	85	agccacacccctggagctagcaaaaaaaaaaaaaaaaaaaaaaaaaaag catatgactaaa aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaacgcgtatcgatgatcgcggcc gcatacagcagcaattggcaagctgcttacatagaactcgcggcgattggc atgcccgccttaaaatttttattttattttctttcttttcgaatcggat tttgttttaataatttcaaaaaaaaaaaaaaaaaaaaaaaaaaacgcgtgat aacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaacgt aaaaaggccgcgttgctggcgtttttccataggctccgccccctgacgag catcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggacta taaagataccaggcggtttccccctggaagctccctcgtgcgctctcctgtt ccgaccctgcccgttacccgataacctgtccgcctttctcccttcgggaagc gtggcgctttctcatagctcacgctgtaggtatctcagttcgggtgtaggtc gttcgtccaaagctgggctgtgtgcacgaacccccgttcagcccgaccgc tgcgccttatccggtaaactatcgtcttgagtcacaaccggtaagacacgac ttatcgccactggcagcagccactggtaacaggattagcagagcgaggtat gtaggcggtgctacagagttcttgaagtggtggcctaactacggctacact agaagaacagttatttggtatctgcgctctgctgaagccagttaccttcgga aaaaagatttgtagctcttgatccggcaaaacaaaccacgcgtggtagcggt tttttgttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaa gatcctttgatctttctacggggtctgacgctcagtggaacgaaaaactca cgtaagggatttttggtcatgagattcaaaaaggatcttcacctagatc ctttaaatataaaatgaagttttaaatcaatctaaagtatatat agtagtaa acttggctcgacagttaccaatgcttaatcagtgaggcaccttatctcagcg atctgtctatttctgttcacatagtttgctgactccccctgctgtagata actacgatacgggagggttaccatctggccccagtgctgcaatgataccg cgagacccacgctcaccggctccagatttatcagcaataaaccagccagcc ggaaggccgagcgcagaagtggctcctgcaactttatccgcctccatccag ctattaatgttgcgggaagctagagtaagtagttcgccagttaatagt ttgcgcaacgttgttgccatttgctacaggcatcgtggtgtcagctcgtcg tttggtatggcttcatcagctccggttcccaacgatcaaggcgagttaca tgatccccatgttgtagcaaaaagcggttagctcctcgggtcctcgatc gttgtcagaagtaagttggccgagtggttatcactcatggttatggcagca ctgcataattctctactgtcatgccatccgtaagatgctttctgtgact ggtgagtaactcaaccaagtcattctgagaatagtgtagtcggcgacggaggt tgctcttgcccgcgctcaatcgggataataccgcccacatagcagaact ttaaaagtgtcatcattggaaaaacgttcttcggggcgaaaaactctcaagg atcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaac tgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaaa ggaaggcaaaatgccgcaaaaagggaataaggcgacacgggaatgttga atactcactcttcttcttcaatatttgaagcatttatcagggttat tgtctcatgagcggatacatatttgaatgtatttagaaaaataaacaata gggttccgcgcacatttccccgaaaagtgccacctgagctctaatacgcac tcactatggacatttgctctgacacaaactgtgttcactagcaacctcaa acagacaccatgaattacatccctacgcaaacgttttacggggcgagatgg

TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
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Nucleic Acid Template, Sam002- mouse IL-12, kanamycin resistance) (SVP-327)- the GOI (mouse IL-12) is bolded. The GOI can be substituted with one or more different GOIs.	86	aaaaaaaaaacgcggaagagccgcgtgataacgcaggaagaacatgtgag caaaaggccagcaaaaggccaggacccgtaaaaaaggccgcgttgcggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtcagagggtggcgaaacccgcagaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgaccctgccgcttacgggat acctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcac gctgtagggtatctcagttcgggtgtaggtcgttcgctccaagctgggctgtg tgcacgaacccccgttcagcccgaccgctgcgccttatccggtaactatc gtcttgagtcacacccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggatgtaggcggtgctacagagttct tgaagtgggtggcctaactacggctacactagaagaacagtttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacaccccgctggtagcggtgggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatcttttcta cgggggtctgacgctcagtggaacgaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaaatgaa gttttaaatcaatctaaagtatatatgagtaacctgaggctatggcagggc ctgcgcccccgacgttgggtgcgagccctgggccttcacccgaacttgggg ggtgggggtggggaaggaagaaacgcgggcgtattggccccaatggggctc tcggtgggggtatcgacagagtgccagccctgggaccgaaccccgctttat gaacaaacgacccaacccgctgcgttttatctgtctttttatgcgctca tagcgcggttccttcgggtattgtctccttcggtgtttcagttagctcc ccctagggtggcggaagaactccagcatgagatccccgcgtcgaggatca tccagccggcgtcccggaacacgattccgaagcccaacctttcatagaagg cggcggtggaatcgaaatctcgtgatggcaggttggcgctcgttggtcgg tcatctcgaaacccagagtcctcgtcagaagaactcgtcagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgatccgtaaacgacaggaa cgggtcagcccatcgcccgcaagctctcagcaatcacgggtagccaa cgctatgtcctgatagcgatccgccacaccagccggccacagtcgatgaa tccagaaaagcggccattttccaccatgatattcggaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcatgctcgcttgagcctggc gaacagttcggctggcgagccctgatgctcttcgtccagatcatcctg atcgacaagaccggcttccatccgagtagctgctcgtcgatgcgatgttt cgcttggtggtcgaaatgggcaggtagccggatcaagcgtatgcagccgag cattgcatcagccatgatggatactttctcggcaggagcaaggttgagatga caggagatcctgccccggcacttcgcccatagcagccagtccttcccgc ttcagtgacaacgtcgagcacagctcgcaaggaacgcccgtcggtggccag ccacgatagccgctgcctcgtcttgagttcattcagggcaccggacag gtcggtcttgacaaaaagaacgggcgcctcgtcgctgacagccggaacac gcggcatcagagcagcagattgtctgtgtgcccagtcatagcgaataag cctctccacccaagcggcgggagaacctgcgtgcaatccatcttgttaaat catgcgaaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggctccaaaaagcctcctcactactctggaatagctcagaggccgagg cggcctcggcctctgcataaataaaaaaattagtcagccatggggcgag aatgggcggaactgggaggagttagggcgggatgggaggagttagggcg ggactatgggtgctgactaattgagatgcatgctttgcatactctgcctg ctggggagcctggggactttccacacctggttgctgactaattgagatgca tgctttgcatactctgcctgctggggagcctggggactttccacacctta actgacacacattccacagctggttctttccgctcaggactcttctcttt tcaatattattgaagcatttatcaggggtattgtctcatgagcggatacat atttgaatgtatttagaaaaataaacaaatagggggtccgcgcacatttcc ccgaaaagtgccacctgagctctaatacagctcactataggataggcgagg catgagagaagccagaccaattacctacccaaatggagaaagttcacgt tgacatcgaggaagacagccattcctcagagctttgcagcggagcttccc cgagttgagggtagaagccaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggagggtgaccc atccgacacgatccttgacattggaagtgcgcccgcgcgcagaatgtattc taagcacagatcattgtatctgtccgatgagatgtgcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaagggaataac tgataagggaattggacaagaaaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagtcgtgtgc ctacgaaggcgaagtgcgtgtttaccaggatgtatcgcggttgacggacc gacaagtcctctatcaccaagccaataagggagttagagtcgctactggat aggtttgacaccacccctttatgtttaagaacttggtggagcatatcc atcatactctaccaactgggcccagcaaacctgttaacggctcgtaacat

TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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Nucleic Acid Template Sam002- human IL-12, kanamycin resistance (SVP-315)- the GOI (human IL-12) is bolded. The GOI can be substituted with one or more different GOIs.	87	aaaaaaaaacgcggaagagccgctgataacgcaggaagaacatgtgag caaaaggccagcaaaaggccaggaaacgtaaaaaggccgcttgcgtggcgt ttttccataggtccgccccctgacgagcatcacaaaaatcgacgctcaa gtcttgagtcacaaccggtaagacacgaacttatcgccactggcagcagcca ctggtaacaggatttagcagagcagaggtatgtaggcgggtgctacagagtct acctgtccgcttctctcccttcgggaagcgtggcgttctctcatagctcac gctgtaggatctcagttcgggtgtaggtcggttcgctccaagctgggctgtg tgcacgaacccccgttcagcccgaccgctgcgccttatccggttaactatc gtcagaggtggcgaaccccgacaggactataaagataaccaggcgtttcccc ctggaagctccctcgctgcgctctcctgttcgcagccctgccgcttacggat tgaagtgggtggcctaactacggctacactagaagaacagtatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacaaacacccgctggtagcgggtgggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctca cgggggtctgacgctcagtggaacgaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaaatgaa gttttaaatcaatcaaaagtatatatgagtaacctgaggctatggcagggc ctgcgcccccgactgtggctgcgagccctgggcttcacccgaacttgggg gggtgggtggggaaaaaggaagaaacgcgggcgtattggccccaatggggct tcggtggggtatcgacagagtgccagccctgggaccgaaccccgctttat gaacaaacgacccaacacgctgcgttttattctgtctttttattgcgctca tagcgcgggttccctccggtattgtctccttcggtgttcagttagcctcc ccctagggtgggcaagaactccagcatgagatccccgcgctggaggatca tccagccggcgtcccgaaaaacgattccgaagcccaacctttcatagaagg cggcggtgggaatcgaaatctcgatggcaggttggcgctcgttggctcgg tcatttcgaaacccagagctcccgctcagaagaactcgtcaagaaggcgaat gaaggcgatgcgctgcgaatcgggagcggcgataccgtaaagcacaggaa cgggtcagcccatcgccgccaagctctcagcaatatcacgggtagccaa cgctatgtcctgatagcagatccgccacacccagccggccacagtcgatgaa tccagaaaaagcggccattttccaccatgatattcggaagcaggcatcgcc atgggtcacgacagatcctcgccgtcgggcatgctcgcttgagcctggc gaacagttcggtggcgagccctgatgctcttcgctccagatcatcctg atcgacaagaccggcttccatccgagtagctgctcgtcgatgcatgttt cgcttggtggtcgaaatgggcaggtagccggtcaagcgtatgcagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgcccatagcagccagtccttcccgc ttcagtgacaacgtcgagcacagctgcgcaaggaaaccccgctcggtggccag ccacgatagccgcgctgcctcgtcttgagcttcaatcagggcacccggacag gtcggtcttgacaaaaagaacggggcgcctcgcgtgacagccggaacac ggcgcatcagagcagcagattgtctgtgtgcccagtcatagcgaatag cctctccacccaagcggccggagacactgcgtgcaatccatcttgttcaat

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		catgcgaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggcctccaaaaagcctcctcactacttctggaatagctcagagggcggagg cggcctcggcctctgcataaataaaaaaattagtcagccatgggcgagg aatggcggaactggcgaggtagggcgggatggcgaggtagggcg ggactatgggtgctgactaattgagatgcatgctttgcatacttctgcctg ctggggagcctggggactttccacacctgggtgctgactaattgagatgca tgctttgcatacttctgcctgctggggagcctggggactttccacacctaa actgacacacattccacagctgggtctttccgcctcaggactcttctcttt tcaatattattgaagcatttaccaggggtattgtctcatgagcggatacat atttgaatgtatttagaaaaataaacaataagggggtccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgaactcactataggataggcgcg catgagagaagccagaccaattacctaaccataatggagaaagtccacgt tgacatcgaggaagacagccattcctcagagctttgcagcggagcttccc gcagtttgaggtagaagccaagcaggtcactgataatgacctgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggagggtggacc atccgacacgatccttgacattggaagtgcgcccgcgcagaaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtgcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaagggaataac tgataagggaattggacaagaaaatgaaggagctggccgcgctcatgagcga ccctgacctggaactgagactatgtgcctccacgacgacgagctcgtgtcg ctacgaagggaagtcgctgtttaccaggatgtatccgcggttgacggacc gacaagtctctatcaccaagccaataaggagtagagtcgctactggat aggctttgacaccaccttttatgtttaagaacttggtggagcatatcc atcactactctaccaactgggcccagcaaacctgttaacggctcgtaacat aggcctatgcagctctgacgttatggagcggtcacgtagaggatgtccat tcttagaaagaagtatttgaaccatccaacaatgttctatctctgttg ctcgaccatctaccacgagaagagggaacttactgaggagctggcactgcc gtctgtatttcaactacgtggcaagcaaaattacacatgtcgggtgtgagac tatagttagttgcgacgggtacgtcgttaaaagaatagctatcagtcagg ccgtgatgggaagcctcaggctatgctgctacgatgcaccgcgagggtat ctgtgctgcaagtgcagacacattgaacggggagagggtctcttttcc cgtgtgcacgtatgtgcagctacattgtgtgaccaaatgactggcactact ggcaacagatgtcagtcggacgacgcgcgcaaaactgctgggtgggctcaa ccagcgtatagtcgtcaacggtcgcaccagagaaacacaaatccatgaa aaattaccttttgccgtagtgcccaggcatttgctaggtgggcaaggga atataagggaagatcaagaagatgaaaggccactaggactacgagatagaca gttagtcaggggtgtgtgggcttttagaaggcacaagataaacatctat ttataagcgcccgatacccaaacatcatcaaatgaacagcgatttcca ctcattcgtgctgccaggataggcagtaacacattggagatcgggctgag aacaagaatcaggaaaaatgttagaggagcacaaggagccgtcacctctcat taccgcccaggagcgtacaagaagctaaagtcgcgagccgatagggtcaagg gggtcgtgaagccaggaggtgcgcgcagctctaccacctttggcagctga gttgaggagcccactctggaagccgatgtcgacttgatgttacaaggagc tggggcccgtcagtgagacacccctcgtggttgataaagggtaccagctac cgatggcgaggacaagatcggtcttaccgtgtgctttctccgcaggctgt actcaagagtgaaaaaattatcttgcatccacctctcgtgaaacaagtcat agtgataaacacactctggccgaaaaggcggttatgccgtggaaacatacca tggtaaagtgtggtgccagaggacatgcaatacccgccaggactttca agctctgagtgaaagtgccaccattgtgtacaacgaacgtgagttcgtaaa caggtaacctgcaccatattgccacacatggaggagcgtgaacactgatga agaatattacaaaactgtcaagcccagcgacgacggcgaataacctgta cgacatcgacaggaacagtgccgtcaagaaagaactagtcactgggctagg gctcacaggcgagctggtggatcctcccttccatgaattcgctacgagag tctgagaacacgaccagccgtccttaccagtaaccaacatagggggtga tgccgtgccaggatcaggcaagtctggcatcataaaagcgcagctaccaca aaaagatctagtggtgagcgccaagaaagaaaactgtgcagaaattataag ggacgtcaagaaaatgaaagggtggacgtcaatgccagaaactgtggactc agtgtcttgaaatggatgcaaacaccccgtagagacctgtatattgacga agcttttgcttgtcatgcaggtactctcagagcgtcatagccattataag acctaaaaaggcagtgctctgcggggtcccaaacagtcgggtttttttaa catgatgtgctgaaagtgcattttaaccacgagatttgacacaaagtctc ccacaaaagcatctctcgccgttgcaactaaatctgtgacttcgggtcgtctc aaccttggtttacgacaaaaaaatgagaacgacgaatccgaaagagactaa gattgtgatgacactaccggcagtaaccaacctaaagcaggacgatctcat tctcacttggttcagagggtgggtgaagcagttgcaaatagattacaagg caacgaaataatgacggcagctgcctctcaagggtgacccgtaaaagggtg gtatgccgttcggtaacaggtgaatgaaatcctctgtacgacccacctc agaacatgtgaacgtcctactgacccgcacggaggacccgatcgtgtggaa aacactagccggcgacccatggataaaaaactgactgccaagtaccctgg gaatttcaactgccacgatagaggagtggaagcagagcatgagccatcat gaggcacatcttgagagaccggacctaccgagcttccagaataaggc

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		aaacgtgtgttgggccaagcctttagtgccgggtgctgaagaccgctggcat agacatgaccactgaacaatggaacactgtggattatTTTgaaacggacaa agctcactcagcagagatagtattgaaccaactatgcgtgaggttcttttg actcgatctggactccggctctattttctgcacccactgttcggttatccat taggaataatcactgggataaactccccgtcgctaactgtacgggctgaa taaagaagtggctcgtcagctctctcgcagggtacccacaactgcctcgggc agttgccactggaagagtctatgacatgaacactggtagactgcgaatta tgatccgcgcataaacctagtacctgtaaacagaagactgcctcatgctt agtcctccaccataatgaacaccacagagtgactttcttcattcgtcag caaattgaagggcagaactgtcctgggtggtcggggaaaagtgtccgtccc aggcaaatgggtgactgggtgtcagaccggcctgaggctaccttcagagc tcggctggatttaggcacccagggtgtagtgcctaaatagacataatatt tgtaatgtgaggaccccatataaatccatcactatcagcagtgtagaaga ccatgccattaaagcttagcatgttgaccaagaaagcttgtctgcatctgaa tcccgcggaacctgtgtcagcatagggtatgggtacgctgacagggccag cgaaagcatcattgggtctatagcgcggcagttcaagttttcccggtatg caaaccgaaatcctcacttgaagagacggaagtctgtttgtattcattgg gtacgatcgcaaggccgtacgcacaaattcttacaagctttcatcaacctt gaccaacatttatacagggtccagactccagaagccggatgtgacacctc atatcatgtggtgcgaggggatattgccacggccaccgaaggagtgattat aatgctgctaacagcaaggacaacctggcgagggggtgtgcggagcgct gtataagaaattcccggaagcctcgatttacagccgatcgaagtaggaaa agcgcgactgggtcaaagggtgcagctaaacatatcattcatgccttaggacc aaacttcaacaaagtTTCggagggtgaagggtgacaaacagtTggcagaggc ttatgagtcactcgctaagattgtcaacgataacaattacaagtcagtgc gattccactgttgtccaccggcatctttccgggaacaaagatcgactaac ccaatcattgaaccatttgctgacagctttagacaccactgatgcagatgt agccatatactgcagggacaagaatgggaaatgactctcaaggaaagcagt ggctaggagagaagcagtgaggagatagcatatccgacgactcttcagt gacagaacctgatgcagagctggtaggggtgcatccgaagagtctttggc tggaaggaagggtacagcacaaagcgtggcaaaactttctcatatttgga agggaaccaagtTtcaccaggcgcccaaggatagcagaaattaatgccat gtggcccggtgcacaggaggccaatgagcaggtatgcattatctcctcg agaaagcatgagcagtttaggtcgaaatgcccgtcgaaagtcggaagc ctccacaccacctagcagctgccttgctgtgcatccatgccatgactcc agaaagagtacagcgctaaaagcctcaggtccagaacaaattactgtgtg ctcatcctttccattgcccgaagtatagaatcactgggtgtgcagaagatcca atgctcccagcctatattgtttctcaccgaaagtgcctgcgtatattcatcc aaggaaagtattcgtggaaacaccacggtagacgagactccggagccatc ggcagagaaccaatccacagagggggacacctgaacaaaccacctataac cgaggatgagaccaggactagaacgcctgagccgatcatcatcgaaggagga agaagaggatagcataagtttgctgtcagatggcccgaccaccaggtgct gcaagtgcaggcagacattcacgggcgcgcctctgtatctagctcactcgt gtccattcctcatgcacccgactttgatgtggacagtttatccatacttga cacctggagggagctagcgtgaccagcggggcaacgtcagccgagactaa ctcttacttcgcaaaagagttaggagttctggcgcgacccggtgcctgcgc tcgaacagttatcaggaacctccacatcccgctccgcgcacaagaacacc gtcacttgcacccagcagggcctgctcgagaaccagcctagtttccacccc gccaggcgtaaatagggtgatcactagagaggagctcgaggcgcttacc gtcacgcactcctagcagggtcggtctcgagaaccagcctgggtctccacccc gccaggcgtaaatagggtgattacaagagaggagttgaggcgcttcgtagc acaacaacaatgacgggttgatgcgggtgcatacatctttcctccgacac cggtcaagggcatttacaacaaaaatcagtaaggcaaacgggtgctatccga agtggtgttgagaggaccgaattggagatttcgtatgccccgcgcctcga ccaagaaaaagaagaattactacgcaagaattacagttaaatccacacc tgctaacagaagcagataccagtcaggaagggtggagaacatgaaagccat aacagctagacgtattctgcaaggcctagggcattattgaaggcagaagg aaaagtgagtgctaccgaacctgcattcctgttcctttgtattcacttag tgtgaaccgtgccttttcaagcccaagggtcgagtggaagcctgtaacgc catgttgaaagagaaatttccgactgtggcttcttactgtattattccaga gtacgatgcctatttgacatgggtgacggagcttcagtgctgcttagacac tgccagtttttgccctgcaagctgcgcagctttccaaagaaacactccta tttggaaaccacaatacgatcggcagtgcttcagcgatccagaacacgct ccagaacgtcctggcagctgccacaaaagaaattgcaatgtcacgcaaat gagagaattgcccgtattggattcggcggtctttaatgtggaatgcttcaa gaaatagcgtgtataaatgaatttgggaaacgtttaaagaaaaccccat caggcttactgaagaaaacgtggtaaatacattaccaaattaaaaggacc aaaagctgctgctcttttgcgaagacacataatttgaatatgttgaggga cataccaatggacaggtttgttaattggacttaaagagagacgtgaaagtga tcagggaacaaaacatactgaagaacggcccaagggtacaggtgatccaggc tgccgatccgctagcaacagcgtatctgtgcggaatccaccgagagctgggt

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		taggagattaaatgCGgtcctgcttcCGaacattcatacactgtttgatat gtcggctgaagactttgacgctattatagccgagcacttccagcctgggga ttgtgttctggaaactgacatcgctcggttgataaaagtgaggacgacgc catggctctgaccgcttaatgatctggaagacttaggtgtggacgcaga gctgttgacgctgattgaggcggttttcggcgaaatttcatcaatacattt gcccactaaaactaaatttaaattcggagccatgatgaaatctggaatgtt cctcacactgtttgtgaacacagtcattaacattgtaatcgcaagcagagt gttgagagaacggctaacggatcaccatgtgcagcattcattggagatga caatatcgtaaaggagtgcaaatcggacaaattaatggcagacaggtgcgc cacctggttgaatatggaagtcaagattatagatgctgtggtggcgagaa agcgcccttattctgtggagggtttattttgtgtgactccgtgaccggcac agcgtgcggtgtggcagacccctaaaaaggctgtttaagcttggcaaac tctggcagcagacgatgaacatgatgatgacaggagaagggcattgcatga agagtcaacacgctggaaccgagtggtattctttcagagctgtgcaaggc agtagaatcaaggatatgaaaccgtaggaaacttccatcatagttatggccat gactactctagctagcagtggttaatcattcagctacctgagaggggcccc tataactctctacggctaacctgaatggactacgacatagtctagtccgcc aagttcgaaggcgcgctctagagccaccat ggactggacctggcgaaatc tggtcttgggttgcgcgcgtacagggactcaccgaatatgggagctgaaga aggacgtgtatgtcgaggctggactggtaccagatgctcctggcgaaa tgggtggttttaacatgtgataccccgaggaggacggcaccacatggactc tggaccagagtcttgaggtgctggggtccggcaagactctgacaatccagg ttaaggagttcggcgacgcaggacagtacacttgtcacaaggagggtgagg tgctttctcacagcctgttctgctctcacaagaagggaagcgggtatttgg caaccgacatctcacaaggaccagaaggagccccaaaaaagaccttctga gatgtgagggccaagaattacagcggtagattcacttgttgggtggctcaca ccatatccacagacttgacctcagtgctaaaagttcacgaggagctcag atcctcaaggcgttacctgtggcgacgcagcgtgtccgcagaaagagtca ggggagacacaagaatacaggtactctgtcgagtgccaggaggattccg cctgtccggccgcagaggagtctttacctattgaggtgatggctcgatgcg tgcacaagcttaagtacgagaattacacatcaagtttttctcaccgcgaca tcattaaacctgatccaccaagaacctgcagctcaagcctctgaagaata gcaggcaggtcgaggttaagctgggagtaacctgataacctgggtccaccccc acagttatttccagcctcaccttctgctcgaagtcagggaagagcaaga gagagaagaaggataggggttccacagataagacttcagctactgtgatct gcagaaagaatgcgtctatctctgtgcgagcacaagacaggtactacagtt ctagctggagcgagtgggcatcagtcacctgcagtggtggcggaagcggag ggggcagcggaggtgggagcggaggagcaggaacctccagttgctacac ctgacccgggaatgtttccatgcctccaccattcccagaatctcctccgag ccgtgtccaatatgctgcaaaaggctcggcagaccttgagggtttacctct gcacctcagaagaaatcgatcatgaggatatcacaaggataagacgagca ctgttgaggcatgcctgccctggagctaaactaagaatgagctttgctga acagcaggagacttcttctattaccaaaggtagctgtcttgcagcagga agacatcttttatgatggccctgtgtctatctagcatatatgaagacctga agatgtaaccaggtggaattcaaaacctgaatgctaagcttctcatggatc ccaagaggcaaatcttctggaccagaatatgcttgctgtcatagatgaac tgatgcaggcgttgaattttaacagcgagacgggtgcctcaaaaagctcac tggagaacctgattttataaaacgaagatcaagctgtgtattttactac acgcctttagaatccgcgctgttaccatcgacagagtaatgtcctaactaa atgcttcatagcatcgatgatatcgcgccgcatacagcagcaattggcaa gctgcttacatagaactcgcgcgattggcatgccgccttaaaatttttat tttatttttctttcttttccgaatcggtatttggtttttaataatttcaaaa aaaaaaaaaaaa

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Second nucleic acid construct template, modified mRNA encoding with capsids/envelop from SIN, Amp resistance (SVP-464) - the viral structural protein of SIN is bolded. The sequence can be substituted with one or more different viral structural proteins.	88	gctcgcctttctgtgtgtccaatcttctattaaaggttccttgttccttaag tccaaactactaaactgggggatattatgaagggccttgagcatctggattc tgccataataaaaaacatttattttcattgcaaagccacacccctggagctag caaaaaaaaaaaaaaaaaaaaaaaaaaaaaagcatatgactaaaaaaaaaa aaa aaaaaaaaaacgcgtatcgatgatatcgcgccgcatacagcagcaattggc aagctgcttacatagaactcgcgccgatggcatgcccgccttaaaatttt atttatttttcttttcttttccgaatcggattttgtttttaaattttcaa aaaaaaaaaaaaaaaaaaaaaaaaaaaaaacgcgtataacgcaggaaagacatgt gagcaaaagggccagcaaaagggcaggaaacgtaaaagggccgcgttgctgg cgtttttccataggtccgccccctgacgagcatcacaaaaatcgacgct caagtccagaggtggcgaacccgcacaggactataagataaccagggcttcc ccccgggaagctccctcggtcgctctcctgttccgaccctgcgccttaccg gatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagct cacgctgtaggatctcagttcggtgtaggctcgctccaagctgggct gtgtgcacgaacccccggttcagcccgaccgctgcgccttatccggttaact atcgctcttgagtcacacccggtaagacacgacttatcgccactggcagcag ccactggtaacaggattagcagagcgaggtatgtaggcgggtgctacagagt tcttgaagtgggtggcctaactacggctacactagaagaacagttttggtat tctcgctctgctgaagccagttaccttcggaaaaagagtttggtagctctt gatccggcaaacacacccacgcgtggtagcgggttttttggttgcaagcagc agattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttcta cgggggtctgacgctcagtggaacgaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaaatgaa gttttaaatcaatctaaggtatataatgagtaaaccttggtctgacagttacc aatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcat ccatagttgctgactccccgctcggttagataactacgatacgggagggtc taccatctggccccagtgctgcaatgataccgcgagaccacgctcaccgg ctccagatttatcagcaataaacccagccagccggaagggccgagcgcagaa gtggtcctgcaactttatccgcctccatccagctctataatgttgccggg aagctagagtaagtagttcgccagttaatagtttgccgaacggttggtgcca ttgtacagggcatcggtgtcagcgtcgtcggttggtatgggttcattca gctccggttcccaacgatcaaggcgagttacatgatccccatgtgtgca aaaaagcgggttagctccttcgggtcctccgatcgttgtcagaagtaagttgg ccgcagtggtatcactcatggttatggcagcactgcataattctcttactg tcatgccatccgttaagatgctttctgtgactgggtgagtaactcaaccaagt cattctgagaatagtgatgcggcgacccaggttgctcttgccggcgctcaa tacgggataatccgcgcacatagcagaactttaaaagtgtcatcattg gaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagat ccagttcgatgtaacccactcgtgcacccaactgatcttcagcatctttta ctttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaa aaaagggtaataagggcgacacggaaatgtgaatactcatactcttctt ttcaatattatgaagcatttatcagggttattgtctcatgagcggataca tatttgaatgtatttagaaaaataacaaataggggttcgcgcacatttc cccgaagtgcccactgagctctaatcagactcactataggataggacat ttgcttctgacacaaactgtgttactagcaacctcaaacagacaccatgaa tagaggaaatattaaatgctgagcgcgcgccttccccgcgcgcacagc tatgtggaggcctaggcgaaggaggcaggcggcacccactgcccgcgctaa tgggttggcatcccaattcaacagctaacacagctgtcagtgccctagt cattggtcaagccaccgcacctcaacaaacaggccacgcccgcctaccacg ccctaagaagcagtcctccgaagcaaccacgaatgctaaaaagacaaaacc gcaggagaaacaaggagaaacaacccgcaaaacgaagccgggttaagagaca gcgtatggcactaaaattgatgctgacaggactctttgacgtgaacaacga agccgggtgacgtaataaggacacgcctcgtatggaaggcaaaagtgtgaa gccacttcacgtgaaaggggtcatagaccacccgggttttggcaaaagctgaa atttaccagtgatcagcgtacgacatggagtttcgcccagcttccagccag tatgaaaagcgaagcattttgcatacaccagcagcaccctgaagggttcta caactggcatcatggtgcccgtgacgtacagcggcggtcgctttacagtccc gcgcggagttgggggaaaaggcgacgtggaagaccgatcatggataatac aggttaagtggtgcgcgatagtgctgggcggcgctgacgaaggcgccgcac tgctctctccgtcgtcacttggaaacagcaaaaggaaagaccatcaaacgac tcgggaagggaactgaggaatggcttcgcagcaccattagtcacgggtatgtg cttctttgagaatgtatcgttcccggtgcagtcgttcgcctacgtgctatcc ccgaaaaaccatctcgtgccccttgacatcctggaggagaaagctgaacaacga cgctacgataacctatttagcggcagtggtgaagtgcaggacgtccggctg gaacaaacgcagcatcacgcagcatttccctgaccagccgctacgtggg cacttgctcgtattgcccatactgaacccgtcttcagcccaattaaagat cgagcaagtttgggacgagggcgacgagcatcaatacgaatacagacatc cgccagttcgggtacgacaaagtggggcagcagcgtcaacaagtacag gtacatgtcaattgaccaggaccatgcctgaaagaaggctctatggatca

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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Nucleic Acid Template, Sam002-full plasmids of self-amplifying mRNA encoding mouse wild type IL-18, Amp resistance) (SVP-305)- the GOI (mouse IL-18) is bolded. The GOI can be substituted with one or more different GOIs.	90	aaaaaaaaaacgctgataaacgcaggaagaacatgtgagcaaaaggccag caaaaggccaggaaccgtaaaaaggccgcttgcgtggcgcttttccatagg ctccgccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtgg cgaaaccgcagagactataaagataaccaggcgctttcccctggaagctcc ctcgtgcgctctcctgttcgacacccctgcgccttaccggatacctgtccgc tttctccttcgggaagcgtggcgctttctcatagctcacgctgtaggtat ctcagtcggtgtaggctcgttcgctccaagctgggctgtgtgcacgaaccc ccggttcagcccgacccgctgcgccttaccggtaactatcgtcttgagtc aaccggtaagacacgacttatcgccactggcagcagccactggtaacagg attagcagagcagaggtatgtaggcggtgctacagagttcttgaagtggtgg cctaactacggctacactagaagaacagtatattgggtatctgcgctctgctg aagccagttaccttcggaaaaagagttggtagctcttgatccggcaaaaca accaccgctggtagcggttttttggttgcaagcagcagatcacgcgcaga aaaaaggatctcaagaagatcctttgactcttttctacggggtctgacgct cagtggaacgaaaactcacgttaagggtatttgggtcatgagattatcaaaa aggatcttcacctagatccttttaattaaaaatgaagttttaaatcaatc taaaagtatatagtagtaaaacttggtctgacagttaccaatgcttaactcag gaggcacctatctcagcagatctgtctatttcgttcattccatagttgcctga ctccccgtcgttagataaactacgatacgggaggggttaccatctggcccc agtctgcgaatgataccgcgagacccacgctcaccggctccagatttatca gcaataaacagccagccggaaggcccgagcgcagaaagtggtcctgcaact ttatccgctccatccagctctatttattgttgccgggaagctagagtaagt agttcgccagttaatagtttgccgaacggttggtgccattgctacaggcatc gtggtgtcacgctcgtcgtttgggtatggcttcattcagctccgggttcccaa cgatcaaggcgagttacatgatcccccatgtgtgcaaaaaagcggttagc tccttcgggtcctccgatcgttgtcagaagtaagttggccgcagtggtatca ctcatggttatggcagcactgcataattctcttactgtcatgccatccgtga agatgcttttctgtgactggtgagtagtcaaccaagtcattctgagaatag tgtatgcggcgacaggttgctcttgccggcgctcaatacgggataaatacc gcgccacatagcagaactttaaaagtgcctcatcattggaaaacggttcttcg ggcgcaaaaactctcaaggatcttaccgctgtttgagatccagttcagatgtaa cccactcgtgcacccaactgatcttcagcatctttactttcaccagcgtt tctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaggggaataagg gcgacacggaatgttgaatactcatactcttctcttttcaatattattga agcatttatcagggttattgtctcatgagcggatacatatttgaatgtatt tagaaaaataaacaatagggttccgcgcacatttcccgaaaagtgcga cctgagctctaatacgaactcactataggataggcggcgcatgagagaagcc cagaccaattacctaaccacaaatggagaaagttcacgttgacatcgaggaa gacagcccatctcctcagagcttttcagcggagcttcccgagtttagagta gaagccaagcaggtcactgataatgaccatgctaatgccagagcgttttcg catctggcttcaaaactgatcgaaacggaggtggaccatccgacacgac cttgacattggaagtgcgccccccgcagaatgtattctaaagcacaagtat

TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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Description	SEQ ID NO.	Sequence
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Nucleic Acid Template- Sam002-full plasmids of self-amplifying mRNA encoding human wild type IL-18, kana resistance) (SVP-479) - the GOI (human IL-18) is bolded. The GOI can be substituted with one or more different GOIs.	92	aaaaaaaaaacgcggaagagccgcgtgataacgcaggaagaacatgtgag caaaaggccagcaaaaggccaggacccgtaaaaggccgcgttgctggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtgagaggtggcgaaacccgacaggactataagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgaccctgccgcttacggat acctgtccgcttctcctcctcggaagcgtggcgcttctcatagctcac gctgtaggatctcagttcgggtgtaggtcgttcgctccaagctgggctgtg tgcacgaacccccgttcagcccgaccgctgcgcttatccggtaactatc gtcttgagtcacacccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtctc tgaagtgggtggcctaactacggctacactagaagaacagtatttggtatct gcgctcgtcgaagccagttacctcggaagagagtggtagctcttgat ccggcaaacacaccccgctggtagcggtgggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctca cggggctcagcgtcagtggaacgaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaataaaaaatgaa gttttaaatcaatctaaagtatatatgagtaacctgaggctatggcagggc ctgcgcccccgacgttggtgcgagccctgggccttcacccgaacttgggg ggtggggtggggaaggaagaaacgcgggcgtattggccccaatggggctc tcggtggggtatcgacagagtgccagccctgggaccgaacccccgcgttat gaacaaacgacccacacccgtgcgttttatctgtcttttatgcgctca tagcgcggttccctccggtattgtctcctccggtgttccagttagcctcc ccctagggtgggcgaagaactccagcatgagatccccgcgtggaggatca tccagccggcgtcccggaacacgattccgaagcccaacctttcatagaagg cggcggtggaatcgaaatctcgtgatggcaggttggcgctcgttggtcgg tcatctcgaaccccgaggtcccgctcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgatccgtaaaagcagcaggaa cgggtcagcccatcgcccgcaagctctcagcaatatcacgggtagccaa cgctatgtcctgatagcgatccgccacacccagccggccacagtcgatgaa tccagaaaagcggccattttccacatgatattcggaagcagggcatcgcc atgggtcacgacgagatcctcgccgtcgggcatgctcgcttgagcctggc gaacagttcggtggcgagccctgatgctcttcgtccagatcatcctg atcgacaagaccggttccatccagtagtgcgtcgtcgtcgtatgcatggtt cgcttggtggtcgaaatgggcaggtagccggtcaagcgtatcgagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgccccatagcagccagtccttcccgc ttcagtgacaacgtcgagcacagctgcgcaaggaaacgcccgtcggtggccag ccacgatagccgcgtgcctcgtcttcaggttcatcagggcaccggacag gtcggtcttgacaaaaagaacgggcgcctcgtcgtgacagccggaaacac gcggcatcagagcagcgtatgtctgtgtgcccagtcagccgaatag cctctccacccaagcggccggagaaacctgcgtgcaatccatcttgttcaat catcgcaaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggccctcaaaaaagcctcctcactactctggaatagctcagagccgaggg cggcctcggcctctgcataaataaaaaaattagtcagccatggggcgag aatgggcggaactgggcggagttaggggcgggatgggcggagttagggcg ggactatgggtgctgactaatgagatgcatgctttgcatactctgcctg ctggggagcctggggactttccacacctgggtgctgactaattgagatgca tgctttgcatactctgcctgctggggagcctggggactttccacaccccta actgacacacattccacagctgggtctttccgctcaggactcttctcttt tcaatattattgaagcatttatcagggttatgtctcatgagcggatacat atttgaatgtatttagaaaaataaacaataagggggtccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgactcactataggataggcggcg catgagagaagccagaccaattacctacccaaatggagaaagtccagct tgacatcgaggaagacagccattcctcagagctttgcagcggagcttccc cgagtttagggtagaagccaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggaggtggaccc atccgacacgatccttgacattggaagtgcgcccgcgcgagaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtgcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaaggaaataac tgataaggaattggacaagaaaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagtcgtgtgc ctacgaaggcgaagtgcgtgttaccaggatgtatcgcggttgacggacc gacaagtctctatcaccaagccaataaggaggttagagtcgctactggat aggttttgacaccacccctttatgtttaagaacttggtggagcatatcc atcatactctaccaactgggcccagcgaacccgtgttaacggctcgtaacat aggcctatgcagctctgacgttatggagcggtcacgtagagggatgtccat tcttagaaagaagtatttgaaccatccaacaatgtctatctctgttgg ctcgaccatctcacacgagaagaggacttactgaggagctggcacctgcc

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

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Nucleic Acid Template, Sam002-full plasmids of self-amplifying mRNA encoding human wild mutant IL-18, kana resistance (SVP-408)- the GOI (human variant IL-18) is bolded. The GOI can be substituted with one or more different GOIs.	93	aaaaaaaaaacgcggaagagccgcgtgataaacgcaggaaagaacatgtgag caaaaggccagcaaaaggccagggaacccgtaaaaaggccgcgttgctggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtgagaggtggcgaaacccgacaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgaccctgccgcttacgggat acctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcac gctgtagggtatctcagttcgggtgtaggtcgttcgctccaagctgggctgtg tgcacgaacccccgttcagcccgaccgctgcgccttatccggtaactatc gtcttgagtcacacccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtctc tgaagtgggtggcctaactacggctacactagaagaacagtatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacaccccgctggtagcggtgggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctca cggggctctgacgctcagtggaacgaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaatgaa gttttaaatcaatctaaagtatatatgagtaacctgaggctatggcagggc ctgcgcgcccgacgttggtgcgagccctgggccttcacccgaacttgggg ggtggggtggggaaggaagaaacgcgggcgtattggccccaatggggctc tcggtggggtatcgacagagtgccagccctgggaccgaaccccgctttat gaacaaacgacccacacccgtgcgttttatctgtcttttatgtcgtca tagcgcggttccttcgggtattgtctccttcggtgtttcagttagcctcc ccctagggtggcggaagaactccagcatgagatccccgcgtcgaggatca tccagccggcgtcccggaacacgattccgaagcccaacctttcatagaagg cggtgggtggaatcgaaatctcgtgatggcaggttgggcgctcgttggtcgg tcatctcgaaacccagagtcgccgtcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgatccgtaaacgacaggaa cgggtcagcccatcgcccgcaagctcttcagcaatatcacgggtagccaa cgctatgtcctgatagcgatccgccacaccagccggccacagtcgatgaa tccagaaaagcggccattttcacccatgatattcggaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcatgctcgcttgagcctggc gaacagttcggctggcgagccctgatgctcttcgtccagatcatcctg atcgacaagaccggcttccatccgagtagctgctcgtcgatgcgatgttt cgcttggtggtcgaaatgggcaggtagccggatcaagcgtatgcagccgag cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgcccatagcagccagtccttcccgc ttcagtgacaacgtcgagcacagctgcgcaaggaaacgcccgtcggtggccag ccacgatagccgcgtgcctcgtcttgagttcattcagggcaccggacag gtcggtcttgacaaaaagaacgggcgccctcgctgacagccggaacac gcggcatcagagcagcgatgtgtgtgtgcccagtcatagcgaataag cctctccacccaagcggcgggagaaacctgcgtgcaatccatcttgttaaat catcgcaaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggctccaaaaagcctcctcactactcttggaatagctcagagggcgagg cggcctcggcctctgcataaataaaaaaattagtcagccatggggcgag aatgggcggaactgggaggagttagggcgggatgggaggagttagggcg ggactatgggtgctgactaattgagatgcatgctttgcatactctgcctg ctggggagcctgggactttccacacctggttgctgactaattgagatgca tgctttgcatactctgcctgctggggagcctgggactttccacaccccta actgacacacattccacagctgggtcttttcgctcaggactcttctcttt tcaatattattgaagcatttatcaggggtattgtctcatgagcggatacat atttgaatgtatttagaaaaataaacaaatagggggtccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgaactactataggataggcgagg catgagagaagccagaccaattacctacccaaatggagaaagtccagct tgacatcgaggaagacagccattcctcagagctttgcagcggagcttccc cgagttgaggtagaagccaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggagggtgaccc atccgacacgatccttgacattggaagtgcgcccgcgcgcagaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtgcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaagggaataac tgataagggaattggacaagaaaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagctcgtgtgc ctacgaaggcgaagtgcgtgtttaccaggatgtatcgcggttgacggacc gacaagctctctatcaccaagccaataagggagtagagtcgctactggat aggtttgacaccacccctttatgtttaagaacttggtggagcatatcc atcatactctaccaactgggcccagcaaacctgttaacggctcgtaacat

TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

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Nucleic Acid Template, sam002-mouse IL-12-IRES- mouse mutant IL18, kana resistance (SVP-525)- the GOI (mouse IL-12) is bolded; the GOI (mouse variant IL-18) is italicized and underlined; and the IRES is bolded and italicized. The GOI can be substituted with one or more different GOIs and the IRES is removed.	94	aaaaaaaaaacgcggaagagccgctgataacgcaggaaagacatgtgag caaaaggccagcaaaaggccaggacccgtaaaaaggccgcttgcctggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtacagagtgaggcgaacccgacaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgacccctgccgcttacggat acctgtccgcttctcctcctcggaagcgtggcgcttctcatagctcac gctgtaggatctcagttcggtgtaggctcgttcgctccaagctgggctgtg tgcacgaacccccgttcagcccgaccgctgcgccttatccggttaactatc gtcttgagtcacacccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttct tgaagtgggtggcctaactacggctacactagaagaacagtatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacaccccgctggtagcggtgggtttttgtttgcaagcagc agattacgcgcagaaaaaaaggatctcaagaagatcctttgatctttctca cgggggtctgacgctcagtggaacgaaaaactcacgttaagggattttgggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaatgaa gttttaaatcaatctaagtatatatgagttaacctgaggctatggcagggc ctgcgcccccgacgttggtgcgagccctgggcttcacccgaacttgggg gggtgggtggggaaaaaggaagaaacgcgggcgtattggccccatggggct tcggtgggtatcgacagagtgccagccctgggaccgaacccccgctttat gaacaaacgacccacacccgctgcttttatctgtctttttatgctgctca tagcgcggttcttccggtattgtctccttcggtgtttcagttagcctcc ccctagggtgggcgaagaactccagcatgagatccccgcgctggaggatca tccagccggcgtcccgaaaaacgattccgaagcccaacctttcatagaagg cggcggtggaatcgaaatctcgtgatggcaggttggcgctcgttggctcgg tcatctcgaaccccgaggtcccgctcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgatccgttaagcagcaggaa gcggtcagcccatcgccgcaagctctcagcaatatcacgggtagccaa cgctatgtcctgatagcgatccgccacaccagccggccacagtcgatgaa tccagaaaaagcgccattttccacatgatattcggaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcatgctcgcttgagcctggc gaacagttcggtggcgagccctgatgctcttcgtccagatcatcctg atcgacaagaccggcttccatccgagtaagtgctcgtcgtatgcatggtt cgcttggtggtcgaaatgggcaggtagccggtcaagcgtatgcagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgccccatagcagccagtccttccgc ttcagtgacaacgtcgagcacagctgcgcaaggaaaccccgctcggtggccg ccacgatagccgcgctgcctcgtcttcaggttcatcagggcacccggacag gtcggtcttgacaaaaagaacgggcgcctcgtcgtgacagccggaacac gcggcatcagagcagccgattgtctgtgtgcccagtcagccgaatag cctctccacccaagcgccgggagaaactcgtgcgtgcaatccatcttgtcaat catgcgaaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggctccaaaaaagcctcctcactacttctggaaatagctcagagccgagg cggcctcggcctctgcataaataaaaaaattagtcagccatggggcggaag aatgggcggaactgggcggagttaggggcgggatgggcggagttagggcg ggactatgggtgctgactaatgagatgcatgctttgcataactctgcctg ctggggagcctggggactttccacacctggttgcgtgactaatgagatgca tgctttgcataactctgcctgctggggagcctggggactttccacaccccta actgacacacattccacagctgggtctttccgctcaggactcttctttt tcaatattattgaagcatttatcaggggtattgtctcatgagcggatatac atttgatgtatttagaaaaataaacaataaggggttccgcgcacatttcc ccgaaaagtgccacctgagctctaatacagctcactataggataggcgggc catgagagaagccagaccaattacctaaccacaaatggagaaagttcacgt tgacatcgaggaagacagccattcctcagagctttgcagcggagcttccc cgagttgaggtagaagccaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggaggtggaccc atccgacacgatccttgacattggaagtgcgcccgcggcagaaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtgcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaaggaaataac tgataaggaattggacaagaaaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtcctccacgacgacgagtcgtgtcg ctacgaagggaagtgcgtgtttaccaggatgtatccgcggttgacggacc

TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Nucleic Acid Template, sam002-mouse IL-12-IRES-mouse mutant IL18, kana resistance (SVP-526) - the GOI (mouse IL-12) is bolded ; the GOI (mouse IL-18) is <i>italicized</i> and underlined; and the IRES is bolded and <i>italicized</i> . The GOI can be substituted with one or more different GOIs and the IRES is removed.	95	aaaaaaaaaacgcggaagagccgcgtgataacgcaggaaagaacatgtgag caaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtcagaggtggcgaaacccgacaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgacccctgccgcttacggat acctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcac gctgtaggtatctcagttcgggtgtaggtcgcttccgctccaagctgggctgtg tgcacgaacccccggttcagcccgacgcgtgcgccttatccggtaaatctc gtcttgagtcacaaccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggatttagcagagcgaggtatgtaggcggtgctacagagtctt tgaagtggtggcctaactacggctacactagaagaacagtaatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacacccgctggtagcgggtggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctta cggggtctgacgctcagtggaacgaaaaactcacgttaagggtatttggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaaatgaa gttttaaatcaatctaagatataatgagtaacctgaggctatggcagggc ctgcgcgcccgacgttggtgcgagccctgggccttcacccgaacttgggg ggtggggtggggaaggaagaaacgcgggcgtattggcccaatggggctc tcggtggggtatcgacagagtgcacgcccctgggacgcaaccccgctttat gaacaaacgaccccaacacccgtgcgttttattctgtcttttattgcccgtca tagcgcgggttccttcgggtattgtctccttcgctgtttcagttagcctcc ccctagggtgggcgaagaactccagcatgagatccccgcgctggaggatca tccagccggcgtcccgaacacgattccgaagcccaactttcatagaagg cggcgggtggaatcgaaatctcgtgatggcagggttgggcgtcgcttggtcgg tcatttcgaacccagagtcgccgtcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcgcgataccgtaaaagcacgaggaa gcggtcagcccatcgcgcgccaagctcttcagcaatatcaggggtagccaa cgctatgtcctgatagcgatccgcacacccagccggccacagtcgatgaa tccagaaaaagcggccttttccaccatgatattcggcaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcgatgctcgcttgagcctggc gaacagttcggctggcgagcgcctgatgctcttcgctccagatcatcctg atcgacaagaccggcttcacatccgagtaagtgctcgctcgatcgatggtt cgcttggtggtcgaaatgggaggttagccgatcaagcgtatgcagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgcccggcacttcgcccataagcagccagtccttcccgc ttcagtgacaacgtcagacacagctgcgcaaggaaacccgctcgctggccag ccacgatagccgcgctgcctcgtcttcgagttcatcaggggcaccggacag gtcggtcttgacaaaaagaacggggcgccctgcgctgacagccggaacac ggccgcatcagagcagccgattgtctgttggtgccagtcatagcgcaatag cctctccacccaagcggccggagaacctcggtgcaatccatcttgttcaat catgcgaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggccctcaaaaaagcctcctcactacttctggaatagctcagaggccgagg cggcctcggcctctgcataaataaaaaaattagtcagccatggggcgagg aatgggcggaactgggcggaggttagggcgggatgggcggagttagggcg ggactatggttgctgactaattgagatgcatgctttgcatacttctgcctg ctggggagcctggggactttccacacctggttgctgactaattgagatgca tgctttgcatacttctgcctgctggggagcctggggactttccacacctca actgacacacattccacagctggttctttccgctcaggactcttctttt tcaatattattgaagcatttatcagggttatgtctcatgagcggatatac attggaatgtatttagaaaaataaacaataagggttccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgaactcactataggataggcggcg catgagagaagccagaccaattacctaaccacaaatggagaaagtccagct tgacatcgagggaagacagccattcctcagagctttgcagcggagctccc gcagtttgaggtagaagcgaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggaggtggaccc atccgacacgatccttgacattggaagtgcgcccgcccgagaaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaaggaaataac tgataaggaaatggacaagaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagctcgtgtcg ctacgaagggcaagtgcgtgtttaccaggatgtatcgcgggttgacggacc gacaagtctctatcaccaagccaaataaggaggttagagtcgctactggat aggctttgacaccaccccttttatgtttaagaacttggtggagcatatcc atcatactctaccaactgggcccagcaaacccgtgttaacggctcgtaacat aggcctatgcagctctgacgttatggagcgggtcacgtagagggtatgctcat tcttagaaagaagtatttgaacacatccaacaatgttctatctctgttg ctcgaccatctaccacgagaagagggaacttactgaggagctggcactgccc gtcgtatctcacttactggtggaagcaaaatacacatgtcgggtgtgagac tatagttagttgcgacgggtacgtcgtaaagaatagctatcagtcagg

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Nucleic Acid Template, sam002-human IL-12-IRES-human wild type IL-18, kana resistance (SVP-640)-the GOI (human IL-12) is bolded ; the GOI (human IL-18) is italicized and underlined ; and the IRES is bolded and italicized . The GOI can be substituted with one or more different GOIs and the IRES is removed.	96	aaaaaaaaaacgcggaagagccgcgtgataacgcaggaaagaacatgtgag caaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgcggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtcagaggtggcgaaacccgacaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttcgcgacctgccgcttacggat acctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcac gctgtaggtatctcagttcgggtgtaggtcgcttccgctccaagctgggctgtg tgcacgaacccccggttcagcccgacgcgtgcgccttatccggtaaatctc gtcttgagtcacaaccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtctt tgaagtggtggcctaactacggctacactagaagaacagtaatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacacccgctggtagcgggtggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctta cggggtctgacgctcagtggaacgaaaaactcacgttaagggttttggtca tgagattatcaaaaaggatcttcacctagatccttttaaatataaaatgaa gttttaaatcaatctaagatataatgagtaacctgaggctatggcagggc ctgcgcccccgacgttggtgcgagccctgggccttcaccgaaactgggg ggtggggtggggaaggaagaaacgcgggcgtatggccccaatggggctc tcggtggggtatcgacagagtgcacgccccgggacgcaaccccgctttat gaacaaacgaccccaacacccgtgcgttttatctcgtctttttattgcccgtca tagcgcgggttccttcgggtatgtctccttcggtgttcagttagcctcc ccctagggtgggcgaagaactccagcatgagatccccgcgctggaggatca tccagccggcgtcccggaaaacgattccgaagcccaaccttcatagaagg cggcgggtggaatcgaaatctcgtgatggcagggtgggcgctcgcttggtcgg tcatttcgaacccagagtcgccgtcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgataccgtaaaagcacaggaa gcggtcagcccatcgcgcgcaagctcttcagcaatatcaggggtagccaa cgctatgtcctgatagcgatccgcacacccagccggccacagtcgatgaa tccagaaaaagcggccattttccaccatgatattcggcaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcgatgctcgcttgagcctggc gaacagttcgggtggcgagcgcctgatgctcttcgctccagatcatcctg atcgacaagaccggcttccatccgagtaagtgctcgctcgatcgatggtt cgcttggtggtcgaaatgggaggttagccggtatcaagcgtatgcagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgcccataagcagccagtccttccgcg ttcagtgacaacgtcagacacagctgcgcaaggaaacccgctcgctggccag ccacgatagccgcgctgcctcgtcttcgagttcatcaggggcaccggacag gtcggtcttgacaaaaagaacggggcgccccctgcgctgacagccggaacac ggccgcatcagagcagccgattgtctgttggtgccagtcatagcgcaatag cctctccacccaagcggccggagaacctgcgtgcaatccatcttgttcaat catgcgaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggccctcaaaaaagcctcctcactacttctggaatagctcagagggcagag cggcctcggcctctgcataaataaaaaaattagtcagccatggggcgagag aatgggcggaactgggcggaggttagggcgggatgggcggagttagggcg ggactatggttgctgactaattgagatgcatgctttgcatacttctgcctg ctggggagcctggggactttccacacctggttgctgactaattgagatgca tgctttgcatacttctgcctgctggggagcctggggactttccacacctca actgacacacattccacagctggttctttccgctcaggactcttctcttt tcaatattattgaagcatttatcagggttatgtctcatgagcggatatacat attggaatgtatttagaaaaataaacaataagggttccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgaactcactataggataggcggcg catgagagaagccagaccaattacctacccaaaatggagaaagtccagct tgacatcgagggaagacagccattcctcagagctttgcagcggagcttccc gcagtttgaggtagaagcgaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggaggtggaccc atccgacacgatccttgacattggaagtgcgcccgcccgagaaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaaggaaataac tgataaggaaatggacaagaaatgaaggagctggccgcccgtcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagctcgtgtcg ctacgaagggaagtgcgtgtttaccaggatgtatcgcgggttgacggacc gacaagtctctatcaccaagccaaataaggaggttagagtcgctactggat aggccttgacaccaccccttttatgtttaagaacttggtggagcatatcc atcatactctaccaactgggcccagcaaacccgtgttaacggctcgtaacat aggcctatgcagctctgacgttatggagcgggtcacgtagagggtatgctcat tcttagaaagaagtatttgaaccatccaacaatgttctatctctgttggtg ctcgaccatctaccacgagaagagggaacttactgaggagctggcactgccc gtcgtatctcacttactggtggaagcaaaatacacatgtcgggtgtgagac tatagttagttgcgacgggtacgtcgtaaagaatagctatcagtcagg

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Nucleic Acid Template, sam002-human IL-12-IRES- human mutant IL-18, kana resistance (SVP-641)- the GOI (human IL-12) is bolded; the GOI (human variant IL-18) is italicized and underlined; the IRES is bolded and italicized. The GOI can be substituted with one or more different GOIs and the IRES is removed.	97	aaaaaaaaaacgcggaagagccgcgtgataacgcaggaaagaacatgtgag caaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgt ttttccataggctccgccccctgacgagcatcacaaaaatcgacgctcaa gtcagaggtggcgaaacccgacaggactataaagataccaggcggttcccc ctggaagctccctcgtgcgctctcctgttccgacccctgccgcttacggat acctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcac gctgtaggtatctcagtcgggtgtaggtcgcttccgctccaagctgggctgtg tgcacgaacccccggttcagcccgacgcgtgcgccttatccggtaaatctc gtcttgagtcacaaccggtaagacacgacttatcgccactggcagcagcca ctggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtctt tgaagtggtggcctaactacggctacactagaagaacagtaatttggtatct gcgctctgctgaagccagttaccttcggaaaaagagtggtagctcttgat ccggcaaacacacccgctggtagcgggtggttttttggttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatcttttcta cggggtctgacgctcagtggaacgaaaaactcacgttaagggtatttggtca tgagattatcaaaaaggattctcacctagatccttttaaatataaaatgaa gttttaaatcaatctaagatataatgagtaacctgaggctatggcagggc ctgcgcgcccgacgttggtgctgcgagccctgggccttcacccgaacttggg ggtgggtggggaaggaagaaacgcgggcgtattggccccaatggggtc tcggtggggtatcgacagagtgcacgcccctgggacgcaaccccgctttat gaacaaacgacccaacacccgtgcgttttattctgtcttttattgcccgtca tagcgcgggttccttcgggtattgtctccttcgctgtttcagttagcctcc ccctagggtggcggaagaactccagcatgagatccccgcgctggaggatca tccagccggcgtcccggaaaacgattccgaagcccaactttcatagaagg cggcgggtggaatcgaaatctcgtgatggcagggtgggcgctcgcttggtcgg tcatttcgaacccagagtcgccgtcagaagaactcgtcaagaaggcgata gaaggcgatgcgctgcgaatcgggagcggcgataccgtaaaagcacgaggaa gcggtcagcccatcgcgcgccaagctcttcagcaatatcacgggtagccaa cgctatgtcctgatagcgatccgccacacccagccggccacagtcgatgaa tccagaaaaagcggccattttccaccatgatattcggcaagcaggcatcgcc atgggtcacgacgagatcctcgccgtcgggcgatgctcgcttgagcctggc gaacagttcggctggcgcgagccctgatgctcttcgctccagatcatcctg atcgacaagaccggcttccatccgagtaagtgctcgctcgatcgatggtt cgcttggtggtcgaaatgggaggttagccggtatcaagcgtatgcagccgccc cattgcatcagccatgatggatactttctcggcaggagcaaggtgagatga caggagatcctgccccggcacttcgcccataagcagccagtccttcccgcc ttcagtgacaacgtcagacacagctgcgcaaggaaacgcccgtcgtggccag ccacgatagccgcgctgcctcgtcttcgagttcattcaggggcaccggacag gtcggtcttgacaaaaagaacggggcgccctgcgctgacagccggaacac ggccggtatcagagcagccgattgtctgtgtgcccagtcatagcgcaatag cctctccaccaagcggccggagaacctcgtgcaatccatcttgttcaat catgcgaacgatcctcatcctgtctcttgatcgatctttgcaaaagccta ggccctcaaaaaagcctcctcactacttctggaatagctcagagggcagag cggcctcggcctctgcataaataaaaaaattagtcagccatggggcggag aatgggcggaactgggcggaggttagggcgggatgggcggagttaggggag ggactatgggtgctgactaattgagatgcatgctttgcatacttctgcctg ctggggagcctggggactttccacacctggtgctgactaattgagatgca tgctttgcatacttctgcctgctggggagcctggggactttccacacctca actgacacacattccacagctggttctttccgctcaggactcttctcttt tcaatattattgaagcatttatcagggttatgtctcctatgagcggatatac attgaaatgatttagaaaaataaacaaatagggggtccgcgcacatttcc ccgaaaagtgccacctgagctctaatacgaactcactataggataggcggcg catgagagaagccagaccaattacctaaccacaaatggagaaaagttcacgt tgacatcgagggaagacagccattcctcagagctttgcagcggagcttccc gcagtttgaggtagaagccaagcaggtcactgataatgaccatgctaatgc cagagcgttttcgcatctggcttcaaaactgatcgaaacggaggtggaccc atccgacacgatccttgacattggaagtgcgcccgcccgagaaatgtattc taagcacaagtatcattgtatctgtccgatgagatgtcggaagatccgga cagattgtataagtatgcaactaagctgaagaaaaactgtaaggaaataac tgataaggaaatggacaagaaatgaaggagctggccgcgctcatgagcga ccctgacctggaaactgagactatgtgcctccacgacgacgagtcgtgtcg ctacgaagggaagtcgctgtttaccaggatgtatcgcgggttgacggacc gacaagtctctatcaccaagccaaataaggaggttagagtcgctactggat aggctttgacaccaccccttttatgtttaagaacttggctggagcatatcc atcatactctaccaactgggcccagcaaacccgtgttaacggctcgtaacat aggcctatgcagctctgacgttatggagcgggtcacgtagagggtatgctcat tcttagaaaagaagtatttgaaccatccaacaatgttctatctctgttg ctcgaccatctaccacgagaagagggaacttactgaggagctggcactgccc gtcgtatctcacttactgtggcaagcaaaattacacatgtcggtgtgagac tatagttagttgcgacgggtacgtcgtaaaagaatagctatcagtcagg

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
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Nucleic Acid Template, sam002-Luciferase-Amp resistance (SVP-062)-the GOI (Luciferase, SEQ ID NO: 25) is bolded. The GOI can be substituted with one or more different GOIs.	98	aaaaaaaaaacgcgtgataacgcaggaaagacatgtgagcaaaaggccag caaaaggccaggaaccgtaaaaaggccgcgttgctggcgcttttccatagg ctccgccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtgg cgaaaccgcagagactataaagataccaggcgctttcccctggaagctcc ctcgtgcgtctcctgttccgacctgcgccttaccggataacctgtccgcc tttctccttcgggaagcgtggcgctttctcatagctcacgctgtaggtat ctcagttcggtgtaggctcgttcgctccaagctgggctgtgtgcacgaaccc cccgttcagcccgaccgctgcgccttaccggtaactatcgtcttgagtc aaccggtaagacacgacttatcgccactggcagcagccactggtaacagg attagcagagcaggtatgtaggcgggtgctacagagttcttgaagtgggtg cctaactacggctacactagaagaacagatattgggtatctgcgctctgctg aagccagttaccttcggaaaaagagttggtagctcttgatccggcaaa accaccgctggtagcggttttttgtttgcaagcagcagatcacgcgcaga aaaaaaggatctcaagaagatcctttgactctttctacggggctgcagct cagtggaacgaaaaactcacgttaagggtatttggtcatgagattataaaa aggatcttcacctagatccttttaaatataaaatgaagttttaaatcaatc taaagtatatatgagtaaaacttggtctgacagttaccaatgcttaactcag gaggcacctatctcagcagatctgtctatttcgttcacatagttgctga ctccccgtcgtgtagataaactacgatacgggaggggttaccatctggcccc agtgtgcaatgataccgcgagacccacgctcaccggctccagatttatca gcaataaaccagccagccgaaggccgagcgcagaagtggtcctgcaact ttatccgctccatccagctctattaatgttgccgggaagctagagtaagt agttcgccagttaatagtttgcgcaacgttggtgccattgctacaggcatc gtggtgtcacgctcgtcgtttggtatggctcattcagctccgggtcccaa

TABLE 16-continued

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TABLE 16-continued

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TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		catactgaagaacggcccaaggtacaggtgatccaggctgccgatccgcta gcaacagcgtatctgtgcggaatccaccgagagctggtaggagattaaat gcggtcctgcttccgaacattcatacactgtttgatattgctggctgaagac tttagcgtattatagccgagcacttccagcctggggattgtgttctggaa actgacatcgcgctcggttgataaaagtgaggacgacgccatggctctgacc gcgttaatgatcttggaagacttaggtgtggacgcagagctgttgacgctg attgaggcggttctcgcgcaaatctcatcaatacatttgccactaaaact aaatttaaattcggagccatgatgaaatctggaatgttctcacactgttt gtgaacacagtcattaacattgtaatcgcaagcagagtgttgagagaacgg ctaaccggatcccatgtgcagcattcattggagatgacaatatcgtgaaa ggagtcaaatcggacaaattaatggcagacaggtgcgccacctgggtgaat atggaagtcaagattatagatgctgtggtggcgagaaaagcgcttatttc tgtggaggggttattttgtgtgactccgtgaccggcacagcgtgccgtgtg gcagacccctaaaaaggctgttaagcttggcaaacctctggcagcagac gatgaacatgatgatgacaggagaaggcattgcatgaagagccaacacgc tggaaaccgagtggttattcttcagagctgtgcaaggcagtagaatcaagg tatgaaaccgtaggaacttccatcatagttatggccatgactactctagct agcagtggttaaatcattcagctacctgagaggggccccataactctctac ggctaacctgaatggactacgacatagtctagtcgcccaagttcgaggcg cgctctagagccaccatggaggacgctaaaaatatcaagaagggtcccg tcctttctatccactcgaggacggtaccgctggtagcaactgcacaaggc tatgaaacgggtatgctcttgtgcccaggcaccatagctttcccgacgccc catagaggtggacattacttacgcagagtaactttgagatgtctgtccggct ggcagaggcaatgaaaagatatggcctcaacacaaaccataggatagtggt ttgcagcgagaattctcttcagttttttatgcccgtcctcggggctctttt catcgcgctggctgtcgccccgcgaacgacatatcaacgagcagagagct gctgaacagcatgggaatcagtcagcctacagtggttttgtatcaagaa gggcttgcaaaagatccttaacgtacagaaaaagctccctattattcagaa gatcattataatggattcaaaagaccgactatcagggattccagtcfaatgta tactttcgtcacttcccacttctctcgggttaatgaatacgaacttcgt tccgaaagtttcgatcgggacaaaactattgcactcatcatgaactctct tgggtccactgggtgcctaaagggtgtgcacttcccacaggactgcacg tgtaaggttctctcagcagcagagatcccatattcggttaaccaaactatcc tgataccgctatctctctgtagtgcctttccaccatggattcggcatgtt caccactctgggttacttgatatgtgggttcgcgctggctccttatgtacag atttgaaaggagagctcttttgaggagcttgaggactacaaaatccaaag tgctctctcgtccccacattgttctcattcttcgccaaatctactctcat cgacaagtacgatttgagcaacttgcaagaaattgctctcgggggggctcc actctccaaagaagtgcggcgaagcgttgctaaaagggtccacttgccagg cattcgccagggtatggccttactgagacaacctcagctattctcattac ccctgaggggtgacgataagcctgggtgcgctcggttaaggtagtaaccatttt cgaggcaaaagggtggtcgacttgataccggaaaaaacctcggtgtgaatca gagaggcgaattgtgtgtcaggggtccaatgatcatgagcgggtacgtcaa taatccgaggccactaacgcttgattgataaagatgggtggcttcattc aggtgatatagcatactgggatgaggatgaacattttctcattgtcgatag gctcaaaagcctgatcaagtataagggttaocaaagtggccctgcggaact cgaatctatactgctgcaacatccaaacatttttgacgctggcgtggccgg actgccgatgatgatgctgggtgaactccctgcgcgcttgctgctcctgga acatggtaagaccatgacagaaaaggagattgtagattacgtgggttcaca agtgactacagctaaagaaattgcgcgggggtgtcggtttcgtggacgagg acccaagggcctcactgggaagctcgacgcccagaaaaatccgagaaatcct gattaaagcaaaaggcggaagattgcagtgtagcatcgatgatcgcg cgggcgcatacagcagcaattggcaagctgcttacatagaactcgcgcgga ttggcatgcgccttaaaatttttattttttttttttttttttttttccgaat cggattttgtttttaattttcaaaaaaaaaaaaaaaaaa

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Nucleic Acid Template, modified mRNA-eGFP-Amp resistance (SVP-355) - the GOI (Luciferase, SEQ ID NO: 25) is bolded . The GOI can be substituted with one or more different GOIs.	99	gctcgctttcttctgctgtccaatttctattaaaggttcctttgttcctaag tccaactactaaactgggggatattatgaaggcccttgagcatctggattc tgctaataaaaaacatttattttcattgcaaagccacaccctggagctag caaaaaaaaaaaaaaaaaaaaaaaaaaaaaagcatatgactaaaaaaaaa aa aaaaaaaaaacgcgtatcgatgatatcgcgccgcatacagcagcaattggc aagctgcttacatagaactcgcgccgattggcatgccgcctaaaaattttt atttatttttcttttcttttccgaatcggattttgttttaattttcaa aaaaaaaaaaaaaaaaaaaaaaaaaacgcgtgataaacgcaggaaagacatgt gagcaaaagccagcaaaagccaggaaccgtaaaaagccgcgttgctgg cgttttccataggctccgccccctgacgagcatcacaaaaatcgacgct caagtcagaggtggcgaaccgccagaggactataaagataccaggcgcttc cccttggaagctccctcgctgcgctctcctgttccgaccctgccgcttaccg gatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagct cagctgtagggtatctcagttcgggtgtaggtcgttcgctccaagctgggct gtgtgcagcaacccccggttcagcccgaccgctgcgccttatccggtaact atcgctcttgagtcacacccggttaagacacgacttatcgccactggcagcag ccactggtaacaggatttagcagagcgaggtatgtaggcggtgctacagagt tcttgaaagtggtggcctaactacggctacactagaagaacagatttggtat tctcgctctgctgaagccagttaccttcggaaaaagagttggtagctctt gatccggcaaacacacccgctggtagcggttttttggttgcaagcagc agattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttcta cggggtctgacgctcagtggaacgaaaaactcaggttaagggtatttggtca tgagattatcaaaaaggatcttcacctaagatccttttaataaaaaatgaa gttttaatacaatcaaatgataatagagtaaaacttggtctgacagttacc aatgcttaatacagtgaggcacctatctcagcgatctgtctatttcgttcat ccatagttgctgactccccgctcggtgtagataactacgatacgggagggct taccatctggcccgagtgctgcaatgatccgcgagacccacgctcaccgg ctccagatttatcagcaataaacccagccagccggaagggccgagcgcagaa gtggtcctgcaactttatccgcctccatccagctctaatatggtgcccggg aagctagagtaagtagttcgccagtttaatagtttgcccaacggttggtgca ttgtacagggcatcggtgtcagcgtcgtcggttggtatggcttcattca gctccggttcccaacgatcaaggcgagttacatgatccccatggtgtgca aaaaagcgggttagctccttcgggtcctccgatcggtgtcagaagtaagtgg ccgagtggtatcactcatggttatggcagcactgcataatctcttactg tcatgccatccgtaagatgcttttctgtgactgggtgagtaactcaaccaagt cattctgagaatagtgatgcggcgacagagttgctcttgcccggcgctcaa tacgggataataccgcgccacatagcagaactttaaaagtgtcatcattg gaaaacggttcttcggggcgaaaaactctcaaggatcttaccgctgttgagat ccagttcgatgtaacccactcgtgcacccaactgatcttcagcatctttta ctttaccagcgtttctgggtgagcaaaaacaggaaggcaaatgccgcaa aaaaggggaataagggcgacacggaatggtgaatactcataactctccttt ttcaatattatgaagcatttatcagggttattgtctcatgagcggataca tatttgaaatgtatttagaaaaataaacaataagggttccgcgcacatttc cccgaaggtgccacctgagctctaatacgaactcactataggacatttgct tctgacacaactgtgttcactagcaacctcaaacagacacc atggtagca agggcgaggagctgttcacgggggtgggtgccatcctggctgagctggacg gcgacgtaaaacggccacaagttcagcgtgtccggcgaggcgaggcgatg ccactacggcaagctgacctgaagttcatctgcacacccggcaagctgc ccgtgccctggccacccctcgtgaccacctgacctacggcgtgagtgct tcagccgctaccccgaccacatgaagcagcagactttctcaagtcgcgca tgccgaaggctacgtccaggagcgcaccatcttctcaaggacgacggca actacaagaccgcgcgaggtgaagttcgaggcgacacccctggtgaacc gcacgagctgaaggccatcgacttcaaggaggacggcaacatcctggggc acaagctggagtacaactacaacagccacaacgtctatatcatggccgaca agcagaagaacggcatcaagggtgaacttcaagatccgccacaacatcgagg acggcagcgtgcagctcgcgcgaccactaccagcagaaccccccatcgggc acggccccgtgctgctgcgcgacaaccactacctgagcaccacgtcgcgc tgagcaaaagaccacaacgagaagcgcgatacatgggtcctgctggagtg tgacgcgcgcgggatacctctcgcatggacgagctgtacaagtag

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
Nucleic Acid Template, modified mRNA-Luciferase-(SVP-062)-the GOI (Luciferase, SEQ ID NO: 25) is bolded. The GOI can be substituted with one or more different GOIs.	100	gctcgctttcttctgtgtccaatttctattaaaggttccttgttccctaag tccaactactaaactgggggatattatgaagggccttgagcatctggattc tgccataataaaaaacatttattttcattgcacacccctggagctag caaaaaaaaaaaaaaaaaaaaaaaaaaagcatatgactaaaaaaaaa aaaaaaaaaacgcgtatcgtatgatcgcggcgcatcacgagcaattggc aagctgcttacatagaactcgcggcgattggcatgccgccttaaaatttt atttatttttcttttctttccgaatcggattttgtttttaatatttcaa aaaaaaaaaaaaaaaaaaaaaaaaaacgcgtataacgcaggaaagacatgt gagcaaaaggccagcaaaaggccaggaaccgtaaaaggccgcgttgctgg cgtttttccataggtccgccccctgacgagcatcacaaaaatcgacgct caagtgcagggtggcgaacccgacaggaactataagataccaggcggttc ccccggaagctcctcgtgcgctctcctgttccgaccctgcgccttaccg gatacctgtcgcctttctcccttcgggaagcgtggcgctttctcatagct cacgctgtaggatctcagttcggtgtaggctcgttcgctccaagctgggt gtgtgcagcaacccccgttcagcccgaccgctgcgccttatccgtaact atcgcttgagtcacacccggtgaagacaggaacttatcgccactggcagcag ccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagt tcttgaagtgggtgacctaacgagctacactagaagaacagttttggtta tctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctctt gatccggcaaaccaaccacgcgtggtagcggttttttgtttgcaagcagc agattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctca cggggtctgacgctcagtggaacgaaaaactcacgttaagggattttggtca tgagattatcaaaaaggatcttcacctagatccttttaataaaaaatgaa gttttaatacaatctaagtatatatgagttaacttggtctgacagttacc aatgcttaatacagtgaggcacctatctcagcgatctgtctatttcgttcac ccatagttgctgactcccgctcgtgtagataactacgatacgggaggggt taccatctggccccagtgctgcaatgatccgcgagaccacgctcaccgg ctccagatttatcagcaataaacccagccagccggaagggccgagcgcagaa gtgctcctgcaactttatccgcctccatccagctctataattgttgccggg aagctagagtaagtagttcgccagttaatagtttgcgcaacggttgttgcca ttgtacagggcatcgtggtgtcacgctcgtcgtttggtatgggttcattca gctcgggttcccaacgatcaaggcgagttacatgatccccatgttgtgca aaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttgg ccgcagtggtatcactcatggttatggcagcactgcataattctcttactg tcatgccatccgttaagatgctttctgtgactggtgagtaactcaaccaagt cattctgagaatagtgatgcggcgacccaggttgctccttgcccgcgctcaa tacgggataataccgcgccacatagcagaactttaaaagtgtcatcattg gaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagat ccagttcgatgtaacccactcgtgcacccaactgatcttcagcatctttta ctttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaa aaaagggaataaggggcgacacggaatgtgaataactcatactcttcttt ttcaatattatgaagcatttatcagggttattgtctcatgagcggataca tatttgaatgtatttagaaaaataacaaataggggttcgcgcacatttc cccgaagtgccacctgagctctaatcagactcactataggacatttgct tctgacacaactgtgttcactagcaacctcaaacagacaccatggaggacg ctaaaaatacaagaagggtcccgctcctttctatccactcgggaaggta ccgctggtgagcaactgcacaaggctatgaacgggtatgctcttggtccag gcaccatagctttcaccgacgcccacatagagggtggacattacttaacgag agtaactttgagatgtctgtccggtggcagaggcaatgaaaagataaggcc tcaacacaaaccataggatagtggtttgcagcgagaattctctcagtttt tatgccgctcctcggggtctctttcatcggcggtgctgtcgcccccgcca acgacatatcaacgagcgagagctgtgtaacagcatgggaatcagtcagc ctacagtggtttttgtatcaagaagggttgcaaaagatccttaacgtac agaaaaagctccctattatcagaagatcattataatggattcaaaagaccg actatcagggttccagtcactgtatctttogtacttcccacctctctc ctgggtttaatgaatacgaacttcgttcccgaaagtctcgatcgggacaaaa ctattgcactcatcatgaactctctgggtccactgggtgacctaaagggtg tgcgacttccccacaggaactgcattgttaagggtctctcgcgcacagatc ccatattcggtaacccaaatcattcctgataccgctatcctctctgtagtc ctttccaccatggattcggcatgttcaccactctgggttacttgatattgtg ggttcgcggtggtccttatgtacagatttgaagaggagctctttttgagga gcttgcaggactacaaaatccaaagtgtctctcgtctccacacattgttct cattcttcgcaaatctactctcatcgacaagtaacgatttgagcaacttgc acgaaattgtcttcgggggggtccactctccaaagaagtcggcgaaagcgg ttgctaaaagggtccacttgcaggcatcgcgaggggtatggccttactg agacaacctcagctattctcattacccctgagggtgacgataagcctgggtg ccgtcggtaaggtagtaccatttttcgaggcaaaagggtggtcagcttgata ccggaacaaacctcggtgtgaatcagagaggcgaaattgtgtgtcagggttc caatgatcatgagcggttacgtcaataatcccgaggccactaacgccttga

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		ttgataaagatgggttggttcattcaggtgatatagcatactgggatgagg atgaacattttcttcatgtcgataggctcaaaagcctgatacagataaagg gttaccagtggtcccttgcgaactcgaatctatactgctgcaacatccaa acatttttgacgctggcgtggcggactgccgatgatgatgctgggtgaac tccctgcgcgcttgtcgtcctggaaacatggtaagaccatgacagaaaagg agattgtagattacgtggcttcacaagtgaactacagctaagaaattgcgcg gggtgtcgttttctgtggacgaggtacccaagggcctcactgggaagctcg acgccagaaaaatccgagaaatcctgatataagcaagaagggcggaaaga ttgcagtgtag
GOI, human IL18-mutant- the leading peptide is bolded.	101	atggacatgagagtcaccagctcaactgcttgggtcttcttctgctgtggttg cgcggtgcccagatgtc acttcgggaagtggagtcacagttgagtggtatt cgcaacctcaatgatcaggtgtgttcatagaccagggtaacccgccactt tttgaagatatgaccgatagcgattgcgcgcgacaacgccccaggaccatc ttcataataaagtgcgtacggagactcaccggcccgggggaaggctgttaca atccgtaaaagtgcgagaagatctccacgcttcttgcgaaaaataagatt atcagttttaaagaaatgaaccctcctgacaacataaaagatacgaatca gacatcatcttcttccagcgctcagttcctggacatgataataagatgcag ttcgagtcacttagctatgaagggtattttctggcttgtgagaaagaacgg gatttgtttaaaacttatactgaagaaggaggatgaacttgggtgatcggctc attatgtttaccgtacaaaatgaggac
GOI, human IL18-the leading peptide is bolded.	102	atggacatgagagtcaccagctcaactgcttgggtcttcttctgctgtggttg cgcggtgcccagatgtc acttcgggaagtggagtcacagttgagtggtatt cgcaacctcaatgatcaggtgtgttcatagaccagggtaacccgccactt tttgaagatatgaccgatagcgattgcgcgcgacaacgccccaggaccatc ttcataataaagtgcgtacggagactcaccggcccgggggaaggctgttaca atccgtaaaagtgcgagaagatctccacgcttcttgcgaaaaataagatt atcagttttaaagaaatgaaccctcctgacaacataaaagatacgaatca gacatcatcttcttccagcgctcagttcctggacatgataataagatgcag ttcgagtcacttagctatgaagggtattttctggcttgtgagaaagaacgg gatttgtttaaaacttatactgaagaaggaggatgaacttgggtgatcggctc attatgtttaccgtacaaaatgaggac
IRES	103	acgttactggccgaagcgcgttggaataaggccggtgtgctgttgtctata tgttattttccaccatattgccgtcttttggcaatgtgagggcccggaac ctggccctgtcttcttgacgagcattcctaggggtcttccctctcgcca aaggaaatgcaaggtctgttgatgtcgtgaaggagcagttcctctggaag cttcttgaagacaacaacgtctgtagcgaccttgcaggcgagcggaaacc ccccacctggcgacaggtgcctctgcggccaaaagccacgtgataaagata cacctgcaaaagggcgacaaacccagtgccacgttgtgagttggatagttg tggaaaagagtc aaatggctctcctcaagcgtattcaacaaggggtgaaagg atgccagaaaggtaccccatgtatgggatctgatctggggcctcggtgca catgctttacatgtgttttagtcgaggttaaaaaaacgtctaggccccccga accacggggacgtgggttttcttggaaaaacacgataata
Upper linker of IRES	104	gcgatcgcgctagcaacgggtttccctctagcgggatcaattccgcccccc cccta
Down linker of IRES	105	acgcgttgccacc
mouse mutant IL-18	106	atgacctcccggttgtgagggactggctgctgctatgctgggtgctgct gctgtgagtggtggcacacttcggacgactccattgtactacagccgttata cgaaatataaatgatcaggtgtgtttgtagacaaaaggcagcccggtgtt gaagacatgactgatattgatcagagtgccctctgagccccagactagactg attatatatgcttatggcgactctagagctagagggaaagcagtaactctg tcagtgaaagacagcaagatgtctaccctgtcttgcaagaataaaataatc tcatttgaggagatggacccccctgaaaaacatagatgacatccaaagtgac ctgatattcttccagaagcgggtacccggctcataacaaaatggagttcgag tcaagcctctatgagggctcacttctcgctgcaaaagggaagatgacgcc ttcaaattgatcttgaaaaaaaagacgagaatggtgataagctgtgcatg ttcacactcactaatttgcatcagtc
mouse wild type IL-18	107	atgacctcccggttgtgagggactggctgctgctatgctgggtgctgct gctgtgagtggtggcaactttgggagactccactgtaccactgcccgttata cggaatataaacgatcaggtcctctttagagacaagagacagccagtttcc gaggacatgactgatatagatcagagtgccctctgagcctcaaacctcggtc ataatatacatgtataaggatagcgaggttcggggacttgccgtgactttg tcctgtaaaagactcaaaaatgagcacctgtcatgcaaaaataagataatt tcttctgaggaaatggacctcccgagaacattgatgatatacagtcaggac

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		ttgatattttccagaaacgggttcccggtcataacaaaatggagtttgag tcttcattgtacgagggtcatttttggcatgccagaaggaggatgatgcc ttcaaacgtgatttgaagaaaaagacgagaacggagacaagtctgtcatg ttacacttaccacccctcatcaatcc
Capsids/envelop from SIN virus	108	atgaatagaggaatatttaacatgctgagccgcgcgccttccccgcctcc acagctatgtggaggcctaggcgaaggaggcaggcggcaccactgcccgc cgtaatgggttggcatcccaattcaacagctaaccacagctgtcagtgcc ctagtcattgggtcaagccacccgacctcaacaaccaaggccacgcctga ccacgcctaaagaagcagtcctccgaagcaaccacccaatgctaaaaagaca aaaccgcaggagaaacaggagaaacaaccgcgaacacgaagcgggttaag agacagcgtatggcactaaaattgatgctgacaggactctttgacgtgaac aacgaagccggtgacgtaataggacacgcctcgctatggaaggcaaatg atgaagccacttcacgtgaaaggggtcatagaccacccgggtttggcaaa ctgaaatttaccagtcacagcgtacgacatggagtgcgccagcttcca gccagtatgaaaagcgaagcatttgcatacaccagcgagcaccctgaaggc ttctacaactggcatcatgggtgcccgtgcagtagcggcggtcgctttaca gtcccgcgcggaggttgggggaaaaaggcgacagtggaagaccgatcatggat aatcacggtaaggtgggtgcgcatagtgctgggcggcgctgacgaagcgcc cgactgctctctccgtcgctcacttggaacagcaaggaaagaccatcaaa acgactccgaagggactgagggaatggctgcagcaccattagtcaaggct atgtgcttcttgagaatgtatcggtcccggtgcagtcgtccgctacgtgc tattcccgaaaaccatctcgtgccccttgacatcctggaggagaacgtgaac aacgacgcctacgatacctatttagcggcagtggtgaagtgcaggacgtcc ggctcggaacaaacgcagcatcaccgcagatttcacctgaccagcccgtagc ctgggcacttgctcgattgcccataactgaacccctgctcagcccaatt aagatcgagcaagtttgggacgaggcggacgacggatcaatacgaatacag acatccgcccagttcggtgacgacaaagtggggcagcagacgtcaacaag tacaggtacatgtcaattgaccaggacatgccgtgaaagaaggctctatg gatcatatcaagatcagcacatccgggcccgtgtaggcggcttcaatcaca aggtacttcttgcgtgcaaatgcccacgggagatagcgttacggttagc ataaccggcggcactatggccacccctcgtacattagcagctaaaataagg caaaaattcgtaggacgtgagaagtatgacctccccagcacacggaaaag aacatcccctgcaccgtgatgatcgcttaaggagacgtcagccggatcac atcccatgcacaggcctggacctcatgcctatacctcatacctagaggct acctcaggaaaaatttacgcgaacaccaccatcaggaaagaacatcacgtac gagtgaagtgtggcgattataaaactgcgaccgttccgtcagaacagag atcgaggatgtactgccaataaaacagtgctggcgctacaagagcgaccag accaagtgggtgtcaactcgccgatctaatcaggcacgctgaccacaca gccaaggtgaaattccacttcccttcaagccctacctcagtagatgcctg gtaccactggcacacgagccgactgtaattcatgggttcaaacacataaagc ctccatttagacacagaccaccaaccttgctcaccattagaagactgggc gagaacccgaacgcagctcggaaggatctcaggaaaaacagttaggaaat ttactgtgcacagagacgggctagagtatatatggggcaatcacgaaccg gtgagagtatatgcacaagagtcggcccccggtgacctcacggatggcca cacaaaataatcaacactactatcaacgccaccagatatacactagcttg gctgtagcagcttccgtgtggcggtgatagtcggcctggccaccgcctg atgtgtatctgcagagcgcgacgggagtgctaaccccgtagcgttagcc ccgaatgcagtcacccgcagctccttggccttattgtgctgcttccgtcc gccaacgccgaatcttcaccgcacagatgggatacctatggtcgaacagt caaacatttttctgggtgcagctgtgcatccccctggcagtagtaatagta ctcgtacgctgctgttcgtgctgcctcccttttttagtggttgcaggcgcc tacctggcgaaggttagacgccttcgaacatgcgaccactgtcccaaatgtg ccgaagatccgtataaagctttgggtcgagagagccggatcacgcccgcct aacttggaatatctgttatgtcttcagaggctccctccctcgattccacca gaatcacatcacctgcaaaactccactatcgtgccttccccgaaagtga tgctgcggttccccctgagtgccaacccgcttcgcatgcgcgactacaactgc aaagtatttggcgcgcttaccctttcatgtggggtggagcgcaatgttt tgccagcagcgagaacagccagatgagtgaggcgatgtgcagctatcgcc gattgtgcccttgatcacgcacaggcggtgaaagtgcacacagccgcaatg aaggtcggaactacgcatttggtatgggaacaccactagcctcctggatgcg tacgtaaacggagtcacgcaggccacatctaaagacctgaaagtcatgtcc ggacccatctcgtccgcgtttcccccttttgactctaaagtctgtgatcac cgagggtcgtgtacaattacgattttcagagtagcgggcaatgaaacgg gggtccctttggggacattcaagctagctctttgacgggcaacgacctgat gccagcacccgacatacgtttgcttaaacccgcagcaaaagtgtgcagctt ccctatacccaagcagcatcaggatttgaaatttggaagaacaaactccgga agaccactgcaagagacggccccatttggctgaagattgcccgtcaacccct ctgaaggccgtggatgttccctatgggaacatcccgatctcgattgatc ccccacgcgccttatttagaccatccgatgcgccaatttatctccgaagt aagtcgaagtcagcgactgcacatactctgctgacttcggaggcatggcc

TABLE 16-continued

SEQUENCES:		
Description	SEQ ID NO.	Sequence
		accctacagtatgtgtctgaccgagagggccagtgcccagtgccattcccat tcaagtaccgccacactccaggaatcaactgttcacgtcttgaaaaaggg gcgacgacagtgcaacttcagcacagctagccctcacgccagctttgtggtc tcattatgcggaagaagacgacctgcaccgcaggatgcaaaccacctgtc ggccacatagtcagcacacccacaagaaggaccaagagttccagaccgcc gtctccaggacatcatggagctggctcttcgcgctgttcggaggagcttcg tccttagtaatcatgggactgttaattctctcctgcagcatgatgtcact aacaccagaaga

[0257] All of the foregoing sequences, components, molecules and methods, and any and all combinations thereof are non-limiting examples and aspects of the present disclosure.

[0258] In addition, it is to be understood that any particular aspect of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Since such aspects are deemed to be part of the whole of the present disclosure, any part of the whole

disclosure may be excluded even if the exclusion is not set forth explicitly herein.

[0259] It is to be understood that while the present disclosure has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the present disclosure, which is defined by the scope of the appended claims. Other aspects, advantages, and alterations are within the scope of the following claims.

SEQUENCE LISTING

Sequence total quantity: 108
 SEQ ID NO: 1 moltype = DNA length = 38
 FEATURE Location/Qualifiers
 source 1..38
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1
 taacctgaat ggactacgac atagtctagt ccgccaag 38

SEQ ID NO: 2 moltype = DNA length = 92
 FEATURE Location/Qualifiers
 source 1..92
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 2
 gaacttccat catagttatg gccatgacta ctctagctag cagtgttaaa tcattcagct 60
 acctgagagg ggccctata actctctacg gc 92

SEQ ID NO: 3 moltype = DNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 3
 atggactacg acatagtcta gtccgccaag 30

SEQ ID NO: 4 moltype = DNA length = 939
 FEATURE Location/Qualifiers
 source 1..939
 mol_type = other DNA
 organism = Mus musculus

SEQUENCE: 4
 atgtgggagc ttgaaaaaga cgtctatgta gtagaagtgg actggacacc tgatgctect 60
 ggcgagacag ttaacctcac atgcgatacc cctgaggaag atgatatcac ctggacttct 120
 gaccagagac acgggggtgat tgggagcggc aaaaccctga cgatcactgt gaaggagttt 180
 ctggacgccc gccagtatac ctgtcacaaag gggggggaga cctgagtcga tagccacctg 240
 ttgctgcaca agaaggagaa tggcatctgg tctacagaga tctgaagaa ctttaagaac 300
 aagaccttcc tgaagtgtga agcaccaaac tacagtggtc gctttacctg cagctggctg 360
 gtccaaagaa acatggacct gaaattttaa ataaagagta gctcttcgag tectgattcc 420
 agggccgtga cgtgcggcat ggcaagcctt tcagccgaaa aagtcacgct ggatcagcga 480
 gactatgaga agtacagcgt tagctgtcag gaggacgtaa cttgcccgcac tgccgaggag 540
 actctgccca tagagctcgc tctggaggcc aggcagcaga acaaatatga gaattacagc 600
 actagtttct ttattagaga catcatcaaa cccgaccac ccaagaatct gcagatgaag 660
 ccgctgaaga atagtcaggt cgaggtttcc tgggaatata cagattcatg gtccactccg 720

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cattcttatt tttccttaaa attctttgtt aggattcagc ggaaaaaaga aaagatgaaa 780
gagacggagg aaggggtgcaa ccagaagggg gccttctctg tggaaaagac aagcactgag 840
gtccaatgta aggggtgggaa cgtttgcgtg caggctcagg atcgctacta caacagcagt 900
tgctctaagt gggcctgcgt accttgctgc gtcaggagt 939

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```

SEQ ID NO: 5      moltype = DNA length = 579
FEATURE          Location/Qualifiers
source           1..579
                 mol_type = other DNA
                 organism = Mus musculus

```

```

SEQUENCE: 5
aggggtgatcc cagtgtctgg gccggcccct tgcttgtctc aatccagaaa cctcctcaag 60
accactgacg atattgtaaa gactgcccga gagaagctaa aacactactc ttgtacagct 120
gaagatatag accatgagga tataaacacg gaccagacct ctactctgaa aacctgtctg 180
cctcttgagc tgcacaagaa cgagtcctgt ctggctaccc gcgaaacctc aagcacaacc 240
agaggtagtt gctgcccccc acaaaagaca tcgcttatga tgaccttggt tctgggatct 300
atztatgagg acctgaagat gtaccaaact gagttccagg caataaatgc tgctctccag 360
aatcacaaat atcaacaaat catccttgat aaggggatgc tggtcgcaat cgacgagctc 420
atgcaaatgc tgaaccacaa tggggaaacc ctacggcaga aaccaccggt gggagaggcc 480
gacccttacc gtgttaaaat gaagttgtgt attcttttgc atgcattctc tacaagagtc 540
gttaccatca atcgcgctat ggggtacctg tcatcagcc 579

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SEQ ID NO: 6      moltype = DNA length = 426
FEATURE          Location/Qualifiers
source           1..426
                 mol_type = other DNA
                 organism = Mus musculus

```

```

SEQUENCE: 6
gggtacctgt catcagccgg cggtagtgtt ggtgggagcc acaagtcctc cccccagggt 60
ccggatccgc tcttgatcag actgagacat ctgattgata ttgtcgagca gttgaagatc 120
tatgagaatg acctcgatcc tgagttactg agtgcccac aggacgtaa agggagctgt 180
gaacacgcgc catttgcttg ttttcagaag gccaaagctg aaccttctaa tcccgggaat 240
aacaaaaatt tcattatcga tctcgtcgcg cagctgaggg ggcgacttcc tgcacggcgg 300
ggggggaaaa agcaaaagca catcgcaaa tgctcctcat gcgactctta cgagaaacgt 360
acccttaagg agttccttga aagactcaaa tgggtgctgc aaaagatgat ccaccagcat 420
ctcagc 426

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```

SEQ ID NO: 7      moltype = DNA length = 918
FEATURE          Location/Qualifiers
source           1..918
                 mol_type = other DNA
                 organism = Homo sapiens

```

```

SEQUENCE: 7
atatgggagc tgaagaagga cgtgtatgtc gtggagctgg actggtagcc agatgctcct 60
ggcgaaatgg tggttttaac atgtgatacc cccgaggagg acggcatcac atggactctg 120
gaccagagtt ctgaggtgct ggggtccggc aagactctga caatccaggt taaggagtgc 180
ggcgacgcag gacagtacac ttgtcacaag ggaggtgagg tgctttctca cagcctgttg 240
ctgctccata agaaggaaga cgttatgttg tcaaccgaca tcctcaagga ccagaaggag 300
ccccaaaaaa agactcttct gagatgtgag gccaaagaat acagcggtag attcacttgt 360
tggtggctca ccaccatatt cacagacttg accttcagtg tcaaaagtgc acgagggagc 420
tcagatcctc aaggcggtac ctgtggcgca gcgacgctgt ccgcagaaag agtcagggga 480
gacaacaagg aatacagtag ctctgtcgag tgccaggagg attccgcttg tccggccgca 540
gaggagtctt tacctattga ggtgatggtc gatgccgtgc acaagcttaa gtacgagaat 600
tacacatcaa gttttttcat ccgcgacatc attaaacctg atccaccaa gaacctgcag 660
ctcaagcctc tgaagaatag caggcaggtc gaggtaagct gggagtatcc tgatacttgg 720
tccaccccc acagttatgt cagcctcacc ttctgcgtcc aagtcagggg aaagagcaag 780
agagagaaga aggatagggt gtccacagat aagacttcag ctactgtgat ctgcagaaag 840
aatgcgtcta tctctgtgag agcacaagac aggtactaca gttctagctg gagcagtggt 900
gcatcagtc cctgcagt 918

```

```

SEQ ID NO: 8      moltype = DNA length = 591
FEATURE          Location/Qualifiers
source           1..591
                 mol_type = other DNA
                 organism = Homo sapiens

```

```

SEQUENCE: 8
aggaacctcc cagttgctac acctgacctg ggaatgttcc catgcctcca ccattcccag 60
aatctcctcc gagccgtgtc caatatgctg caaaaggctc ggcagacctt ggagttttac 120
ccttgccact cagaagaagt cgatcatgag gatatacaca aggataagac gagcactgtt 180
gaggcatgcc tgccccctga gctaactaag aatgagtctt gcctgaacag caggggagact 240
tctctcatta ccaacggtag ctgtcttgcc agcaggaaga catcttttat gatggccctg 300
tgtctatcta gcataatga agacctgaag atgtaccagg tggaaattcaa aacctgaat 360
gctaagcttc tcattggtac caagaggcaa atcttctctg accagaatat gcttgctgtc 420
atagatgaac tgatgcaggg gttgaatttt aacagcgaga cggtgcctca aaaaagctca 480
ctggaagaac ctgattttta taaaacgaag atcaagctgt gtattttact acacgccttt 540
agaatccgcg ctgtttacat cgacagagta atgtctacc taaatgtctc a 591

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SEQ ID NO: 9          moltype = DNA length = 393
FEATURE              Location/Qualifiers
source                1..393
                     mol_type = other DNA
                     organism = Homo sapiens

SEQUENCE: 9
caggacaggc atatgatccg gatgcggcag ctgatcgata ttgtagacca gttgaagaat 60
tatgtgaacg acttagtgcc ggaattcctc cccgcccccg aggcgtgga gactaattgt 120
gagtggctct cattctcatg ctccaaaaa gcacagctga agagtgccaa tacgggcaat 180
aacgaaagga tcatcaatgt aagtataaag aagttaaaac gcaaaccgcc cagtaccaac 240
gctggacgca ggcaaaaaca cagggtgaca tgcccctcgt gtgattcgta cgaaaaaaaa 300
cctccaaagg aattcctgga aaggttcaag tccttattac agaaatgat tcaccagcac 360
ctgagtagta ggacccacgg atccgaagac tcc                                     393

SEQ ID NO: 10         moltype = DNA length = 589
FEATURE              Location/Qualifiers
source                1..589
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 10
tttccatagg ctccgcccc ctgacgagca tcacaaaaat cgacgtcaa gtcagaggtg 60
gcgaaacccg acaggactat aaagatacca ggcgtttccc cctggaagct cctcgtgctg 120
ctctcctggt cgcacctgac cgcttaccgg atacctgtcc gcctttctcc cttcggggaag 180
cgtggcgctt tctcatagct cacgctgtag gtatctcagt tcggtgtagg tcgttcgctc 240
caagctgggc tgtgtgcacg aaccccccg tccagcccgac cgtcgcgct tatccggtaa 300
ctatcgtctt gagtccaacc cggtaagaca cgacttatcg ccactggcag cagccactgg 360
taacaggatt agcagagcga ggtatgtagg cgggtgctaca gagtcttga agtggtggcc 420
taactacggc tacactagaa gaacagtatt tggatatctg gctctgctga agccagttac 480
cttcggaaaa agagttggta gctcttgatc cggcaaaaac accaccgctg gtagcgggtg 540
ttttttgtt tgcaagcagc agattacgcg cagaaaaaaa ggatctcaa 589

SEQ ID NO: 11         moltype = DNA length = 1752
FEATURE              Location/Qualifiers
source                1..1752
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 11
atgagtaacc tgaggctatg gcagggcctg ccgccccgac gttggctgcg agccctgggc 60
cttcaccgca acttggggggg tggggtgggg aaaaggaaga aacgcgggcg tattggcccc 120
aatgggggtc cgttgggggta tcgacagagt gccagccctg ggaccgaacc ccgcttttat 180
gaacaaacga cccaacacgg tcggttttat tctgtctttt tattgccgtc atagcgcggg 240
ttccttcogg tattgtctcc ttccgtgttt cagttagcct cccctagggg tgggcgaaga 300
actccagcat gagatccccg cgctggagga tcatccagcc ggcgtcccgg aaaacgattc 360
cgaagcccaa cctttcatag aaggcggcgg tggaaatcga atctcgtgat gccaggttgg 420
gcgtcgcttg gtcggtcatt tcgaacccca gagtcccgc cagaagaact cgtcaagaag 480
gcgatagaag gcgatgcgct gcgaatcggg agcggcgata ccgtaaagca cagggaagcg 540
gtcagcccat tcgcccgaac gctcttcagc aatatacagg gtagcccaac ctatgtcctg 600
atagcgatcc gccacaccca gccggccaca gtcgatgaat ccagaaaagc ggccattttc 660
caccatgata ttccggcaagc aggcacgcgc atgggtcacg acgagatcct cgcgctcggg 720
catgctcgcc ttgagcctgg cgaacagttc ggctggcgcg agccctgat gctcttcgtc 780
cagatcatcc tgatcgatag gaccggcttc catccgagta cgtgctcgct cgatgcgatg 840
tttcgcttgg tggtcgaatg ggcaggtagc cggatcaagc gtatgcagcc gccgcattgc 900
atcagccatg atggataact tctcggcagg agcaaggtga gatgacagga gatcctgccc 960
cggcaacttc cccaatagca gccagtcctc tcccgttca gtgacaacgt cgagcacagc 1020
tgcgcaagga acgcccgtcg tggccagcca cgatagccgc gctgcctcgt cttgcagttc 1080
attcagggca ccggacaggt cggctcttgac aaaaagaacc gggcgccctc gcgctgacag 1140
ccggaacacg gcggcatcag agcagccgat tgtctgttgt gccagtcac agccgaatag 1200
cctctccacc caagcggcgg gagaacctgc gtgcaatcca tcttgttcaa tcatgcgaaa 1260
cgatcctcat cctgtctctt gatcgatctt tgcaaaagcc taggcctcca aaaaagcctc 1320
ctcactactt ctggaatagg acaggggcgc ggcctctgca taaataaaaa 1380
aaattagtca gccatggggc ggagaatggg cggaaactggg cggagttagg ggcgggatgg 1440
gcggagttag gggcgggact atgggtgctg actaattgag atgcatgctt tgcatacttc 1500
tgctgtctgg ggagcctggg gactttccac acctgggtgc tgactaattg agatgcatgc 1560
tttgataact tctgctgctt ggggagcctg gggactttcc acaccctaac tgacacacat 1620
tccacagctg gttctttccg cctcaggact cttccttttt caatattatt gaagcattta 1680
tcagggttat tgtctcatga gcggatacat atttgaatgt atttagaaaa ataaacaaat 1740
aggggttccg cg                                     1752

SEQ ID NO: 12         moltype = DNA length = 988
FEATURE              Location/Qualifiers
source                1..988
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 12
atgagtaaac ttggtctgac agttaccaat gcttaatcag tgaggcacct atctcagcga 60

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tctgtctatt tegtctatcc atagttgcct gactccccgt cgtgtagata actacgatac 120
gggaggggctt accatctggc cccagtgtct caatgatacc gcgagaccca cgctcaccgg 180
ctccagattt atcagcaata aaccagccag ccggaaggcg cgagcgcaga agtggctctg 240
caactttatc cgctccatc cagttctatta attgttgccg ggaagctaga gtaagtagtt 300
cgccaggttaa tagtttgccg aacgttggtt ccattgtctac aggcctcgtg gtgtcacgct 360
cgtcgttttg tatggcttca ttacgtcccg gtccccaacg atcaaggcga gttacatgat 420
cccccatgtt gtgcaaaaaa gcggttagct ccttcgggtc tccgatcggt gtcagaagta 480
agttggccgc agtggttatca ctcatgggta tggcagcact gcataattct cttactgtca 540
tgccatccgt aagatgcttt tctgtgactg gtgagtactc aaccaagtca ttctgagaat 600
agtgtatgcg gcgaccgagt tgctcttgcc cggcggtcaat acgggataat accgcgccac 660
atagcagaac tttaaaagtg ctcatcattg gaaaacgttc ttccggggcga aaactctcaa 720
ggatcttacc gctgttgaga tccagtccga tgtaaccac tcgtgcaccc aactgatctt 780
cagcatcttt tactttcacc agcgtttctg ggtgagcaaa aacaggaagg caaaatgccg 840
caaaaaaggg aataaggcg acacggaaat gttgaatact catactcttc ctttttcaat 900
attattgaag catttatcag ggttattgtc tcatgagcgg atacatattt gaatgtattt 960
agaaaaataa acaaataggg gttccgcg 988

```

```

SEQ ID NO: 13      moltype = DNA  length = 24
FEATURE
source            Location/Qualifiers
                  1..24
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 13
taatacgact cactatagga tagg 24

```

```

SEQ ID NO: 14      moltype = DNA  length = 19
FEATURE
source            Location/Qualifiers
                  1..19
                  mol_type = other DNA
                  organism = Escherichia coli

```

```

SEQUENCE: 14
taatacgact cactatagga 19

```

```

SEQ ID NO: 15      moltype = RNA  length = 39
FEATURE
source            Location/Qualifiers
                  1..39
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 15
cggcgcatga gagaagccca gaccaattac ctacccaaa 39

```

```

SEQ ID NO: 16      moltype = RNA  length = 26
FEATURE
source            Location/Qualifiers
                  1..26
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 16
ttcgaaggcg cgctcttaga gccacc 26

```

```

SEQ ID NO: 17      moltype = RNA  length = 40
FEATURE
source            Location/Qualifiers
                  1..40
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 17
gcgggcgcatg agagaagccc agaccaatta cctacccaaa 40

```

```

SEQ ID NO: 18      moltype = RNA  length = 50
FEATURE
source            Location/Qualifiers
                  1..50
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 18
acatttgctt ctgacacaac tgtgttcact agcaacctca aacagacacc 50

```

```

SEQ ID NO: 19      moltype = RNA  length = 7487
FEATURE
source            Location/Qualifiers
                  1..7487
                  mol_type = other RNA
                  organism = unidentified

```

```

SEQUENCE: 19
atggagaaaag ttcacgttga catcgaggaa gacagcccat tcctcagagc ttgcagcgg 60
agcttccccg agtttgaggt agaagccaag caggtcactg ataatgacca tgctaagtc 120
agagcggtttt cgcactctggc ttcaaaactg atcgaaacgg aggtggaccc atccgacacg 180
atccttgaca ttggaagtg gcgcgcccgc agaatgtatt ctaagcacia gtatcattgt 240
atctgtccga tgagatgtgc ggaagatccg gacagattgt ataagtatgc aactaagctg 300

```

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aagaaaaact	gtaaggaat	aactgataag	gaattggaca	agaaaaatgaa	ggagctggcc	360
gcggtcatga	gcgacctga	cctggaaaact	gagactatgt	gcctccacga	cgacgagtcg	420
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SEQ ID NO: 20 moltype = DNA length = 57
 FEATURE Location/Qualifiers
 source 1..57
 mol_type = other DNA
 organism = unidentified

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SEQ ID NO: 21 moltype = RNA length = 138
 FEATURE Location/Qualifiers
 source 1..138
 mol_type = other RNA
 organism = unidentified

SEQUENCE: 21
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 ttttgttttt aatatttc 138

SEQ ID NO: 22 moltype = RNA length = 154
 FEATURE Location/Qualifiers
 source 1..154
 mol_type = other RNA
 organism = unidentified

SEQUENCE: 22
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SEQ ID NO: 23 moltype = DNA length = 1566
 FEATURE Location/Qualifiers
 source 1..1566

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mol_type = other DNA
organism = Homo sapiens

SEQUENCE: 23
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SEQ ID NO: 24      moltype = DNA length = 720
FEATURE           Location/Qualifiers
source            1..720
                  mol_type = other DNA
                  organism = Aequorea victoria

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SEQ ID NO: 25      moltype = DNA length = 1653
FEATURE           Location/Qualifiers
source            1..1653
                  mol_type = other DNA
                  organism = unidentified

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SEQUENCE: 25
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SEQ ID NO: 26 moltype = RNA length = 3768
 FEATURE Location/Qualifiers
 source 1..3768
 mol_type = other RNA
 organism = Venezuelan equine encephalitis virus

SEQUENCE: 26

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source            1..31
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SEQUENCE: 29
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SEQ ID NO: 30      moltype = DNA length = 31
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source            1..31
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FEATURE           Location/Qualifiers
source            1..45
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SEQUENCE: 32
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SEQ ID NO: 33      moltype = DNA length = 45
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SEQUENCE: 33
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SEQUENCE: 34
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 organism = synthetic construct

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 organism = synthetic construct

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FEATURE           Location/Qualifiers
source            mol_type = other RNA
                  organism = synthetic construct

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SEQ ID NO: 44      moltype = RNA length = 1614
FEATURE           Location/Qualifiers
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                  mol_type = mRNA
                  organism = Homo sapiens

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                  organism = Homo sapiens

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FEATURE           Location/Qualifiers
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                  mol_type = other DNA
                  organism = Mus musculus

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cgaatggaga actaccggaa tctgaaactt ccagaaatgc tgaccttcaa gttctacctc 240
cctaagcagg ctactgaact taagattttg caatgtcttg aagatgaact gggcccactt 300
cgacacgtat tggatctgac ccagtcaaaa tcctttcaat tggaggacgc agagaacttc 360
ataagcaaca tcagggtcac cgtgggtgaag ctcaaggggt ctgacaatac atttgaatgt 420
caatttgatg atgagtcagc caccgtcgtc gatttttttg ggagatggat cgcattctgc 480
cagagcataa tttctaccag ccttcagtag 510

```

```

SEQ ID NO: 47      moltype = DNA length = 405
FEATURE           Location/Qualifiers
source            1..405
                  mol_type = other DNA
                  organism = Homo sapiens

```

```

SEQUENCE: 47
atgtatcgaa tgcaattgct gtccctgcatt gcgctctctc tggcgctgtt caccaattca 60
aactgggtta acgtaatatc tgatttgaaa aaaatcgaag acctcataca gagtatgcac 120

```

-continued

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atagatgcta cgctgtacac agaaagcgac gtacaccga gctgtaaggt cactgccatg 180
aagtgccttc ttctogaact ccaagttatc agtctogaat ccgggggatgc gtcaatccat 240
gacactgttg agaattctcat tattctcgcg aacaactcac tcagctcaaa tgggaacgtc 300
actgaatccg gctgcaagga atgcgaggaa ttggaagaga agaatttaa agagtctctg 360
cagagtttcg ttcattattgt tcaaatgttt atcaacacaa gttag 405

```

```

SEQ ID NO: 48      moltype = DNA length = 405
FEATURE           Location/Qualifiers
source            1..405
                  mol_type = other DNA
                  organism = Mus musculus

```

```

SEQUENCE: 48
atgtattcaa tgcagcttgc cagttgtgtc acactcactc ttgtctctgt ggtaatttcc 60
aattggatcg atgtccggta tgatttggag aagatcgaat ctctcatcca aagcatacac 120
attgacacca ctttgtatag tgatagcgat ttccacccat cttgtaaggt gaccgccatg 180
aattgccttc tgttggaact tcaagttata cttcacgagt attcaaacat gacactgaat 240
gaaactgtcc gaaacgtact ctaccttgct aactccaccc ttctcatctaa taagaacgta 300
gctgagagcg gatgtaagga gtgtgaggag ctggaagaga agacatttac tgaatttctt 360
cagagtttta tccgcatagt gcaaatgttt attaacacct cctag 405

```

```

SEQ ID NO: 49      moltype = DNA length = 660
FEATURE           Location/Qualifiers
source            1..660
                  mol_type = other DNA
                  organism = Homo sapiens

```

```

SEQUENCE: 49
atgtatcgaa tgcatttgc gtctctgcat gcgctctctc tggcgcttgt caccaatttc 60
aactgggtta acgtaaatatc tgatttggaa aaaatcgaa acctcataca gtagtatgcac 120
atagatgcta cgctgtacac agaaagcgac gtacaccga gctgtaaggt cactgccatg 180
aagtgccttc ttctogaact ccaagttatc agtctogaat ccgggggatgc gtcaatccat 240
gacactgttg agaattctcat tattctcgcg aacaactcac tcagctcaaa tgggaacgtc 300
actgaatccg gctgcaagga atgcgaggaa ttggaagaga agaatttaa agagtctctg 360
cagagtttcg ttcattattgt tcaaatgttt atcaacacaa gtggaggggg gtcagggggg 420
ggctcaggcg gcggcagtg gggcagcaca acttgtccac ctccgatgag tgtggaacat 480
gctgacattt gggtaaaagag ttactcactc tattcacgag agcgatacat ctgtaaatag 540
gggtttaaaa gaaaagccgg tacatcctct cttactgaat gcgtactcaa taaggcgacc 600
aacgtcgcgc actggaccac tccttccttg aaatgcattc gcgaccgggc tctcgtttag 660

```

```

SEQ ID NO: 50      moltype = DNA length = 660
FEATURE           Location/Qualifiers
source            1..660
                  mol_type = other DNA
                  organism = Homo sapiens

```

```

SEQUENCE: 50
atgtaccgca tgcactttt gtcttgcatt gcacttagct tggcactcgt cactaatttc 60
attacatgcc ctccgcctat gtctgtggag cagcgagaca tctgggtaaa gtcttactcc 120
ctctactcaa gagaacgata tatctgtaat tcagggttta agcgaaaagc aggaacctct 180
tcaactgacag aatgcgtgct taataaggcg acgaacgttg ctcatgtgac tacacctagt 240
ttgaagtgtg ttcgggagtg ggcactcgta ggaggggggt cagggggttg ctcaggcgcc 300
ggcagtgagg gcacgaattg ggtaaacgta attagcgatc tgaagaagat cgaggacttg 360
atacaaaagta tgcataatcga tgcactctct tataccgagt ctgattgata cccctcttgt 420
aaagtaacag cgaatgaagt cttttgttg gagttgcaag tgataagcct ggaagtgagg 480
gatgcgtcca tccacgatac cgtcgagaaac ttgataatac tggccaacaa cagtctctct 540
agtaacggta acgtaaccga gagcgggtgt aaagaatgag aagaattgga agagaaaaat 600
atcaagaagt tccttcagtc atttgtgcat atagtgcaga tgtttatcaa cacatcctag 660

```

```

SEQ ID NO: 51      moltype = DNA length = 663
FEATURE           Location/Qualifiers
source            1..663
                  mol_type = other DNA
                  organism = Mus musculus

```

```

SEQUENCE: 51
atgtattcaa tgcagcttgc cagttgtgtc acactcactc ttgtctctgt ggtaatttcc 60
aattggatcg atgtccggta tgatttggag aagatcgaat ctctcatcca aagcatacac 120
attgacacca ctttgtatag tgatagcgat ttccacccat cttgtaaggt gaccgccatg 180
aattgccttc tgttggaact tcaagttata cttcacgagt attcaaacat gacactgaat 240
gaaactgtcc gaaacgtact ctaccttgct aactccaccc ttctcatctaa taagaacgta 300
gctgagagcg gatgtaagga gtgtgaggag ctggaagaga agacatttac tgaatttctt 360
cagagtttta tccgcatagt gcaaatgttt attaacacct ccggaggggg gtcagggggg 420
ggctcaggcg gcggcagtg gggcagcgga actacatgtc cactcccggt cagtatcgaa 480
catgcagata tcagagttaa aaattattca gtgaacagca gggagcgcta cgtgtgtaat 540
agtggtttta aaaggaaagc cggcacttct acattgattg aatgcgttat caataagaat 600
actaacgtag ctcaactggac cacaccctct ctgaagtgtg tccgagaccc ttcacttgcc 660
tag 663

```

```

SEQ ID NO: 52      moltype = DNA length = 663

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-continued

FEATURE
source

Location/Qualifiers
1..663
mol_type = other DNA
organism = Mus musculus

SEQUENCE: 52

atgtattcaa	tgcagcttgc	cagttgtgtc	acactcactc	ttgtcctgct	ggtaatttcc	60
ggaactactt	gccaccacc	ctcagtcgtc	gagcacgctg	atataagagt	taaaaactat	120
tctgtaaatt	ctagggaaag	atatgtatgt	aacagcgggt	tcaaaaggaa	ggcagggaca	180
tcaaccctta	tgcagtgctg	cattaacaaa	aatacaaacg	tgcgccattg	gaccactcca	240
tcactcaaat	gtatacgaga	ccctagcttg	gcaggagggg	ggtcaggggg	tggtcagggc	300
ggcggcagtg	ggggcagcaa	ttggatagat	gtccggtagc	acctcgagaa	gatcgagtca	360
cttattcaat	ctatacatat	tgacactacc	ctgtatactg	acacgcagct	tcaccaagt	420
tgtaaagtca	ctgctatgaa	ttgctttttg	ctcgagttgc	aggctcctct	gcatgagtat	480
tctaataatga	ctttgaatga	aaccgtcaga	aatgtcttgt	atcttgctaa	cagtaccctt	540
agctccaaca	agaatgttgc	cgaatcaggt	tgtaagaagt	gcgaggaggt	ggaagagaaa	600
acctttaccg	aatttcttca	atcattcctc	agaattgtac	aaatgttcat	caatacttca	660
tag						663

SEQ ID NO: 53

FEATURE
source

Location/Qualifiers
1..540
mol_type = other DNA
organism = Mus musculus

SEQUENCE: 53

atgacctccc	ggcttgtgag	ggtagctggc	gctgctatgc	tggtggctgc	tgctgtgagt	60
gtggcaaat	ttgggagact	ccactgtacc	actgccgtta	tacggaatat	aaacgatcag	120
gtcctctttg	tagacaagag	acagccagtt	ttcgaggaca	tgactgatat	agatcagagt	180
gcctctgagc	ctcaaacctc	gctcataata	tacatgtata	aggatagcga	ggttcgggga	240
cttgccgtga	ctttgtccgt	gaaagactca	aaaatgagca	ccctgtcatg	caaaaataag	300
ataattttct	tgcaggaaat	ggaccctccc	gagaacattg	atgatatcca	gtccgacttg	360
atatttttcc	agaaacgggt	tcccggtcat	aacaaaatgg	agtttgagtc	ttcattgtac	420
gagggtcatt	ttttggcatt	ccagaaggag	gatgatgcct	tcaaaactgat	tttgaagaaa	480
aaagacgaga	acggagacaa	gtctgtcatg	tttacactta	ccaaccttca	tcaatcctag	540

SEQ ID NO: 54

FEATURE
source

Location/Qualifiers
1..540
mol_type = other DNA
organism = Mus musculus

SEQUENCE: 54

atgacctccc	ggcttgtgag	ggtagctggc	gctgctatgc	tggtggctgc	tgctgtgagt	60
gtggcacact	tgggagcact	ccattgtact	acagccgtta	tacgaaatat	caatgatcag	120
gtgttgtttg	tagacaaaag	gcagcccggt	tttgaagaca	tgactgatat	tgatcagagt	180
gcctctgagc	cccagactag	actgattata	tatgcttatg	gcgactctag	agctagaggg	240
aaagcagtaa	ctctgtcagt	gaaagacagc	aagatgtcta	ccctgtcttg	caagaataaa	300
ataatctcat	ttgaggagct	ggacccccct	gaaaacatag	atgacatcca	aagtgacctg	360
atattctttc	agaagcgggt	acccgggtcat	aacaaaatgg	agttcgagtc	aagcctctat	420
gagggtcact	ttctgcctcg	ccaaaaggaa	gatgacgcct	tcaaatgat	cttgaaaaaa	480
aaagacgaga	atggtgataa	gtctgtcatg	ttcacactca	ctaatttgca	tcagtcatag	540

SEQ ID NO: 55

FEATURE
source

Location/Qualifiers
1..1257
mol_type = other DNA
organism = Mus musculus

SEQUENCE: 55

atgaatccat	ccgcagctgt	aattttctgc	cttatcctcc	tggggctgag	tggaacccaa	60
ggaatccccc	tcgctcgaa	agtagctgct	aactgtatcc	acatagacga	tgccccggtt	120
cgaatgaggg	ctatcggaag	gctcgaaata	atacccgctc	ctctctcatg	cccccgcgta	180
gaaataatag	caactatgaa	gaagaacgac	gagcaacgat	gcttgaaccc	agaaagcaag	240
actatcaaga	acttgatgaa	agcttttagc	caaaagcggg	gtaaacgagc	ccccggggga	300
ggcggctccg	gcggaggagg	gagtggtggc	ggtggatccg	gaggcgggtg	gagtagccct	360
gtgattagga	atgcacgatg	cagctgtata	agcacctccc	ggggcactat	tcactataag	420
agccttaaa	atctcaagca	gtttgcacca	tctccaaatt	gtaacaaaac	tgagattatt	480
gctacactga	agaaacgggg	ccaaacctgc	ctggaccagc	atagtgccaa	tgtaaaaaag	540
cttatgaagg	aatgggagaa	aaaaatcagt	caaaagaaaa	agcaaaaacg	cggtaaaaaa	600
caccagaaaa	acatgaagaa	tcggaaacca	aaaaccccac	agtcaggagg	aaggagtcga	660
aaaaacaacg	ggggagcgcg	aagtggcgcg	gggggcagcg	gtggcggggg	atcaggcgcg	720
gggggcaggt	tccttatgtt	caaacaaggg	cgggtgcctg	gtataggggc	aggtatgaaa	780
gcagtgaaga	tggtctgaat	cgaaaaagca	agtgtgatct	accctagtaa	tggtctgtgat	840
aaagttgaag	tatttgtgac	tatgaaagct	cacaagagac	agcgggtgtt	ggatcctaga	900
tccaaacaag	cccgactcat	catgcaagcc	atagaaaaag	agaacttctc	gcgaaggcag	960
aatatgggtg	gaggtggaag	cggtagcgcg	ggtagcgcg	ggggaggtag	tggtgggggc	1020
ggtagcgtaa	ctagtgtctg	accagaggag	tccgatgggg	atctctcttg	tgctgtcggt	1080
aagaccatat	catccgggat	tcaccttaag	cacattacca	gcctcgaagt	gatcaaaagt	1140
ggtaggcatt	gtgcctgtgc	ccagctcata	gccacattga	agaatgggag	gaaaatttgc	1200

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cttgatagac aggcaccatt gtataagaaa gtcattaaaa aaattcttga atcatag 1257

SEQ ID NO: 56 moltype = RNA length = 3765
 FEATURE Location/Qualifiers
 source 1..3765
 mol_type = other RNA
 organism = Semliki Forest virus

SEQUENCE: 56

atgaattaca tccctacgca aacgttttac gggcgccagat ggcgacctcg accgcgcgct 60
 cccccctggc ctttgcaagc tacgcccgtg gcccccgtgg tcccagactt tcaggcacag 120
 caaatgcagc agcaactcat cagcgccgta aatgcgctga caatgagaca gaacgcaatt 180
 gctcctgcta ggctcccaa accaaagaag aagaagacaa ccaaaccaaa gccgaaaacg 240
 cagcccaaga agatcaacgg aaaaacgcag cagcaaaaga agaaagacaa gcaagccgac 300
 aagaagaaga agaaaccggg aaaaagagaa agaattgtgca tgaagattga aaatgactgt 360
 atcttcgaag tcaaacacga aggaagggtc actgggtacg cctgcctggg gggcgacaaa 420
 gtcatgaaac ctgcccacgt gaaaggagtc atcgacaacg cggacctggc aaagctagct 480
 ttcaagaaat cgagcaagta tgaccttgag tgtgccaga taccagtcca catgaggctg 540
 gatgcctcaa agtacacgca tgagaagccc gagggacact ataactggca ccacggggct 600
 gttagctaca cgggaggtga gttcactata cgcacaggag cgggcacaaac gggagacagt 660
 ggccggccca tctttgacaa caagggggagg gtatgcgcta tcgtcctggg cggggcccaac 720
 gagggctcac gcacagcact gtcggtgggc acctggaaca aagatatggt gactagagt 780
 acccccgagg ggtcggaaga gtggtccgccc ccgctgatta ctgcoatgtg tgtccttgcc 840
 aatgctacct tcccggtgct ccagcccccg tgtgtacctt gctgctatga aaacaacgca 900
 gagggccacac tacggatgct cgaggataac gtggataggc cagggtacta cgacctcctt 960
 caggcagcct tgacgtgcc aaacgggaaca agacaccggc gcagcgtgtc gcaacacttc 1020
 aacgtgtata agcctacacg cccctacatc cgcgtactgc cgcactgcgg agcagggcac 1080
 tcgtgtcata gccccgtagc aattgaagcg gtcagggtccg aagctaccga cgggatgctg 1140
 aagattcagt tctcggcaca aattggcata gataagagtg acaatcatga ctacacgaag 1200
 ataaggtacg cagacgggca cgcatttgag aatgcgctcc gggtcatctt gaaggtagcc 1260
 acctccggag actgtttcgt ccatggcaca atgggacatt tcatactggc aaagtgccca 1320
 ccgggtgaat tctgcaggc ctcgatccag gacaccagaa acgcggtccg tgcttcgaga 1380
 atacaatatc atcatgacct tcaaccgggt ggtagagaaa aatttacaat tagaccacac 1440
 tatggaaaag agatcccttg caccacttat caacagacca cagcgaagac cgtggaggaa 1500
 atcgacatgc atatgcgccc agatacgcgg gacaggacgt tgctatcaca gcaatctggc 1560
 aatgtaaaaga tcacagtcgg aggaagaaga gtgaaataca actgcacctg tggaaaccgga 1620
 aacgttggca ctactaatc ggacatgacg atcaaacagt gtctaataga gcagtgccac 1680
 gtctcagtga cggaccataa gaaatggcag ttcaactcac ctttgcctcc gagagccgac 1740
 gaaccggcta gaaaaggcaa agtccatata ccatccctcg tggacaacat cacatgcaga 1800
 gttccaatgg cgcgcgaacc aacgcgtatc caccgcaaaa gagaagtgc actgcacctt 1860
 caccagatc atccacgct cttttccctac cgcacactgg gtgaggaccg gcagtatcac 1920
 gaggaatggg tgacagcgcc ggtggaacgg accatacccg taccagtggg cgggatggag 1980
 taccactggg gaaacaacga ccagtgagg ctttgggtcc aactcaccac tgaagggaaa 2040
 ccgcacggct ggcgcgatca gatgtacag tactactatg ggctttaccg ggcgctaca 2100
 gtatccgagg tcgtcgggat gagcttactg gcgttgatat cgtatcttcg gtcgtgtac 2160
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 gttccgtgga cgtcggggat actctgctgc gcccccgagg cgcacgcagc tagtgtggca 2280
 gagactatgg cctactgtg ggacaaaac caagcgttgt tctgggtgga gtttgccggc 2340
 cctgttgctc gcactcctcat catcacgtat tgcctcagaa acgtgctgtg ttgctgtaag 2400
 agcctttctt ttttagtgtt actgagcctc ggggcaaccg ccagagctta cgaacattcg 2460
 acagtaatgc cgaacgtggt ggggttcccg tataaggctc acattgaaag gccaggatat 2520
 agccccccta ctttgagat gcaggttgtt gaaaccagcc tcgaaccaac ccttaatttg 2580
 gaatacataa cctgtcatgt caagacggtc gtcccgctgc cgtacgtgaa gtgctgcggc 2640
 gctcagagt gctccactaa agagaagcct gactaccaat gcaaggttta cacaggcgtg 2700
 taccggttca tgtggggagg ggcataatgc ttctgcgact cagaaaaacac gcaactcagc 2760
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 catacagcat cgctgaaggg caaagtgagg gttatgtacg gcaacgtaaa ccagactgtg 2880
 gatgtttacg tgaacggaga ccatgcgctc acgatagggg gtactcagtt catattcggg 2940
 ccgctgtcat cggcctggac cccgttcgac aacaagatag tcgtgtacaa agacgaagt 3000
 ttcaatcagg acttcccgcc gtacggatct gggcaaccag ggcgcttcgg cgacatccaa 3060
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 tcaccggcca tgggtccatgt accgtacaca cagacacctt cagggttcaa atattggcta 3180
 aaggaaaaag ggacagccct aaatacgaag gctccttttg gctgccaaat caaaacgaac 3240
 cctgtcaggg ccatgaaact cgcgctggga aacatccctg tctccatgaa ttgctgtac 3300
 agcgccttta cccgcattgt cgaggcgccg accatcattg acctgacttg cacagtggct 3360
 acctgtacgc actcctcgga ttccgcgccg gtcttgacac tgacgtacaa gaccgacaaa 3420
 aacggggact gctctgtaca ctgcactct aacgtagcta ctctacagga gccacagca 3480
 aaagtgaaga cagcaggtta ggtgacctta cacttctcca cggcaagcgc atcaccttct 3540
 tttgtggtgt cgctatgcag tgctaggggc acctgttcag cgtcgtgtga gccccgaaa 3600
 gaccacatag tcccatatgc ggctagccac agtaacgtag tgtttccaga catgtcgggc 3660
 accgcactat catgggtgca gaaaatctcg ggtggtctgg gggccttcgc aatcggcgct 3720
 atcctggtgc tgggtgtggt cacttgcatt gggctccga gataa 3765

SEQ ID NO: 57 moltype = RNA length = 3738
 FEATURE Location/Qualifiers
 source 1..3738
 mol_type = other RNA

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organism = Sindbis virus
SEQUENCE: 57
atgaatagag gaatatattaa catgctgagc cgccgcccc tccccgcccc cacagctatg 60
tggaggccta ggcgaaggag gcaggcgcca ccactgcccc cccgtaattgg gttggcatcc 120
caaatccaac agctaaccac agctgtcagt gccctagtca ttggtcaagc caccgcacct 180
caacaaccaa ggcacgcccc gctaccacgc cctaagaagc agtccccgaa gcaaccaccg 240
aatgctaaaa agacaaaacc gcaggagaaac aaggagaaac aaccgcgaaa accgaagccg 300
ggtaagagac agcgtatggc actaaaattg atgctgacag gactctttga cgtgaacaac 360
gaagccgggtg acgtaatagg acacgcccctc gctatgggaag gcaaaagtgt gaagccactt 420
cacgtgaaag ggggtcataga ccaccgggtt ttggcaaaag tgaattttac caagtcata 480
gcgtacgaca tggagttcgc ccagcttcca gccagtatga aaagcgaagc atttgcatac 540
accagcgagc accctgaagg cttctacaac tggcatcatg gtgcccgtga gtacagcggc 600
ggtcgcttta cagtcgcccg cggagttggg ggaaaaaggc acagtggag accgatcatg 660
gataatacag gtaagggtgt cgcgatagtg ctggcgcgcg ctgacgaagg cgcccgcact 720
gctctctccg tcgtcacttg gaacagcaaa ggaaagacca tcaaacgcac tccggaaggg 780
actgaggaat ggtctgcagc accattagtc acggctatgt gctcttttga gaattgtatc 840
ttcccgctga gtcgtccgct tacgtgctat tcccgaaaac catctcgtgc ccttgacact 900
ctggaggaga acgtgaacaa cgacgcctac gataccttat tagcggcagt gttgaagtgc 960
aggacgtccg gtcggaacaa ccacgcatc accgacgatt tcacctgac cagcccgta 1020
ctgggcactt gctcgtattg ccatcact gaacctgtct tcagcccaat taagatcgag 1080
caagtttggg acgaggcgga cgacggatca atacgaatac agacatccgc ccagttcggg 1140
tacgacaaaa gtggggcagc agacgtcaac aagtacaggt acatgtcaat tgaccaggac 1200
catgccgtga aagaaggctc tatggatcat atcaagatca gcacatccgg gccctgtagg 1260
cggcttcaat cacaaaggta cttctgtctt gcgaaatgcc caccgggaga tagcgttacg 1320
gttagcataa ccggcgccac tatggccacc tctgtacat tagcacgtaa aataaggcaa 1380
aaattcgtag gacgtgagaa gtagacactt ccccgagcag acggaagaa catcccctgc 1440
accgtgtatg atcgtcttaa ggagacgtca gccggatata tcaccatgca caggcctgga 1500
cctcatgcct atacctcata cctagaggct acctcaggaa aaatttacgc caaacacca 1560
tcaggaaaag acatcacgta cgagtgtaa gttggcgatt ataaaactgc gaccgtttcc 1620
gtcagaacag agatcgcgag atgtactgcc ataaaacagt gcgtggcgta caagagcgac 1680
cagaccaagt ggggtgtcaa ctgcgccgat ctaatcaggc acgtgacca cacagcccaa 1740
ggtaaatcc acttctctt caagcctacc ttcagtacat gcctggtacc actggcacac 1800
gagccgactg taattcatgg gttcaaacac ataagcctcc atttagacac agaccacca 1860
acctgtctca cactagaagg actggcgag aaaccggaac cgacgtcgga gtggatctca 1920
ggaaaaacag ttagggaatt tactgtcgac agagacgggc tagagtatat atggggcaat 1980
cacgaaccgg tgagagtata tgacaaagag tccgcccccg gtgacctca cggatggcca 2040
cacaaaaata ttcaacacta ctatcaacgc caccagtat acactagctt ggctgtagca 2100
gcttccgctg tggcggtgat agtcggcctg gccacggcgg tgatgtgtat ctgcagagcg 2160
cgacgggagt gcctaacccc gtacgctgta gccccgaatg cagtcacccc gacgtccttg 2220
gccttattgt gctgcgttcg gtccgccaac gccgaatctt tcaccgacac gatgggatac 2280
ctatggtcga acagtcaaac atttttctgg gtgcagctgt gcacccctt ggcatgtatg 2340
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organism = synthetic construct
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source             Location/Qualifiers
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cacgagattt gcacacaagt cttccacaaa agcatctctc gccgttgctc taaatctgtg 420
acttcggtcg tctcaacctt gttttacgac aaaaaaatga gaacgacgaa tccgaaagag 480
actaagattg tgattgacac tacgggcagt accaaacctc agcaggacga tctcattctc 540
acttgtttca gaggggtggg gaagcagttg caaatagatt acaaggcaca cgaataaatg 600
acggcagctg cctctcaagg gctgaccctg aaaggtgtgt atgcggttcg gtacaagggtg 660
aatgaaaaac ctctgtacgc acccacctca gaacatgtga acgtcctact gacccgcagc 720
gaggaccgca tcgtgtggaa aacactagcc ggcgacccat ggataaaaaa actgactgcc 780
aagtaccctg ggaatttcac tgccacgata gaggagtggc aagcagagca tgatgccatc 840
atgaggcaca tcttgtagag accggaccct accgacgtct tccagaataa ggcaaacgtg 900
tgttgggcca aggttttagt gccggtgtgt aagaccgctg gcatagacat gaccactgaa 960
caatggaaca ctgtggaatt ttttgaacg gacaaagctc actcagcaga gatagtattg 1020
aaccacatgc gcgtgagggt ctttggactc gatctggact ccggtctatt ttctgcacc 1080
actgttcctg tatccattag gaataatcac tgggataact cccgctgcgc taacatgtac 1140
gggctgaa 1148

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SEQ ID NO: 63      moltype = RNA length = 1169
FEATURE
source             Location/Qualifiers
                   1..1169
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 63
gaataatcac tgggataact ccccgctgcc taacatgtac gggctgaata aagaagtgg 60
ccgtcagctc tctcgtaggt acccacaact gcctcgggca gttgccactg gaagagtcta 120
tgacatgaac actggtacac tgcgcaatta tgatccgcgc ataaacctag tactgtaaa 180
cagaagactg cctcatgctt tagtctctca ccaaatgaa caccacaga gtgacttttc 240
ttcattctgc agcaaatgta agggcagaac tgtcctggtg gtcggggaag agttgtccgt 300
cccaggcaaa atggttgact ggttgtcaga ccggcctgag gctaccttca gagctcggct 360
ggatttaggc atcccagggt atgtgcccac atatgacata atatttgta atgtgaggac 420
cccatataaa tactcatcact atcagcagtg tgaagaccat gccattaagc ttagcatgtt 480
gaccaaaaaa gcttgtctgc atctgaatcc cggcggaacc tgtgtcagca taggttatgc 540
ttacgctgac agggccagcg aaagcatcat tgggtgctata gcgcggcagt tcaagttttc 600
ccgggtatgc aaaccgaaat cctcacttga agagacggaa gttctgtttg tattcattgg 660
gtacgatcgc aaggcccgta cgcacaattc ttacaagctt tcatcaacct tgaccaacat 720
ttatacaggt tccagactcc acgaagccgg atgtgcaccc tcatatcatg tgggtgcagg 780
ggatattgcc acggccaccg aaggagtgtat tataaatgct gctaacagca aaggacaacc 840
tggcggaggg gtgtgaggag cgctgtataa gaaattcccg gaaagcttcg atttacagcc 900
gatgaagta ggaaaagcgc gactgggtcaa aggtgcagct aaacatatca ttcattgcc 960
aggaccaaac ttcaacaagg ttccggaggt tgaaggtagc aaacagttgg cagaggctta 1020
tgagtccatc gctaagattg tcaacgataa caattacaag tcagtacgga ttccactgtt 1080
gtccacggcg atcttttcg ggacaaaaga tcgactaacc caatcattga accatttgct 1140
gacagcttta gacaccactg atcgagatg 1169

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SEQ ID NO: 64      moltype = RNA length = 1149
FEATURE           Location/Qualifiers
source            1..1149
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 64
attgaacccat ttgtgcacag ctttagacac cactgatgca gatgtagcca tatactgcag 60
ggacaagaaa tgggaaatga ctctcaagga agcagtggtc aggagagaag cagtggagga 120
gatatgcata tccgacgact ctccagtgac agaacctgat gcagagctgg tgagggtgca 180
tccgaagagt tctttggctg gaaggaaggg ctacagcaca agcgatggca aaactttctc 240
atatttgaa gggaccaagt ttccaccaggc ggccaaggat atagcagaaa ttaatgccat 300
gtggcccggt gcaacggagg ccaatgagca ggtatgcatg tatatcctcg gagaaagcat 360
gagcagattt aggtcgaaat gccccgtcga agagtgggaa gcctccacac cactagcac 420
ctgctcttgc ttgtgcatcc atgccatgac tccagaaaga gtacagcgcc taaaagcctc 480
acgtccagaa caaattactg tgtgctcatc ctttccattg ccgaagtata gaatcactgg 540
tgtgcagaag atccaatgct cccagcctat attgttctca ccgaaagtgc ctgcgtatat 600
tcatccaagg aagtatctcg tggaaacacc accggtagac gagactccgg agccatcgcc 660
agagaaccac tccacagagg ggacacctga acaaccacca cttataaccg aggatgagac 720
caggactaga acgctgagc cgatcatcat cgaagaggaa gaagaggata gcataagttt 780
gctgtcagat ggcccgaccc accaggtgct gcaagtgcag gcagacattc acgggcccgc 840
ctctgtatct agctcatctt ggtccattcc tcatgcatcc gactttgatg tggacagtgt 900
atccatactt gacaccttgg agggagctag cgtgaccagc ggggcaacgt cagccgagac 960
taactcttac ttcgcaaaaga gtatggagtt tctggcgaga ccggtgacct cgctcgaac 1020
agtattcagg aacctccac atcccgctcc gcgcacaaga acacgctcac ttgcaccag 1080
cagggcctgc tcgagaacca gcttagtttc caccgcgcca ggcgtgaata ggggtgatcac 1140
tagagagga 1149

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SEQ ID NO: 65      moltype = RNA length = 1198
FEATURE           Location/Qualifiers
source            1..1198
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 65
agaaccagcc tagtttccac cccgccaggc gtgaataggg tgatcactag agaggagctc 60
gaggcgctta ccccgctcac cactcctagc aggtcgggtc cgagaaccag cctgggtctc 120
aaaccgcgag gcgtaaatag ggtgattaca agagaggagt ttgaggcggt cgtagcaca 180
caacaatgac ggtttgatgc ggttgcatat atcttttctt ccgacacccg tcaagggcat 240
ttacaacaaa aatcagtaag gcaaacgggt ctatccgaag ttggtgttga gaggaccgaa 300
ttggagattt cgtatgcctc gcgcctcgac caagaaaaag aagaattact acgcaagaaa 360
ttacagttaa atccccacac tgctaacaga agcagatacc agtccaggaa ggtggagAAC 420
atgaaagcca taacagtagt acgtattctg caaggcctag ggcattattt gaaggcagaa 480
ggaaaagtgg agtgcctacc aacctgcat cctgttcctt tgtattcatc tagtgtgaac 540
cgtgcctttt caagcccccA ggtcgcatg gaagcctgta acgcatgtt gaaagagAAC 600
tttcgcatgt tggcttctta ctgtattatt ccagagtacg atgcctattt ggacatgggt 660
gacggagctt catgctgctt agacactgcc agtttttgcg ctgcaaaagt gcgcagcttt 720
ccaaagaaac actcctattt ggaacccaca atacgatcgg cagtgccttc agcgatccag 780
aacacgctcc agaacgtcct ggcagctgcc acaaaaagaa attgcaatgt cagcgaatg 840
agagaattgc ccgtattgga ttccggcgcc tttaatgtgg aatgcttcaa gaaatatgag 900
tgtaataatg aatattggga aacgttttaa gaaaaccca tcaggcttac tgaagaaaac 960
gtggtaaatt acattaccac attaaaagga ccaaaagctg ctgctctttt tcggaagaca 1020
cataatttga atatgttgca ggacatacca atggacaggt ttgtaatgga cttaaagaga 1080
gacgtgaaag tgactccagg aacaaaacat actgaagAAC ggcccaaggt acagggtgatc 1140
caggctgccc atccgctagc aacagcgtat ctgtgcggaa tccaccgaga gctgggtta 1198

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SEQ ID NO: 66      moltype = RNA length = 1099
FEATURE           Location/Qualifiers
source            1..1099
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 66
gctagcaaca gcgtatctgt gcggaatcca ccgagagctg gttaggagat taaatgcggg 60
cctgtcttcg aacattcata cactgtttga tatgtcggct gaagactttg acgctattat 120
agccgagcac ttccagcctg gggattgtgt tctggaaact gacatcgctg cgtttgataa 180
aagtgaggac gacgccatgg ctctgacccg gttaatgatt ctggaagact taggtgtgga 240
cgagagctg ttgacgctga ttgaggcgcc tttcggcgaa atttcatcaa tacatttgcc 300
cactaaaact aaattttaat tcggagccat gatgaaatct ggaatgttcc tcacactgtt 360
tgtaacaca gtcatttaac ttgtaatcgc aagcagagtg ttgagagAAC ggctaaccgg 420
atccacctgt gcagcattca ttggagatga caatatcgtg aaaggagtca aatcggacaa 480
attaatggca gacaggtgag ccacctggtt gaatatggaa gtcaagatta tagatgctgt 540
gggtggcgag aaagcgctt atttctgtgg aggggttatt ttgtgtgact ccgtgaccgg 600
cacagcgtgc cgtgtggcag accccctaaa aaggctgttt aagcttggca aacctctggc 660
agcagacgat gaacatgatg atgacaggag aagggcattg catgaagagt caacacgctg 720
gaaccgagtg ggtattcttt cagagctgtg caaggcagta gaatcaaggt atgaaaccgt 780
aggaacttcc atcatagtta tggccatgac tactctagct agcagtggtta aatcattcag 840
ctacctgaga ggggccccta taactctcta cggtcaacct gaatggacta cgacatagtc 900

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tagtccgcca agttcgaagg cgcgcctcta gagccaccat gaccgagtag aagcccacgg 960
tgccgctcgc caccgcgcgac gacgtcccca gggccgtacg caccctcgcc gccgcgttcg 1020
ccgactaccc cgcacacgcgc cacaccgtcg atcccgaccg ccacatcgag cgggtcaccg 1080
agctgcaaga actcttcct 1099

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SEQ ID NO: 67      moltype = DNA length = 593
FEATURE           Location/Qualifiers
source            1..593
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 67
cacgcgcgtc gggctcgaca tcggcaaggt gtgggtcgcg gacgacggcg ccgcggtggc 60
ggtctggacc acgcgggaga gcgtcgaagc gggggcgggtg ttcgccgaga tcggcccgcg 120
catggccgag ttgagcgggt cccggctggc cgcgcagcaa cagatggaag gcctcctggc 180
gccgcaccgg cccaaggagc ccgcgtgggt cctggccacc gtccgcgtct cgcgccacca 240
ccagggcaag ggtctgggca gcgcgctcgt gctccccgga gtggaggcgg ccgagcgcgc 300
cggggtggcc gccttcctgg agacctccgc gcccgcgaac ctccctctct acgagcggct 360
cggcttcacc gtcaccgcgc acgtcgaggt gcccgaaagg ccgcgcacct ggtgcatgac 420
ccgcaagccc ggtgcctgac atcgatgata tcgcggccgc atacagcagc aattggcaag 480
ctgcttatat agaactcgcg gcgattggca tgccgcctta aaatttttat tttatttttc 540
ttttcttttc cgaatcggtat ttgttttta atatttcaaa aaaaaaaaaa aaa 593

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SEQ ID NO: 68      moltype = DNA length = 988
FEATURE           Location/Qualifiers
source            1..988
                  mol_type = other DNA
                  organism = Streptomyces alboniger

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SEQUENCE: 68
atgagtaaac ttggtctgac agttaccaat gcttaatcag tgaggcacct atctcagcga 60
tctgtctatt tcgttcaccc atagttgcct gactccccgt cgtgtagata actacgatac 120
gggaggggctt accatctggc ccagtgctg caatgatacc gcgagaccca cgctcaccgg 180
ctccagattt atcagcaata aaccagccag ccggaaggcg cgagcgcaga agtggctcctg 240
caactttatc cgctccatcc cagtctatta attgttgccg ggaagctaga gtaagtagtt 300
cgccaggttaa tagtttgccg aacgttggtg ccattgctac aggcacgtg gtgtcacgct 360
cgtcgttttg tatggcttca ttcagctccg gttcccaacg atcaaggcga gttacatgat 420
cccccatggt gtgcaaaaaa gcggttagct ccttcgggtcc tccgatcggt gtcagaagta 480
agttggccgc agtggttatca ctcatggtta tggcagcact gcataattct cttactgtca 540
tgccatccgt aagatgcttt tctgtgactg gtgagtactc aaccaagtca ttctgagaat 600
agtgtatcgc gcgaccgagt tgctcttgcc cggcgctcaat acgggataat accgcgccac 660
atagcagaac tttaaaagtg ctcatcattg gaaaacgttc ttcggggcga aaactctcaa 720
ggatcttacc gctgttgaga tccagttcga tgtaacccac tcgtgcaccc aactgatctt 780
cagcatcttt tactttcacc agcgtttctg ggtgagcaaa aacaggaagg caaaatgccg 840
caaaaaaggg aataaggggc acacggaat gttgaatact catactcttc ctttttcaat 900
attattgaag catttatcag gggtattgtc tcatgagcgg atacatattt gaatgtattt 960
agaaaaataa acaaataggg gttccgcg 988

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SEQ ID NO: 69      moltype = RNA length = 578
FEATURE           Location/Qualifiers
source            1..578
                  mol_type = other RNA
                  organism = Homo sapiens

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SEQUENCE: 69
gctgactgtg acttagttgc attacaccct ttcttgacaa aacctaaact gcacagaaaa 60
cacgatgaga ttggcatggc tttatttgtt tttgtttttg tttgtttgtt tgttttggtt 120
tgactcagga tttaaaaact ggaacgggtg aggtgacagc agttgggttg agcagacatc 180
ccccaaagtt ctgcaatgtg gccgagaact ttgattgtac attgttcttt ttttaatggt 240
cattccaaat atcgtgagat gcattgttac aggaagtccc ttgcctccta aaagccaccc 300
cacttctctc taaggagaaat gcccagatcc tctcccgagt ccacacacgg gaggtaatag 360
cattgctttt gggtaaatga tataatgcaa aaatctttta atctttgcct taatactttt 420
ttattttttt aattttgaat gatcagcctt tgtgactccc cttttttgtc acccaacttg 480
agatgtatga aggtcttttc tctccctcgg agtgggtgga ggcagccagg gcttaactgt 540
acactaaact gagaccagtt gaataaaagt gcatacca 578

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SEQ ID NO: 70      moltype = RNA length = 414
FEATURE           Location/Qualifiers
source            1..414
                  mol_type = other RNA
                  organism = Homo sapiens

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SEQUENCE: 70
catcacattt aaaagcatct cagcctacca tgagaataag agaaagaaaa tgaagatcaa 60
aagcttattc atctgttttt ctttttcgtt ggtgtaaagc caacaccctg tctaaaaaac 120
ataaattttt ttaatcattt tgctcttttt ctctgtgctt caattaataa aaaatggaaa 180
gaatctaata gagtggtaca gcactgttat ttttcaaaga tgtgttgcta tctgaaaat 240
tctgtagggt ctgtggaagt tccagtgttc tctcttattc cacttcggta gaggatttct 300
agtttcttgt gggctaatta aataaatcat taatactctt ctaagttatg gattataaac 360
attcaaaata atattttgac attatgataa tcttgaataa aagaacaaaa acca 414

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SEQ ID NO: 71      moltype = RNA length = 90
FEATURE           Location/Qualifiers
source            1..90
                  mol_type = other RNA
                  organism = Homo sapiens

SEQUENCE: 71
accgcccagaga ccgcgctccgc cccgcgagca cagagcctcg cctttgcoga tccgcccgcc 60
gtccacacccc gccgccagct caccatggat                                     90

SEQ ID NO: 72      moltype = RNA length = 41
FEATURE           Location/Qualifiers
source            1..41
                  mol_type = other RNA
                  organism = Homo sapiens

SEQUENCE: 72
ctagcttttc tctctgtgca accccacacg cctttggcac a                                     41

SEQ ID NO: 73      moltype = RNA length = 9305
FEATURE           Location/Qualifiers
source            1..9305
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 73
gcggcgcatg agagaagccc agaccaatta cctacccaaa atggagaaa ttcacgttga 60
catcgaggaa gacagcccat tctcagagc tttgcagcg agcttcccgc agtttgaggt 120
agaagccaa gaggctcactg ataataacca tgctaatagcc agagcgtttt cgcactctggc 180
ttcaaaactg atcgaaacgg aggtggaccc atccgacacg atccttgaca ttggaagtgc 240
gcccccccgc agaattgtatt ctaagcacaa gtatcattgt atctgtccga tgagatgtgc 300
ggaagatccg gacagattgt ataagtatgc aactaagctg aagaaaaact gtaaggaaat 360
aactgataag gaattggaca aaaaaatgaa ggagctggcc gccgtcatga gcgaccctga 420
cctggaacct gagactatgt gccctccacg cgacgagtcg tgtcgctacg aagggcaagt 480
cgctgtttac caggatgtat acgcggttga cggaccgcga agtctctatc accaagccaa 540
taaggaggtt agagtgcctt actggatagg ctttgacacc accccttcta tgtttaagaa 600
cttggtctga gcatatccat catactctac caactggggc gacgaaacgg tgttaacggc 660
tcgtaacata ggctatgca gctctgacgt tatggagcgg tcacgtagag ggaatgtccat 720
tcttagaaag aagtatttga aaccatccaa caatgttcta tctctgtgtg gctcgacct 780
ctaccacgag aagagggact tactgaggag ctggcacctg ccgtctgtat ttcaactacg 840
tggcaagcaa aattacacat gtcggtgtga gactatagtt agttgcgacg ggtacgtcgt 900
taaaagaata gctatcagtc caggcctgta tgggaagcct tcaggctatg ctgctacgat 960
gcaccgcgag ggtattctgt gctgcaaaat gacagacaca ttgaaacggg agaggggtctc 1020
ttttcccgct tgcacgtatg tgcagctac attgtgtgac caaatgactg gcatactggc 1080
aacagatgtc atgtcggagc agcgcacaaa actgctgggt gggctcaacc agcgtatagt 1140
cgtcaacggt cgcacccaga gaaacaccaa taccatgaaa aattaccttt tgcccgtagt 1200
ggcccaggca tttgctaggt gggcacaagg atataaggaa gatcaagaag atgaaaggcc 1260
actaggacta cgagatagac agttagtcat ggggtgttgt tgggcttcta gaaggcacaa 1320
gataacatct atttataagc gcccggtatc ccaaaccatc atcaaaagtg acagcgattt 1380
ccactcattc gtgctgcccga ggataggcag taacacattg gagatcgggc tgagaacaag 1440
aatcaggaaa atgttagagg agcacaaggga gccgtcacct ctcattacgc cggaggacgt 1500
acaagaagct aagtcgcgag ccgatgaggg taaggaggtg cgtgaagccg aggagtttgcg 1560
cgcagctcta ccactcttgg cagctgatgt tgaggagccc actctggaag ccgatgtcga 1620
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acactctggc cgaaaagggc gttatgccgt ggaaccatac catggtaaa tagtggtgcc 1860
agagggacat gcaatacccg tccaggactt tcaagctctg agtgaaaagt ccaccattgt 1920
gtacaacgaa cgtgagttcg taaacaggta cctgcacct attgccacac atggaggagc 1980
gctgaacact gatgaagaat attacaaaac tgtcaagccc agcgagcacg acggcgaata 2040
cctgtacgac atcgacagga aacagtgcgt caagaaagaa ctagtactg ggctagggct 2100
cacagggcga ctggtggatc ctcccttcca tgaattcgcc tacgagagtc tgagaacacg 2160
accagccgct ctttaccacg taccacacat aggggtgtat ggcgtgccag gatcaggcaa 2220
gtctggcatc attaaaagcg cagtcaccaa aaaagatcta gtggtgagcg ccaagaaaga 2280
aaactgtgca gaaattataa gggacgtcaa gaaaatgaaa gggctggacg tcaatgccag 2340
aactgtggac tcagtgtctt tgaatggatg caaacacccc gtagagaccc tgtatatgta 2400
cgaaagctttt gcttgtcatg caggtaactct cagagcgctc atagccatta taagacataa 2460
aaaggcagtg ctctgcgggg atcccaaaaca gtgcggtttt ttaaacatga tgtgcctgaa 2520
agtgcatttt aaccacgaga tttgcacaca agtcttccac aaaagcatct ctgcgcgttg 2580
cactaaatct gtgacttcgg tctgtccaac cttgttttac gacaaaaaaa tgagaacgac 2640
gaatccgaaa gagactaaga ttgtgattga cactaccggc agtaccacaa ctaagcagga 2700
cgatctcatt ctactctgtt tcagaggggtg ggtgaagcag ttgcaaatag attacaaagg 2760
caacgaaata atgacggcag ctgcctctca agggctgacc cgtaaagggt tgtatgccgt 2820
tcggtaacaag gtgaatgaaa atcctctgta cgcacccacc tcagaacatg tgaacgtcct 2880
actgaccgcg acggaggacc gcactcgtgt gaaaacaccta gccggcgacc catggataaa 2940
aacactgact gccaaagtacc ctgggaattt cactgccacg atagaggagt ggcaagcaga 3000
gcatgatgcc atcatggagc acatcttggg gagaccggac cctaccgacg tcttccagaa 3060
taaggcaaac gtgtgttggg ccaaggcttt agtgccggtg ctgaagaccg ctggcataga 3120

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catgaccact	gaacaatgga	acactgtgga	ttattttgaa	acggacaaag	ctcactcagc	3180
agagatagta	ttgaaccaac	tatgctgtgag	gttcttttgga	ctcgatctgg	actccggtct	3240
atthttctgca	cccactgttc	cgttatccat	taggaataat	cactgggata	actccccgtc	3300
gcctaactatg	tacgggctga	ataaagaagt	ggtcctgcag	ctctctcgca	ggtacccaca	3360
actgcctcgg	gcagtttccca	ctggaagagt	ctatgacatg	aacactggta	cactgcgcaa	3420
ttatgatccg	cgcataaacc	tagtacctgt	aaacagaaga	ctgcctcatg	ctttagtcct	3480
ccaccataat	gaacacccac	agagtgaact	ttcttcattc	gtcagcaaat	tgaagggcag	3540
aactgtcctg	gtggtcgggg	aaaagttgtc	cgtcccaggc	aaaatggttg	actggttgtc	3600
agaccggcct	gaggtacact	tcagagctcg	gctggattta	ggcatcccag	gtgatgtgcc	3660
caaatatgac	ataatatttg	ttaatgtgag	gaccccatat	aaataccatc	actatcagca	3720
gtgtgaagac	catgocatta	agcttagcat	gttgaccaag	aaagcttgct	tgcatctgaa	3780
tcccggcgga	acctgtgtca	gcataggtta	tggttacgct	gacagggcca	gcgaaagcat	3840
cattgggtgct	atacgcgggc	agttcaagtt	ttcccgggta	tgcaaaccca	aatcctcact	3900
tgaagagagc	gaagttctgt	ttgtattcat	tgggtacgat	cgaaggccc	gtacgcacaa	3960
ttcttacaag	ctttcatcaa	ccttgaccaa	catttataca	ggttccagac	ttccacgaagc	4020
cggatgtgca	cctcatatc	atgtggtgcg	aggggatatt	gccacggcca	ccgaaggagt	4080
gattataaat	gctgctaaac	gcaaaaggaca	acctggcgga	ggggtgtgcg	gagcgctgta	4140
taagaaattc	ccggaaagct	tcgatttaca	gccgatcgaa	gtaggaaaag	cgcgactggt	4200
caaagggtgca	gctaaccata	tcattcatgc	cgtaggacca	aaattcaaca	aagtttcgga	4260
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FEATURE

Location/Qualifiers

source

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taaggcgac acggaatgt tgaatactca tactcttctt tttcaatat tattgaagca 2160
tttatcaggg ttattgtctc atgagcggat acatatttga atgtatttag aaaaataaac 2220
aaataggggt tccgcgcaca tttccccgaa aagtgccacc tgagctctaa tacgactcac 2280
tataggacat ttgcttctga cacaactgtg ttcactagca acctcaaaac gacacctagg 2340
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tgaccacctt gacctacggc gtgcagtgct tcagccgcta ccccgaccac atgaagcagc 2580
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aggacgacgg caactacaag acccgcgcgc aggtgaagtt cgagggcgac accctgggtga 2700
accgcatcga gctgaaggcg atcgacttca aggaggacgg caacatctcg gggcacaagc 2760
tggagtacaa ctacaacagc cacaacgtct atatcatggc cgacaagcag aagaacggca 2820

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tcaaggtgaa	cttcaagatc	cgccacaaca	tcgaggacgg	cagcgtgcag	ctcgcgcgacc	2880
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tgagcaccca	gtccgcctcg	agcaaaagacc	ccaacgagaa	gcgcgatcac	atggtcctgc	3000
tggagtctcg	gaccgcccgc	gggatcactc	tcggcatgga	cgagctgtac	aagtag	3056

SEQ ID NO: 100 moltype = DNA length = 3989
 FEATURE Location/Qualifiers
 source 1..3989
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 100

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taaaactggg	gatattatga	agggccttga	gcactctggat	tctgcctaat	aaaaaacatt	120
tattttcatt	gcaaaagccac	accctggagc	tagcaaaaaa	aaaaaaaaaa	aaaaaaaaaa	180
aaaagcatat	gactaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	240
aaaaaaaaaa	aaaaaaaaaa	aaaacgcgta	tcgatgatat	cgcgcccgca	tacagcagca	300
attggcaagc	tgtttacata	gaactcgcgg	cgattggcat	gccgccttaa	aattttttatt	360
ttatttttct	tttcttttcc	gaatcggatt	ttgtttttta	tattttcaaa	aaaaaaaaaa	420
aaaaaaaaaa	aacgcgtgat	aacgcaggaa	agaacatgtg	agcaaaaggc	cagcaaaagg	480
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tgttgagatc	cagttcgatg	taaccacact	gtgcacccaa	ctgatcttca	gcacttttta	2040
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tcggttaagg	agtaccattt	ttcgaggcaa	agggtggtcg	cttggatacc	ggaaaaacc	3480
tcggtgtgaa	tcagagaggg	gaattgtgtg	tcaggggtcc	aatgatcatg	agcggttacg	3540
tcaataatcc	cgaggccaact	aacgccttga	ttgataaaga	tggttggctt	cattcagggtg	3600
atatagcata	ctgggatgag	gatgaacatt	tcttctattg	cgataggctc	aaaagcctga	3660
tcaagtataa	gggttaccac	gtggccctcg	ccgaactcga	atctatactg	ctgcaacatc	3720
caaacatttt	tgacgtggc	gtggccggac	tgccccgatg	tgatgctggt	gaactccctg	3780
ccgcgctgtg	cgctcctggaa	catggtaaga	ccatgacaga	aaaggagatt	gtagattacg	3840
tggcttcaca	agtgactaca	gctaagaaat	tcgcccgggg	tgtcgttttc	gtggacgagg	3900

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tacccaagg	cctcactgg	aagctcgacg	ccagaaaaat	ccgagaaatc	ctgattaaag	3960
caaaagaagg	cggaagatt	gcagtgtag				3989

```

SEQ ID NO: 101      moltype = DNA length = 537
FEATURE            Location/Qualifiers
source              1..537
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 101
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gtgttggtta tagaccaggg taaccgcccc ctttttgaag atatgaccga tagcgattgc 180
cgcgacaaac cccccaggac catcttcata ataagtgcgt acggagactc acgggcccgg 240
gggaaggctg ttacaatatc cgtaaagtgc gagaagatct ccacgctttc ttgcgaaaat 300
aagattatca gttttaaaga aatgaaccct cctgacaaca taaaagatag gaaatcagac 360
atcatcttct tccagcgctc agttcctgga catgataata agatgcagtt cgagtcactc 420
agctatgaag ggtattttct ggcttgtag aaagaacggg atttggttaa acttatactg 480
aagaaggagg atgaacttgg tgatcggctc attatgttta ccgtacaaaa tgaggac 537

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SEQ ID NO: 102      moltype = DNA length = 537
FEATURE            Location/Qualifiers
source              1..537
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 102
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gtgttggtta tagaccaggg taaccgcccc ctttttgaag atatgaccga tagcgattgc 180
cgcgacaaac cccccaggac catcttcata ataagtgcgt acggagactc acgggcccgg 240
gggaaggctg ttacaatatc cgtaaagtgc gagaagatct ccacgctttc ttgcgaaaat 300
aagattatca gttttaaaga aatgaaccct cctgacaaca taaaagatag gaaatcagac 360
atcatcttct tccagcgctc agttcctgga catgataata agatgcagtt cgagtcactc 420
agctatgaag ggtattttct ggcttgtag aaagaacggg atttggttaa acttatactg 480
aagaaggagg atgaacttgg tgatcggctc attatgttta ccgtacaaaa tgaggac 537

```

```

SEQ ID NO: 103      moltype = RNA length = 550
FEATURE            Location/Qualifiers
source              1..550
                    mol_type = other RNA
                    organism = unidentified

```

```

SEQUENCE: 103
acgttactgg ccgaagccgc ttggaataag gccggtgtgc gtttgtctat atgttatttt 60
ccaccatatt gccgtctttt ggcaatgtga gggcccgga accctggccct gtcttcttga 120
cgagcattcc taggggtctt tccccctcgc ccaaaggaat gcaaggtctg ttgaatgtcg 180
tgaagggaagc agttcctctg gaagcttctt gaagacaaac aacgtctgta gcgacccttt 240
gcaggcgagcg gaacccccca cctggcgaca ggtgcctctg cggccaaaag ccacgtgtat 300
aagatacacc tgcaaaaggcg gcacaacccc agtgccacgt tgtgagttgg atagtgtggt 360
aaagagtcac atggctctcc tcaagcgtat tcaacaaggg gctgaaggat gcccgagaagg 420
taccocattg tatgggatct gatctggggc ctcggtgcac atgctttaca tgtgtttagt 480
cgagggttaa aaaacgtcta ggccccccga accacgggga cgtgggtttc ctttgaaaaa 540
cacgataata                                     550

```

```

SEQ ID NO: 104      moltype = RNA length = 57
FEATURE            Location/Qualifiers
source              1..57
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 104
gcgatcgcgc tagcaacggt tcccccttag cgggatcaat tccgcccccc cccccta 57

```

```

SEQ ID NO: 105      moltype = RNA length = 57
FEATURE            Location/Qualifiers
source              1..57
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 105
gcgatcgcgc tagcaacggt tcccccttag cgggatcaat tccgcccccc cccccta 57

```

```

SEQ ID NO: 106      moltype = DNA length = 537
FEATURE            Location/Qualifiers
source              1..537
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 106
atgacctccc ggcttgtag ggtactggct gctgetatgc tgggtggctg tgctgtgagt 60
gtggcacact tcggacgact ccattgtact acagccgtta tacgaaatat caatgatcag 120

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gtgtgtgttg	tagacaaaag	gcagcccggtg	tttgaagaca	tgactgatat	tgatcagagt	180
gcctctgagc	cccagactag	actgattata	tatgcttatg	gcgactctag	agctagaggg	240
aaagcagtaa	ctctgtcagt	gaaagacagc	aagatgtcta	ccctgtcttg	caagaataaa	300
ataatctcat	ttgaggagat	ggacccccct	gaaaacatag	atgacatcca	aagtgaacctg	360
atattctttc	agaagcgggt	acccggtcat	aacaaaatgg	agttcgagtc	aagcctctat	420
gagggtcact	ttctcgctcg	ccaaaaggaa	gatgacgcct	tcaaattgat	cttgaaaaaa	480
aaagacgaga	atggtgataa	gtctgtcatg	ttcacactca	ctaatttgca	tcagtca	537

SEQ ID NO: 107 moltype = DNA length = 537

FEATURE Location/Qualifiers

source 1..537

mol_type = other DNA

organism = Mus musculus

SEQUENCE: 107

atgacctccc	ggcttgtgag	ggtactggct	gctgctatgc	tggtggctgc	tgctgtgagt	60
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gtctcttttg	tagacaagag	acagccagtt	ttcgaggaca	tgactgatat	agatcagagt	180
gcctctgagc	ctcaaacctcg	gtctcataata	tacatgtata	aggatagcga	ggttcgggga	240
cttgccgtga	ctttgtccgt	gaaagactca	aaaatgagca	ccctgtcatg	caaaaataag	300
ataatttctt	tcgaggaat	ggacccctcc	gagaacattg	atgatatcca	gtccgacttg	360
atatttttcc	agaaacgggt	tcccggtcat	aacaaaatgg	agtttgagtc	ttcattgtac	420
gagggtcatt	ttttggcatt	ccagaaggag	gatgatgcct	tcaaactgat	tttgaagaaa	480
aaagacgaga	acggagacaa	gtctgtcatg	tttacactta	ccaaccttca	tcaatcc	537

SEQ ID NO: 108 moltype = RNA length = 3735

FEATURE Location/Qualifiers

source 1..3735

mol_type = other RNA

organism = Sin Nombre orthohantavirus

SEQUENCE: 108

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caaattcaac	agctaaccac	agctgtcagt	gccctagtca	ttggtcaagc	caccgcacct	180
caacaaccaa	ggcaccgccc	gctaccacgc	cctaagaagc	agtcgccgaa	gcaaccaccg	240
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ggtaagagac	agcgtatggc	actaaaattg	atgctgacag	gactctttga	cgtgaacaac	360
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ggtcgcttta	cagtcgccgg	cggagttggg	ggaaaaggcg	acagtggaa	accgatcatg	660
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cctcatgcct	atacctcata	ccttagaggct	accctaggaa	aaattttacg	caaacaccaca	1560
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gagccgactg	taattctatg	gttcaaacac	ataagcctcc	atttagacac	agaccacca	1860
acctgtctca	ccattagaag	actggcgag	aaaccggaac	cgacgtcgga	gtggatctca	1920
ggaaaaacag	ttaggaaatt	tactgtcgac	agagacgggc	tagagtatat	atggggcaat	1980
cacgaaccgg	tgagagtata	tgacacaagag	tccgcccccg	gtgacctcca	cggatggcca	2040
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gactacaact	gcacaagtatt	tgccgcgctt	taccctttca	tgtgggggtg	agcgcaatgt	2700
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acatctaaag	acctgaaagt	cattgcccga	cccatctcgt	cgcggttttc	cccttttgac	2940
tctaaagtgc	tgatacaccg	agggtcctg	tacaattacg	attttccaga	gtacggggca	3000
atgaaaccgg	gtgcctttgg	ggacattcaa	gctagctctt	tgacgggcaa	cgacctgatt	3060
gccagcaccg	acatacgttt	gcttaacccc	gcagcaaaga	atgtgcacgt	tcctataacc	3120
caagcagcat	caggatttga	aatttggaag	aacaactccg	gaagaccact	gcaagagacg	3180
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agtaccgcca	cactccagga	atcaactgtt	cacgtcttgg	aaaaaggggc	gacgacagt	3480
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ggaggagctt	cgtccctagt	aatcatggga	ctgttaatct	tctcctgcag	catgatgtct	3720
actaacacca	gaaga					3735

1. A composition for cascade amplifications of therapeutic payloads (CATP) comprising:

- a first type nucleic acid construct encoding a self-amplifying mRNA (sa-mRNA) or multiple self-amplifying mRNA (sa-mRNA) encoding at least one gene of interest (GOI) or a plurality of GOIs;
- a second type nucleic acid construct encoding an mRNA encoding at least one virus structural protein; and
- at least one payload delivery system,

wherein the at least one payload delivery system is a non-viral payload delivery system and

wherein the payload is at least one nucleic acid construct.

2. The composition of claim 1, wherein the at least one virus structural protein encodes a viral capsid protein, a viral envelope protein, or a combination thereof.

3. The composition of claim 1, wherein the first nucleic acid construct or the second nucleic acid construct is a mRNA molecule.

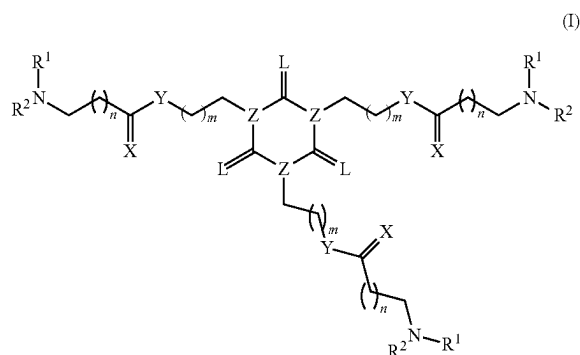
4. The composition of claim 1, wherein the first nucleic acid construct or the second nucleic acid construct is a DNA construct.

5. The composition of claim 1, wherein the first type nucleic acid construct is a sa-mRNA molecule or multiple sa-mRNA molecules.

6. The composition of claim 4, wherein the second nucleic acid construct is a mRNA molecule.

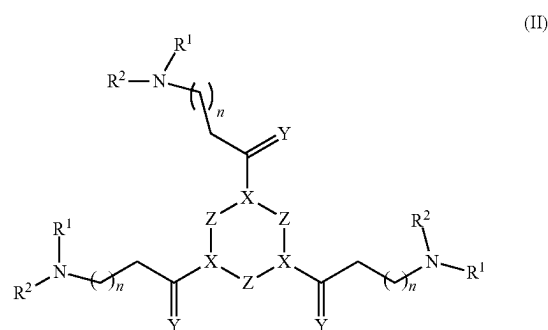
7. The composition of claim 1, wherein the non-viral payload delivery system is a lipidoid, a polymer, a core-shell nanoparticle, a nanoparticle mimic, a lipid nanoparticle (LNP), a polymeric nanoparticle, a micelle, liposome, an exosome, a virus-like particle (VLP), a lipoplex, a polyplex, a lipopolyplex, or a microbe.

8. The composition of claim 7, wherein the LNP comprises an ionizable lipid compound selected from Formulae I, II, IV, V, IV, and VII, wherein Formula I is



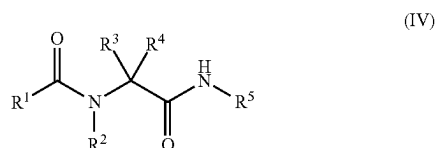
or a salt or isomer thereof;

wherein Formula II is



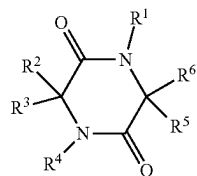
or a salt or isomer thereof;

wherein Formula IV is

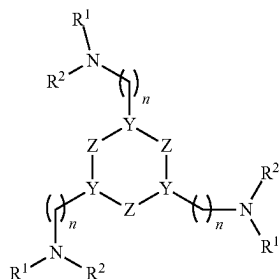


or a salt or isomer thereof.

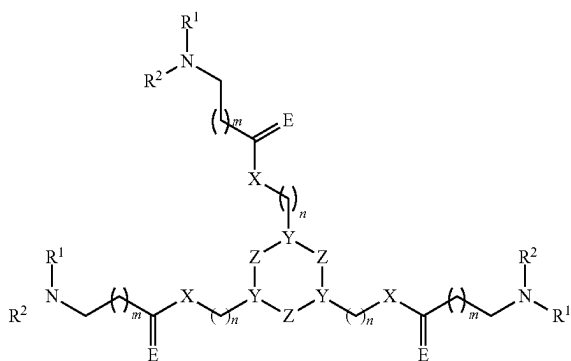
wherein Formula V is



or a salt or isomer thereof;
wherein Formula VI is



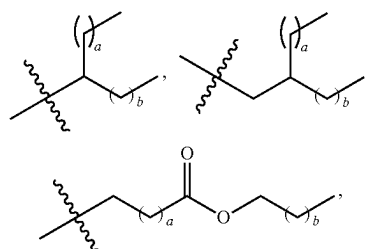
or a salt or isomer thereof;
wherein Formula VII is



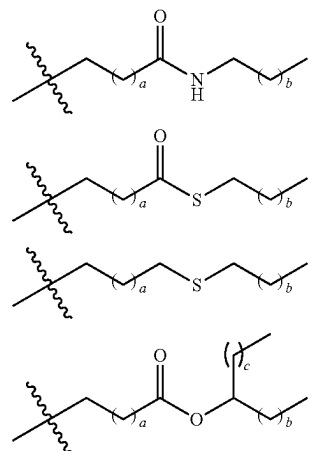
or a salt or isomer thereof;
wherein, each m and n are independently an integer from 0-10;

each R¹, R², R³, R⁴, R⁵ and R⁶ are independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,

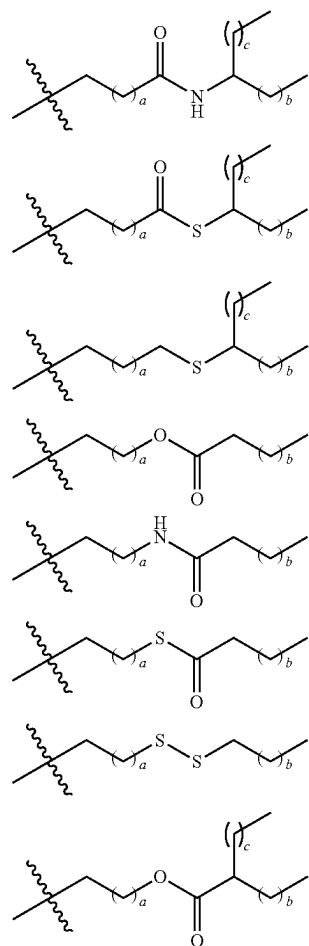
(V)



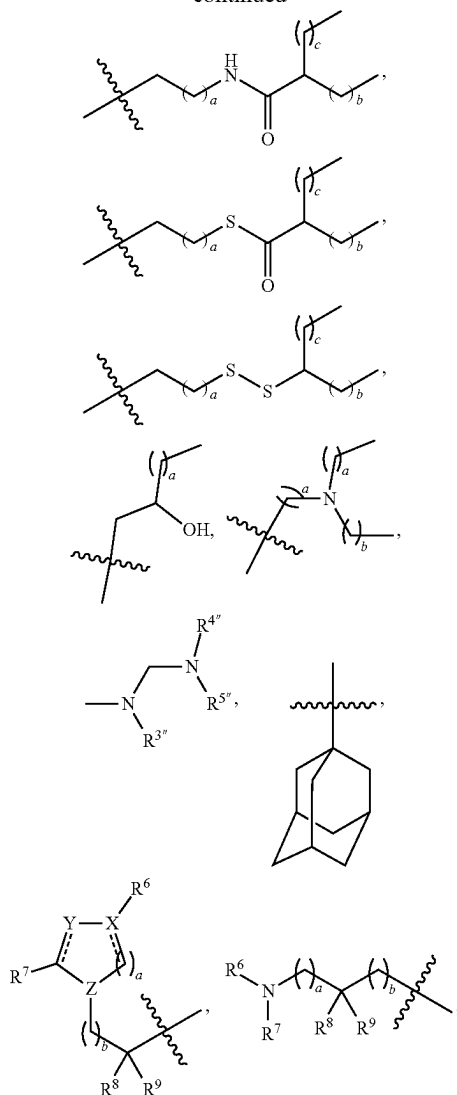
(VI)



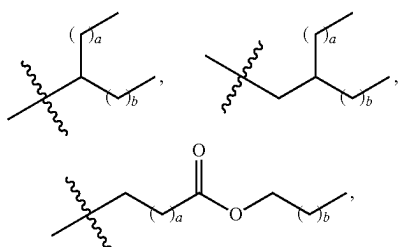
(VII)



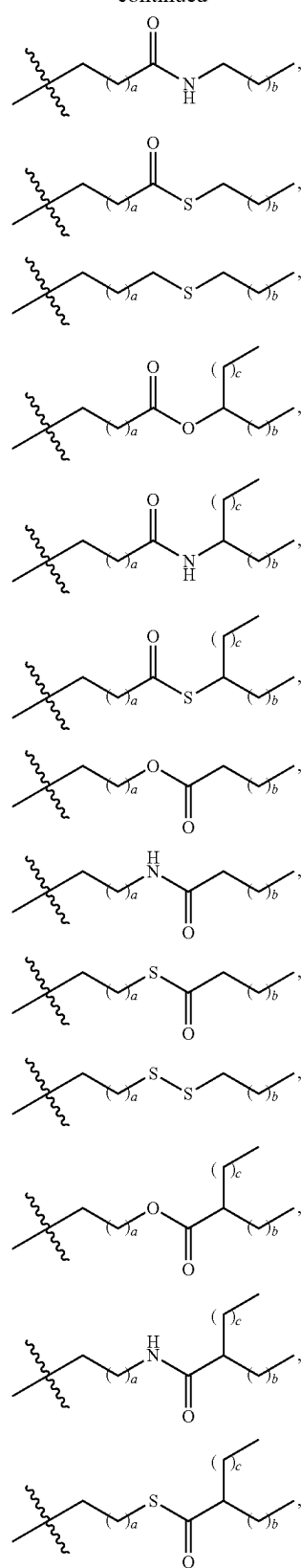
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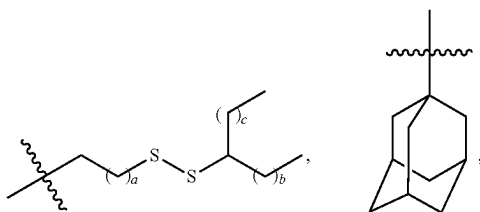
a, b and c are each independently an integer from 0-24;
 each R^6 , R^7 , R^8 and R^9 are independently selected from H, C1-C24 alkyl, C1-C24 alkenyl, C1-C24 alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



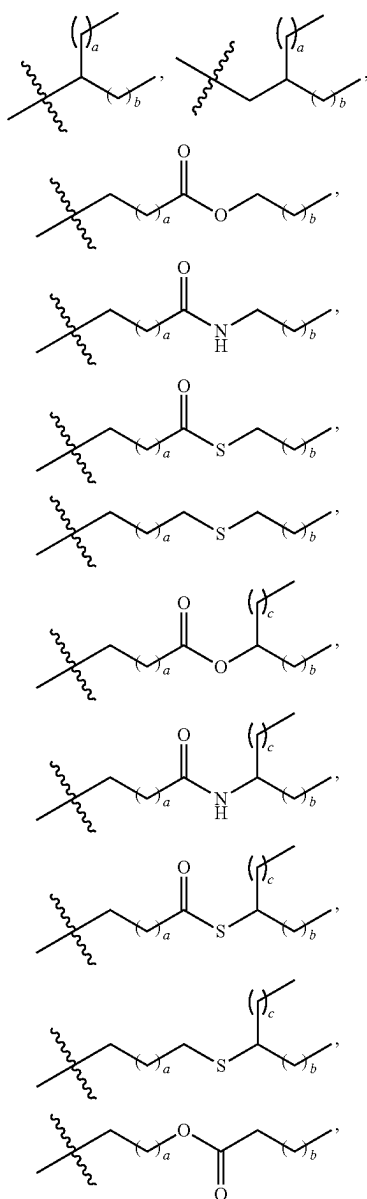
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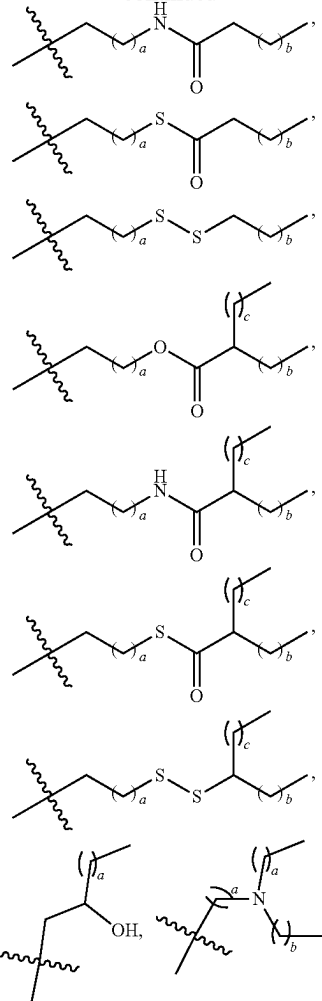
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each R^3 , R^4 and R^5 are independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



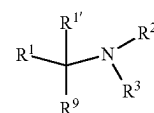
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a, b and c are each independently an integer from 0-24; each X is independently selected from CH_2 , NH, O, or S; each Y is independently selected from CH_2 , NH, O, or S; each Z is independently selected from CH, N, CH_2 , NH, O, or S; and

each E is independently selected from CH_2 , NH, O, or S.

9. The composition of claim 7, wherein the LNP comprises an ionizable lipid compound selected from Formula III, wherein Formula III is



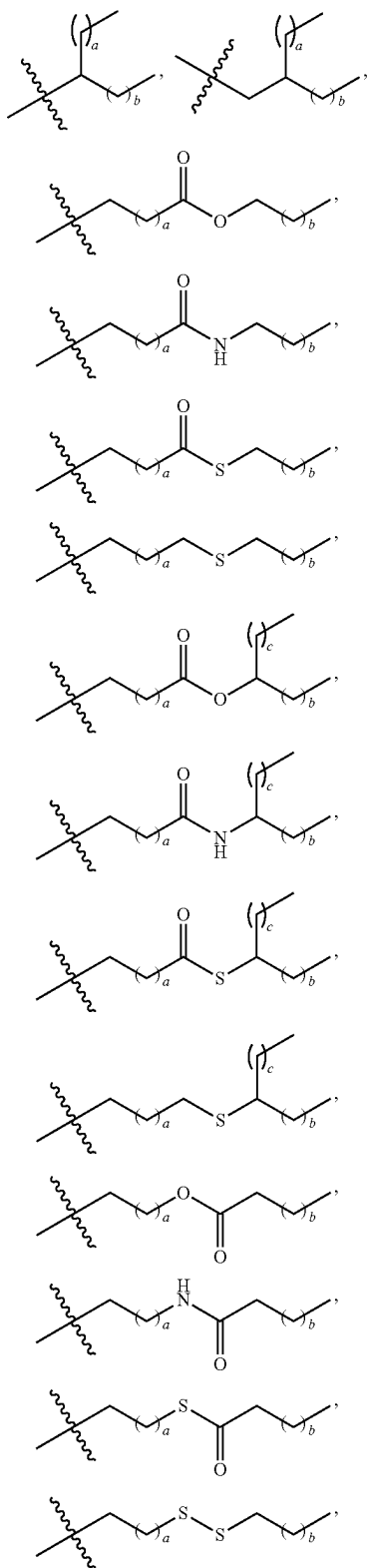
(III)

or a salt or isomer thereof, wherein

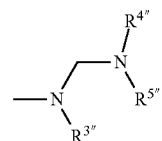
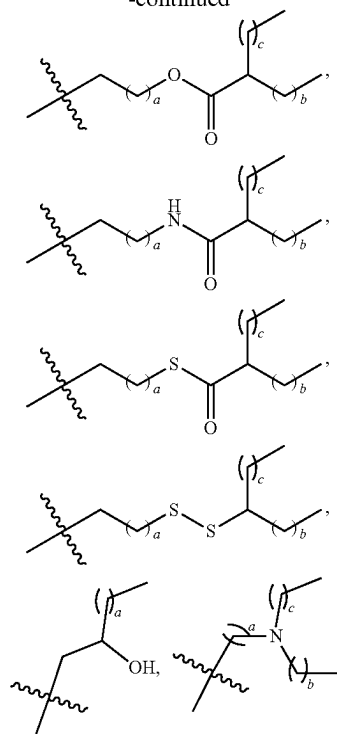
each n is independently an integer from 0-10;

each R^1 , $R^{1'}$, R^2 and R^3 are independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl,

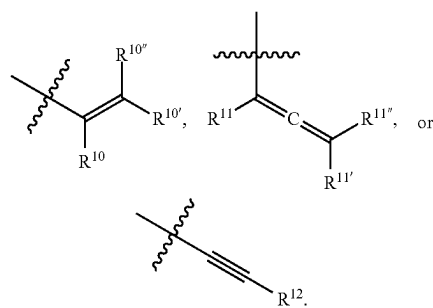
substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



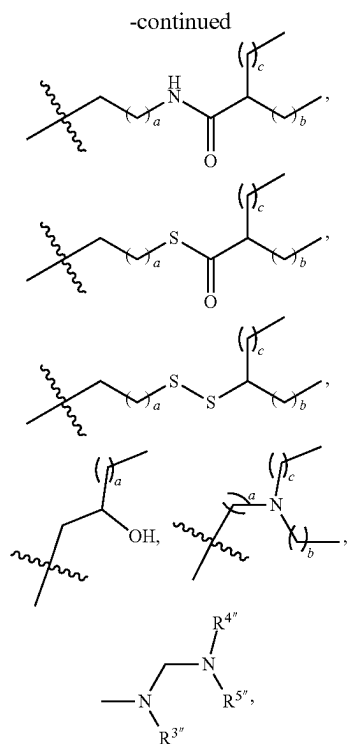
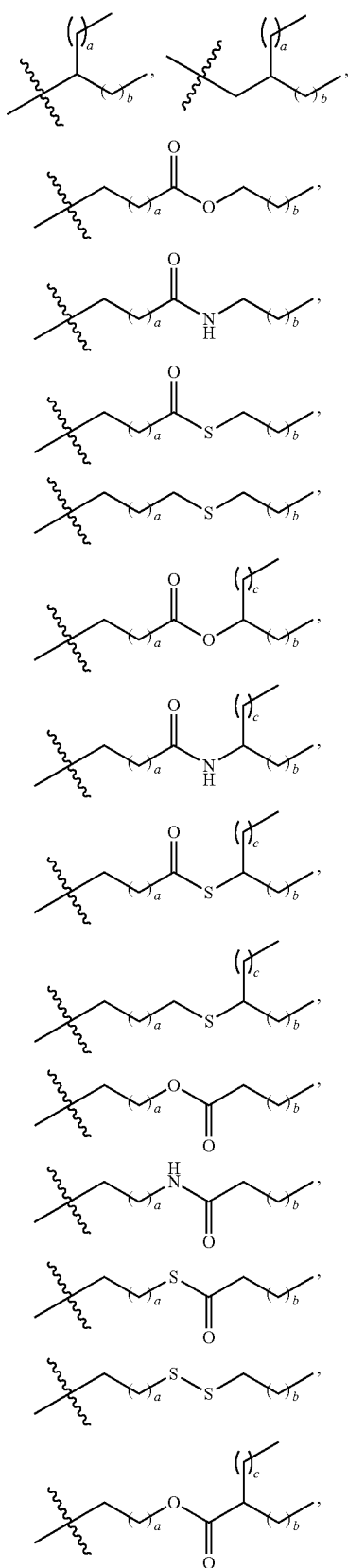
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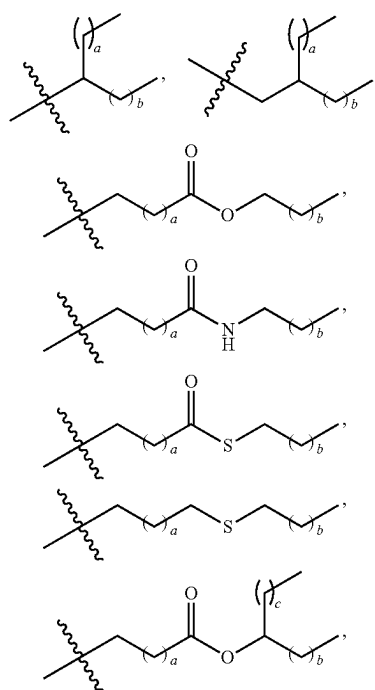
a, b and c are each independently an integer from 0-24,
each R^9 is independently selected from

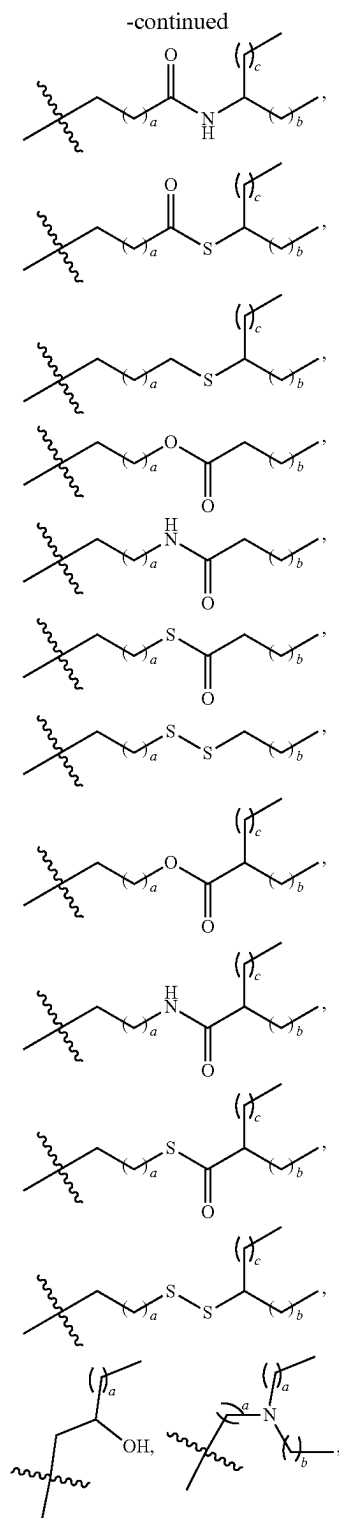


wherein each R^{10} , $R^{10'}$, $R^{10''}$, R^{11} , $R^{11'}$, $R^{11''}$, and R^{12} are independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



a, b and c are each independently an integer from 0-24;
 $R^{3''}$, $R^{4''}$, and $R^{5''}$ are independently selected from H,
 C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted
 alkyl, substituted alkenyl, substituted alkynyl,
 substituted acyl, substituted carbocyclyl, substituted
 heterocyclyl, substituted aryl, substituted heteroaryl,





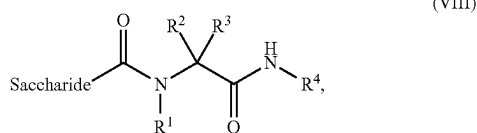
a, b and c are each independently an integer from 0-24.

10. The composition of claim 7, wherein the LNP comprises an ionizable lipid compound, wherein the ionizable lipid is (Ia), (Ib), (Iia), (Iib), (IIIa), (IIIb), (IIIc), (IIId), (IIIe), (IIIf), (IIIg), (IIIh), (IIIi), (IIIk), (IIIl), (IIIm), (IIIn), (IIIo),

(IIIp), (IIIq), (IIIr), (IIIs), (IIIt), (IIIu), (IIIv), (IIIw), (IIIx), (IIIy), (IIIz), (III2), (III3), (III4), (IVa), (IVb), (IVc), (IVd), (IVe), (IV), (IVg), (IVh), (IVi), (IVk), (IVl), (IVm), (IVn), (IVo), (IVp), (IVq), (IVr), (IVs), (IVt), (IVu), (IVv), (IVw), (IVx), (IVy), (IVz), (IVaa), (IVab), (IVac), (IVad), (IVae), (Va), (Vb), (VIa), (VIb), (VIc), (VIIa), (VIIb), (P501C1), (P501G3), (P501G6), (P501C1), (P501G3), (P501G6), (HP9E9), (HP9G3), (HP9G4), (HP9G6), (HP9G7), (HP9G8), (HP9G9), (HP9G10), (HP10G6), (HP11E7), (HP11E8), (HP11G4), (HP12F4), (HP12F6), (HP12G7), (HP13B9), (HP13C6), (HP13F10), (HP13G8), (HP15A4), (HP15C2), (HP15C4), (HP15E3), (HP15E4), (HP15E6), (HP15E7), (HP15E8), (HP15G5), (HP16B6), (HP16C4), (HP16C6), (HP16G6), (HP17E7), (HP17G6), (HP18E7), (HP19E2), (HP19E6), (HP19E8), (HP19G6), (HP19G7), (HP19G8), (HP19G9), (HP19G10), (HP20A10), (HP20F7), (HP20F8), (HP20F10), (HP20G3), (HP20G10), (P502B9), (P502D5), (P502D10), (P502F10), (P502H5), (P502H7), (P502H9), (P502H10), (P502H11), (P503B7), (P503B9), (P503B10), (P503H7), (P503H10), (P504B10), (P504D4), (P504D5), (P504D7), (P504D9), (P504D11), (P504F9), (P504F10), (P504H4), (P504H7), (P504H9), (P505B4), (P505B7), (P505B9), (P505B10), (P505C10), (P505D4), (P505D7), (P505D9), (P505D10), (P505D11), (P505H7), (P505H9), (P506D7), (P506D10), (P509D5), (P509D7), (P509D9), (P509D10), (P510B2), (P510B6), (P510D1), (P510D2), (P510D3), (P510D4), (P510D5), (P510D6), (P510D7), (P510D9), (P510D10), (P510D11), (P510D12), (P510F7), (P510H3), (P510H4), (P510H5), (P510H7), (P510H11), (P510H12), (P511D1), (P511D2), (P511D3), (P511D7), (P511D11), (P511D12), (P511F5), (P511F6), (P511F7), (P511F11), (P511F12), (P511H1), (P511H2), (P511H3), (P511H5), (P511H6), (P511H7), (P511H9), (P511H11), (P511H12), (P512A9), (P512A10), (P512A11), (P512A12), (P513A9), (P513A10), (P514A1), (P514A2), (P514A3), (P514B1), (P514B2), (P514A7), (P514A9), (P514A10), (P514B8), (P514B9), (P515A11), (P515A12), (P515B10), (P515B11), (P515B12), (P516A7), (P516A8), (P516A9), (P516B7), (P516B9), (P517A7), (P518A7), (P518A9), (P518A10), (P518B7), (P518B8), (P518B9), (P518B11), (P519A9), (P519B7), (P519B8), (P519B9), (P520B7), (P520B8), (P520B9), (P520B10), (P521B9), (P521B10), (P521B11), (P521B12), (P522B9), (P522B10), (P522B11), (P522B12), (P536B3), (P536B4), (P537B2), (P537B3), (P537B4), (P538B2), (P538B3), (P538B4), (P538C2), (P540B2), (P540B3), (P540B4), (P541B2), (P541B3), (P541B4), (P541B9), (P542B2), (P542B3), (P542B4), (P542B9), (P543B3), (P543B4), (P544B3), (P544B4), (P551B8), (P551B9), (P554B9), (P554B10), (P556A11), (P556B3), (P556B9), (P558A11), (P558B9), (P559A11), (P559B3), (P559B4), (P559B9), (P559B10), (P560B3), (P560B4), (P560B5), (P560B9), (P560B10), (P560B11), (P561A11), (P561B4), (P561B5), (P561B9), (P562A11), (P562B3), (P562B4), (P562B5), (P562B9), (P563A5), (P563A11), (P563B3), (P563B4), (P563B5), (P563B9), (P563B10), (P563B11), (P564A11), (P564B2), (P564B3), (P564B4), (P564B5), (P564B9), (P564B10), (P565B3), (P565B4), (P565B5), (P565B9), (P565B10), (P565B11), (P566B3), (P566B4), (P566B10), (P569A9), (P569A10), (P569B10), (P570A8), (P570A9), (P570A10), (P571A8), (P571A9), (P571B2), (P572A9), (P572A10), (P572B2), (P573A8), (P573A9), (P573A10), (P573B2), (P573B4), (P573B9), (P573B10), (P574A8), (P574A9), (P574A10), (P574B2), (P574B3), (P574B9), (P574B10),

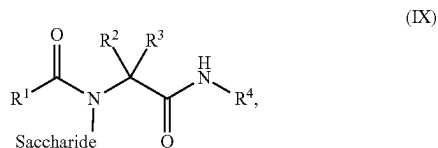
(P575A9), (P575A10), (P575B9), (P575B10), (P576A4), (P576A8), (P576A9), (P576A10), (P576B3), (P576B4), (P576B9), (P576B10), (P577B3), (P577B4), (P584B3), (P584B4), (P585B3), (P586B3), (P586B4), (P589B3), (P589B4), (P590B4), (P591B2), (P591B3), (P591B4), (P591E4), (P592A4), (P592B2), (P592B3), (P592B4), (P592B5), (P592E3), (P593B2), (P593B3), (P593B4), (P593B5), (P593E2), (P593E9), (P593H3), (P593H4), (P594B10), (P594B5), (P595B2), (P595B3), (P595B4), (P595B5), (P595E2), (P595E3), (P595E4), (P595H4), (P596B2), (P596B3), (P596B4), (P596B5), (P596E3), (P596E4), (P596H3), (P597A4), (P597B2), (P597B3), (P597B4), (P597B5), (P597B6), (P597D4), (P597E2), (P597E3), (P597E4), (P597H4), (P598B1), (P598B2), (P598B3), (P598B10), (P598B5), (P598E2), (P598E3), (P598E4), (P598H3), (P598H4), (P599B2), (P599B3), (P599B4), (P617F12), (P618H5), (P623A5), (P623B8), (P625A6), (P625A9), (P625F8), (P625F9), (P625H8), (P627E10), or salt or isomer thereof.

11. The composition of claim 7, wherein the LNP comprises a modular lipid compound selected from Formulae VIII-IX, wherein Formula VIII is



or a salt or isomer thereof;

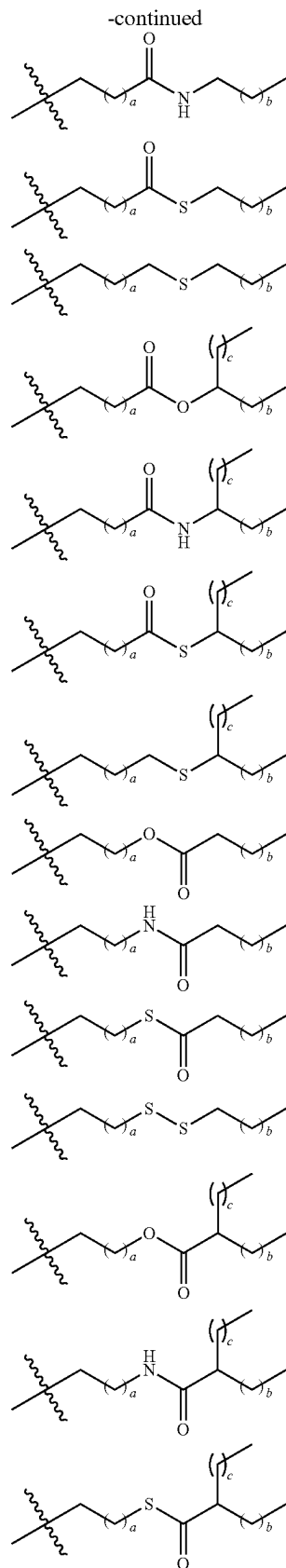
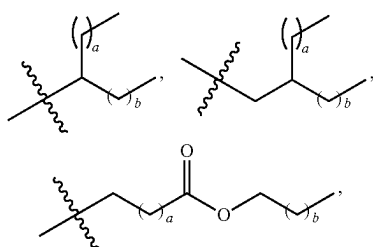
wherein Formula IX is

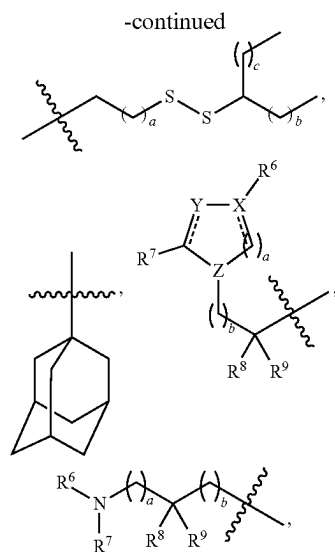


or a salt or isomer thereof;

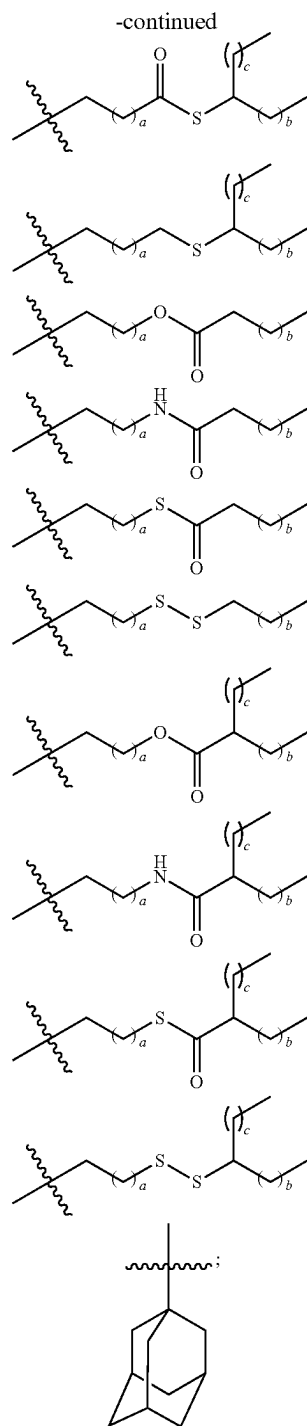
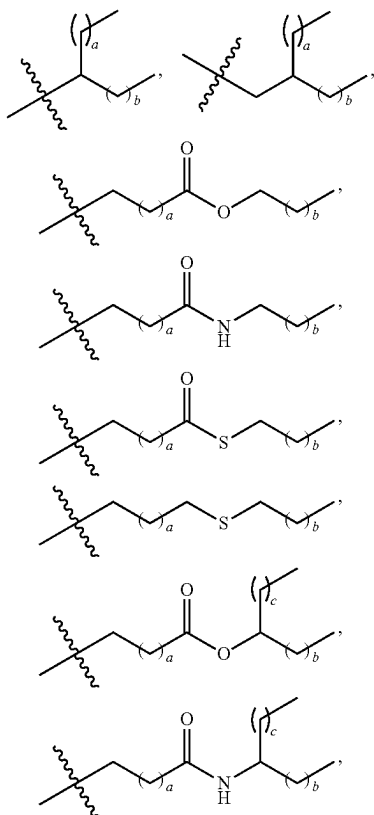
wherein

each of R¹, R², R³ and R⁴ is independently selected from H, C1-C24 alkyl, C1-C24 alkenyl, C1-C24 alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG),



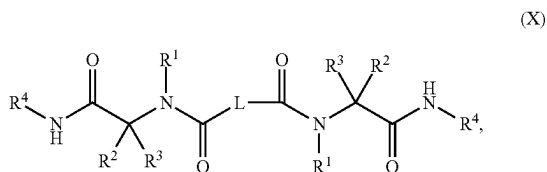


a, b and c are each independently an integer from 0-24;
each of R^6 , R^7 , R^8 and R^9 is independently selected from H, C1-C24 alkyl, C1-C24 alkenyl, C1-C24 alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,



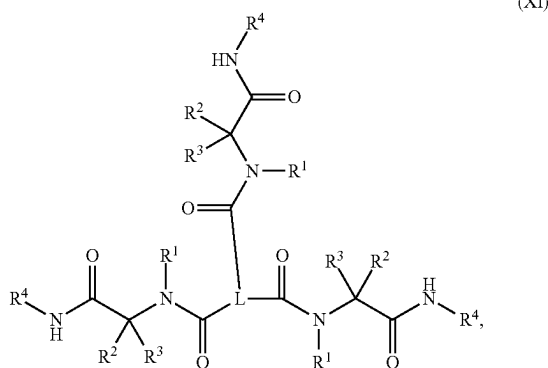
a, b and c are each independently an integer from 0-24;
each X is independently selected from CH or N;
each Y is independently selected from CH_2 , NH, O, or S;
each Z is independently selected from CH or N;
each saccharide is independently selected from monosaccharides, disaccharides, oligosaccharides, and polysaccharides.

12. The composition of claim 7, wherein the LNP comprises a modular lipid compound selected from Formulae X-XX, wherein Formula X is



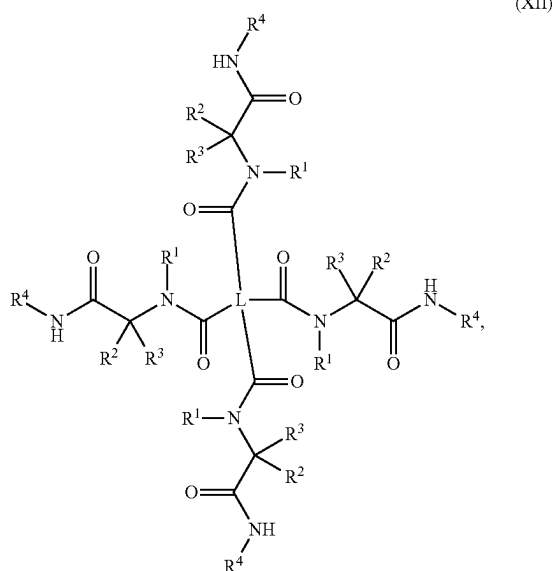
or a salt or isomer thereof;

wherein Formula XI is



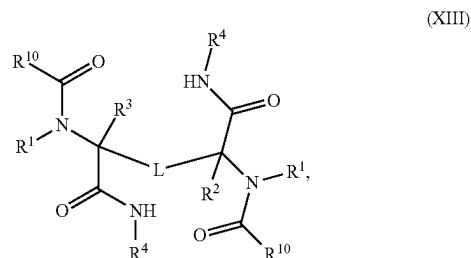
or salt or isomer thereof;

wherein Formula XII is



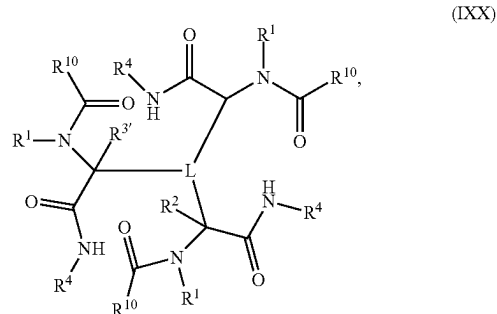
or salt or isomer thereof.

wherein Formula XIII is



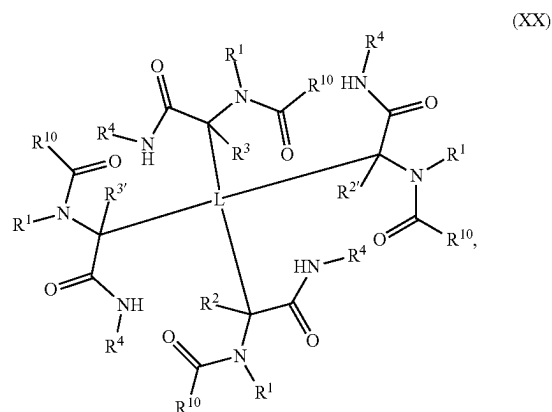
or salt or isomer thereof;

wherein Formula IXX is



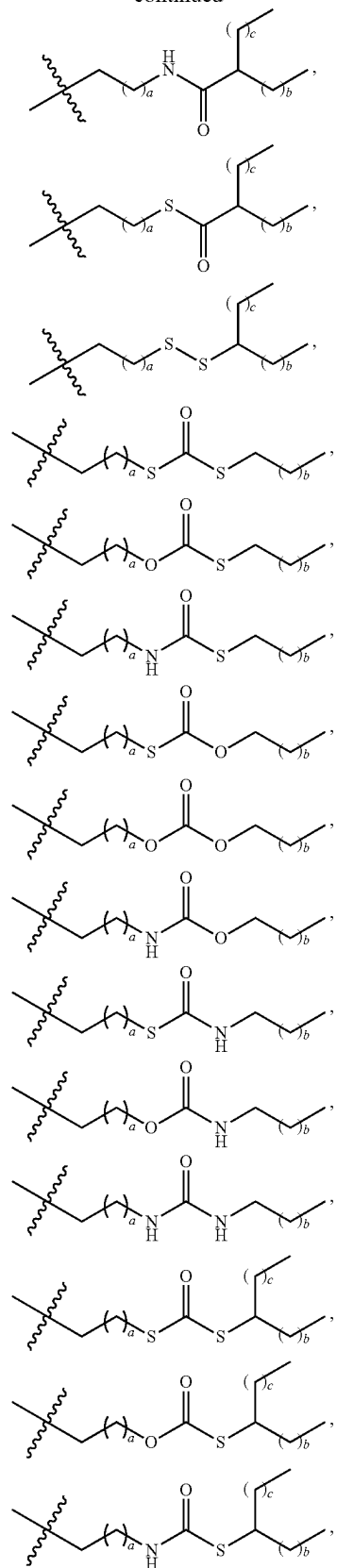
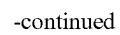
or a salt or isomer thereof,

wherein Formula XX is

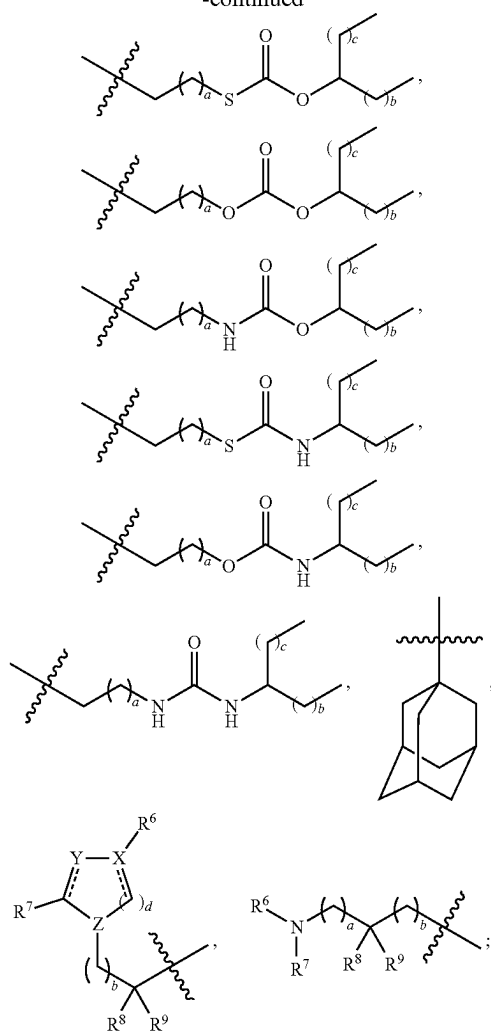


or a salt or isomer thereof;

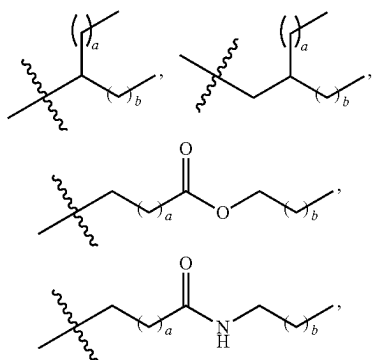
wherein each R^1 , R^4 , and R^{10} , is independently selected from C_2 - C_{24} alkyl, C_2 - C_{24} alkenyl, C_2 - C_{24} alkynyl, substituted C_2 - C_{24} alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG),



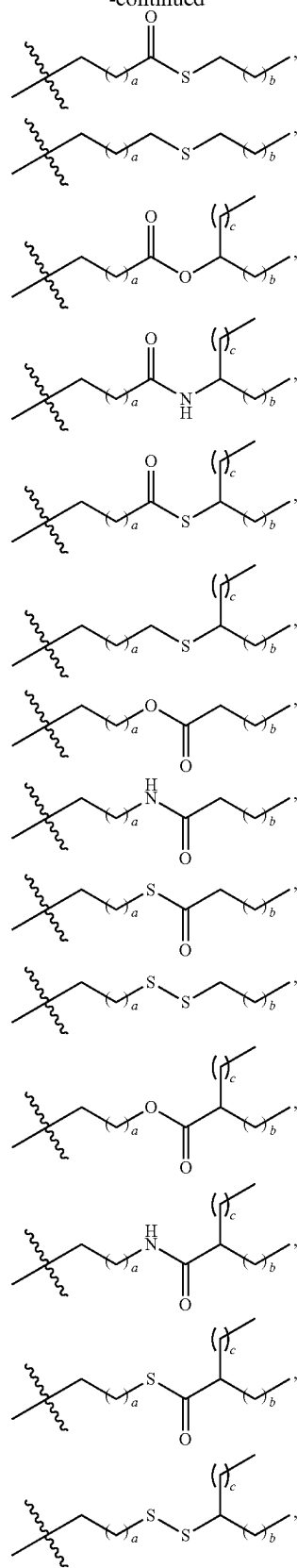
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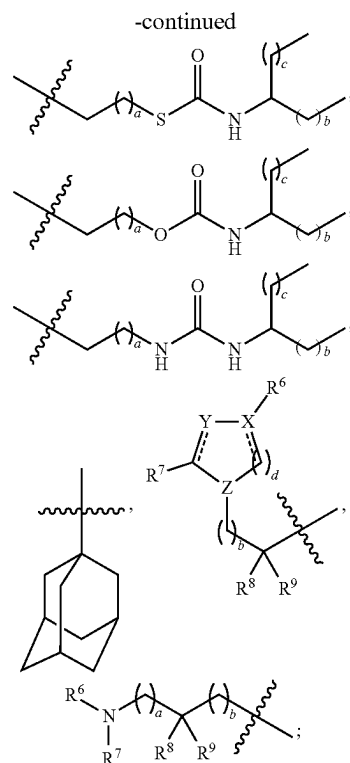
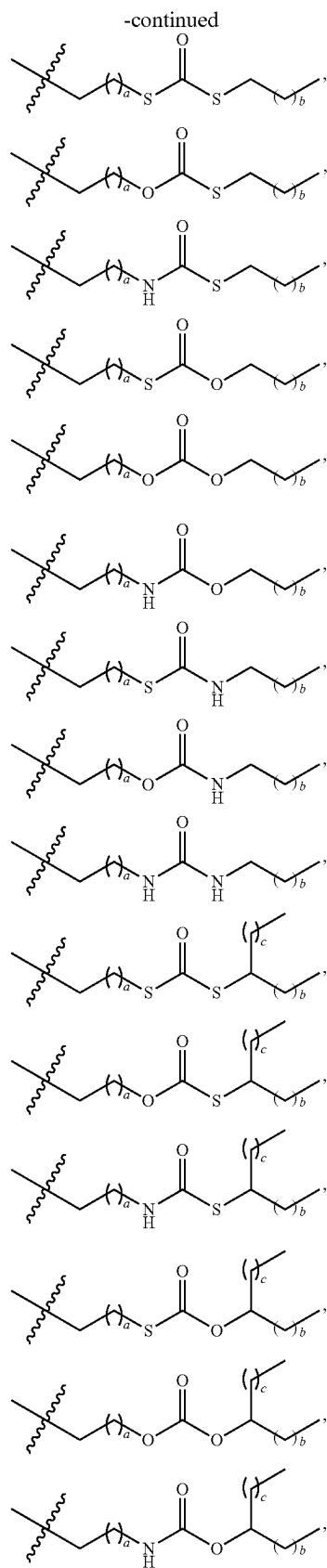


wherein each R^2 , $R^{2'}$, R^3 and $R^{3'}$ is independently selected from H, C_1 - C_{24} alkyl, C_1 - C_{24} alkenyl, C_1 - C_{24} alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG) and

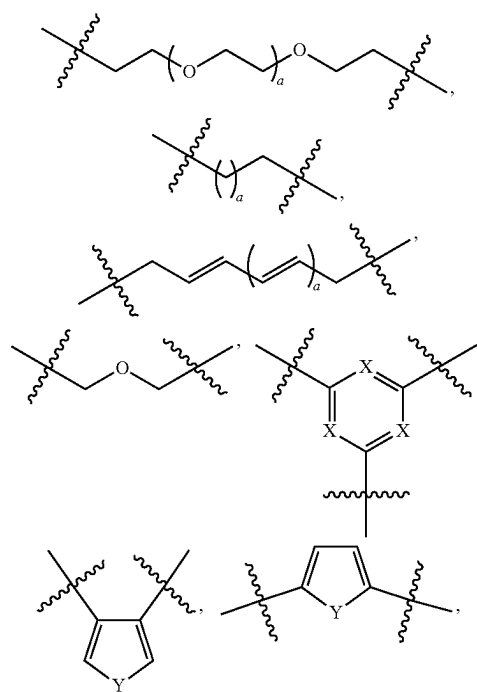


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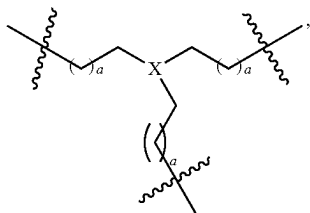
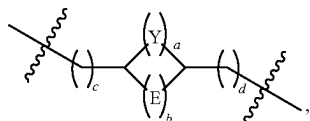
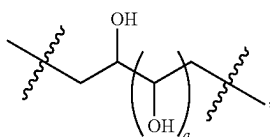
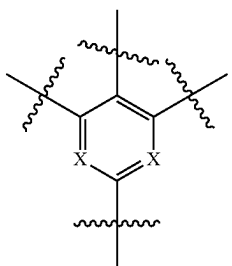
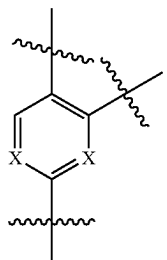
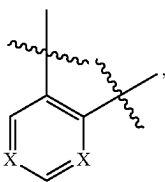
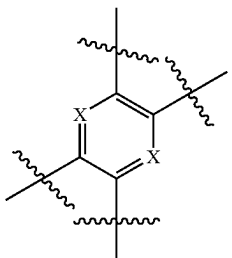
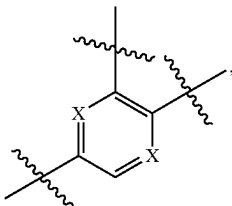
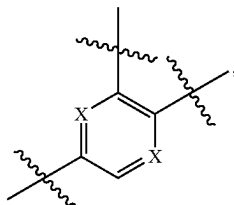
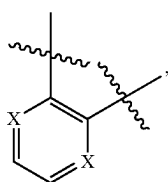
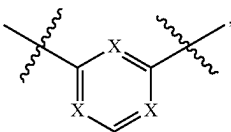
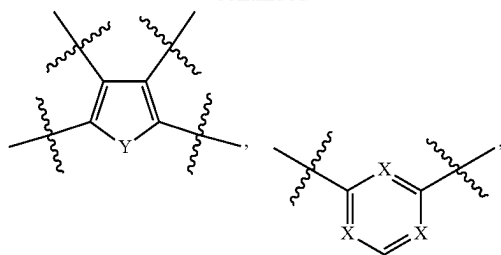




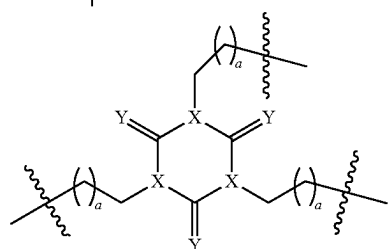
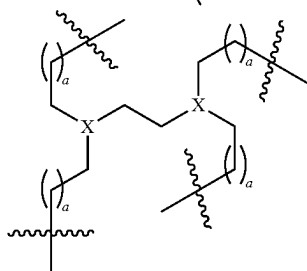
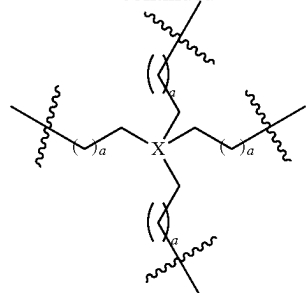
wherein each L is independently selected from alkyl, alkenyl, alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, poly(ethylene glycol) (PEG) and



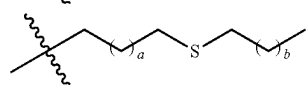
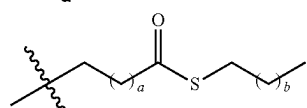
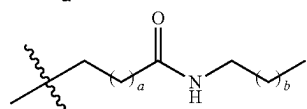
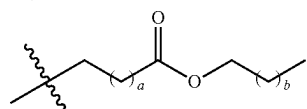
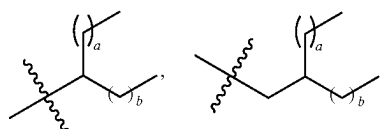
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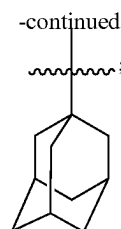
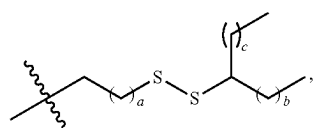
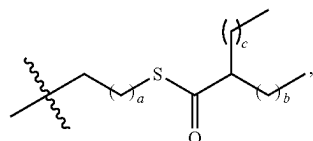
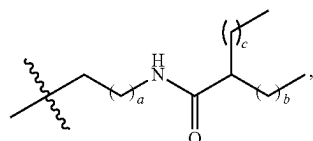
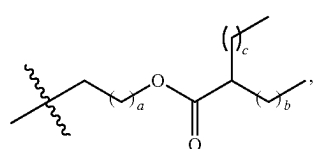
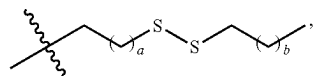
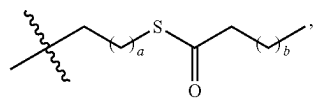
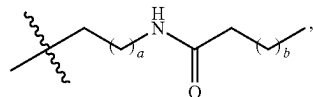
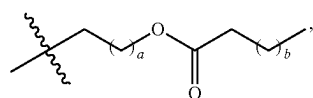
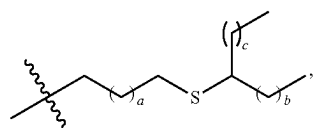
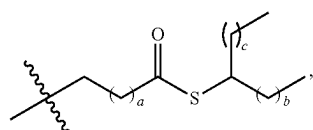
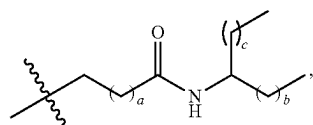
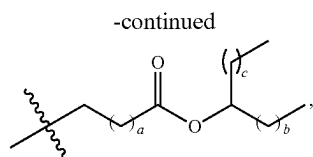


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wherein R⁶, R⁷, R⁸ and R⁹ are each independently selected from H, C₁-C₂₄ alkyl, C₁-C₂₄ alkenyl, C₁-C₂₄ alkynyl, substituted alkyl, substituted alkenyl, substituted alkynyl, substituted acyl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl, substituted carbocyclyl, substituted heterocyclyl, substituted aryl, substituted heteroaryl,





a, b, c and d are each independently an integer from 0-24;
each E is independently selected from CH₂, NH, O, or S;
each X is independently selected from CH or N;
each Y is independently selected from CH₂, NH, O, or S;
and

each Z is independently selected from CH or N.

13. The composition of claim 7, wherein the LNP comprises a modular lipid compound, wherein the modular lipid is M1, M2, M3, M4, M5, M6, M7, M8, M9, M10, M11, M12, M13, M14, M15, M16, M17, M18, M19, M20, M21, M22, M23, M24, M25, M26, M27, M28, M29, p1_E2, P1_F11, P1_C6, P1_E8, or a salt or isomer thereof.

14. The composition of claim 1, wherein the first type nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[5' UTR]-[nsP 1-4]-[SGP]-[GOI]-[3' UTR]-[Poly A]

wherein,

5' UTR is a 5' untranslated region,

nsP is a plurality of non-structural replicase domain sequences,

SGP is a subgenomic promoter,

GOI is one gene of interest or a plurality of genes of interest,

Y UTR is a 3' untranslated region, and

Poly A is a poly A tail.

15. The composition of claim 1, wherein the first type nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[5' UTR]-[nsP]-[ORF]-[3' UTR]-[PolyA]

wherein

5' UTR is a 5' untranslated region,

nsP is at least one non-structural replicase domain sequence,

ORF is two or more open reading frames,

Y UTR is a 3' untranslated region, and

Poly-A is a 3' poly-adenylated tail (poly-A tail),

wherein the ORF comprises at least two SGPs, each operably linked to at least one GOI.

16. The composition of claim 1, wherein the first type nucleic acid construct comprises an operably linked nucleic acid sequence comprising from 5' to 3':

[5' UTR]-[nsP]-[SGP]-[L]-[GOI]-[L]-[3' UTR]-[PolyA]

wherein

5' UTR is a 5' untranslated region,

nsP is a plurality of non-structural replicase domain sequences,

L is a linker independently selected from a sequence comprising any one of SEQ ID NOs: 27-37,

SGP is a subgenomic promoter,

GOI is one or more genes of interest,

3'UTR is a 3' untranslated region, and

Poly-A is a poly-A tail.

17. The composition of claim 1, wherein the first type nucleic acid construct is a sa-mRNA, wherein the sa-mRNA

is SEQ ID NO: 41, 42, 73, 74, 75, or 76 comprising at least one substitution selected from:

- a) a substitution of C with A at position 2623;
- b) a substitution of T with C at position 3309;
- c) a substitution of A with C at position 3493;
- d) a substitution of A with G at position 3867;
- e) a substitution of A with G at position 4674;
- f) a substitution of G with A at position 5795; or
- g) a substitution of A with G at position 5897,

wherein position numbers are counted from the first type nucleotide of the sa-mRNA.

18. A pharmaceutical composition comprising the composition of claim **1** and a pharmaceutically acceptable carrier.

19. A method of treating a subject having a tumor or cancer, comprising administering to a subject the pharmaceutical composition of claim **18**.

20. A method of inducing apoptosis of a cancerous cell or tumor, comprising contacting the cancerous cell with the pharmaceutical composition of claim **18** such that the first nucleic acid construct and at least one virus structural protein replicate within said cancerous cell, to express the GOI, wherein the first nucleic acid replication, expression of the GOI and the at least one virus structural protein replication within the cancerous cell results in cell death.

21. A method of delivering a GOI encoding a therapeutic gene product to cancerous cells in vivo comprising contacting the cancerous cells with the pharmaceutical composition of claim **18**.

22. A method of effecting in vivo synthesis of an oncolytic defective viral particle comprising contacting a cancerous cell with the pharmaceutical composition of claim **18** such that the first nucleic acid construct and the at least one virus structural protein replicate within said cancerous cell and express the GOI.

23. A method of inducing production of 1) a therapeutic payload 2) a pseudoviral particle and 3) sa-mRNA in a cell comprising contacting a cancerous cell with the pharmaceutical composition of claim **18** such that the first type nucleic acid construct and the second type nucleic construct produce the at least one virus structural protein, the sa-mRNA, and the GOI within said cancerous cell, and

wherein the at least one virus structural protein form the pseudoviral particle, which encapsulates the sa-mRNA.

24. A method of increasing expression of a polypeptide encoded by a GOI in a cell comprising contacting the cell with the pharmaceutical composition of claim **18** such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said cell, and the first nucleic acid construct expresses the GOI within said cell,

wherein the at least one virus structural protein forms a pseudoviral particle that encapsulates one or more of the sa-mRNA, and

wherein expression of the polypeptide is increased by at least 100-fold compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

25. A method of increasing expression of a polypeptide encoded by a GOI in a cell comprising contacting the cell with the pharmaceutical composition of claim **18** such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and

the sa-mRNA within said cell, and the first nucleic acid construct expresses the GOI within said cell,

wherein the at least one virus structural protein forms a pseudoviral particle, which encapsulates one or more of the sa-mRNA, and

wherein expression of the polypeptide is increased by 50-fold, 100-fold, 200-fold, 300-fold, 400-fold, 500-fold, or more compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

26. A method of increasing CD8 T cell in a tumor draining lymph node or spleen comprising contacting a tumor with the pharmaceutical composition of claim **18** such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within said tumor to produce the at least one virus structural protein and the sa-mRNA, and the first nucleic acid construct expresses the GOI within said tumor,

wherein the at least one virus structural protein forms a pseudoviral particle, which encapsulates one or more of the sa-mRNA, and

wherein CD8 T cell count is increased by 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, or more compared to contacting the cell with the pharmaceutical composition in the absence of the second nucleic acid construct.

27. A method of treating a subject with a payload delivery system encapsulating a mRNA encoding at least one gene of interest (GOI) or a plurality of GOIs and minimizing immunogenicity of the payload delivery system comprising administering the pharmaceutical composition of claim **18** to the subject such that the first nucleic acid construct and the second nucleic construct produce the at least one virus structural protein and the sa-mRNA within a cell of the subject, and the first nucleic acid construct expresses the GOI within said cell,

wherein the at least one virus structural protein forms a pseudoviral particle, which encapsulates one or more of the sa-mRNA, and wherein said non-viral payload delivery system has reduced immunogenicity compared to viral payload delivery systems.

28. A method of producing a mRNA comprising:

a) contacting cells with a nucleic acid template encoding two expression units, the nucleic acid template comprising:

- i) an origin of replication sequence (Ori);
- ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter; and
- iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter,

wherein the first expression unit encodes a selectable marker and the second expression unit encodes a mRNA;

b) selecting cells that express the selectable marker;

c) subculturing the selected cells to obtain a population of cells that express the selectable marker;

d) propagating the population of cells to increase the copy number of nucleic acid template replicates comprising the second expression unit encoding the mRNA; and

e) cleaving the nucleic acid template replicates between the first expression unit and the second expression unit to produce the mRNA.

29. A nucleic acid molecule comprising any one or combination of SEQ ID NOs in Table 16.

30. A method of treating a subject having a tumor or cancer, comprising administering to a subject a composition comprising an IL-12 and an IL-18 or IL-12 and an mutant IL-18.

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